Maui Nui Landscape Conservation Plan

Maui, Molokaʻi, Lānaʻi, and Kahoʻolawe

Hawaiʻi Division of Forestry and Wildlife (DOFAW)

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This document is subject to periodic review and revision

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Executive Summary

Maui Nui Landscape Conservation Plan

HAWAIʻI DIVISION OF FORESTRY AND WILDLIFE

Maui Nui is home to an extraordinary concentration of species found nowhere else on Earth, and to one of the most severe extinction crises in the nation. The **Maui Nui Landscape Conservation Plan (MNLCP)** is the first islands-wide, landscape-scale roadmap showing where conservation actions can most effectively prevent extinctions, recover at-risk species, and restore native ecosystems across Maui, Molokaʻi, Lānaʻi, and Kahoʻolawe.

442	163	240	100+
native species at risk of extinction across Maui Nui	face imminent extinction within five years without emergency action	at-risk species not yet on state or federal endangered species lists	Land managers, scientists, and species experts contributed their expertise

The Challenge

With less than 0.2 percent of the nation's land area, Hawaiʻi accounts for 29 percent of its imperiled species (USFWS 2026). More than 60 percent of Hawaiʻi's native ecosystems have been destroyed or highly degraded, and many native species now survive only in small, fragmented populations in high-elevation refugia or fenced areas. Invasive predators, ungulates, weeds, disease, fire, and climate change continue to drive declines.

Conservation capacity in Hawaiʻi has grown substantially, and many high-quality plans already exist—recovery plans, the State Wildlife Action Plan, and site management plans. But these plans are not integrated at the landscape or management level. Managers must reconcile dozens of documents with different assumptions and scales while working under limited budgets and staffing. The central question is not simply that 442 species are at risk, but *where* each species should be recovered, *what* actions will do the most good, and *when* they must happen, so that the collective needs of all species are met as efficiently as possible. Answering that for 442 species without an integrated framework is simply not feasible.

What the MNLCP Does

Led by DOFAW Maui with USGS, USFWS, the University of Hawai‘i at Mānoa and Hilo, the Bishop Museum, and other land managers and scientists, the MNLCP is built around three integrated goals:

1. **Prevent extinctions** of the most imperiled native plants and animals.
2. **Recover all at-risk species** to healthy, self-sustaining population levels.
3. **Protect and restore native ecosystems** so they provide habitat for all Maui Nui biodiversity and sustain clean water, cultural values, and local economies.

It uses an Integrated Multi-Scale Conservation Planning approach that links three analyses into one decision-support system: spatial prioritization (“where to manage,” led by USGS), action prioritization through a modified Priority Threat Management process (“what to do,” led by UH Mānoa), and ecosystem prioritization (led by UH Hilo). Seven expert Taxonomic Teams shaped every stage, and DOFAW land managers translated the results into implementation guidance (“when to do it”). The plan is designed for a 20-year implementation period and is delivered as a living system, including the Adaptive Conservation Planner, a spatial tool managers can use to build annual work plans and update priorities as conditions, knowledge, and funding change.

Who Needs Help: 442 At-Risk Species

Experts identified 442 native species at risk of extinction within the next 20 years without significant intervention. Only 202 are currently protected under state or federal endangered species laws; the remaining 240 were identified by species experts because of very small or declining populations, restricted ranges, or climate vulnerability. Of the 442, 163 could be lost within five years without immediate emergency action.

Taxonomic group	At risk	Listed	Imminent extinction	Primary threats	Highest-priority actions
Forest birds	4	3	2	Mosquito-borne avian malaria; habitat loss	Landscape-scale mosquito suppression; translocation; habitat protection
Seabirds	3	3	0	Rats, mongoose, and cats; habitat loss; lights, power lines, and turbines	Predator-exclusion fencing; predator control; habitat management
Waterbirds	5	5	0	Wetland loss; introduced predators; hybridization with mallards (kōloa maoli)	Wetland habitat management; predator control; solve hybridization, then reintroduce
Plants	197	175	102	Habitat loss; ungulates; invasive weeds, rats, and invertebrates; fire	Emergency seed collection and propagation; ungulate removal; weed and rat control; fire management

Taxonomic group	At risk	Listed	Imminent extinction	Primary threats	Highest-priority actions
Terrestrial snails	49	3	49	Rosy wolfsnail (*Euglandina*), rats, Jackson's chameleon	Predator-proof exclosures (kīpuka kāhuli); ex situ facility on Maui; new predator-control tools
Terrestrial arthropods	172	8	9	Habitat loss; invasive ants and wasps; host-plant decline	Emergency threat abatement; ungulate removal; surveys to fill data gaps
Aquatic invertebrates	11	4	1	Habitat loss; invasive fish, frogs, and shrimp	Protect the only known *Megalagrion nesiotes* site; instream flows; invasive species control
Hawaiian hoary bat (ʻōpeʻapeʻa)	1	1	0	Wind turbine collisions; habitat loss	Minimize wind-energy mortality; protect forests and wetlands; develop population monitoring
Total	**442**	**202**	**163**		

Table 1. At-risk species by taxonomic group, number listed under state and federal endangered species laws, number facing imminent extinction, and the primary threats and highest-priority actions identified in the plan.

Where to Act: Recovery Footprints

For each taxonomic group, the plan maps the planning units (about 2 km^2 each) that contain the existing or restorable habitat needed to reach recovery goals. Combined across all groups:

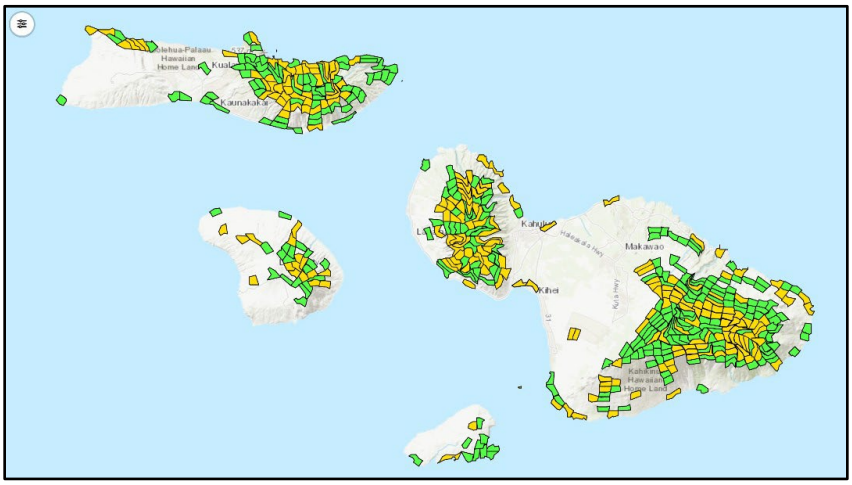


Figure 1. Combined recovery footprint for all taxonomic groups (green) and priority areas (yellow).

- **The full recovery footprint** covers 492 planning units and provides habitat for 299 at-risk species.
- **The priority footprint** narrows this to 216 planning units, about 43 percent of the full footprint, while still supporting habitat for 267 at-risk species. These are the most important places to start.
- **Most of the footprint is already conservation-oriented land** (about 66 percent), and about half already has some ungulate fencing. The footprints span all four islands,

centered on upland Haleakalā, West Maui, East Molokaʻi, and Lānaʻihale, and generally have above-average climate resilience.

- **143 species still lack the data** needed to map recovery habitat, mostly terrestrial arthropods. The plan sets out targeted surveys so these species can be added over time.

What to Do: Priority Conservation Actions

Working with about 100 managers and scientists, the Action Prioritization Team evaluated the expected benefit, feasibility, and cost of 27 conservation strategies (17 single-focus strategies and 10 combinations) for 17 species groups. The target is at least a 50 percent probability that each group persists with viable, self-sustaining populations over 20 years. Key findings:

- **No single strategy is enough.** Implementing two complementary combination strategies together—S25 (plant- and invertebrate-focused) and S27 (bird- and bat-focused)—comes closest to meeting every group's needs and puts all 17 individual actions to work.
- **Doing nothing, or only what is done now, is not enough.** Without action, persistence is very low for most groups. Current efforts help, especially for seabirds and waterbirds, but feasible new actions significantly improve outcomes for every group.
- **Foundational habitat work pays off across groups.** Ungulate fencing and removal, weed control, predator control, and wetland management benefit many species at once, and some actions (such as fencing paired with weed control) deliver far more together than alone.
- **New tools are the single most important investment.** Future tools for rare plant and invertebrate conservation (S13) ranked in the top three actions for 13 of the 17 species groups. Forest birds depend on new tools to control mosquito-borne disease, and snails depend on landscape-scale control of the rosy wolfsnail.
- **Some groups are within reach with proven methods.** Seabirds, waterbirds, and the Hawaiian hoary bat can reach the 50 percent goal without relying on unproven technologies. Plants remain the hardest group to secure, even with full implementation.

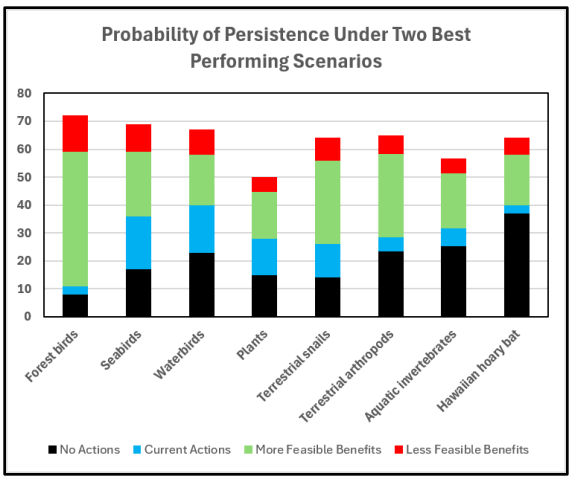


Figure 2. Expected probability of persistence by taxonomic group under the two best-performing strategies (S25 and S27): no action (black), current actions (blue), additional benefits from feasible actions (green), and less-feasible benefits (red).

When to Act: Emergency First, Then Sequenced Recovery

The 163 species facing imminent extinction require immediate emergency measures: ex situ seed banking and captive propagation, very small predator-proof exclosures, and intensive protection of the last wild populations. Examples include emergency propagation for 102 plants, a network of kīpuka kāhuli exclosures and an ex situ laboratory for all 49 at-risk snails, and protection of the only known Maui Nui population of the damselfly *Megalagrion nesiotes*. Beyond emergencies, Taxonomic Teams recommended an implementation sequence for each group: begin with the highest-benefit actions and foundational steps (fencing, ungulate removal, weed control) that set the stage for later reintroductions, while pursuing cost-effective, field-ready actions as opportunities arise.

Protecting and Restoring Native Ecosystems

Healthy native ecosystems keep additional species from becoming at risk and give recovering species room to expand. The plan maps ten ecosystems and several special features (anchialine pools, lava tubes, bogs, and cliffs), and combines habitat quality with land management status to identify opportunities. Some ecosystems are nearly gone: lowland dry forest, once the most extensive type, retains only about two percent of its original area. Montane wet forest and subalpine areas remain largely intact and are core habitat for the remaining forest birds. Ecosystem priorities are:

1. **Maintain** managed, fenced high-quality habitat.
2. **Initiate** ungulate management on unfenced high-quality conservation lands, such as windward Haleakalā, two areas of Molokaʻi, and Lānaʻihale.

3. **Engage** landowners and partners where high-quality habitat lies outside conservation status, including eastern Molokaʻi, above Makawao, and Kahikinui.
4. **Restore** degraded habitats that matter most for species recovery, connectivity, diversity, and climate resilience.

What Success Requires

The MNLCP reframes Maui Nui's biodiversity crisis not as an insurmountable problem but as a finite, solvable challenge. Realizing it will require leadership and partners to:

- **Fund emergency actions now** for the 163 species that could be lost within five years.
- **Invest in new tools** for landscape-scale control of mosquitoes, predatory snails, rats, and other key threats, and for captive rearing and propagation of at-risk plants and invertebrates.
- **Sustain and expand foundational management** in the recovery footprint: fencing, ungulate removal, weed and predator control, wetland and instream-flow management, and fire prevention.
- **Build partnerships** with private landowners, communities, and agencies to promote recovery and habitat management.
- **Protect, manage, and restore** native ecosystems to provide for ecosystem functions and biodiversity.
- **Fill data gaps** through targeted surveys for the 143 species that cannot yet be mapped.
- **Use the plan as a living system**, with the Adaptive Conservation Planner guiding annual work plans and budgets, and monitoring and adaptive management tracking progress over the 20-year implementation period.

Together, these steps give DOFAW and its partners a clear, science-based pathway to ensure that Maui Nui's native species and ecosystems not only persist but thrive once again.

1. Introduction

The Maui Nui Landscape Conservation Plan (MNLCP) has three complementary objectives. The first is to prevent the extinction of the most at-risk native plants and animals. The second is to restore all at-risk species to self-sustaining, healthy population levels. The third is to restore Maui Nui's native ecosystems to provide habitat for at-risk species as well as for other native species.

This plan focuses on the conservation of native terrestrial and freshwater biodiversity across the landscapes of Maui Nui, encompassing the islands of Maui, Molokaʻi, Lānaʻi, and Kahoʻolawe. These islands support an extraordinary diversity of native species and ecosystems, the vast majority of which occur nowhere else on Earth. Maui Nui spans high volcanic peaks; mesic, wet, and dry forests; subalpine shrublands; wetlands; streams; and coastal dunes—habitats that collectively sustain hundreds of rare, threatened, and endangered species.

The analytic scope of the MNLCP extends across all lands but focuses needed conservation actions, when possible, on those lands already managed or set aside for conservation or covered by conservation agreements with private landowners.

a. Purpose and Need

i. General

Global biodiversity is being lost at rates unprecedented in human history, with profound consequences for ecosystem services, economies, food security, and human well-being. Island ecosystems are among the most vulnerable to these losses, and the Hawaiian Islands exemplify the severity of this crisis. The archipelago's extraordinary evolutionary heritage—shaped over tens of millions of years of geographic isolation—produced one of the highest concentrations of endemic species on Earth. Yet today, hundreds of native Hawaiian species have already gone extinct and hundreds more are at risk of extinction. Maui Nui alone harbors 442 native plant and animal species that experts consider to be at risk of becoming extinct in the near future. The loss of these species represents not only the erosion of biological diversity, but the

degradation of cultural heritage, ecosystem function, and economic well-being.

The Hawaiian Islands are among the most isolated inhabited islands on Earth, lying roughly 2,100 miles from the nearest continent. They are environmentally diverse—ranging from dry tropical coasts to wet rainforests to alpine summits—and have an exceptionally stable climate with relatively small annual temperature fluctuations. Prior to human arrival, Hawai‘i’s ecosystems evolved in near-total isolation, giving rise to remarkable levels of plant and animal endemism: approximately 90 percent of terrestrial species are found nowhere else (Allison 2003). Polynesian settlement around a thousand years ago began the first major human-caused transformation of these systems, introducing agriculture, non-native species, and fire that initiated the decline of native lowland ecosystems and the loss of countless species, including almost all of Hawai‘i’s flightless birds.

The pace and scale of ecological change intensified dramatically following Western contact in the late eighteenth century, as large-scale ranching, plantation agriculture, commercial forestry, and urban development combined to destroy native forests, drain wetlands, divert streams, and introduce aggressive invasive species and predators. Fire regimes intensified, erosion increased, and remaining native habitats became increasingly fragmented.

These threats cause a diversity of impacts ranging from large-scale impacts affecting entire ecosystems (like introduced ungulates degrading wet forests or global warming causing sea level rise that inundates coastal habitats) to impacts on taxonomic groups (like introduced predatory snails on native snails or introduced avian malaria on forest birds) to impacts on individual populations (like localized mortality of bats by wind turbines or housing developments on rare plant habitat).

The cumulative result is that Hawai‘i has one of the highest extinction rates in the world. With less than 0.2 percent of the land area of the United States, the Hawaiian Islands now account for 29 percent of the nation’s imperiled species (USFWS 2026). Many native species persist only in small, highly fragmented populations, often restricted to high-elevation refugia or to intensively managed, fenced areas. More than 60 percent of Hawai‘i’s native ecosystems have been destroyed or are highly degraded (Jacobi et al. 2017). The persistence of remaining biodiversity depends largely on sustained,

active management—underscoring both the severity of past land-use change and the urgency of coordinated landscape-scale conservation action.

ii. Specific

The Division of Forestry and Wildlife (DOFAW), under the authority of Chapter 195D, Hawai‘i Revised Statutes, is charged with conserving the State’s native species and ecosystems. This mission is carried out in close partnership with federal agencies, non-governmental organizations, private landowners, and communities across the islands. Over recent decades, conservation capacity in Hawai‘i has grown substantially, with increased funding, expanded protected lands, and more intensive management efforts. Despite this progress, many species continue to decline, including hundreds with fewer than 50 individuals remaining in the wild. Current conservation approaches are not keeping pace with the scale, complexity, and urgency of Hawai‘i’s biodiversity crisis.

Many high-quality management plans exist to guide conservation action in Hawai‘i, including species recovery plans, the State Wildlife Action Plan, site-specific management plans, and broader strategic frameworks. While these plans are excellent at providing essential scientific guidance, they are not integrated at the landscape or management level. Managers must reconcile dozens of documents, each with different assumptions, scales, and objectives, while operating under limited budgets and staffing. This fragmentation obscures opportunities for integrated, multi-species strategies, limits the ability to prioritize actions, and slows progress toward preventing extinction and restoring ecosystem integrity.

Consider, for example, planning for even a relatively small conservation reserve such as the Makawao Forest Reserve (MFR). Although MFR contains suitable habitat for many at-risk species, no planning framework exists to help managers determine which of those species should be recovered within the reserve, how many individuals are needed to contribute to long-term recovery, which areas within the reserve provide the greatest conservation value for each species, or which complementary management actions will improve cost-effective outcomes for different suites of species.

The challenge is not simply that Maui Nui supports 442 species at risk. Rather, it is determining where across the landscape each species should be recovered so that the collective recovery needs of all species are met. Every planning unit differs in habitat quality, climate resilience, species composition,

and its contribution to overall recovery objectives. Some planning units may contribute to the recovery of only a few species, while others may simultaneously advance recovery for dozens. The objective is therefore not to simply identify the best sites for individual species, but the complementary network of planning units that together achieves recovery objectives for all species.

Expecting managers to derive this network by independently reviewing dozens of recovery plans and other planning documents for each management area, while determining where each of the 442 Maui Nui species should be recovered across the landscape, is simply not feasible without an integrated analytical framework.

The MNLCP was developed to address these challenges directly. Led by DOFAW Maui in collaboration with a broad coalition of public and private agencies, organizations, landowners, and technical experts, the MNLCP combines spatial prioritization identifying where to focus land management, action prioritization to focus management on the most important conservation actions to undertake, and ecosystem management to benefit all Maui Nui biodiversity. The collaboration of state, federal, and non-profit partners using this new approach represents a fundamental shift in how conservation is planned and implemented in Maui Nui. It provides, for the first time, an islands-wide, comprehensive, landscape-scale roadmap showing where conservation actions can most effectively prevent extinctions, recover species, and restore ecosystems. By integrating species recovery needs, management actions, and ecosystem priorities into a single decision-support system, the plan enables conservation practitioners to act more strategically and less reactively. Rather than addressing species one by one or site by site, the plan integrates population-level recovery objectives for all species at risk into a spatially explicit, climate-resilient plan that identifies where conservation actions can achieve the greatest collective benefit.

Of the 442 native Maui Nui species identified for special management in this plan, 202 are federally or state-listed, while the remaining 240 have extremely small population sizes, very restricted distributions, anticipated negative impacts from human-caused climate change, or declining population trends that place them at elevated risk of extinction. Preventing extinctions and achieving stable or increasing population levels requires coordinated actions—including the exclusion of non-native ungulates (such as goats, deer,

and cattle), invasive weed control, predator and pest management, habitat restoration, fire prevention, disease mitigation, and, in some cases, ex situ propagation and reintroduction into the wild. These actions must be implemented at appropriate spatial scales and sustained over time. The MNLCP provides a unified framework to make this coordination possible.

b. Applications and Use

The MNLCP is both a decision-support framework and a practical roadmap for implementation. It provides DOFAW and partners with the information and tools needed to guide cost-effective, coordinated conservation across multiple species and landscapes. The plan enables land managers to:

- Prioritize actions based on effectiveness and efficiency,
- Estimate and plan for costs across species and sites,
- Identify lands that should be managed under long-term protection,
- Design integrated work plans that deliver multiple conservation benefits at individual sites, and
- Coordinate actions across jurisdictions using shared data.

Beyond day-to-day management, the plan serves as a strategic roadmap for the broader conservation community, articulating the full set of actions required to prevent extinctions and recover all at-risk species in Maui Nui. By organizing complex information into a coherent, spatially explicit plan, the MNLCP reframes the biodiversity crisis not as an insurmountable problem, but as a finite and solvable challenge—one that can be addressed through science-based prioritization, collaboration, and sustained commitment. The plan also supports engagement with landowners and communities, informs policy and planning processes, justifies financial investment, identifies critical research needs, and provides accountability mechanisms through performance metrics and adaptive management.

c. A Path Forward

The overarching goals of the MNLCP are to prevent further extinctions, recover species currently at risk, and stabilize ecosystems so they can continue to provide habitat for all native plants and animals as well as essential public benefits—clean water, food security, cultural value, and natural beauty—for future generations. Hawai‘i’s native species and habitats are intricately linked to the state’s ecological health, biocultural heritage, and economic prosperity. These ecosystems form the foundation of cultural practices, natural resource

management, and local economies, and their loss diminishes all who depend on them.

The MNLCP is not intended to only be a one-time, static reference document. Instead, it provides a flexible, manager-driven, and expert-informed decision-support system to help land and species managers coordinate conservation actions, identify priority areas for management, recommend implementable actions, establish performance metrics, and provide a foundation for adaptive management, budgeting, staffing, and partnership development. The chapters that follow describe the plan's planning framework, spatial prioritization, action strategies, ecosystem-based approaches, and implementation tools. Together, they present a clear pathway forward—one grounded in science, informed by place, and driven by a shared commitment to ensure that Hawai‘i's native species and ecosystems not only persist, but thrive once again.

2. Guiding Principles and General Methods

a. Three Integrated Goals

The MNLCP is organized around three interrelated conservation goals that, taken together, will protect the full breadth of Maui Nui's biodiversity:

- Prevent extinctions.
- Restore all currently at-risk species to healthy, self-sustaining population levels.
- Protect all remaining native ecosystems and restore degraded systems to provide adequate habitats for all native Maui Nui biodiversity.

b. Planning Frameworks and Approaches

The Maui Nui Landscape Conservation Plan represents an initial application of an Integrated Multi-Scale Conservation Planning (IMCP) approach that combines species recovery planning, spatial prioritization, threat-based action prioritization, ecosystem restoration planning, and adaptive implementation across spatial, ecological, and management scales into a unified decision-support system. This approach provides the overall planning strategy for all aspects of the MNLCP, with spatial prioritization and action prioritization efforts aligning their own approaches under the IMCP approach (Figure 1).

Teams of experts on Hawaiian plants, animals, ecosystems, and land management assisted in guiding and developing the plan. These expert teams were, in turn, supported by analytic teams composed of research scientists from the U.S. Geological Survey (USGS), U.S. Fish and Wildlife Service (USFWS), University of Hawai'i at Mānoa (UH- Mānoa), University of Hawai'i at Hilo (UH-Hilo), and the Bishop Museum. DOFAW and these teams worked together to produce this plan to recover all 442 Maui Nui species currently at risk of extinction, to protect and restore key native ecosystems, and to do so in the most effective and efficient manner possible.

The MNLCP is built around a systematic IMCP approach designed to link species-level recovery needs to landscape-scale priorities and on-the-ground implementation. At its foundation are species recovery targets: when possible, each at-risk species in Maui Nui is assigned a population-level recovery target based on abundance, distribution (across islands and mountains), and viability criteria, derived from existing recovery plans and supplemented by expert input.

Species distribution modeling coupled with known species occurrences, ecosystem integrity maps, published literature, and expert knowledge provides the spatial context for recovery planning. Some taxonomic groups used expert knowledge to directly identify and map the habitat areas needed for species recovery (called recovery footprints). Other groups used spatial prioritization algorithms and expert reviews to integrate habitat requirements with multiple other management-relevant factors such as accessibility, ongoing management actions, climate resilience, or fire risk to identify a network of areas (recovery footprints) that can efficiently meet habitat requirements for all at-risk species, as well as important management needs.

Ecosystem conservation planning complements species-level recovery by identifying ecosystems that are most critical to long-term biodiversity maintenance. Decision-support tools—including spatial data layers, priority rankings, and threat–cost matrices—translate analysis into action, supporting annual work planning, budget development, and interagency coordination. Monitoring and adaptive management provide the feedback mechanism that allows the framework to evolve over time, with metrics for species recovery, threat reduction, and ecosystem condition embedded throughout the plan.

c. Procedural Methods

The expansive scope of this project does not lend itself to a species-by-species or one-size-fits-all approach. Instead, a variety of methodological approaches were needed to properly identify and address Maui Nui's conservation needs. In addition, since a project of this scale and complexity has not been attempted before, several new or highly modified approaches have been employed. These key approaches and analytic methods are discussed below.

d. Integrated Multi-Scale Conservation Planning

The MNLCP is not a spatial prioritization exercise or a standalone framework. It is a state-led, expert-engaged, multi-component plan that deliberately integrates several conservation planning approaches into a single, flexible decision-support system. The defining characteristics of the IMCP approach are:

- State-led and practitioner-driven, rather than researcher-directed or top-down prescriptive mandates.
- Expert-engaged, with specialists contributing qualitative judgment through open-ended, collaborative review.
- Integrated conservation across multiple scales:

    - Biologically, from populations to species, entire taxonomic groups and ecosystems,
    - Spatially, from islands to mountains (mauna) to management sites, and
    - Temporally, from emergency actions for species facing imminent extinction to mid- and long-term actions aimed at recovering species or restoring ecosystems.
- Intentional integration of three subordinate planning components:
    - *Spatial prioritization* using a comprehensive planning process that identifies where on the landscape to focus conservation efforts to achieve explicit recovery targets.
    - *Action prioritization* to identify the most effective conservation actions and the most efficient way to implement them for synergistic effects.
    - *Ecosystem prioritization* to identify opportunities for large-scale conservation that benefits all of Maui Nui's plants and animals.
- Decision-support orientation, providing flexible guidance to managers.
- Living products such as decision-support applications and implementation handbooks that can be updated as conditions, knowledge, and available resources change.
- Transparent and adaptable, with both embedded in plan development and decision processes, not solely in analytical tools.
- Partnership-based, acting through collaboration with respect for local communities and cultures.
- Solution-oriented, promoting a culture of finding solutions to overcome challenges.

e. How the Planning Processes Fit Together

The MNLCP uses Integrated Multi-Scale Conservation Planning (IMCP) as the overall planning method. IMCP sits above, and intentionally encompasses, the individual analytical components used in the plan:

- IMCP is the overarching planning approach.
- Spatial prioritization is implemented through the Transparent, Flexible, Expert-engaged Systematic Conservation Planning (TFE-SCP; Fortini et al. 2025) process.
- Quantitative prioritization algorithms are implemented via OptimTFE, the site-selection prioritization tool used within the TFE-SCP process.

- Conservation actions are evaluated through a modified Priority Threat Management (PTM) process.
- Ecosystem prioritization provides complementary landscape-scale conservation.

This structure ensures that spatial tools are understood as components of a broader planning system rather than as the framework itself (Figure 1). It also ensures that the on-the-ground knowledge of dozens of subject matter experts is integrated throughout the planning process.

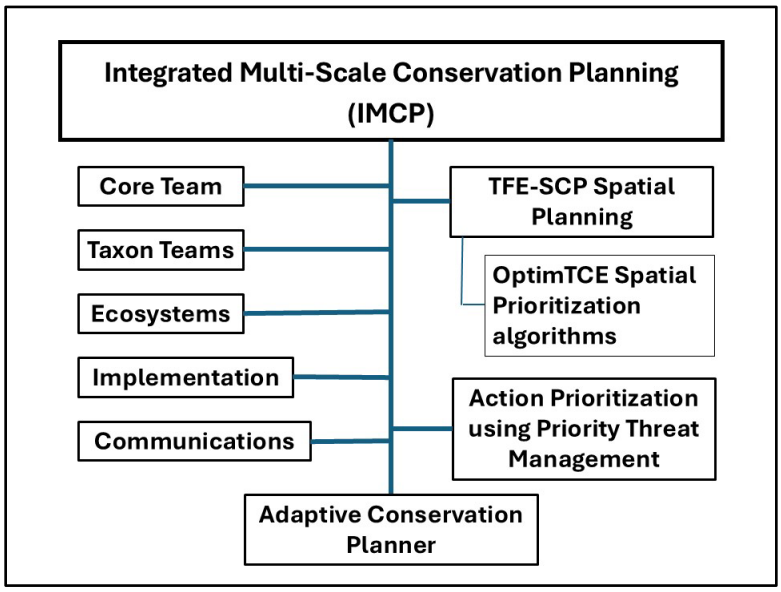


Figure 1. Relationships of MNLCP teams with included planning processes and algorithms.

3. Planning Teams and Their Strategic Initiatives

a. Importance

The MNLCP is an ambitious effort using a multi-scale conservation planning approach. This approach combines traditional teams overseeing plan development and communications, subject matter specialists and implementation-oriented land managers, and more specialized analytic teams. These latter teams used custom-designed prioritization processes to identify where to focus land management, what the most important conservation actions are for species and ecosystems, and the restoration and protection opportunities for key ecosystems. Each team played an important and strategic role in MNLCP development and implementation.

b. Taxonomic Teams (DOFAW-led)

Teams of species specialists were the irreplaceable backbone of the planning effort. They provided input on which at-risk species to include in the plan and their distributions, habitat requirements, conservation needs, and management feasibility of actions. DOFAW coordinated these teams and participated throughout the planning effort. UH-H, USGS, and USFWS participated in teams and provided additional analyses and information. Seven taxonomic teams were established to focus on seven key taxonomic groups; the Hawaiian hoary bat, the only native terrestrial mammal in the Hawaiian Islands, is the eighth taxonomic group but did not have its own taxonomic team.

i. **Forest birds**
ii. **Seabirds**
iii. **Waterbirds**
iv. **Plants**
v. **Snails**
vi. **Terrestrial arthropods**
vii. **Aquatic invertebrates**

These teams were especially important in six key tasks:

i. **Identifying the at-risk species** that needed to be included in the MNLCP,
ii. **Establishing targets** for restoring species to self-sustaining healthy populations using a variety of outputs from the spatial, action, and ecosystem prioritization efforts,
iii. **Identifying known locations** of populations,

iv. **Selecting areas** that provide appropriate habitat for each species,
v. **Determining** the most important conservation actions to implement for preventing extinctions and reaching recovery, and
vi. **Identifying species** that might be facing an imminent threat of extinction in the next five years.

c. Spatial Prioritization for Species (USGS-led)

This effort led by the Spatial Prioritization Team addresses "*where to manage"* for the recovery of all Maui Nui at-risk species. The MNLCP uses the USGS-developed Transparent, Flexible, Expert-engaged Systematic Conservation Planning (TFE-SCP) process (Fortini et al. 2025, Fortini et al. in prep a, and see Appendix 1 for detailed information). The TFE-SCP is a comprehensive planning process with distinct stages, each supported by custom-built tools and each involving input from biological experts and land managers. It emphasizes transparency in how decisions were made, flexibility to adapt to different data and management contexts, and deep expert engagement at every stage. Field biologists and taxon specialists were involved throughout, shaping inputs, criteria, and final products (not just end-stage validation).

The final products are contained within the Adaptive Conservation Planner (Conservation Planner) and consist of spatially explicit and easily updatable recovery footprints and metadata on species, habitats, and threats, which together provide snapshots of where to focus conservation actions for each species group.

i. **Process Pathways.** Two process pathways were used for spatial conservation planning, depending on the number of species in a taxonomic group and the amount of information available for determining habitat needs. Both build on existing species-level and related planning documents available for the species at risk.
    - Expert-led pathways (forest birds, seabirds, waterbirds, aquatic invertebrates) were used when there were fewer focal species and when expert knowledge could provide spatial priorities. Specialists identified priority areas directly through iterative web-map review.
    - Algorithmic pathways (plants, terrestrial arthropods, terrestrial snails) were used when spatial or habitat prioritization dealt with numerous species and potentially had both important differences in habitat quality and key management-oriented considerations like site accessibility or the presence

of ongoing land management. The Maui Nui landscape was overlain with a network of 1,400 medium-sized planning units (ca. 2 km^2) based on watersheds, natural barriers, and elevational bands. The OptimTFE spatial prioritization algorithm was used to generate hundreds of thousands of candidate footprints (each meeting species-specific recovery or preventing-extinction targets). Species experts and land managers could then compare spatial footprints across multiple conservation criteria—including footprint area, habitat quality, biodiversity reserve overlap, ungulate management, accessibility, fire risk, and climate resilience. Spatial footprints were then selected that would meet both biological needs and management desires for feasibility.

ii. **The Adaptive Conservation Planner.** A spatially oriented adaptive analytic tool that allows managers to identify cost efficient spatial strategies for the management and recovery of species at risk and to adaptively adjust those strategies based on outcomes and new information. The Conservation Planner serves as an essential tool for managers charged with conservation of biodiversity and aids in developing annual work plans and exploring different management scenarios.

d. Action Prioritization (UH Mānoa-led)

This effort addresses the "*what to do"* portion of the MNLCP. The Action Prioritization Team used a modified Priority Threat Management process (Carwardine et al. 2019, Camaclang et al. 2021) to identify which management actions provide the greatest benefit for species recovery (Appendix 2). DOFAW and UH-Mānoa convened collaborative workshops in which roughly 100 land managers and scientists contributed professional judgment through open-ended discussions on the expected cost, feasibility, and benefit of different management strategies aimed at recovering 374 of Maui Nui's at-risk plants and animals (see Appendix 2 for details).

The Action Prioritization Team, with the associated species and management experts, used expert judgment and information from published literature, species status assessments, recovery strategies, action plans, and management plans to:

i. **Identify 374 species at risk** of extinction within the next 20 years across the islands of Maui Nui (all of which were also identified by the Taxonomic Teams

as being at risk, though the Taxonomic Teams later identified an additional 68 species),

ii. **Identify key threats** to the survival of at-risk native plants and animals,

iii. **Identify conservation actions to** address those threats,

iv. **Categorize the at-risk species into 17 groups,** based on similarity of responses to threats and management actions,

v. **Quantify the expected benefits** of different conservation strategies—such as ungulate control and predator management, and

vi. **Provide cost estimates** of management options to identify the most cost-effective management solutions.

Taxonomic Teams and species experts further refined the management options for taxonomic groups by identifying the best sequence for implementing actions based on high to low priority and/or the presence of foundational restoration actions that set the stage for later actions (e.g., fencing, ungulate control, weed control, and then outplanting).

e. Ecosystem Prioritization (UH Hilo-led)

The conservation of Hawai‘i’s native ecosystems is an important factor in preventing additional plants and animals from declining to the point that they are at risk of extinction. Protected and restored ecosystems also help Maui Nui’s 442 already at-risk species recover and stay recovered by providing connectivity, buffers from threats, redundancy, and room for population expansion. Some native ecosystems and unique communities have declined to the point that less than two percent of their original area remains.

The MNLCP ecosystem effort focuses on identifying:

i. **Remaining good-quality native ecosystems** and unique community types,

ii. **Impactful conservation strategies** to reach ecosystem goals.

iii. **Conservation opportunities** that take into account:
- Habitat quality,
- The conservation status of lands (conservation-oriented or otherwise), and
- The importance of the area to at-risk species or other values.

f. Implementation (DOFAW-led)

DOFAW land managers, planners, and decision-makers have been engaged throughout the MNLCP planning process and are responsible for initiating

planning, developing management needs and objectives, as well as translating results from the four efforts above into a final cohesive plan that prioritizes overall plan implementation—the "*when to do it*" guidance. This team also oversees development of a Managers' Handbook as a practical guide for DOFAW staff using MNLCP tools and outputs. Importantly, the Implementation Team includes the land managers charged with implementing the MNLCP, along with partner organizations.

g. Core Team (DOFAW-led)

Organized by DOFAW, this team consisted of land managers, key leaders from other teams, and a facilitator. The team provided recommendations and organizational support to DOFAW.

4. Conservation of At-Risk Species

The MNLCP seeks to safeguard the biodiversity of Maui Nui through a suite of conservation actions that protect, manage, and restore native species and ecosystems. The first step in biodiversity conservation planning is identifying which elements of that biodiversity are at risk. Subsequent steps focus on preventing species extinction or decline and then restoring it to healthy, self-sustaining populations in natural ecosystems.

a. Identifying At-Risk Species

Taxonomic Teams and other species experts conducted species-level assessments utilizing expert knowledge, existing endangered species lists, Hawai‘i’s State Wildlife Action Plan, conservation databases, and literature sources to identify those species that are likely to become extinct within the next 20 years without significant management intervention. The primary causes that place species in this category are low numbers of remaining individuals, precipitous declines in numbers or range due to a changing climate, newly introduced invasive species, or habitat loss.

i. **A total of 442 species** of birds, plants, invertebrates, and a bat are identified as being at risk of extinction (Table 1 and Appendix 3).
ii. **State and federal lists** of endangered and threatened species already identified 202 species as being in danger of extinction or threatened with becoming endangered in the future.
iii. **The remaining 240 species** were identified as being at risk by experts in each taxonomic group.

b. Species Facing Imminent Extinction

Of Maui Nui’s 442 at-risk species, 163 are so close to extinction that they may be lost within the next five years without immediate emergency conservation actions.

The emphasis of emergency actions is on effectiveness and quickness of implementation. They frequently involve ex situ actions like captive rearing and seed storage or quasi-in situ actions like very small predator-proof exclosures. The species needing emergency measures are primarily species in the Full Recovery and Interim Recovery categories, but several are also in the Area Survey category (see below). Regardless of their recovery or survey category

status, these species will receive actions aimed at preventing their extinction. These extinction-prevention actions focus on ex situ conservation as well as in situ actions protecting and managing key populations to ensure the species' survival over the course of the MNLCP.

c. Recovery of At-Risk Species

A key goal of the MNLCP is to restore currently at-risk species to healthy, self-sustaining population levels. In many instances, when species reach this goal, they may remain reliant on at least some continuing conservation actions. These "conservation-reliant" species (Goble et al. 2012, Scott et al. 2020) are considered by the MNLCP to no longer be at risk of extinction once they have reached established recovery targets.

The MNLCP provides managers with a flexible roadmap on how to reach recovery for all 442 of the Maui Nui species considered to be at risk of extinction.

Conservation can be most effective if land managers, the people actually doing on-the-ground conservation, know what outcomes they need to reach, where the most important places to work are, and what priority actions they need to implement. To that end, the MNLCP:

i. **Identifies spatially explicit recovery habitat areas** (recovery footprints) whenever possible.
ii. **Identifies potential conservation actions** aimed at addressing key threats and assesses those actions as to how likely they are to actually benefit restoration of each taxonomic group.

Maui Nui's at-risk species vary greatly in the levels of threat facing them and our knowledge of how to protect them from extinction or to recover them to a point that they no longer need species-specific conservation actions. Consequently, our ability to identify conservation needs, appropriate actions, and management capabilities differs among species. Species were categorized into three groups, based on knowledge of their biology, conservation needs, and availability of management techniques.

i. **Interim Recovery Goals:** Developed for species with adequate data on habitat needs, threats, and actions, but with limited known locations for habitat management, long timelines for habitat restoration, or a priority focus on *ex situ* actions due to a lack of methods to control key threats preventing reintroduction efforts (e.g., predators of terrestrial snails). These interim goals

move species conservation forward to a significant degree by creating viable populations, but not enough to secure the species in the long term. Additional populations will be needed to safely remove the species from the at-risk category. These species still have spatially explicit recommendations (recovery footprints) for where to locate recovery populations.

ii. **Full Recovery Goals:** Developed for species with good data on habitat needs, habitat locations, threats, and effective conservation actions. Species have population goals identified that, if reached, are predicted to result in stable or increasing populations with patterns of distribution and abundance such that the species is no longer at risk of extinction and may therefore be removed from the Maui Nui list of at-risk species. Many of these species are federally and State listed with either recovery plans or guidance recovery goals from the Hawai‘i and Pacific Plant Recovery Coordinating Committee. These species have spatially explicit recommendations (recovery footprints) for where to locate recovery populations.

iii. **Insufficient Data to Develop Recovery Goals:** Species lacking adequate information to identify spatially explicit habitat needs require additional surveys or research before useful recovery goals can be identified. For example, many at-risk terrestrial arthropod species lack enough distribution data (usually fewer than five occurrence records) or information on habitat needs to identify where to focus management-intensive conservation actions. For these types of species, the MNLCP sets up a survey and research effort to obtain the data needed to identify appropriate habitat areas and then integrate this into the existing MNLCP. These species do not yet have recovery footprints to spatially focus recovery actions. They appear in the Conservation Planner as data-deficient species. Two types of surveys are planned:

- **Area Surveys** are proposed for species lacking distribution or habitat information (i.e., data-deficient) but they have known host plant affiliations that can be mapped. Area surveys are based on these host plant affiliations and identify planning units with potentially large numbers of species needing surveys. This strategy is likely to be both effective and efficient. These survey areas are not meant to identify where to undertake management actions, but where to potentially find species and habitat that can then be evaluated for inclusion into the MNLCP recovery footprints. The Conservation Planner has a separate map of suggested area survey planning units.

- **Species-specific Surveys** are recommended for species that lack information needed to identify recovery habitat or develop Area Surveys. The MNLCP does not identify areas for these surveys.

Table 1. At-risk species addressed in the MNLCP showing the number of species in each taxonomic group, species protected under State and federal endangered species laws (HRS Chapter 195D and ESA), and the number of at-risk species facing imminent extinction, species with interim recovery or full recovery goals, species targeted for multi-species area surveys, or species-specific surveys.

Taxonomic Group	At-Risk Species	ESA Listing	Facing Imminent Extinction	Recovery		Insufficient Data	
				Interim Recovery	Full Recovery	Area Surveys	Species Surveys
Forest Birds	4	3	2	0	4	0	0
Waterbirds	5	5	0	2	3	0	0
Seabirds	3	3	0	0	3	0	0
Plants	197	175	102	0	195	0	2
Terrestrial Snails	49	3	49	49	0	0	0
Terrestrial Arthropods	172	8	9	31	1	82	58
Aquatic Invertebrates	11	4	1	5	5	0	1
Hawaiian Hoary Bat	1	1	0	1	0	0	0
Total	442	202	163	88	211	82	61

5. How to Use MNLCP Products

a. What the MNLCP provides:

- Emergency actions needed to prevent imminent extinctions. When possible, these actions are identified spatially.
- Recovery actions needed for Maui Nui's at-risk species to reach healthy, self-sustaining population levels. When possible, spatially explicit habitat needs and spatially explicit conservation actions for at-risk species are provided. For species lacking the distributional or other information needed to develop spatially explicit habitat recommendations, those species are targeted for additional surveys and studies to fill data gaps and generate updates to the MNLCP. The intent is to identify habitat needs and conservation strategies for all 442 at-risk species.
- Ecosystem Conservation assessments, actions, and the identification of key areas representing protection needs and conservation opportunities.

b. Emergency Actions to Prevent Extinction

Species experts identified emergency actions or programs to prevent imminent extinctions in forest birds, plants, snails, terrestrial arthropods, and aquatic invertebrates (Table 2). While in situ management is usually preferred, it alone may not prevent extinction when the threat is a cryptic invasive species, disease, extremely low population numbers, or uncontrollable events, like climate change.

Some of these emergency actions are spatially explicit land management actions, while others are research-oriented or have not yet been linked to specific on-the-ground locations. More detailed information on emergency actions is described in the Taxonomic Group treatments.

Table 2. Summary of native Maui Nui species facing imminent extinction and the conservation actions needed immediately.

Taxonomic Group	At-Risk Species	Facing Imminent Extinction	Emergency Preventing Extinction Actions
Forest Birds	4	2	Reduce mosquito/avian malaria transmission in priority zones, translocation to another island
Waterbirds	5	0	
Seabirds	3	0	
Plants	197	102	Collect genetic material, propagation, outplanting, surveys, control local threats
Snails	49	49	Establish snail enclosures, captive propagation, *Euglandina* control research
Terrestrial Arthropods	172	9	Habitat management, fencing
Aquatic Invertebrates	11	1	Protect only known site, survey, prevent invasive species introductions
Hawaiian Hoary Bat	1	0	
Total	442	163	

c. Habitat or Spatial Needs

The MNLCP recovery footprints identify planning units (PUs) that contain existing or restorable habitat needed for at-risk species to reach targeted recovery goals. Priority recovery footprints allow the identification of the most important areas to manage in the near term, with other areas addressed by mid-term or later implementation efforts.

The combined recovery footprint for all eight taxonomic groups covers a significant portion of Maui Nui (Figure 2). This area includes the habitat needed to recover or significantly advance the recovery of all 442 at-risk species except for species lacking distributional or habitat information. The Snail Taxonomic Team determined that all 49 at-risk snail species are facing imminent extinction, and that MNLCP conservation efforts should prioritize preventing these impending losses through a number of exclosures rather than simultaneously addressing broader habitat recovery needs that are not currently possible. Critical recovery actions are still undertaken for snails, particularly research on control of invasive predators and the expansion of supporting infrastructure. This immediate focus on preventing extinctions is an essential first step to advance future recovery.

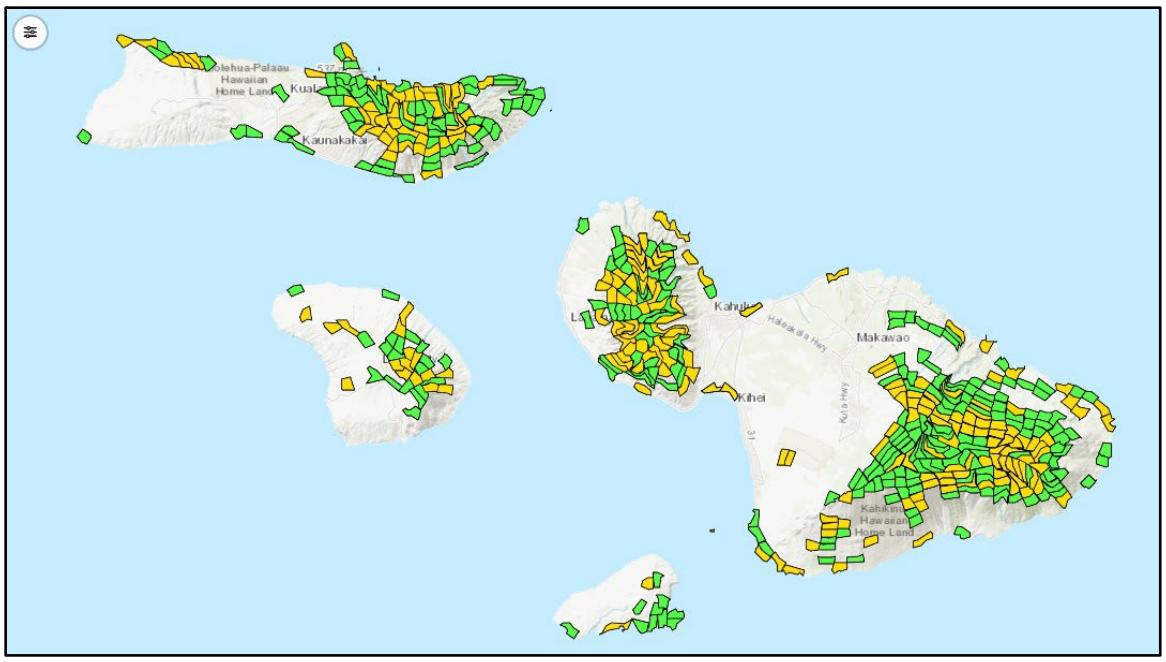


Figure 2. Recovery footprints for all taxonomic groups combined into a single overall footprint (green) with the priority areas for all groups (yellow).

d. Conservation Actions

Prioritization of conservation actions assessed the expected conservation benefits of 27 conservation strategies. These are composed of 17 single-focus strategies and 10 combinations of multiple single strategies. The impact of each strategy was assessed for each species group covered by the MNLCP (Figure 3 and Appendix 2).

The 10 combination strategies (S18-27) used different mixes of individual strategies (Appendix 2) to provide a series of conservation options. Comparing the benefits of these different options provides important insight into the selection of effective conservation strategies.

The conservation action analyses addressed all eight of the MNLCP taxonomic groups and created additional subgroups to obtain a more finely focused assessment of threats and actions for plants (6 subgroups), terrestrial arthropods (3 subgroups), and aquatic invertebrates (3 subgroups) (see Appendix 2). For general discussion purposes, subgroup responses are combined to match the major MNLCP taxonomic groups, but subgroups are addressed individually in the taxonomic group sections.

The probability of persistence for each of the eight taxonomic groups is shown in Figure 3 (and Appendix 2). Persistence is defined as the probability of species groups having viable, self-sustaining populations over a 20-year period. For many Hawaiian species this is a difficult goal to meet. For the MNLCP, the target goal is to get species groups to a minimum 50 percent probability of persistence.

None of the 27 strategies provided a 50 percent or higher probability of persistence for all taxonomic groups. However, implementing the two best-performing strategies—a bird-focused action strategy (S27) and a plant/invertebrate-focused action strategy (S25)—provided the most beneficial outcomes for all eight taxonomic groups (see Appendix 2).

Strategies S25 and S27 are the only two combination strategies that include the following key individual strategies:

- S3: Future ungulate fencing,
- S5: Terrestrial habitat management, and
- S6: Wetland management.

Selecting combination strategies S25 and S27 for the MNLCP results in all 17 of the single-strategy actions being implemented, as noted in Table 3.

Table 3. List of single-focus strategies and the composition of combination strategies S25 and S27. These two strategies provide the best probability of persistence for all eight taxonomic groups. S25 is a more plant- and invertebrate-oriented suite of conservation actions, while S27 is more focused on birds and the bat.

Strategies	Actions	In S25 core	In S27 set
S1	**No Actions**	—	—
S2	**Existing ungulate fences**	Yes	Yes
S3	**Future ungulate fences**	Yes	Yes
S4	**Ungulate control outside fences**	Yes	Yes
S5	**Terrestrial habitat management**	Yes	Yes
S6	**Stream / wetland habitat management**	Yes	Yes
S7	**Invasive vertebrate predator control**	Yes	Yes
S8	**Invasive invertebrate pest control**	Yes	Yes
S9	**Existing predator exclusion fences**	Yes	Yes
S10	**Future predator exclusion fences**	Yes	Yes
S11	**Landscape-scale rodent suppression**	Yes	Yes
S12	**Existing tools for rare plants & inverts**	Yes	—
S13	**Future tools for rare plants & inverts**	Yes	—
S14	**Avian disease prevention**	—	Yes
S15	**Future tools for avian malaria**	—	Yes
S16	**Modification & minimization of infrastructure**	—	Yes
S17	**Vertebrate health recovery & rehabilitation**	—	Yes

At the taxonomic group level, all groups had the potential to reach a 50 percent likelihood of persistence if all actions in the combination strategies (S25 and S27) were implemented and effective (Figure 3; the total benefit of current and future actions; all colors in the figure). However, species experts felt that some actions were likely to be less feasible, so the benefits of those actions were decreased to account for these feasibility issues (Figure 3; red = the benefits with feasibility concerns; the combined green, blue, and black sections represent the benefits most likely to result after MNLCP implementation).

Some general observations from these results (Figure 3 and Appendix 2) indicate that:

- Taking no action to conserve at-risk species results in a very low probability of persistence for most species groups.
- Current actions improve the probability of persistence, but this is limited for many species groups, with current actions being most beneficial to seabirds and waterbirds.
- Feasible actions for each species group significantly improve the probability of persistence.

- Forest birds currently have the lowest persistence probability, even with existing conservation efforts. However, they also have the most opportunity for benefits from conservation actions.
- Plants are the least likely group to reach a 50 percent level of persistence.

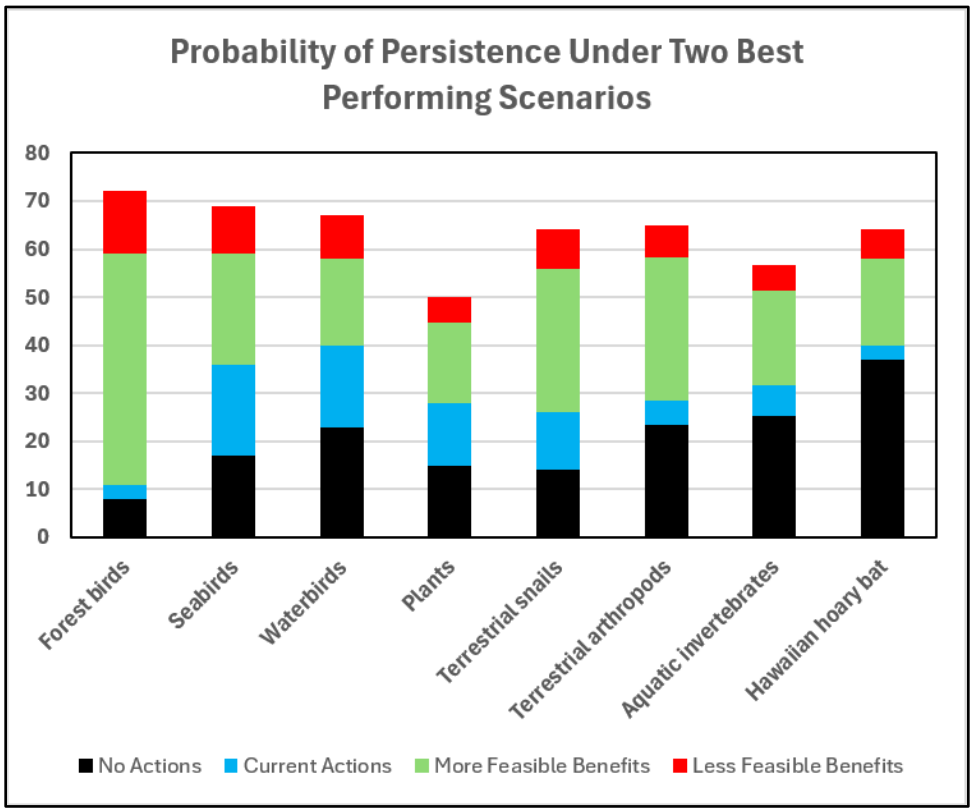


Figure 3. Expected persistence levels for taxonomic groups under no action, current actions only, future actions with only feasible benefits, and future actions assuming maximum feasibility (i.e., all actions are implemented and effective).

The Taxonomic Teams also assessed how to most effectively implement the individual actions (S1-17) contained within the two chosen combination strategies (S25 and S27). The sequencing of these actions can be important not only because some individual strategies provide more benefit than others, but also because some strategies may need to be completed before other strategies can be implemented. For example, before completing a critically important reintroduction task for species recovery, it may be necessary to:

- Identify key threats like presence of diseases or ungulates and how to control them. If the threats do not have established control or mitigative measures, then those will need to be developed.

- Potentially put in fire breaks
- Collect seeds for restoration
- Fence
- Remove ecosystem threats like ungulates
- Remove weeds
- Propagate species for the reintroduction effort
- Reestablish and restore a native community (which could take years)
- Provide short-term irrigation or threat control
- Reintroduce the species

Aquatic systems would have comparable processes, like reestablishing stream flows, removing invasive predators, installing barriers to predators, and captive rearing before doing a translocation.

The sequencing of needed actions at the MNLCP level is less detailed than these examples but is still important to factor in when tasks are sequenced for implementation. Sequencing is addressed in each Taxonomic Group discussion.

e. How MNLCP Products Inform the Conservation of At-Risk Species

Maui Nui's 442 at-risk species are not all equally imperiled. Some are down to a single remaining individual that could die today, while others number in the thousands and may have a decade or more before going extinct. Every one of these species needs action to prevent its loss, and the MNLCP provides recommended actions for all of them—but the urgency of those actions varies widely.

For species not facing imminent extinction, the MNLCP takes a more integrated approach, working at the planning unit and population levels as well as the ecosystem level to identify the habitat these species need and the conservation actions that will advance their recovery over the life of the Plan.

Recovery footprints developed by the Spatial Prioritization and Taxonomic Teams identify the areas likely to sustain the number of populations needed for recovery—in effect, answering "where to focus" conservation efforts. Within these recovery footprints, priority footprints were sometimes developed to flag the most important areas to focus on. For some taxonomic groups, even the priority footprint covers an area that would be very difficult to manage all at once. For these situations, priority footprints were refined further into narrower near-term

work objectives or special actions. Each taxonomic group has at least two levels of spatially explicit habitat delineations:

- Full Recovery Footprint. Provides adequate habitat for recovery
- Priority Recovery Footprint. Identifies the most important recovery areas
- Initial Implementation Footprint. Suggested starting points for implementing conservation efforts

The Terrestrial Plant Footprint, for example, identifies a full recovery footprint (Figure 4a with 317 planning units) and a priority footprint that identifies the habitat needed for three populations of each species (Figure 4b with 155 planning units).

The MNLCP's Conservation Planner helped to assess and narrow the plant priority footprint by:

- Reducing the scope of the species targeted from all 195 at-risk plants with habitat data to just the 101 plants facing imminent extinction that have habitat data (i.e., PEPP species, Figure 4c), then
- Further reducing the number of target species by identifying those PEPP species with three or fewer known occupied planning units (Figure 4d).
- Finally, restricting the footprint to only those species with a single known occupied planning unit (Figure 4e).

The degree of narrowing or focusing implementation is driven by various factors such as the urgency with which actions are needed or the availability of resources. In this example, the initial implementation footprint selected for plants focuses on PEPP species with only a single known occupied planning unit (Figure 4e and Option 5 in Table 4), with the expectation that the next higher level (Option 4) with three or fewer known occupied planning units would be implemented soon after.

Table 4. Comparison of alternatives for narrowing priority areas for initial implementation. See Figure 4 and the text for more detail. Note: one PEPP plant and one non-PEPP plant are not included because they lacked enough information to identify appropriate habitat.

Option	Description	PUs	Area (km^2)	Target Species	PEPP Species	Non PEPP Species	Total Species
1	Full Recovery	317	675	195	101	94	195
2	Priority	155	325.5	195	101	94	161
3	Priority PEPP	98	204	101	101	78	175
4	PEPP 1-3 Pops	39	83	36	68	52	120
5	PEPP Only 1 Pop	12	26	14	21	19	50

The full recovery footprint option provides up to 10 populations of each at-risk plant (195 species) and is the desired end outcome for the MNLCP. Priority and narrower alternatives represent priorities for initial implementation, not long-term recovery.

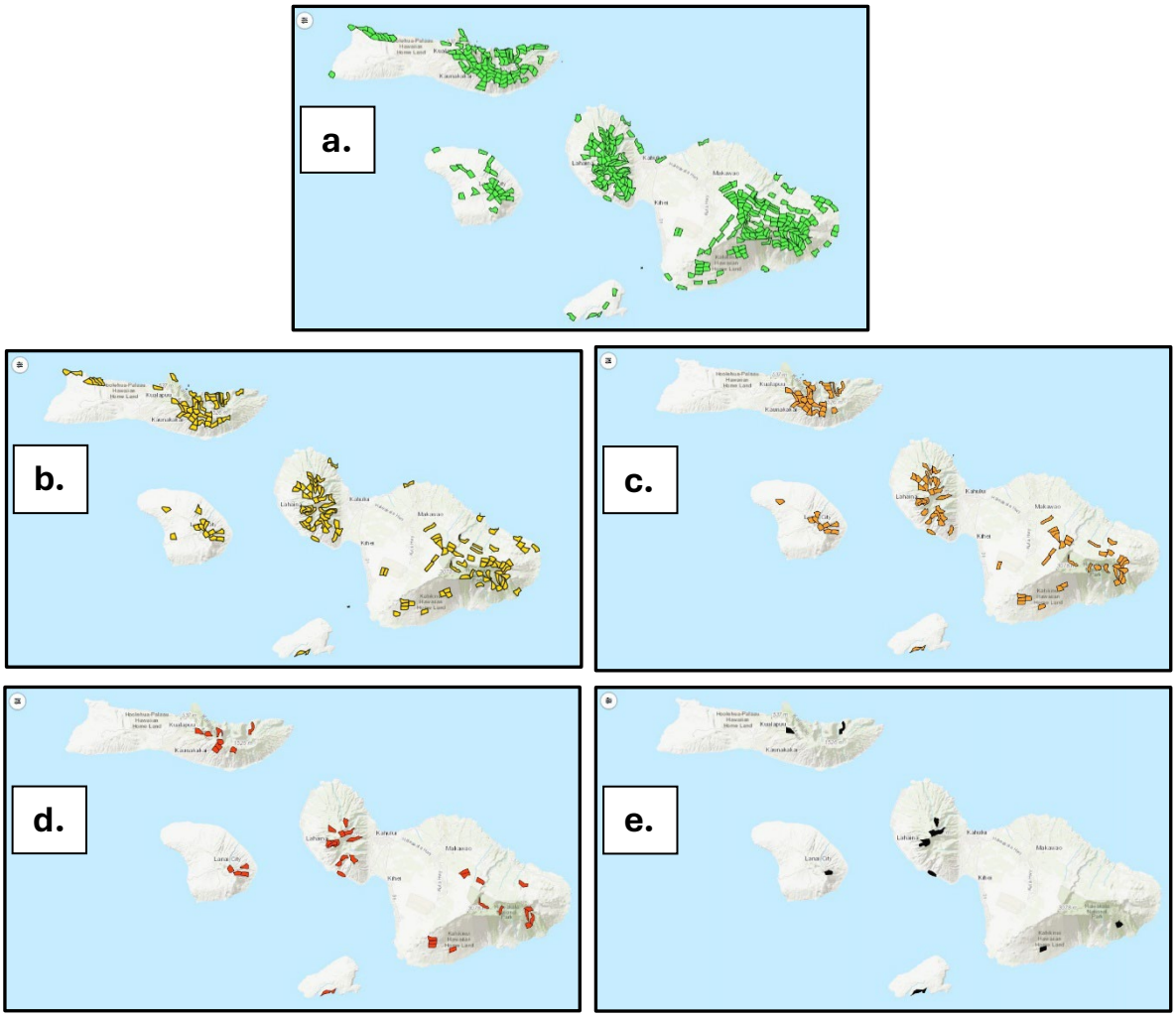


Figure 4. Terrestrial Plant Recovery Footprint: (a) the plant recovery footprint (green), (b) priority habitat areas (yellow), (c) priority habitat areas for PEPP species (orange), (d) priority habitat areas for PEPP species with three or fewer known occupied planning units (red), and (e) priority habitat areas for PEPP species with only a single known occupied planning unit (black).

The Priority Threat Management assessment, led by the Action Prioritization Team, developed recommended suites of actions and strategies aimed at improving the persistence of at-risk species. The assessment found that two combined strategies, S25 and S27, are needed to bring most at-risk species to at

least a 50 percent chance of persistence. This is the “what needs to be done” portion of the MNLCP.

The MNLCP is large, complex, and designed for a 20-year implementation period. The strategies intended for implementation contain numerous actions and sub-actions that need to be implemented across large areas. It is simply not possible to implement this 20-year plan everywhere and all at once.

To help managers implement the MNLCP, species experts examined the suite of actions proposed for each taxonomic group and recommended an implementation sequence to enhance conservation benefits to each taxonomic group. This sequencing, along with the identification of species facing imminent extinction, may help managers better determine “when actions need to be done.”

As a general rule, the actions with the greatest conservation benefit should be initiated first. In some cases, though, actions with lower individual benefit produce a synergistic effect when combined with another action: ungulate fencing paired with weed control, for instance, delivers considerably more benefit than either action alone. Sequencing can also depend on timing rather than benefit alone—some actions, such as habitat restoration, must happen first to set the stage for a later species reintroduction, while a highly cost-effective, immediately actionable medium-priority action may be worth pursuing sooner even if its ultimate benefit is lower than an action that will not be field-ready for several years.

Together, this information helps managers and field teams identify the who, what, when, and where of conservation: “who needs management” comes from the at-risk species lists, “where to focus management” comes from the recovery footprints, “what needs to be done” comes from the Priority Threat Management analyses, and “when actions need to be done” from both the identification of species facing imminent extinction and the sequencing of conservation actions. Combined, they give managers and field teams much of the information needed to successfully restore Maui Nui’s at-risk species.

f. Fine-scale Implementation

In most instances, the MNLCP maps habitat and actions at the level of planning units (PUs) with more detailed habitat selection or the siting of actions to be determined by field teams. The MNLCP also provides implementation tools to help managers and field teams do “within PU” planning at a finer scale. Specifically, the Conservation Planner software will be able to show managers the location of species occurrences, quality habitat, potential recovery sites,

where climatic conditions seem to be changing most rapidly, and where certain microhabitats might be found within a PU.

The information in the MNLCP, when coupled with the associated Conservation Planner implementation tools, is intended to make the MNLCP a living system that can be updated or modified by managers to address changing circumstances and funding levels, unforeseen opportunities, and new information. It also facilitates not only long-range planning but annual implementation decisions as well.

The discussions in each taxonomic group section will provide more information on at-risk species, habitat needs, and conservation actions. Appendices provide detailed information on analytic techniques and methods.

6. Conservation Strategies for Taxonomic Groups

The status, threats, and conservation strategies for each taxonomic group are discussed in detail below and in several appendices.

a. Forest Birds

i. Forest Bird Recovery Strategy

Summary: The Maui Nui forest bird strategy addresses four at-risk birds and focuses on reducing mosquito-borne disease, the primary driver of decline, through integrated pest management and mosquito suppression techniques.

These actions are coupled with intensive monitoring to evaluate effectiveness and guide adaptive management. In parallel, recovery actions include translocation and reintroduction of highly endangered species, protection and restoration of native habitat, and community engagement to build long-term support. Together, this approach aims to stabilize remnant populations, expand birds back into suitable habitat, and create conditions for self-sustaining populations over time.

The Maui Nui efforts are synced with Hawai‘i’s forest bird recovery strategy, which centers on integrated, landscape-scale threat management paired with targeted species interventions.

Planning Process: Review of published recovery plans and expert determination of species at risk, species facing imminent extinction, management needs and locations of emergency actions, recovery goals, recovery footprint, priority footprint, and sequencing of individual conservation strategies.

ii. At-Risk Species

Four birds were identified as at-risk species, with two species facing imminent extinction (Table 5). Table 6 summarizes the current populations of these species.

Kiwikiu (*Pseudonestor xanthophrys*) or Maui parrotbill. State and federally listed as endangered. This insectivorous species was endemic to Maui and

Moloka‘i. Kiwikiu are currently found in a 30 km^2 area in East Maui above 1,500 m in native-dominated rainforest, with an estimated population of fewer than 150 individuals (Maui Forest Bird Recovery Project 2026). There are two non-reproductive individuals in captivity. This species is at very high risk of becoming extinct in the near future.

‘Ākohekohe (*Palmeria dolei*) or crested honeycreeper. State and federally listed as endangered. Although once found in forests of Moloka‘i and Maui, ‘ākohekohe now occupy a small area of mesic and montane forest on Haleakalā, East Maui. The estimated wild population is between 1,200 and 2,000 individuals (Maui Forest Bird Recovery Project 2026). There are no ‘ākohekohe in captivity. This species is at high risk of becoming extinct in the near future.

‘I‘iwi (*Drepanis coccinea*). State listed as endangered on O‘ahu, Moloka‘i, and Lāna‘i; federally listed as threatened throughout its distribution. The species occurs primarily above 1,250 m elevation and was once common in lower elevations throughout the main Hawaiian Islands. The estimated population in Maui Nui is around 50,000 individuals, all on East Maui (Judge et al. 2019).

Maui ‘alauahio (*Paroreomyza montana*) or Maui creeper. Not State or federally listed, but declining. Given the recent precipitous decline in population numbers of numerous other Maui forest bird species, and the perilous state of endangered forest birds, the Maui ‘alauahio was included. The population size is estimated at 99,000, all on East Maui (Judge et al. 2019).

The endangered ‘alalā (*Corvus hawaiiensis*) or Hawaiian crow is an endangered crow native to Hawai‘i Island that has recently been introduced into Maui. It is not addressed in this strategic plan because the ongoing introduction of up to 15 crows into Maui is considered a 5-year pilot effort to determine if ‘alalā can be successfully established in an area without native avian predators (i.e., the ‘io (*Buteo solitarius*) or Hawaiian hawk, which is known to prey on ‘alalā). The ‘io is endemic to Hawai‘i Island.

Table 5. Forest Birds addressed in the MNLCP showing the number of existing populations, species’ protected status (ESA, E=endangered, T=threatened), the number of at-risk species facing imminent extinction, species with interim or full recovery goals, species with insufficient

data to support recovery areas but that are targeted for multi-species area surveys, or basic species-specific surveys.

Forest Birds	Current Populations	ESA Listing	Facing Imminent Extinction	Recovery		Insufficient Data	
				Interim Recovery	Full Recovery	Area Surveys	Species Surveys
Kiwikiu (Maui parrotbill)	1	E	X		3		
ʻĀkohekohe (Crested honeycreeper)	1	E	X		3		
ʻIʻiwi	1	T			3		
Maui ʻalauahio	1				3		
Total (4)		3	2	0	4	0	0

Table 6. Estimated population numbers for Maui Nui at-risk forest birds (individuals).

Forest Birds	E. Maui	W. Maui	Molokai	Lanai	Kahoolawe	Total
Kiwikiu (Maui parrotbill)	150	0	0	0	0	150
ʻĀkohekohe (Crested honeycreeper)	1,768	0	0	0	0	1,768
ʻIʻiwi	50,000	0	0	0	0	50,000
Maui ʻalauahio	99,000	0	0	0	0	99,000

iii. Emergency Actions for Species Facing Imminent Extinction

The Forest Bird Team identified two forest birds at very high risk of near-term extinction: the kiwikiu (Maui parrotbill) and the ʻākohekohe (crested honeycreeper). Both species are threatened primarily by introduced avian malaria, which is transmitted by *Culex* mosquitoes (also introduced). After having relatively stable population levels for a number of years, both species have recently seen significant population declines resulting in rising concern over their survival. Continuing warming trends are likely to promote expansion of *Culex* mosquitoes and further threaten forest birds (Fortini et al. 2015). The Maui Forest Bird Recovery Project focuses on the recovery of forest birds and identified priority actions to prevent the extinction of these two birds.

Kiwikiu (Maui parrotbill). Fewer than 150 remaining individuals. The most important emergency action is to expand the amount of available habitat by reducing mosquitoes and disease transmission in three areas in and adjacent to currently occupied habitat (Figure 5).

ʻĀkohekohe (crested honeycreeper). Under 2,000 remaining individuals. Emergency actions include:

- Expanding available habitat by reducing mosquitoes and disease transmission in the same three areas as for kiwikiu (Figure 5) and
- Establishing a second population through translocation to another suitable habitat area not on Maui.

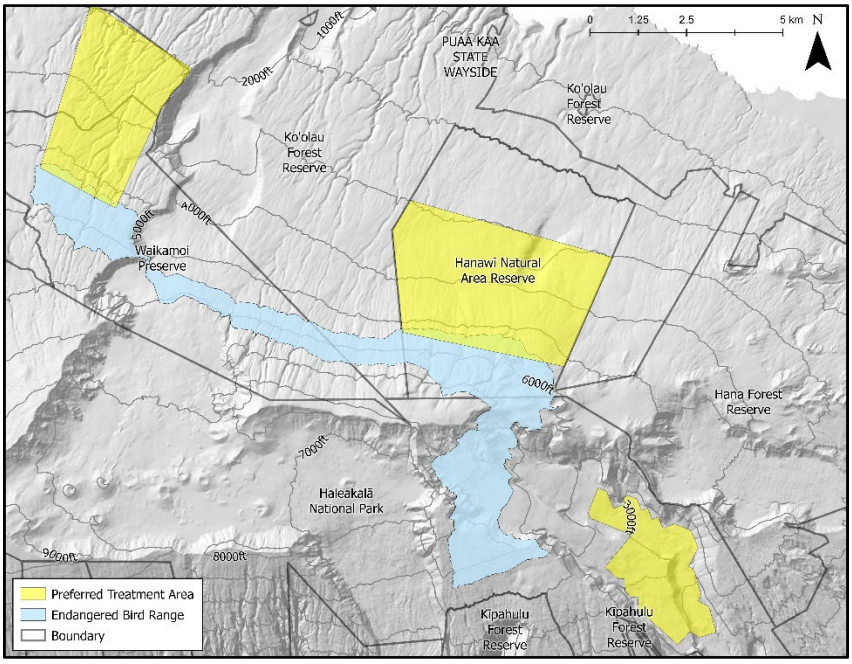


Figure 5. Emergency actions to prevent the extinction of kiwikiu (Maui parrotbill) and ‘ākohekohe (crested honeycreeper). Mosquito control is targeted for three priority areas (yellow) in and adjacent to occupied habitat (blue) needed to prevent extinction.

iv. Recovery Goals

Recovery for all four of Maui Nui’s at-risk forest birds includes having three primary populations, one each on East Maui, West Maui, and Moloka‘i. Target numbers for kiwikiu and ‘ākohekohe are provided for each of the three recovery areas. For ‘i‘iwi and Maui ‘alauahio, the target is a total number of individuals across all three areas (Table 7).

Table 7. Recovery population targets for Maui Nui at-risk forest birds (individuals).

Forest Birds	E. Maui	W. Maui	Molokai	Lanai	Kahoolawe	Total
Kiwikiu (Maui parrotbill)	1,500	500	500	0	0	2,500
‘Ākohekohe (Crested honeycreeper)	4,500	1,500	1,500	0	0	7,500
‘I‘iwi	X	X	X	0	0	50,000
Maui ‘alauahio	X	X	X	0	0	99,000

v. Recovery Habitat

The Forest Bird Team used existing literature, recovery plans, and professional expertise to delineate recovery habitat for the MNLCP. Management of this recovery footprint is necessary to meet the distribution and abundance targets needed to sustain forest bird populations on East Maui, West Maui, and Moloka‘i (Figure 6). The team also identified the priority habitat for near-term conservation actions (Figure 6).

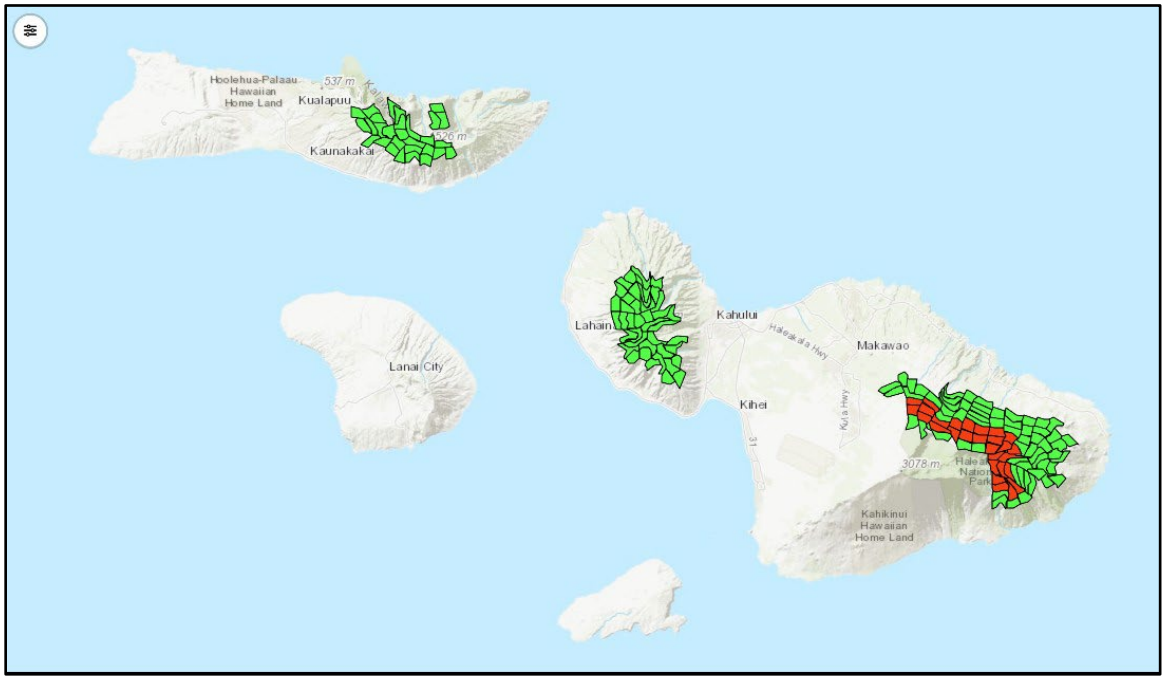


Figure 6. Recovery habitat for all four at-risk Maui Nui forest birds (green). In the case of forest birds, the priority and initial implementation areas identify the same habitat needing management action—specifically mosquito control measures downslope from the population (shown in red).

vi. Conservation Actions

The modified Priority Threat Management process assessed the expected conservation benefits of 27 conservation strategies, including combinations of strategies, across all species groups covered by the MNLCP (Appendix 2).

Forest birds had the lowest chance of persisting without conservation actions of any taxonomic group (Figure 3, No Actions bar (black)) and the lowest persistence with current actions (blue), but they had the highest potential benefit from conservation actions (green); and if all actions are successfully implemented, the highest probability of persistence of any taxonomic group (71.6 percent, red). Only one strategy, S27 (Appendix 2), has a greater than 50 percent persistence probability after feasibility is factored in. Strategy 27 has 14 individual strategies embedded in it. Table 8 shows the most beneficial of those strategies and their individual benefit. To further refine the prioritization and sequencing of conservation actions for forest birds, species experts evaluated and ordered the individual strategies based on their recommended implementation sequence. Development and implementation of effective controls of mosquito-transmitted avian malaria is clearly the most important action to accomplish. Other strategies are also important, especially vertebrate predator control and terrestrial habitat management.

Table 8. Suggested sequencing of Strategy 27 actions to best benefit forest birds. Persistence after feasibility-related reductions is shown for the seven highest sequenced individual strategies. If all actions associated with the combination S27 strategy are implemented, it is anticipated that forest birds will have a 59 percent chance of persisting through the 20-year timeframe of the MNLCP.

Action	Description	Persistence (%)	Sequence
S15	Future Tools to Address Avian Malaria	39	1
S14	Avian Disease Prevention (captive bird facilities, Wolbachia releases, broadcast BTI, manage avian botulism)	28	2
S7	Invasive Vertebrate Predator Control (rats, cats, dogs, mongoose, barn owl, cattle egrets, rabbits)	19	High
S11	Landscape Scale Rat Suppression	17	High
S5	Terrestrial Habitat Management (for at-risk spp)	15	High
S3	Future Ungulate Fencing	14	Medium
S2	Existing Ungulate Fencing	13	Medium

Mosquito control measures associated with S15 and S14 are extremely high priority and should be focused in and around the priority footprint for all four at-risk forest birds. This priority area is the same area identified for extinction-prevention efforts for kiwikiu and ʻākohekohe (Figure 5).

b. Seabirds

i. Seabird Recovery Strategy

Summary: The seabird strategy addresses three at-risk seabirds, none of which are facing imminent extinction. The introduction of mammalian predators, especially rats, mongoose, and cats, along with habitat loss, has greatly reduced the abundance of Maui Nui seabirds. Consequently, the highest priority conservation actions focus on these threats and are:

- The establishment of predator exclusion fencing (which is expected to make the greatest contribution to seabird recovery),
- Invasive vertebrate predator control,
- Terrestrial habitat management, and
- Providing wildlife health and rehabilitation services.

Other important actions include establishment and maintenance of ungulate fencing, and the minimization/modification of infrastructure that negatively impacts seabird survival, such as artificial lights at night, power lines, wind turbines, and roadways.

Planning Process: Expert determination of species at risk, species facing imminent extinction, management needs and locations of emergency actions, recovery goals, recovery footprint, priority footprint, and sequencing of individual conservation strategies.

ii. At-Risk Species

Three seabirds were identified as at-risk species (Table 9), and all three have detailed spatially explicit conservation measures and identified areas for intensive management. Table 10 summarizes the current number of breeding pairs for each species.

ʻUaʻu (*Pterodroma sandwichensis*) or Hawaiian petrel. State and federally listed as endangered. This species is endemic to Hawaiʻi and it nests in high-elevation dry habitats with minimal vegetation (often in lava crevices) and in lower-elevation wet, dense native forests.

ʻAʻo (*Puffinus newelli*) or Newell's shearwater. This species is endemic to Hawaiʻi and is State and federally listed as threatened. Their nesting colonies are typically found on steep mountain slopes within wet to mesic forest habitats.

ʻAkēʻakē (*Oceanodroma castro*) or band-rumped storm petrel. Hawaiʻi's distinct population segment is State and federally listed as endangered. This species is indigenous to Hawaiʻi and nests in natural cavities or burrows across various elevations, though known breeding locations are currently extremely rare.

Table 9. At-risk seabirds addressed in the MNLCP showing the number of existing populations, species' protected status (ESA, E=endangered, T=threatened), the number of at-risk species facing imminent extinction, species with interim or full recovery goals, species with insufficient data to support recovery areas but that are targeted for multi-species area surveys, or basic species-specific surveys.

Seabirds	Current Populations	ESA Listing	Facing Imminent Extinction	Recovery		Insufficient Data	
				Interim Recovery	Full Recovery	Area Surveys	Species Surveys
ʻuaʻu (Hawaiian petrel)	3	E			X		
aʻo (Newell's shearwater)	3	T			X		
ʻakēʻakē (Band-rumped storm petrel)	3	E			X		
Total (3)		3	0	0	3	0	0

Table 10. Current estimates of seabird breeding pairs for Maui Nui.

Seabirds	E. Maui	W. Maui	Molokai	Lanai	Kahoolawe	Total
ʻuaʻu (Hawaiian petrel)	4,000	100-200	1-50	2,000	0	6,101-6,750
aʻo (Newell's shearwater)	50-100	50-100	1-50	1-50	0	102-300
ʻakēʻakē (Band-rumped storm petrel)	50-100	50-100	1-50	50-100	0	151-350

iv. Emergency Actions for Species Facing Imminent Extinction

No seabirds are currently considered to be facing an imminent threat of extinction.

v. Recovery Goals

The MNLCP seeks to roughly double the number of Hawaiian petrel breeding pairs and increase the number of band-rumped storm petrels and Newell's shearwaters by an order of magnitude (Table 11). These targets are apportioned to East Maui, West Maui, Molokaʻi, Kahoʻolawe, and Lānaʻi.

Table 11. Recovery population targets for seabirds (breeding pairs) for Maui Nui.

Seabirds	E. Maui	W. Maui	Molokai	Lanai	Kahoolawe	Total
ʻuaʻu (Hawaiian petrel)	8,000	4,000	1,000	4,000	1,000	18,000
aʻo (Newell's shearwater)	1,000	800	600	600	600	3,600
ʻakēʻakē (Band-rumped storm petrel)	500	400	300	300	300	1,800

vi. Recovery Habitat

The Seabird Team, like the Forest Bird Team, used existing literature, recovery plans, and professional expertise to delineate the recovery footprint needed to sustain seabird populations on all four Maui Nui islands (Figure 7). Offshore islets are included in the recovery footprint regardless of whether they are large enough to be shown in Figure 7. A subset of planning units was selected as a priority area to focus conservation actions.

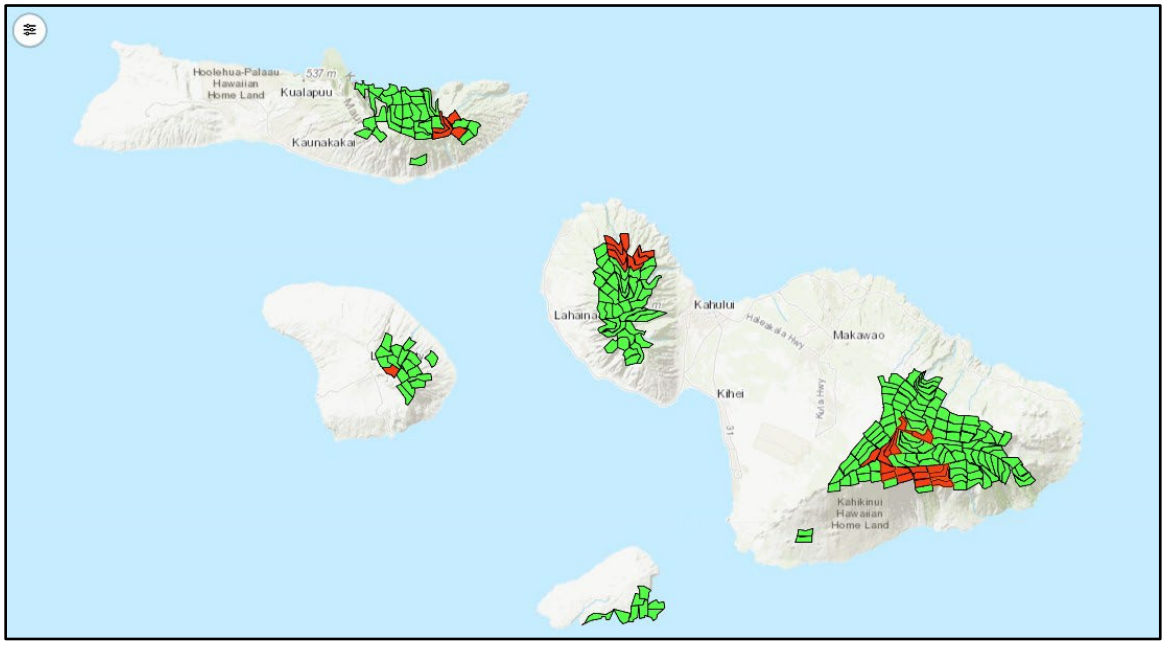


Figure 7. Seabird Recovery Footprint for the three at-risk Maui Nui seabirds (green), with the initial implementation area and priority areas being the same (red).

Additional areas to initiate seabird conservation actions include planning units 3047, 3052, 3053, 4019, 4020, 4021, 4067, 6017, 6195, 6208, and 6209.

vii. Conservation Actions

Like forest birds, seabirds also showed the greatest benefit from the combination strategy S27. Seabirds have a relatively low expectation of persistence without conservation measures (Figure 3, No Actions bar (black)) but, along with waterbirds, have higher-than-average persistence with current conservation efforts (blue). Seabirds show good persistence benefits from feasible conservation actions (green); and if all actions are successfully implemented, very high persistence (69%, red).

Predator control actions like fencing and predator removal rank high as important actions to implement first (Table 12). Programs rescuing injured birds and habitat management are also high priorities.

Table 12. Suggested sequencing of Strategy 27 actions to best benefit seabirds. If all actions are implemented, it is anticipated that seabirds will have a 58 percent chance of persisting through the 20-year timeframe of the MNLCP. Persistence after feasibility-related reductions is shown for the nine highest sequenced individual strategies.

Action	Description	Persistence (%)	Sequence
S10	New Predator Fencing (snails, rodents, mongoose, cats)	48	1
S7	Invasive Vertebrate Predator Control (rats, cats, dogs, mongoose, barn owl, cattle egrets, rabbits)	46	2
S17	Vertebrate Health and Rehabilitation	35	3
S11	Landscape Scale Rat Suppression	42	High
S16	Minimization and Modification of Infrastructure (minimize lighting, modify powerlines and wind turbines, reduce road kill of birds)	38	High
S2	Existing Ungulate Fencing	32	High
S9	Existing Predator Fencing (snails, rodents, mongoose, cats)	31	High
S3	Future Ungulate Fencing	30	High
S5	Terrestrial Habitat Management (for at-risk spp)	29	High

c. Waterbirds

i. Waterbirds Recovery Strategy

Summary: The overall waterbird strategy focuses on five waterbirds, with the emphasis on habitat protection and the control of invasive predators, with some additional species-specific actions as needed:

- For ʻalae keʻokeʻo (Hawaiian coot) and ʻaeʻo (Hawaiian stilt), the priority actions are maintaining suitable habitat—through management of hydrology, sediment, and invasive weeds—and predator control, using predator-proof fencing when cost-effective.
- For nēnē (Hawaiian goose), priorities are predator control, restoration of upland areas, and increasing the population on Molokaʻi.
- For ʻalae ʻula (Hawaiian moorhen), the focus is on protecting and restoring suitable habitat, then reintroducing the species to Molokaʻi.
- For kōloa maoli (Hawaiian duck), the priority action is to develop an effective measure to eliminate the threat posed by hybridization. Once that action is successful it will be feasible to reintroduce kōloa maoli to Maui Nui.

Five Hawaiian waterbirds were likely native to Maui Nui. All five of these species are listed as either endangered or threatened under federal and Hawaiʻi laws (Tables 13 and 14). Two species are probably no longer found on Maui Nui: ʻalae ʻula (Hawaiian moorhen) and kōloa maoli (Hawaiian duck). All of the ducks observed on Maui Nui are probably hybrids with feral mallards. Key threats driving the risk of extinction of waterbirds are habitat loss due to human modifications to wetlands and introduced mammalian predators, specifically mongoose, cats, rats, and domestic dogs. Botulism poisoning is a localized threat in some wetland areas. Hybridization between kōloa maoli and feral mallards is a highly significant threat.

Planning Process: Expert determination of species at risk, species facing imminent extinction, management needs and locations of emergency actions, recovery goals, recovery footprint, priority footprint, and sequencing of individual conservation strategies.

ii. At-Risk Species

‘Alae ke‘oke‘o (Hawaiian coot, *Fulica alai*). Federally and State listed as endangered. Found in wetland ponds and marshes.

‘Ae‘o (Hawaiian stilt, *Himantopus mexicanus knudseni*). Federally and State listed as endangered. Found in shallow lowland wetlands, mudflats, and coastal areas.

Nēnē (Hawaiian goose, *Branta sandvicensis*). Federally listed as threatened and State listed as endangered. Found in lowland and upland grassland and shrubland.

‘Alae ‘ula (Hawaiian moorhen, *Gallinula galeata sandvicensis*). Federally and State listed as endangered. Currently not found on Maui Nui. Its habitat is freshwater wetlands with aquatic plants.

Kōloa maoli (Hawaiian duck, *Anas wyvilliana*). Federally and State listed as endangered. All kōloa-like ducks in Maui Nui are apparently kōloa-mallard hybrids. Its habitat is lowland and upland streams and wetlands.

Table 13. At-risk waterbirds showing the number of existing populations, species’ Endangered Species Act (ESA) status (note nēnē is federally listed as threatened and State listed as endangered, E/T), the number of at-risk species facing imminent extinction, species with interim or full recovery goals, and species with insufficient data to identify recovery habitat. Those species with insufficient data will have either multi-species area surveys or basic species-specific surveys to fill data gaps.

Waterbirds	Current Populations	ESA Listing	Facing Imminent Extinction	Recovery		Insufficient Data	
				Interim Recovery	Full Recovery	Area Surveys	Species Surveys
‘Alae ke‘oke‘o (Hawaiian coot)	3	E			X		
‘ae‘o (Hawaiian stilt)	3	E			X		
nēnē	2	E/T			X		
‘alae ‘ula (Hawaiian moorhen)	0	E		X			
Kōloa maoli (Hawaiian duck)	0	E		X			

Table 14. Current waterbird population estimates for Maui Nui (individuals). There are currently no ‘alae ‘ula on Maui Nui and all kōloa maoli on Maui Nui appear to be kōloa-mallard hybrids.

Waterbirds	Maui	Molokai	Lanai	Kahoolawe	Total
ʻAlae keʻokeʻo (Hawaiian coot)	320	30	14	0	364
ʻAeʻo (Hawaiian stilt)	499	142	43	0	684
Nēnē	580	30	0	0	610
ʻAlae ʻula (Hawaiian moorhen)	0	0	0	0	0
Kōloa maoli (Hawaiian duck)	0	0	0	0	0

iii. Emergency Actions for Species Facing Imminent Extinction

None of the waterbirds are facing imminent extinction, though two species are no longer found in Maui Nui (ʻalae ʻula (Hawaiian moorhen, *Gallinula galeata sandvicensis*) and kōloa maoli (Hawaiian duck, *Anas wyvilliana*)). The ʻalae ʻula requires low-salinity freshwater wetlands, which are rare in Maui Nui, and all ducks resembling kōloa maoli appear to be hybrids with feral mallards.

iv. Recovery Goals

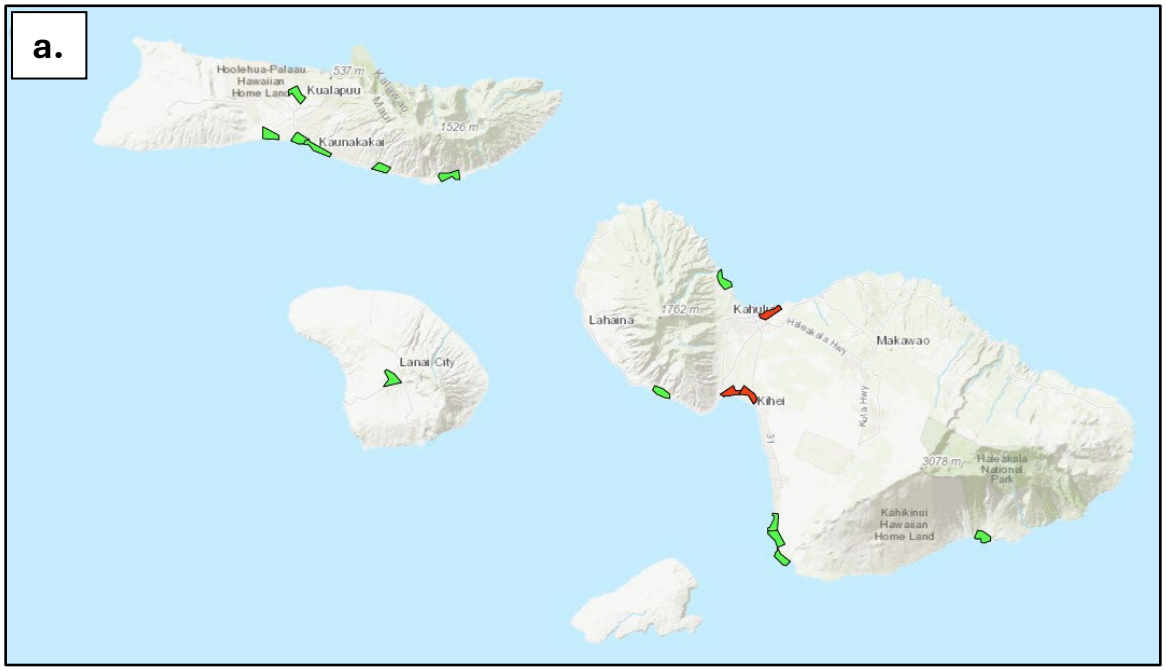
Waterbird conservation in Hawaiʻi has largely been successful and the MNLCP seeks to build upon this success. Suitable wetland areas have been selected for continued waterbird restoration. The recovery footprint for waterbirds includes very important wetlands (and uplands for nēnē) targeted as primary habitat for populations needed to meet recovery requirements and supporting areas to provide additional habitat (Figure 8, Table 15). A subset of wetlands (and uplands for nēnē) in the footprint is identified as priorities for management.

Table 15. Recovery population targets for waterbirds in Maui Nui (individuals). Note: nēnē has a target of two populations on East Maui (one population with 600 and one with 200 individuals). Maui Nui–wide targets for ʻalae keʻokeʻo, ʻalae ʻula, and ʻaeʻo can be spread across any island.

Waterbirds	Maui	Molokai	Lanai	Kahoolawe	Total
ʻAlae keʻokeʻo (Hawaiian coot)	X	X	X	0	750
ʻAeʻo (Hawaiian stilt)	X	X	X	0	750
Nēnē	2 (800)	200	0	0	1,000
ʻAlae ʻula (Hawaiian moorhen)		250			250
Kōloa maoli (Hawaiian duck)	X	250	X	0	250

v. Recovery Habitat

The Waterbird Team and species experts identified key wetlands needed to provide habitats for all five waterbirds. The priority waterbird footprint identifies the priority wetlands to focus initial conservation actions.



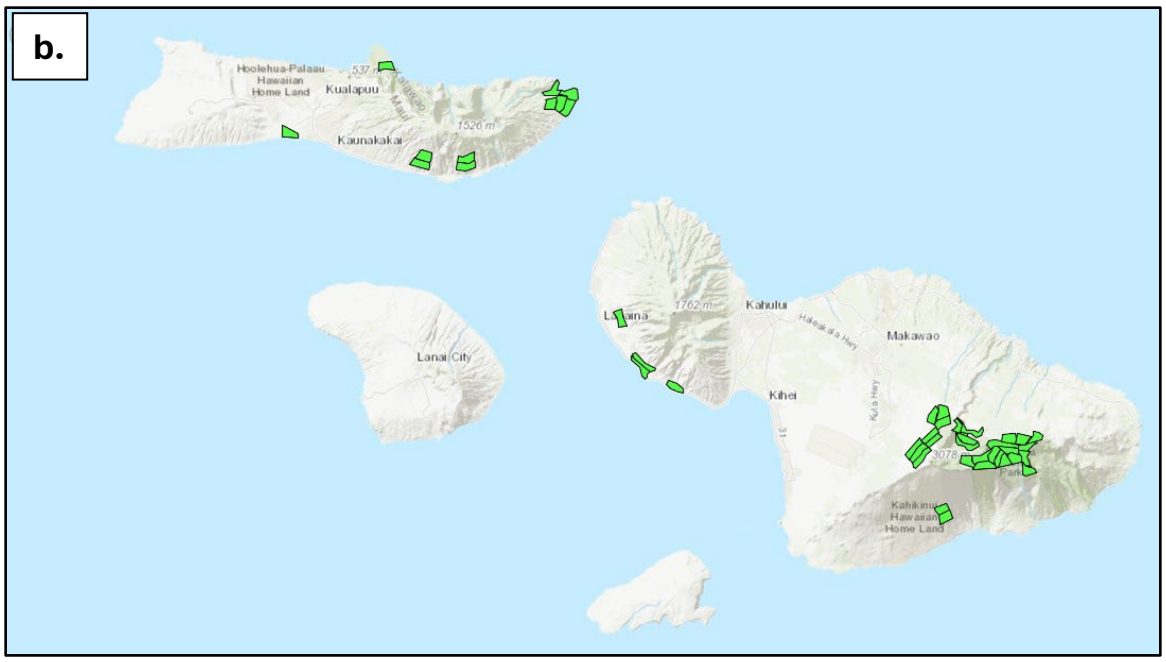


Figure 8. Recovery habitat for: (a) the ʻalae keʻokeʻo (Hawaiian coot), ʻaeʻo (Hawaiian stilt), kōloa maoli (Hawaiian duck), ʻalae ʻula (Hawaiian moorhen); and (b) nēnē (Hawaiian goose). Priority areas and implementation areas are the same (red).

vi. Conservation Actions

In general, waterbirds have responded well to management (Figure 3). For Maui Nui, the most important conservation actions for the ʻalae keʻokeʻo (Hawaiian coot) and ʻaeʻo (Hawaiian stilt) are focused on the control of introduced predators via predator-proof fencing, wetland habitat protection, and habitat management (Table 16). Conservation actions for nēnē focus on restoration, predator control, and reducing human-caused mortality. The ʻalae ʻula (Hawaiian moorhen) is no longer present in Maui Nui and will need to be reintroduced on Molokaʻi. Actions should target management that creates and maintains potential reintroduction sites in good habitat for this species. Since all kōloa-looking ducks appear to be kōloa-mallard hybrids, kōloa maoli (Hawaiian duck) will also need to be reestablished on Maui Nui. However, the reintroduction of kōloa maoli is currently not feasible until appropriate management techniques are developed that can reduce hybridization with feral mallards to an acceptable level.

Table 16. Suggested sequencing of Strategy 27 actions to best benefit waterbirds. Persistence after feasibility-related reductions is shown for the six highest sequenced individual strategies and three additional actions that were added after initial analyses. If

all actions associated with the combination S27 strategy are implemented, it is anticipated that waterbirds will have a 58 percent chance of persisting through a 20-year timeframe.

Action	Description	Persistence (%)	Sequence
S10	New Predator Fencing (snails, rodents, mongoose, cats)	50	1
S7	Invasive Vertebrate Predator Control (rats, cats, dogs, mongoose, barn owl, cattle egrets, rabbits)	49	2
S6	Wetland Habitat Management (for at-risk spp)	38	3
S17	Vertebrate Health and Rehabilitation	38	Medium
S14	Avian Disease Prevention (captive bird facilities, Wolbachia releases, broadcast BTI, manage avian botulism)	34	Medium
S16	Minimization and Modification of Infrastructure (minimize lighting, modify powerlines and wind turbines, reduce road kill of birds)	33	Medium
Add 1	Reintroduce ‘alae ‘ula (Hawaiian moorhen)		Medium
Add 2	Control hybridization between koloa and mallards		Medium
Add 3	Reintroduce koloa maoli (Hawaiian duck)		Medium

The most immediate beneficial actions that need to be undertaken for waterbirds are:

- Additional predator fencing
- Vertebrate predator control (especially cats, dogs, rats, and mongoose)
- Wetland management

Other important tasks include avian disease prevention, providing wildlife health and rehabilitation services, and minimizing the effects of infrastructure. There are also three additional items that need to be added as medium-priority actions: the reintroduction of ‘alae ‘ula (Hawaiian moorhen) on Moloka‘i, development of methods for preventing hybridization of kōloa maoli with feral mallards, and the reintroduction of kōloa maoli (Hawaiian duck; after effective measures are in place to minimize hybridization). It is also important to continue surveillance, planning, and response for avian influenza and other diseases. It would be beneficial to standardize monitoring of waterbird populations at managed sites.

d. Terrestrial Plants

i. Terrestrial Plants Recovery Strategy

Summary: The plant conservation strategy addresses 197 at-risk plants, 102 of which have fewer than 50 individuals remaining in the wild and are facing imminent extinction in the next five years. While plants face many threats, the most important are habitat loss and invasive species. The highest priority conservation actions are:

- Prevent Imminent Extinctions. Emergency actions are needed for 102 plants facing imminent extinction. This includes:
  - Survey,
  - Monitor,
  - Collect seeds and propagules,
  - Control local threats,
  - Propagate and outplant.
- Restore and manage habitats that can support the desired number of healthy self-sustaining populations for all 197 at-risk species. This involves managing the plant recovery footprint to:
  - Remove ungulates from recovery habitat and supporting ecosystem areas.
  - Control key invasive weeds, rats, and invertebrate pests locally until landscape-scale control is possible
  - Maintain 685 existing populations and establish or locate 891 new populations to meet MNLCP goals
  - Develop and implement a fire management strategy to protect the plant recovery footprint and supporting ecosystem footprint
  - Assess ongoing climate changes to determine the need for replacement recovery habitat
- Expand infrastructure to address the need for genetic/seed storage and the production of seedlings for creating or significantly augmenting over 1,500 populations of at-risk plants (with a total of 862,500 or more individual plants or seedlings) to meet MNLCP plant recovery goals. There will also be additional common matrix species needed to undertake general habitat restoration.
- Develop new techniques to address key threats at the landscape level (slugs, black twig borer, coconut rhinoceros beetle, naio thrips, ʻōhiʻa diseases, and ants):
  - Slugs: 101 species
  - Black twig borer: 21 species
  - Coconut rhinoceros beetle: 1 species
  - Naio thrips: 1 species

- Ants: 7 species
- General landscape threats: Rapid ʻŌhiʻa Death, flooding (60 species), landslides and erosion (96 species), drought (79 species), and loss of pollinators (44 species)

Other important actions include integrating species' occurrence and monitoring information into spatially explicit databases and preventing further declines of currently common species by protecting and managing remaining native ecosystems.

Planning process:

- Review of published recovery plans and relevant literature.
- Expert determination of species at risk, species facing imminent extinction, management needs and locations of emergency actions, recovery goals, and the sequencing of individual conservation strategies.
- Spatial prioritization processes involving expert involvement and algorithmic techniques were used to develop and refine the recovery footprint and the priority footprint.
- Action prioritization processes developed suites of potential conservation actions and their potential benefits to taxonomic groups.
- Other agencies provided data and analyses such as climate suitability maps for species or refined habitat quality maps.

ii. At-Risk Species

Appendix 4 provides additional information on the 197 at-risk plants covered by the MNLCP.

iii. Emergency Actions for Species Facing Imminent Extinction

The Plant Team identified 102 species facing imminent extinction in the next five years (Appendix 4). The Plant Extinction Prevention Program (PEPP) was created to prevent the extinction of these plants. PEPP focuses on native species with fewer than 50 remaining individuals in the wild. The program is particularly important to the 102 species needing emergency actions, since all these species have fewer than 50 wild individuals remaining and they are already being addressed by PEPP. PEPP has a standardized set of procedures that tracks the location and status of 685 known sites throughout Maui Nui and undertakes localized actions to address threats to remaining plants from rats, slugs, and invasive weeds. It also collects seeds, cuttings, and other tissue samples for genetic storage or seed banking. When possible, the program also outplants nursery-grown plants back into the wild.

The primary action needed is to expand PEPP/DOFAW capacity on Maui, Molokaʻi, and Lānaʻi to:

- Integrate databases to link stand-alone datasets to spatially explicit data systems and input any legacy hardcopy data into computer databases,
- Survey 95 percent of potential habitat,
- Annually monitor at least one population of each species,
- Control primary local threats (e.g., ungulates, rats, weeds, slugs),
- Collect and maintain ex situ collections from up to 50 maternal lines for each PEPP species (or 90 percent of the remaining live founders), and
- Outplant representatives of all available founders in at least one site.

iv. Recovery Goals

The MNLCP set initial recovery targets for plants through review and consultation with published recovery plans, experts, and the interagency Hawaiʻi and Pacific Plant Recovery Coordinating Committee (HPPRCC). The recovery goals consider multi-island distributions, plant life history characteristics, and the overall size of the species' distribution (Appendix 4).

There is a significant amount of information available on plant distributions, habitat, trends, and needed conservation actions. This includes data on plant populations from Hawaiʻi's Plant Extinction Prevention Program (PEPP), numerous federal recovery plans, the Hawaiʻi State Wildlife Action Plan, the HPPRCC's standardized recovery goals, a pilot study on East Maui using habitat prioritization methods to help identify useful plant conservation strategies, and a recent publication on the potential distribution changes that plants may face as a result of climate change (Fortini et al. 2013).

v. Recovery Habitat

Plants have four characteristics that make them particularly suitable for using algorithmically prioritized habitat selections.

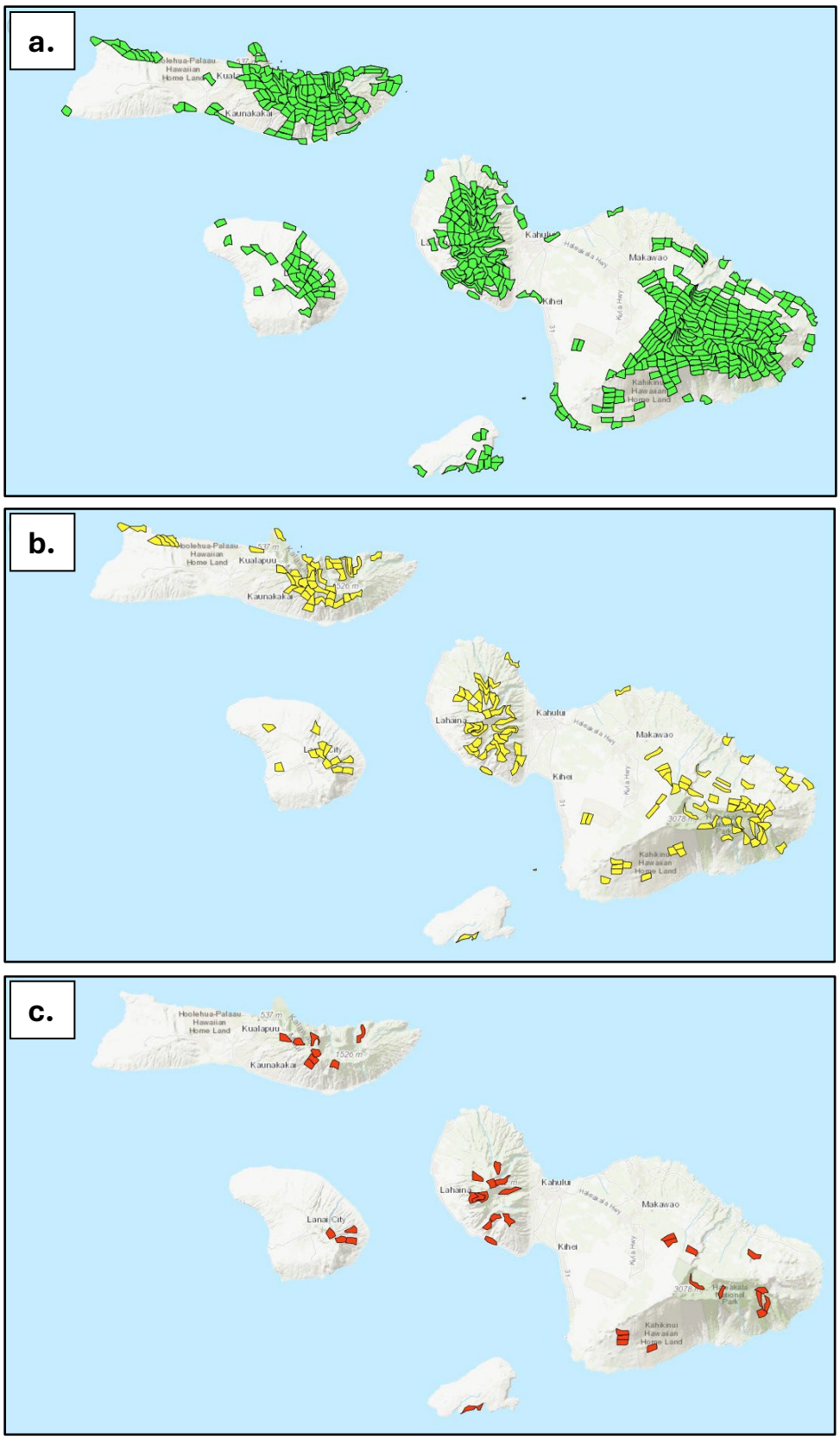
- There are many species in this taxonomic group (197), which makes manual assessments very complex.
- There is a large amount of relevant information on these species (as noted above).
- There are many options, across a broad landscape, for where to potentially locate conservation actions, leading to an extremely large number of potential solutions for managers to choose from.

- There is a dedicated group of species experts to guide the development and review of potential algorithmic solutions.

Development of the Recovery Footprint: The TFE-SCP process (Appendix 1) integrated detailed climate-suitability models, habitat-suitability layers, population targets, logistical constraints, and biogeographical representation requirements across islands and mountain masses to produce high-quality, climate-resilient recovery areas (recovery footprints). These footprints were reviewed extensively to ensure that they provide potential habitat for as many at-risk plants as possible and are logistically feasible to manage. Two species lack sufficient information to determine recovery areas. Consequently, only 195 of the 197 at-risk plant species were included in the recovery habitat analyses.

The process yielded four footprints:

- A recovery footprint of 317 planning units addressing 195 at-risk plants (Figure 9a)
- A priority footprint of 155 planning units that is a smaller subset of the recovery footprint and identifies the planning units needed to support three recovery populations for 195 at-risk plants (Figure 9b)
- A more focused implementation footprint that specifically addresses plants facing imminent extinction with very few known locations (36 PEPP species found in only three or fewer planning units, Figure 9c).
- An extremely narrow footprint restricted to only the 14 plants with a single known occupied or recently occupied planning unit (Figure 9d). The selection of this footprint for initial management was done with the expectation that management would quickly transition to the next higher footprint focused on species with three or fewer occupied planning units.


a.
Hoolehua-Palaau
Hawaiian
Home Land
Kualapuu
Kaunakakai
Kahului
Makawao
Kihei
Kahikinui
Hawaiian
Home Land
b.
Hoolehua-Palaau
Hawaiian
Home Land
Kualapuu
Kaunakakai
Lahaina
Kahului
Makawao
Kihei
Kahikinui
Hawaiian
Home Land
c.
Hoolehua-Palaau
Hawaiian
Home Land
Kualapuu
537 m
Kaunakakai
1526 m
Lahaina
Kahului
Makawao
Lanai City
Kihei
3078 m
Kahikinui
Hawaiian
Home Land

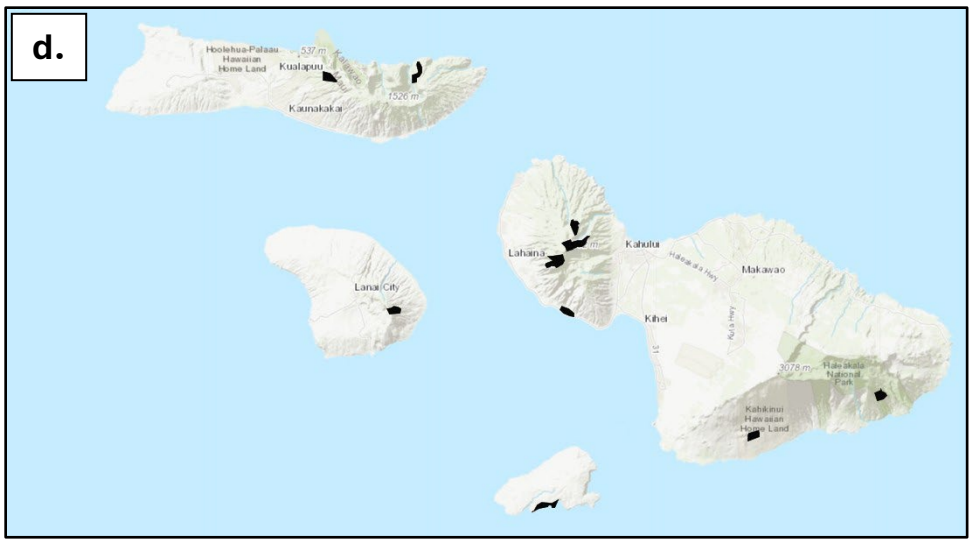


Figure 9. Plant Recovery Footprint (a) for 195 Maui Nui at-risk plants (green), (b) priority areas (yellow), (c) secondary implementation areas (red), and (d) primary implementation areas (black).

Key aspects of footprint development. The plant workflow represents a complete TFE-SCP pathway, from data compilation through algorithmic outputs and expert fine-tuning to a finalized, implementation-ready recovery footprint (see Appendix 1).

- The OptimTFE algorithmic process used the occurrence of known populations, habitat quality, biodiversity-reserve overlap, ungulate-managed areas, and a desire for a compact footprint to select a draft footprint (see Appendix 1).
- The draft footprint potentially provided recovery habitat for 180 of the 195 at-risk plant species and a total of 995 potential populations.
- The expert review portion of the TFE-SCP process identified potential recovery habitat for all 15 of the remaining species needing recovery habitat, with opportunities for an additional 449 potential populations, giving the plant footprint a final total of 1,444 potential plant populations at full buildout.

Within-planning-unit habitat selection. Planning units identify an area (roughly 2 km^2) that should be able to provide enough habitat to support a single population of an at-risk plant species. This is still a large area to search for and find the best location for on-the-ground actions like outplanting of new

populations, finding sites suitable for multiple species, or extending a fence line. To help with this process, several additional tools (Appendix 5) will be provided for managers to focus their field searches for suitable sites by consulting fine-scale GIS layers of species occurrence data, habitat quality, microhabitat assessments in the Find My Microhabitat Tool, GIS polygon layers showing Potential Recovery Sites based on climate suitability and habitat quality, as well as layers showing recent trends in climatic change.

vi. Conservation Actions

General: The composite graph of how plant species may benefit from the selected conservation Strategy 25 (Figure 3) displays the average responses for all plants, with the probability of persistence used as the response to management (this was defined as the probability of species groups having viable, self-sustaining populations over a 20-year time period). These data show that even with extensive plant-oriented management provided by Strategy 25, it is difficult to get plants, as a group, to a 50 percent probability of persistence (Figure 3, combined bar) and that probability drops to 45 percent when adjusted for feasibility (Figure 3, without the less feasible benefits).

The modified Priority Threat Management assessment (Appendix 2) grouped plants into six subgroups (five based on a cluster analysis of threat similarities and one group based on its long-lived life cycle).

The six plant subgroups:

Plant Group 1 (n = 42) represents the highest-threat group, with nearly all species facing all four threats: invasive invertebrates, invasive vertebrates, ungulates, and loss of avian pollinators. Species in this group face the most complex and compounded conservation challenges.

Plant Group 2 (n = 23) is characterized by invasive invertebrates and ungulates but notably lacks threats from invasive vertebrates and loss of avian pollinators. This group may benefit from targeted management of invertebrate pests alongside ongoing ungulate control.

Plant Group 3 (n = 30) is the lowest-threat group, with species primarily affected only by ungulates and largely free from invertebrate, vertebrate, and avian pollinator threats. This group may represent species with comparatively simple management needs.

Plant Group 4 (n = 75) is the largest cluster and is defined by the co-occurrence of invasive invertebrates, invasive vertebrates, and ungulates, but without loss of avian pollinators. This broad group shares a heavy multi-threat burden across many Maui Nui plant species.

Plant Group 5 (n = 13) is a small, distinct group threatened primarily by invasive vertebrates and ungulates, but not by invasive invertebrates or loss of avian pollinators, suggesting a unique threat profile that warrants focused attention.

Plant Group 6 (n = 13) is also a small group identified not by specific threats, but by the long-lived status of its species, which complicates reintroduction efforts and genetic management.

While the six plant groups show some differences in response under combination Strategy 25 (Figure 10), the most beneficial individual strategies for the groups are similar.

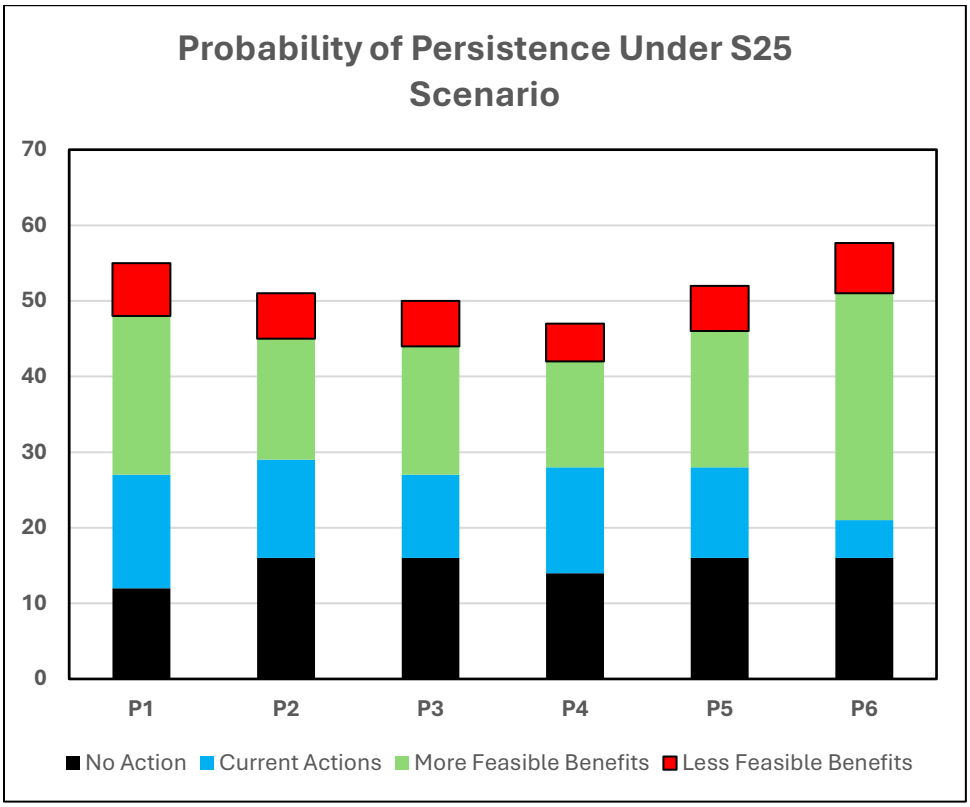


Figure 10. Expected persistence levels for plant groups (P1-6) under no action, current actions only, future actions with more feasible benefits, and future actions assuming maximum feasibility (see text for descriptions of plant groups).

Plant groups come close to reaching a 50 percent persistence threshold only due to synergistic effects of multiple actions, making it important to identify and implement synergistic combinations that get strong benefits, while also scheduling additional actions still needed to see maximum benefits.

Table 17. Suggested sequencing of Strategy 25 actions to best benefit plants. If all actions are implemented, it is anticipated that plants, depending on what subgroup they are in, will have a 42-48 percent chance of persistence (over 20 years).

Action	Description	Persistence (%)	Sequence
S13	Future Tools for Rare Plants and Inverts (Develop ex situ methods, establish invert and plant propagation facilities, develop controls for coconut rhinoceros beetles, black twig borers, and naio thrips)	26	1
S5	Terrestrial Habitat Management (for at-risk spp)	24	3
S11	Landscape Scale Rat Suppression	23	6
S3	Future Ungulate Fencing	23	2
S8	Invasive Invertebrate Predator Control (ants and yellow jackets)	22	5
S7	Invasive Vertebrate Predator Control (rats, cats, dogs, mongoose, barn owl, cattle egrets, rabbits)	21	4

The most important actions to undertake for plants are those actions leading to higher persistence rates (Table 17). However, some of the tasks are not yet implementable and in some instances actions that individually lead to lower persistence rates are foundational in their importance and need to be implemented before other tasks (e.g., ungulate fencing is usually needed before reintroducing plants into the wild). Therefore, a recommended sequencing of action strategies is:

- S13: Future Tools for Rare Plants and Invertebrates (Develop *ex situ* methods, establish invertebrate and plant propagation facilities, develop controls for slugs, coconut rhinoceros beetles, black twig borers, and naio thrips, and return pollination services for the 44 plant species that need avian pollinators). This task also needs to address infrastructure needs, databases, and training for future personnel.
- S3: Future Ungulate Fencing
- S5: Terrestrial Habitat Management (for at-risk species)
- S7: Invasive Vertebrate Predator Control (rats, cats, dogs, mongoose, barn owls, cattle egrets, rabbits)
- S8: Invasive Invertebrate Predator Control (ants and yellowjackets)
- S11: Landscape-scale Rat Suppression

There are caveats to this sequencing.

S10: New predator fencing ranked high only for one subgroup (Group 6). This is one of three strategies that address vertebrate predators like rats. Since this plant group is composed of longer-lived species, it may be more efficient to undertake predator fencing instead of local rat control efforts. S10 may not be necessary if S11 landscape-scale rat suppression is implemented.

Plant experts identified additional tasks that need to be accomplished to better manage PEPP species, specifically:

i. Link current information on the status and trends of populations with a spatially explicit data system. The maintenance and ownership of the database need to be determined.
ii. Develop a strategy for providing pollinators for plant populations needing bird pollinators.

e. Terrestrial Snails

i. Recovery Strategy

Summary: The current situation for terrestrial snails is dire— Hawai‘i once supported at least 750 native land snail species, and only 10–35 percent may still survive (Cowie et al. 2026), with many of these remaining species at risk of extinction. The snail conservation strategy ultimately seeks to restore all the existing species to healthy, self-sustaining populations in the wild.

The strategy addresses 49 at-risk snails, all of which are also facing imminent extinction in the next five years. The most important threat facing terrestrial snails is from introduced predators, especially the rosy wolfsnail (*Euglandina rosea*). The highest priority conservation actions are:

- Prevent Imminent Extinctions. This is the primary focus of the MNLCP's snail conservation strategy. It includes:
  - Establish in situ predator-proof exclosures (kīpuka kāhuli) to maintain at least three populations of each of the 49 species
  - Establish ex situ laboratory facilities on Maui to house all 49 snail species and provide additional ex situ backup populations.
  - Identify areas without *Euglandina* to aid search and rescue efforts aimed at securing a strong genetic representation for each at-risk species
- Restore and manage habitats for healthy self-sustaining populations for all 49 at-risk species. This is currently not feasible due to the lack of landscape-level control methods for introduced predators, especially *Euglandina*, but others as well (e.g., *Oxychilus alliarius*). Kīpuka kāhuli may eventually provide protected areas from which to stage reintroduction efforts across the landscape back into known historic locations.
- Develop new techniques to:
  - Control key threats at the landscape level (*Euglandina*, Jackson's chameleon, and rats)
  - Store genetic material and propagate all at-risk snail species.

Future actions include identifying recovery habitat for each of the 49 snails and then:

- Identifying infrastructure needs like landing zones and staging campsites

- Restoring habitat areas (e.g., removing *Euglandina*, rats, chameleons, and key weeds like strawberry guava).

Planning process:

- Expert determination of species at risk, species facing imminent extinction, management needs and locations of emergency actions, and the sequencing of individual conservation strategies.
- Spatial prioritization processes involving expert involvement and algorithmic techniques were used to develop and refine a system of in situ predator-proof exclosures to provide a safe area for three populations of each species.
- Action prioritization processes developed suites of potential conservation actions and their potential benefits to taxonomic groups.

ii. At-Risk Species

Information on the at-risk terrestrial snails of Maui Nui is found in Appendix 6. This appendix provides information on the number of known wild populations, federal and state regulatory status and species facing imminent extinction.

iii. Emergency Actions for Species Facing Imminent Extinction

The primary focus of the MNLCP with respect to terrestrial snails is on preventing species extinctions, rather than implementing landscape-scale recovery actions. Hawai‘i’s Snail Extinction Prevention Program shares this emphasis on extinction prevention and played a central role in developing the MNLCP snail strategy.

MNLCP will focus on the following objectives:

- Rescue populations of at-risk snails that may be eliminated in the next five years by *Euglandina*, rats, or chameleons and place them into:
  - *Ex situ* captive rearing facilities in laboratories and/or
  - Kīpuka kāhuli, which are small *in situ* snail exclosures with a predator-proof fence that keeps out invasive predators.
- Establish a network of 32 kīpuka kāhuli exclosures that:
  - Protects at least three populations of each species.
  - Prioritizes implementation to protect the most at-risk species first.
  - Locates climate-resilient kīpuka kāhuli across Maui Nui’s climatic regimes in a network that can accommodate all 49 species as cost-effectively as possible.

    - Can potentially serve as protected areas from which to stage reintroduction efforts across the landscape back into known historic locations.
- Establish an *ex situ* captive rearing program on Maui.
- Research landscape-level techniques to control invasive predators, especially *Euglandina rosea*
- Control predators in at-risk populations until snails can be rescued and placed in more secure facilities or protected exclosures (captive rearing or kīpuka kāhuli).
- Survey for additional populations of at-risk species.
- Survey for areas without *Euglandina*.

Efforts to control predators in key snail populations and bring species into captive breeding programs or snail exclosures are already being undertaken by DOFAW staff and the Snail Extinction Prevention Program. The MNLCP will significantly expand these efforts, as well as increase surveys for remnant populations of at-risk species and undertake research on *Euglandina* control.

The MNLCP used an innovative TFE-SCP approach that emphasizes transparency, flexibility to adapt to evolving management needs, and deep expert engagement to develop a network of predator-proof exclosures to protect at least three populations of each of the 49 at-risk snail species (Appendix 6).

Three Key Aspects of Developing the Kīpuka Kāhuli Network:

- Identification of Potential Sites. The OptimTFE algorithmic process used elevation, slope, ruggedness, native habitat status, fire risk, climate resilience, and habitat structure to develop a selection of viable sites for the kīpuka kāhuli network that would protect at least three populations of each of the 49 at-risk snail species (Wilhite et al. in prep a, and see Appendix 1).
- Selection of sites. After extensive expert review, the final network has 32 snail exclosures (4-6 exclosures already planned for construction, with 26 to be constructed in the future) that meet the species-specific habitat needs for each species.
- A priority network. A priority network has also been developed to focus the initial construction on exclosures that can protect those species now found only in captivity and the species closest to extinction (Figure 11).

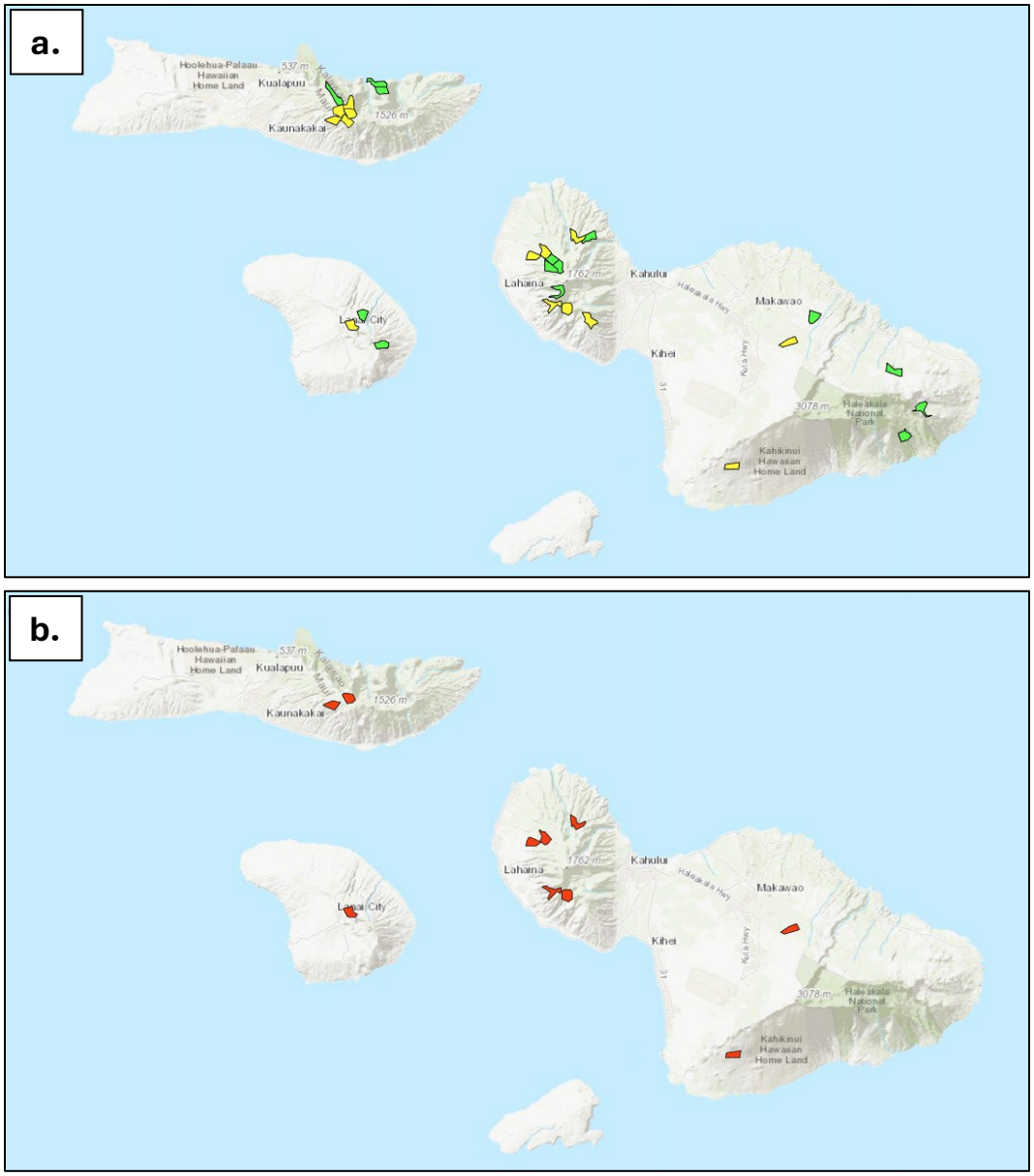


Figure 11. The network of predator-proof snail exclosures. (a) the full network of 32 exclosures (kīpuka kāhuli), with yellow sites being the priority sites to get at least one population of each species in an exclosure and the green sites needed to get three populations per species in exclosures. (b) sites for initial implementation to get the most at-risk species into an exclosure as soon as possible (red).

iv. Recovery Goals

The MNLCP seeks to advance the recovery of terrestrial snails by, first and foremost, preventing the imminent extinction of the remaining snail fauna. This first phase, or interim recovery step, is critical to overall snail conservation. Without it, there will be no snails to recover. As an interim recovery goal, the

MNLCP seeks to maintain three populations of each of the 49 at-risk snail species in predator-proof exclosures (called kīpuka kāhuli) and also in laboratory rearing facilities, if required. A subsequent future step will be to expand populations outside of kīpuka kāhuli.

v. Recovery Habitat

This portion of recovery planning is deferred, but the extinction-prevention efforts focused on kīpuka kāhuli (predator-proof exclosures) are expected to serve as protected areas from which to stage reintroduction efforts across the landscape back into known historic locations. Many of the kīpuka kāhuli are located in close proximity to known snail occurrences, with 39 of the 49 species in the exclosures having a nearest known occurrence of less than 2 km.

vi. Conservation Actions

Figure 3 shows the expected benefits to snails from the best composite action strategy for snails. The survival of terrestrial snails is highly dependent on counteracting a single threat driving the extinction crises, *Euglandina* predation (Table 18). If effective control measures are found and implemented, then prospects for the snails increase to over a 50 percent chance of persistence.

When efforts to protect and restore habitats for snail recovery are undertaken, it will be important to address infrastructure needs, fire management, and biosecurity measures needed to protect recovery sites.

Table 18. Suggested sequencing of Strategy 25 actions to best benefit snails. If all actions are implemented, it is anticipated that snails will have up to a 63 percent chance of persisting through the 20-year timeframe of the MNLCP.

Action	Description	Persistence (%)	Sequence
S10	New Predator Fencing (snails, rodents, mongoose, cats)	44.0	1
S13	Future Tools for Rare Plants and Inverts (Develop ex situ methods, establish invert and plant propagation facilities, develop controls for coconut rhinoceros beetles, black twig borers, and naio thrips)	36.9	2
S8	Invasive Invertebrate Predator Control (ants and yellow jackets)	35.0	4
S11	Landscape Scale Rat Suppression	31.9	6
S7	Invasive Vertebrate Predator Control (rats, cats, dogs, mongoose, barn owl, cattle egrets, rabbits)	31.0	5
S12	Existing Tools for Rare Plants and Inverts	30.5	3

The near-term focus is on preventing extinctions, developing methods to address key threats at a landscape level (e.g., *Euglandina*), and rescuing wild at-risk populations.

Landscape-level habitat management aimed at snail recovery and removal from the at-risk list is deferred to a point after there are viable measures to control *Euglandina*.

A suggested sequencing of actions is provided for the near term and distant future.

Near-term:

1. New predator fencing for snails
2. Future Tools for Rare Plants and Invertebrates (especially improvements in captive rearing, developing landscape-level control methods for *Euglandina*, *Oxychilus alliarius*, chameleons, and rats). This should include developing and maintaining databases on in situ and ex situ populations.
3. Existing Tools for Rare Plants and Invertebrates

Future:

4. Landscape control of invasive invertebrate predators (*Euglandina* and other introduced invertebrates)
5. Localized control of invasive vertebrate predators (rats and chameleons)
6. Landscape-scale rat suppression

f. Terrestrial Arthropods

i. Recovery Strategy

Summary: The Terrestrial Arthropod Team identified 172 species at risk of extinction, 9 of which are facing imminent extinction in the next five years. The most important threats are habitat loss, ecosystem degradation, and predation by or competition with non-native wasps and ants. In addition, insects that are dependent on a specific host plant species can be at risk if their host plant species declines or goes extinct. Since this taxonomic group has thousands of species and high levels of endemism and is generally poorly known, it is likely that many more species may also be at risk of extinction.

The most important conservation actions to undertake for terrestrial arthropods are:

- Prevent Imminent Extinctions. Emergency measures to stabilize the last wild populations of nine terrestrial arthropod species facing imminent extinction.
  - Ensure all nine species have at least one extant or recently occupied site targeted for emergency action (e.g., ant control, protection from fire, outplanting of host plants).
  - Initiate emergency threat abatement at identified sites.
- Initiate detailed review of listed species and well-documented taxa to identify other species facing imminent extinction.
- Restore and manage habitats for arthropod species having adequate information to develop spatially explicit recovery habitats. This involves managing the terrestrial arthropod recovery footprint to:
  - Remove ungulates from recovery habitat
  - Control key invasive weeds, rats, and invertebrate pests
- Establish invertebrate propagation facilities.
- For species currently lacking the occurrence data needed to identify where to locate recovery populations, undertake surveys to fill data gaps and update the recovery footprint to provide recovery habitat.

Planning process:

- Expert determination of species at risk, species facing imminent extinction, management needs and locations of emergency actions, recovery goals, and the sequencing of individual conservation strategies.
- Spatial prioritization processes involving expert involvement and algorithmic techniques were used to:

    - Develop and refine the recovery footprint and the priority footprint for arthropods
    - Identify potential multi-species survey areas for arthropods lacking the information needed to establish recovery habitat
- Action prioritization processes developed suites of potential conservation actions and their potential benefits to taxonomic groups.

ii. At-Risk Species

Of the 172 at-risk terrestrial arthropods, eight are federally listed as endangered. Nine terrestrial insects are currently thought to be facing imminent extinction. The full list of these species is in Appendix 7.

iii. Emergency Actions for Species Facing Imminent Extinction

Nine yellow-faced bee species in the genus *Hylaeus* are facing imminent extinction in the next five years (Table 19). Four of these species are listed under federal and State statutes. A fifth *species, Hylaeus assimulans,* is listed but not currently at imminent risk. The primary actions to prevent extinctions in this group are to undertake habitat protection (fencing) and restoration of native vegetation. Existing fences should be maintained.

Table 19. Yellow-faced bees facing imminent extinction in the next five years. Columns contain the number of populations currently known, the federal and State endangered species status (E), and the conservation actions needed.

Terrestrial Arthropods	# Known Populations	ESA	Facing Imminent Extinction	Emergency Actions
Hylaeus anthracinus	5	E	Yes	Habitat restoration and fencing at known sites.
Hylaeus facilis	1	E	Yes	Habitat restoration and fencing at known sites.
Hylaeus flavipes	2		Yes	Fencing at Olowalu and Kahue. Habitat restoration at known sites.
Hylaeus hilaris	1	E	Yes	Fencing at Ilio Point. Habitat restoration at known sites.
Hylaeus laetus	1		Yes	Habitat restoration and fencing at known sites.
Hylaeus longiceps	3	E	Yes	Fencing at Ilio Point. Habitat restoration at known sites.
Hylaeus paumako	1		Yes	Habitat restoration and fencing at known sites.
Hylaeus psammobius	2		Yes	Fencing at Papanalahoa. Habitat restoration at known sites.
Hylaeus volatilis	1		Yes	Habitat restoration and fencing at known sites.

iv. Recovery Goals

The MNLCP established an initial target of having five healthy, self-sustaining populations for each arthropod species. There was a very wide range of information on the 172 at-risk arthropods. A total of 32 arthropod species had enough occurrence and habitat information to identify recovery habitat. The remaining 140 species had insufficient information to identify recovery habitat.

v. Recovery Habitat

The recovery footprint (Figure 12) for terrestrial arthropods covers the interim recovery habitat needs of 31 species and the full recovery needs of one species (*Manduca blackburnii*). The recovery footprint was algorithmically developed using the TFE-SCP process for species having adequate data to allow for prioritizing habitat (Appendix 1). This group includes native *Drosophila*, *Hylaeus* (yellow-faced bees), and *Manduca* (sphinx moth).

The TFE-SCP and OptimTFE processes used habitat quality, known arthropod occurrences, and alignment with the plant recovery footprint to identify a recovery footprint that included (when feasible) at least five recovery habitat areas for all species with adequate occurrence information.

In addition to the recovery footprint, a priority footprint was developed that identifies areas with populations of the nine arthropod species facing imminent extinction.

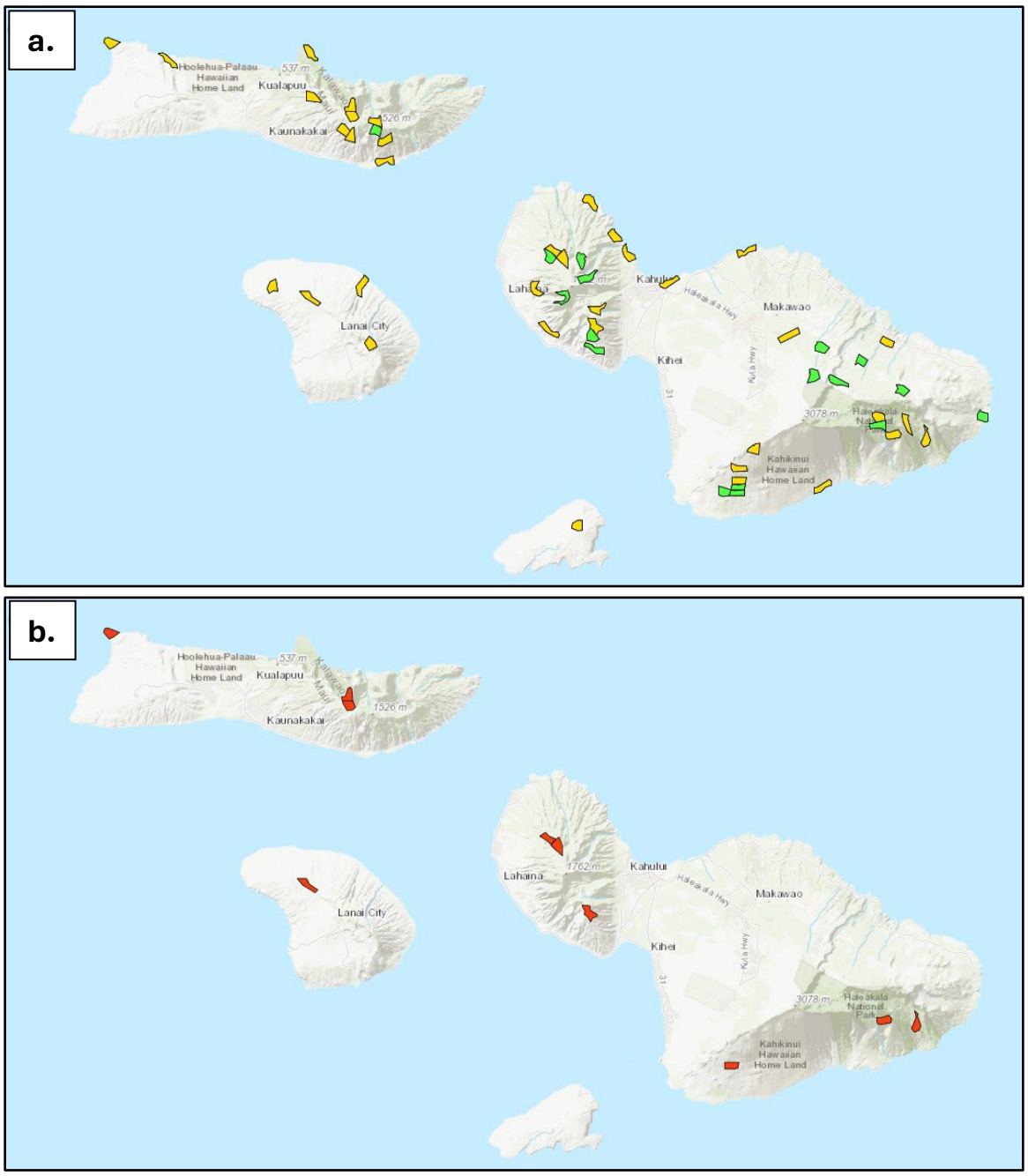


Figure 12. Recovery Footprint for (a) terrestrial arthropods (green), with priority implementation areas (red) and (b) areas selected for initial implementation of conservation actions.

vi. Conservation Actions

General: Figure 3 shows how terrestrial arthropod species, as a group, may benefit from the MNLCP's selected conservation Strategy S25. These data show that with extensive plant- and invertebrate-oriented management provided by Strategy S25, arthropods, as a group, may reach a 65 percent

probability of persistence (Figure 3, combined bar) and that probability drops to 58 percent when adjusted for feasibility (Figure 3, without the less feasible benefits).

The modified Priority Threat Management assessment (Appendix 2) addressed four groups of arthropods (Hylaeus bees, Drosophila, Manduca, and Megalagrion; though this plan groups Megalagrion with Aquatic Invertebrates instead of Terrestrial Arthropods—leaving arthropods with three analyzed groups).

This assessment indicates that arthropods are expected to do well under these management actions, but it is important to look at each group's performance to see if there are significant differences (Figure 13).

All three groups show similar outcomes under Strategy S25 with each having from the mid-50s to mid-60s estimated persistence depending on the feasibility of the actions.

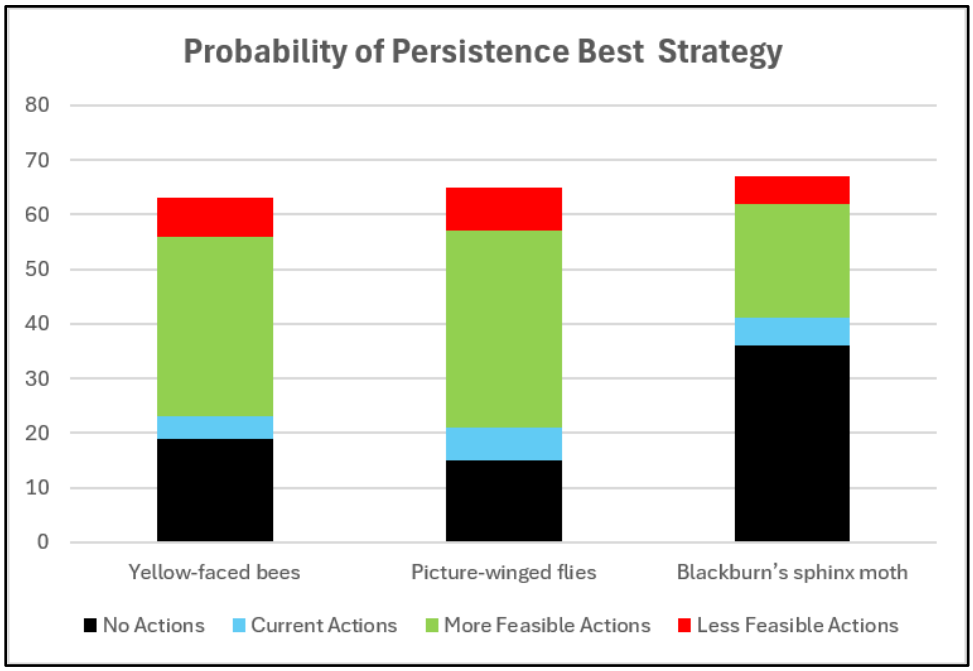


Figure 13. Expected persistence levels for terrestrial arthropod groups under no action, current actions only, future actions with feasible benefits, future actions assuming maximum feasibility.

While the end results seem similar for the three groups, the conservation actions most beneficial to a group have both similarities and differences (Table 20).

- All three groups have their best-performing action as S13: Future tools for rare plants and invertebrates, which includes invertebrate propagation facilities.
- All share three top-five actions:
  - S12: Current tools for rare plants and invertebrates,
  - S8: Invasive Invertebrate Pest Control for ants and yellow jackets,
  - S5: Terrestrial Habitat Management to Support Sensitive Species
- However, each group has one top-five strategy that is not a top five in the other groups:
  - Hylaeus has more benefit from S3 (future ungulate fencing),
  - Drosophila has more from S10 (future predator-proof fences),
  - Manduca has more from S16 (minimization and modification of infrastructure).

Table 20. Benefits of single strategies for yellow-faced bees, picture-wing flies, and Blackburn's sphinx moth; S1 is the Counterfactual (no action option), Rank is the highest to lowest potential benefit and Sequencing is the preferred order for implementation of the five most beneficial action strategies for each group. Green cells are areas of shared priorities among the three groups.

		Yellow-faced bees		Picture-winged flies		Blackburn's sphinx	
Strategy	Description	63% Persistence	Sequence	65% Persistence	Sequence	67% Persistence	Sequence
S1	No Action	19		15		36	
S2	Current Ungulate Fences	25		21		41	
S3	Future Ungulate Fences	27	1	26		41	
S4	Fenceless Ungulate Control	23		22		39	
S5	Habitat Management	34	4	33	2	47	2
S6	Wetland Management	24		20		39	
S7	Vertebrate Predator Control	25		27		40	
S8	Invertebrate Predator Control	40	2	34	4	46	4
S9	Current Predator Fences	21		18		39	
S10	Future Predator Fences	22		31	5	39	
S11	Landscape Rat Control	20		23		39	
S12	Current Plant/Invert Tools	36	3	44	3	42	5
S13	Future Plant/Invert Tools	41	5	44	1	51	1
S14	Current Avian Disease	19		15		38	
S15	Future Avian Disease	19		15		38	
S16	Infrastructure Minimized	19		15		43	3
S17	Vertebrate Rehabilitation	19		15		39	

For Hylaeus bees, species experts recommend starting recovery implementation with ungulate fencing, followed by invasive invertebrate pest control (ants), then continuation of existing tools for rare plants and invertebrates, terrestrial habitat management, and future tools for rare plants and invertebrates. In some areas, impacts from rodents (mice) on host plants may need to be controlled locally.

vii. Surveys for Species with Insufficient Data to Identify Recovery Habitat

As noted in the Recovery Goals section, 140 arthropod species had insufficient information to identify spatially explicit recovery habitat. To assist managers in filling data gaps in distributions and habitat parameters, the Spatial Prioritization Team and Terrestrial Arthropod Team used the TFE-SCP and OptimTFE processes to develop priority maps for at-risk arthropod surveys. The teams used host-plant associations, habitat quality, accessibility, and other criteria to suggest where multi-species surveys should be considered (Figure 14). This is not a recovery footprint, but recommendations on where field surveys are most likely to find new populations of these at-risk arthropods and generate the data needed to develop long-term management plans.

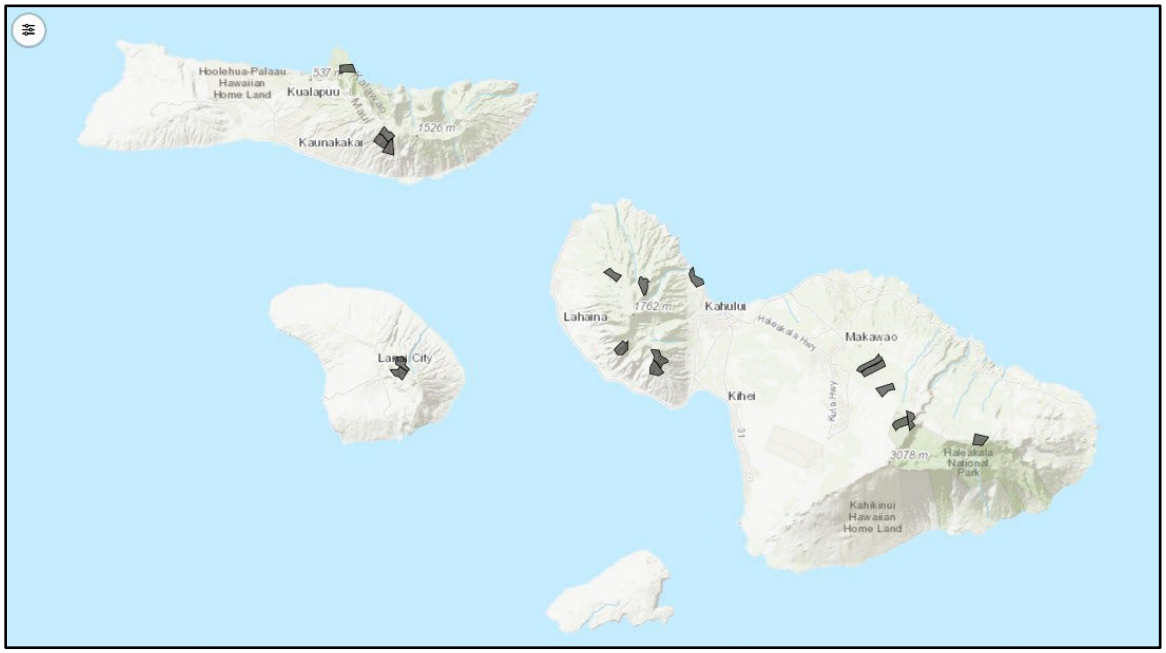


Figure 14. Suggested areas for multi-species area surveys. These areas are thought to have host plants or other features indicating that they may potentially have target species.

g. Aquatic Invertebrates

i. Aquatic Invertebrate Recovery Strategy

Summary: The aquatic invertebrate conservation strategy addresses 11 at-risk species, one of which is facing imminent extinction in the next five years (Table 21). The most important threats facing aquatic invertebrates are habitat loss and predation by aquatic invasive species—both vertebrate species (fish and frogs) and invertebrates (shrimp).

The most important conservation actions are:

- Emergency measures for the only known Maui Nui population of *Megalagrion nesiotes*, including continuing site protection, monitoring the population, and preventing the introduction of aquatic invasive species.
- Restore and manage habitats that can support the desired number of healthy self-sustaining populations for all 11 at-risk species. This involves managing the aquatic invertebrate recovery footprint to:
    i. Establish and maintain minimum instream flows (without increasing the spread of invasive aquatic species)
    ii. Control invasive aquatic species.
    iii. Manage wetland habitats for at-risk species
    iv. Control invasive invertebrate species
    v. Build and maintain ungulate fencing
- Future plant and invertebrate tools

Seven species inhabit anchialine pools and four are more often associated with streams, seeps, or wetlands. Four are protected under federal and Hawai‘i endangered species statutes.

Planning Process: Expert determination of species at risk, species facing imminent extinction, management needs and locations of emergency actions, recovery goals, recovery footprint, priority footprint, and sequencing of individual conservation strategies.

Table 21. At-risk aquatic invertebrates of Maui Nui with information on the number of planning units with a known wild population, Endangered Species Act (ESA) status (E=endangered), status with respect to facing imminent extinction, species with interim or full recovery goals, species targeted for multi-species area surveys, or basic species-specific surveys. Species with an asterisk (*) occur in anchialine pools.

Aquatic Invertebrates	All Known Occupied PUs	ESA Listing	Facing Imminent Extinction	Recovery		Insufficient Data	
				Interim Recovery	Full Recovery	Area Surveys	Species Surveys
Antecaridina lauensis*	1			1			
Calliasmata pholidata*	1			1			
Halocaridina rubra*	7				7		
Lymnaeidea spp.	0						1
Megalagrion nesiotes	2	E	X		4		
Megalagrion pacificum	27	E			46		
Megalagrion xanthomelas	18	E			40		
Metabetaeus lohena*	7				7		
Palaemonella burnsi*	1			1			
Pele ramseyi*	1			1			
Procaris hawaiana*	1	E		1			
Total (11)		4	1	5	5	0	1

ii. At-Risk Species

Eleven species were determined to be at risk of extinction, with one species, *Megalagrion nesiotes*, facing imminent extinction in the next five years (Table 21).

iii. Emergency Actions for Species Facing Imminent Extinction

One aquatic invertebrate was identified as facing an imminent risk of extinction in the next five years, *Megalagrion nesiotes*, the flying earwig Hawaiian damselfly. This species is a dark-colored, weak flier that inhabits deep forest environments, usually resting on vegetation and steep, moist, fern-covered banks. The naiads (immature form) are thought to be terrestrial or semi-terrestrial, living in damp leaf litter. It is listed as endangered and historically found only on Maui and Hawai‘i Island, with a single population on each island. The species is highly susceptible to habitat degradation caused by invasive feral pigs, as well as predation from introduced fish, ants, and other non-native insects. Because it is restricted to such a localized geographic area, natural disasters or climate change heavily threaten its long-term survivability.

Actions Needed: Protect known site from ungulate damage, prevent predation from invasive species, and monitor at least annually.

iv. Recovery Goals

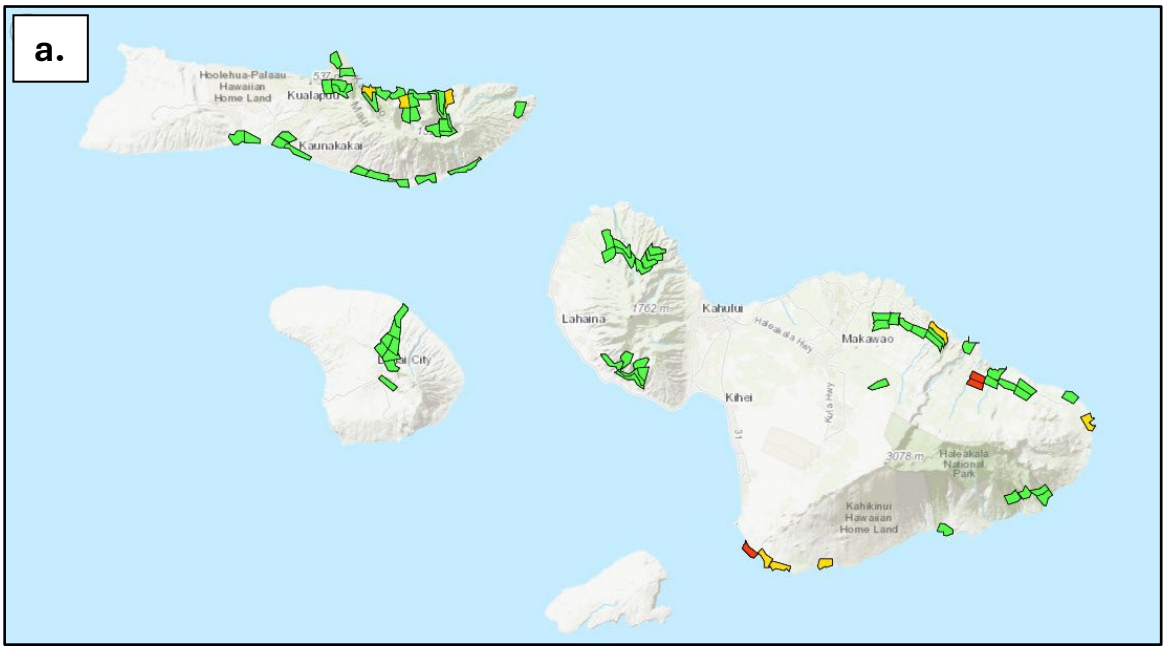
Initial planning for recovery populations tried to reach a recovery goal of 10 populations for each species. These initial targets were modified to take into account multi-island species and available habitat. Five species should be able to reach full recovery goals, five will have interim recovery goals of one population, and one species has no identified recovery area.

v. Recovery Habitat

The Aquatic Invertebrate Team and other species experts used occurrence data, recovery plans, species reports, Hawai‘i’s State Wildlife Action Plan, and scientific literature to identify an aquatic invertebrate recovery footprint for 10 of the 11 species.

The recovery footprint (Figure 15) for aquatic invertebrates covers the interim recovery habitat needs of 5 species, the full recovery needs of 5 species, and is currently lacking information on *Lymnaeidae* spp. (freshwater pond snails).

The aquatic invertebrates are obviously closely tied to streams, wetlands, and anchialine pools. These are mapped in relation to the recovery footprint to emphasize the need for a somewhat different landscape perspective when it comes to protecting and restoring sites for at-risk aquatic species.



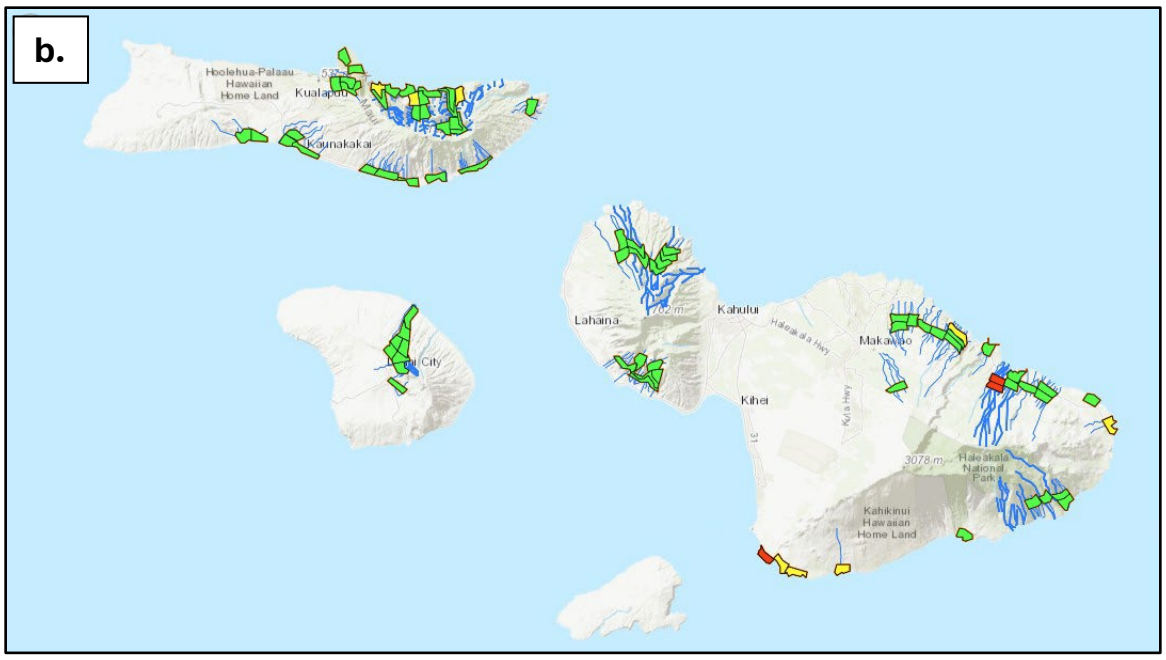


Figure 15. Recovery Footprint for 10 at-risk Maui Nui aquatic invertebrates (green), with priority implementation areas shown in yellow and initial implementation areas in red; without (a) and with (b) associated streams.

vi. Conservation Actions

Figure 3 shows a composite graph of the three groups of aquatic invertebrates that were evaluated using a modified Priority Threat Management process (*Megalagrion*, *Procaris*, and *Lymnaeidae)*. The graph indicates that when combined, aquatic invertebrates may reach an average likelihood of persistence above 50 percent.

As with plants and terrestrial arthropods, it is important to also look at each subgroup to see if there are significant differences in how potential conservation actions may benefit each group (Figure 16, Table 22). Megalagrion and Lymnaeidae did best under the composite Strategy 25, a plant- and invertebrate-centric strategy, while *Procaris* did slightly better under the bird-centric Strategy 27 (one percentage point difference).

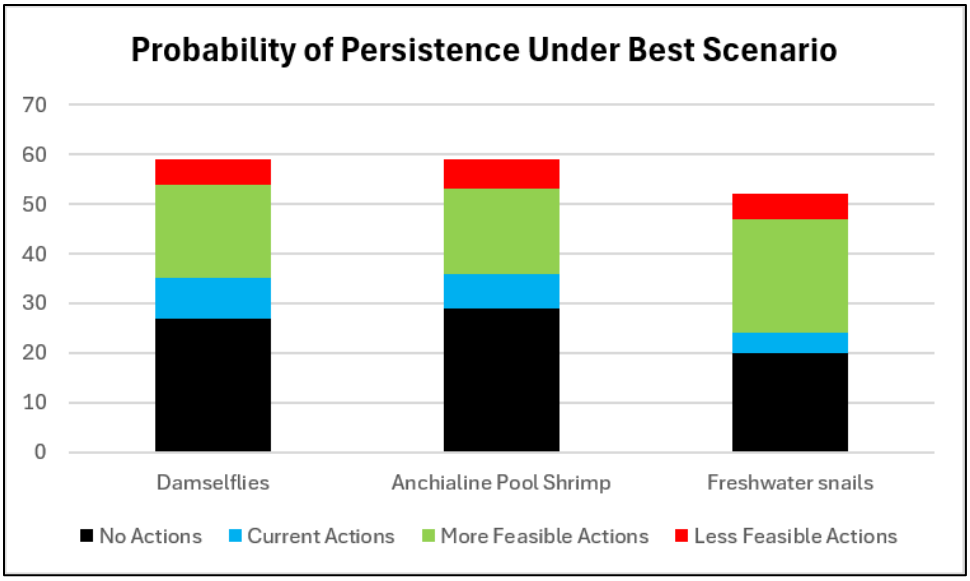


Figure 16. Expected persistence levels for aquatic invertebrate groups under no action, current actions only, future actions with feasible benefits, future actions assuming maximum feasibility.

Table 22. Benefits of single group strategies for *Megalagrion*, *Procaris*, and *Lymnaeidae.* S1 is the Counterfactual (no action option), Persistence is the estimated percent persistence of the species group if that strategy is implemented, the number in the blue heading is the overall estimated persistence if combination strategies S25 and S27 are implemented, Sequence is the suggested order of implementation of the individual strategies (numbers indicate priority order and unranked medium actions).

		Megalagrion		Procaris		Limnaeidae	
Strategy	Description	59% Persistence	Sequence	59% Persistence	Sequence	52% Persistence	Sequence
S1	No Action	27		29		20	
S2	Current Ungulate Fences	31		32		23	
S3	Future Ungulate Fences	34	3	33		25	
S4	Fenceless Ungulate Control	34		31		22	
S5	Habitat Management	36	4	31		27	
S6	Wetland Management	43	1	41	1	30	2
S7	Vertebrate Predator Control	39		31		30	4
S8	Invertebrate Predator Control	38	2	31		33	1
S9	Current Predator Fences	29		30		20	
S10	Future Predator Fences	32		30		20	
S11	Landscape Rat Control	31		30		24	
S12	Current Plant/Invert Tools	47	Medium	30		28	
S13	Future Plant/Invert Tools	51	Medium	33		33	3
S14	Current Avian Disease	27		30		20	
S15	Future Avian Disease	27		30		20	
S16	Infrastructure Minimized	31		33		24	
S17	Vertebrate Rehabilitation	27		30		25	

These three groups have different responses to the various conservation actions.

Procaris shows a narrow range of responses to the different strategies (12-point spread) and a single high benefit strategy (S6) with the others producing only a 1–4 percentage point improvement over a no action alternative.

The Lymnaeidae have a similar 13-point spread but lack a standout strategy. Instead, four strategies are clustered in the 30–33 percent persistence range (S6, 7, 8, and 13).

Megalagrion has a 24-point spread. Megalagrion has three strategies with 10+ point increases (S6, 7, and 8) and two with 20+ point increases over the no action alternative (S12 and 13).

Maintaining individual assessments of these groups may be more appropriate than lumping them into a single aquatic invertebrate strategy.

All three groups need Strategy 6 (Stream/Wetland Habitat Management to Support Sensitive Species), which is arguably the most important strategy to implement first and has two very important tasks for all three groups embedded in it:

- Modify stream barriers/diversions in areas across East Maui where stream flow is reduced and there is potential for lateral entrainment (Figure 17). There is an additional qualifier to discuss at this point. Whenever barriers or diversions are deliberately removed or eliminated due to a structural failure, it is important to assess the need to install a barrier that will allow downstream flows but prevent the upstream migration of aquatic invasive species. It is preferable that existing barriers and diversions on key streams are evaluated and plans developed soon to prevent invasive migrations before structures fail or are removed.
- Remove or prevent the introduction of invasive species from lowland streams below 366 meters (1,200 feet) across Maui Nui. Conduct continuous monitoring to maintain invasive species-free streams and waters (Figure 17).

For Procaris, it is not obvious what other strategies should be implemented next, since the other individual strategies provide only small gains over no action. However, the combined benefits from either composite Strategy 25 or 27 result in over a 50 percent chance of persistence.

For the Lymnaeidae, other strategies for early implementation include:

- S7 (Invasive vertebrate pest control)

- S8 (Invasive invertebrate pest control)
- S13 (Future tools for rare plants and invertebrates)
- It is probably useful to also do foundational actions like S3 (Future ungulate fences) and S5 (Terrestrial habitat management to support sensitive species) to ensure habitat quality.

For Megalagrion, there are several other strategies that seem to be logical choices for early implementation. These include:

- S3 (Future ungulate fences)
- S5 (Terrestrial habitat management to support sensitive species)
- S7 (Invasive vertebrate pest control)
- S8 (Invasive invertebrate pest control)
- S12 (Current tools for rare plants and invertebrates)
- S13 (Future tools for rare plants and invertebrates)

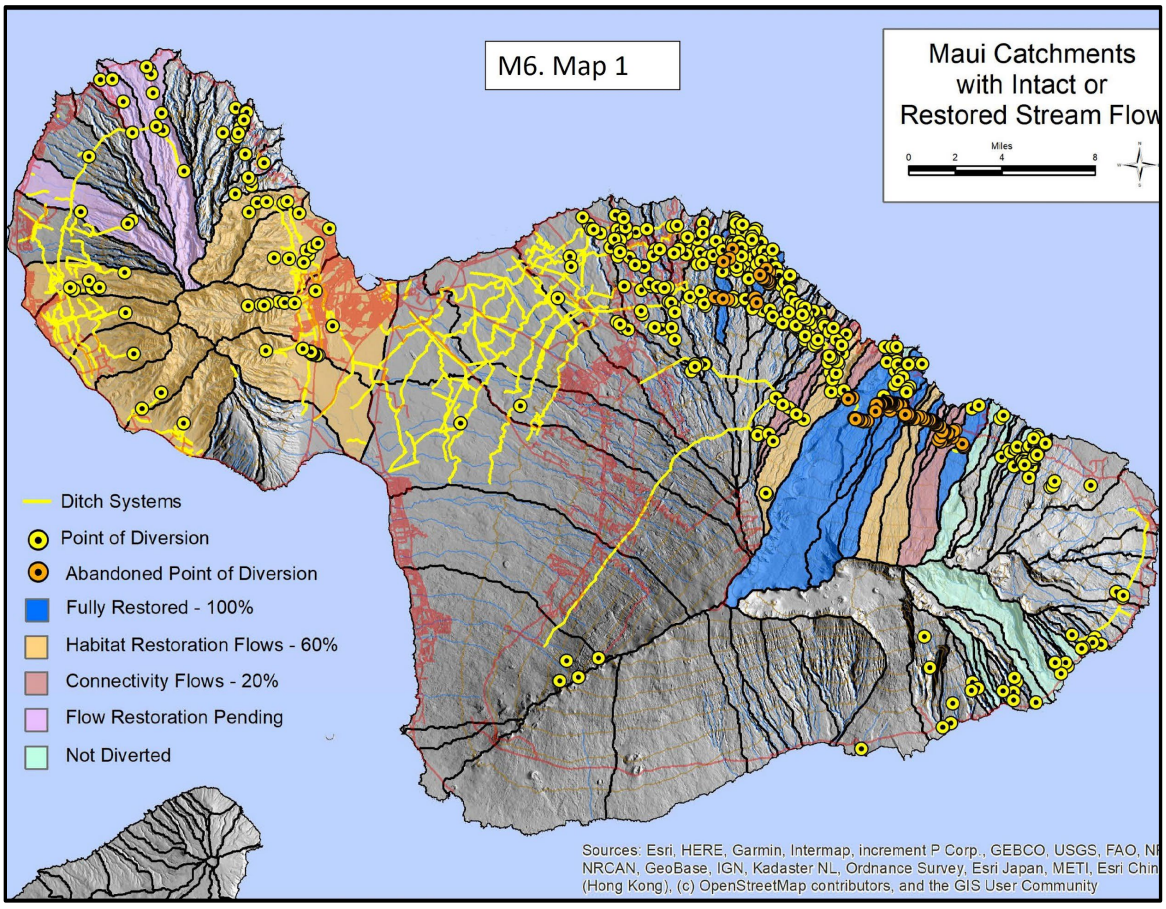


Figure 17. Maui catchments with intact or restored stream flows.

h. Hawaiian Hoary Bat (ʻōpeʻapeʻa)

i. Recovery Strategy

Summary: The Hawaiian hoary bat strategy focuses on reducing or eliminating incidental take of bats by wind turbines, protecting native forests and wetlands, and developing methods to increase bat productivity.

The bat is listed as endangered but is not thought to be facing imminent extinction in the next five years (Table 23). There is no reliable population estimate for bats in Maui Nui. Threats to bats include collisions with wind turbines, habitat loss, and, potentially, pesticide exposure and predation by invasive vertebrate predators.

The MNLCP strategy to conserve the Hawaiian hoary bat involves:

- Preventing the incidental take of bats from various human activities (especially wind energy turbines),
- Protecting forested areas and wetlands included in the MNLCP recovery footprints for forest birds, seabirds, waterbirds, plants, aquatic invertebrates, and terrestrial arthropods,
- Reducing populations of invasive vertebrate predators (especially feral cats and rats) across Maui Nui,
- Developing methods to quantify population trends,
- Identifying actions that can increase productivity in bat populations, and
- Determining whether the presence or absence of flight lanes in forests is an important factor for bats.

Planning Process: Core and Implementation Team determination of species at risk, species facing imminent extinction, recovery goals, recovery footprint, priority footprint, and sequencing of individual conservation strategies.

ii. At-Risk Species

The Hawaiian hoary bat occurs on Maui, Molokaʻi, and Lānaʻi and has been observed on Kahoʻolawe (USFWS 2021). It is not known if these represent three (or four) separate populations or a single metapopulation. While the overall size and trend of bat populations on Maui Nui are not known, there is some indication that bats on the island of Hawaiʻi may be stable or slightly increasing (Gorresen et al. 2013).

Bats typically roost in trees and forage across multiple habitats ranging from native to non-native forests, wetlands, riparian areas, and other habitats. They have been found roosting in ʻōhiʻa (*Metrosideros polymorpha*), pūhala

(*Pandanus tectorius*), coconut palms (*Cocos nucifera*), kukui (*Aleurites moluccana*), kiawe (*Prosopis pallida*), avocado (*Persea americana*), shower trees (*Cassia javanica*), pūkiawe (*Leptecophylla tameiameiae*), fern clumps, eucalyptus (*Eucalyptus* spp.), Cook pine (*Araucaria columnaris*), and Norfolk Island pine (*Araucaria heterophylla*) stands. Their primary food includes nocturnal flying beetles and moths (Jacobs 1999, Todd 2012, and Pinzari et al. 2019).

Table 23. At-risk terrestrial mammals (Hawaiian hoary bat) of Maui Nui with information on the estimated number of individuals, federal and State regulatory status (E = endangered, 1 species), status with respect to facing imminent extinction, species with full recovery or interim recovery (1 species) goals, species targeted for multi-species area surveys, or basic species-specific surveys.

Mammals	Current Populations	ESA Listing	Facing Imminent Extinction	Recovery		Insufficient Data	
				Interim Recovery	Full Recovery	Area Surveys	Species Surveys
Hawaiian Hoary Bat	3	E		X			
Total (1)		1	0	1	0	0	0

iii. Emergency Actions for Species Facing Imminent Extinction

The bat is not at imminent risk of extinction in the next five years.

iv. Recovery Goals

Establishing quantitative recovery goals for the Hawaiian hoary bat is challenging. The interim recovery goal is to maintain the bat at or above its current population level and expand the amount of forest habitat, especially native forests, across Maui Nui.

v. Recovery Habitat

The MNLCP will provide forest and wetland habitats for the Hawaiian hoary bat through the combination of native habitats in the recovery footprints for forest birds, seabirds, waterbirds, terrestrial plants, aquatic invertebrates, and terrestrial arthropods (Figure 18).

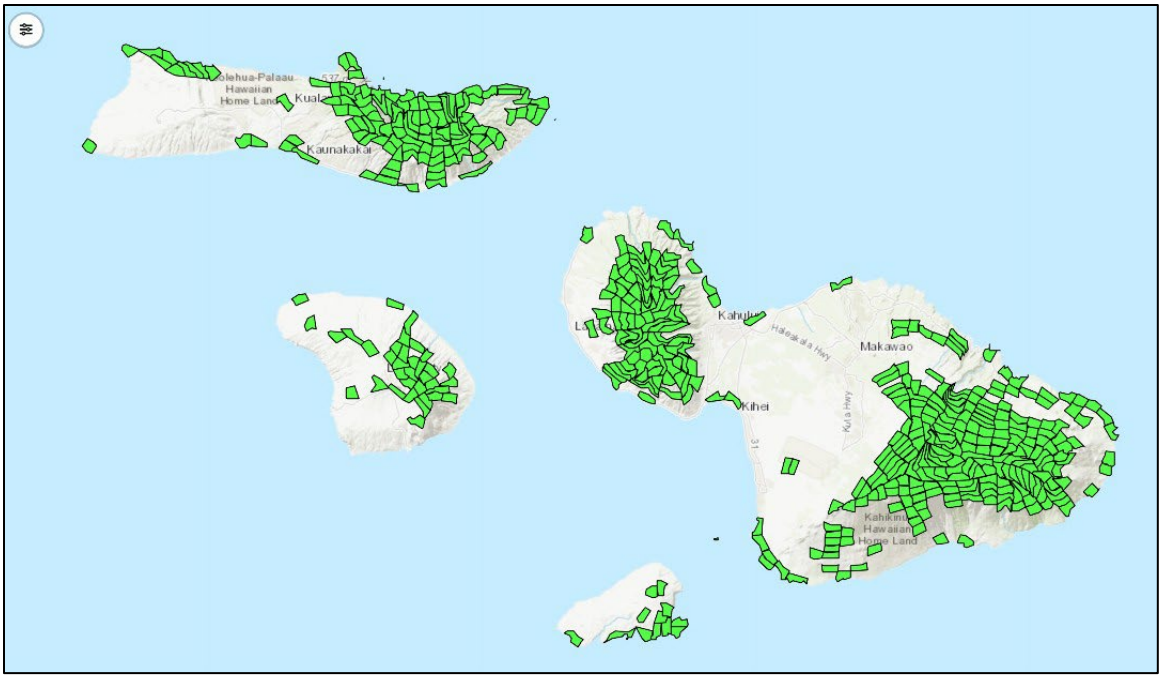


Figure 18. Recovery footprint for the Hawaiian hoary bat (ʻōpeʻapeʻa) composed of the recovery footprints for forest birds, seabirds, waterbirds, plants, aquatic invertebrates, and terrestrial arthropods. There are no priority footprints for the bat.

vi. Conservation Actions

A modified Priority Threat Management process was used to determine how various conservation actions might affect the future persistence of the bat on Maui Nui (Figure 3 and Appendix 2). The bat had the highest baseline persistence of any species assessed (37 percent without conservation actions). It also reached at least a 50 percent probability of persistence in seven of the 27 strategies. Composite Strategy 27, the most bird-centric strategy, appears to provide the highest level of persistence for the Hawaiian hoary bat, since the estimated persistence if this strategy were fully implemented was 63 percent.

The most beneficial of the single strategies is S16 (Minimization and modification of infrastructure) which targets minimization of bat mortality at wind energy farms (Table 24). This is also the most strategic of potential actions because:

- There are no accurate population estimates or trends that can be used for population viability analysis (PVA) to determine how many annual bat deaths a population can withstand before becoming unsustainable.

- No conservation measures have been documented to offset bat wind turbine mortality by reducing other causes of bat mortality or by increasing the reproductive output of a bat population.

Other potentially useful strategies include the foundational strategies S6 (Stream/Wetland Habitat Management to Support Sensitive Species) and S5 (Terrestrial habitat management to support sensitive species), which, while not documented to cause increases in bat populations, do lay the foundation for ecosystem restoration that may increase roosting and foraging opportunities.

Predation from invasive vertebrates like cats and rats may have an important impact on Hawaiian hoary bats. Predation on mainland bats has been shown, but this has not been documented in Hawai‘i. Limited control of predators would occur under S7 (Invasive Vertebrate Pest Control) aimed at the seabird footprint and forest bird footprint, but S11 (Landscape-scale Rat Suppression) would likely have more impact as it occurs over large landscapes.

Table 24. Suggested sequencing of Strategy 27 actions to best benefit Hawaiian hoary bats. If all actions are implemented, it is anticipated that bats will have a 63 percent chance of persisting for 20 years.

Strategy	Description	Persistence (%)	Sequence
16	Minimization and Modification of Infrastructure (minimize lighting, modify powerlines and wind turbines, reduce road kill of birds)	49.2	1
11	Landscape Scale Rat Suppression	48.4	
6	Wetland Habitat Management (for at-risk spp)	48.3	2
7	Invasive Vertebrate Predator Control (rats, cats, dogs, mongoose, barn owl, cattle egrets, rabbits)	46.6	
10	New Predator Fencing (snails, rodents, mongoose, cats)	44.7	
5	Terrestrial Habitat Management (for at-risk spp)	41.9	3
9	Existing Predator Fencing (snails, rodents, mongoose, cats)	41.9	
3	Future Ungulate Fencing	41.2	
2	Existing Ungulate Fencing	40.9	

i. Combined Recovery Footprints Across All Taxonomic Groups

i. **General Observations.** The combined recovery footprint for all eight species groups (Figure 19a) includes 492 planning units covering 1,050 km^2. This footprint provides habitat for 250 at-risk species. There are 143 at-risk species that lack sufficient information to reliably identify habitat and planning units for their recovery and 49 snail species that have locations identified for in situ exclosures, but not recovery habitat. These species will require additional surveys to determine their habitat requirements and result in their eventual inclusion in the recovery footprints.

   From a landscape perspective, the footprints:

   - Are located primarily on lands already oriented towards conservation (65.7 percent),
   - Are located on all four Maui Nui islands, with upland areas of Haleakalā, West Maui, East Moloka‘i, and Lāna‘ihale being the most significant.
   - Already have at least some degree of ungulate fencing (50 percent),
   - Are, on average, either native or a mix of native and non-native species, and
   - Generally, have a greater-than-average level of climate resilience.

ii. **Priority Footprints.** Most of the taxonomic groups also identified priority areas for implementation (Figure 19b). These footprints advance recovery but are not expected to meet recovery goals; they are, however, generally the most important areas to focus MNLCP recovery actions. The combined footprint of priority areas includes 216 planning units covering 457 km^2. It provides habitat for 267 at-risk species. This is about 43 percent of the size of the full recovery footprints.

iii. **Initial Implementation Areas.** Some taxonomic groups had relatively large recovery footprints and priority areas; in those instances, it is useful to focus further attention on where to start implementation efforts (Figure 19c).

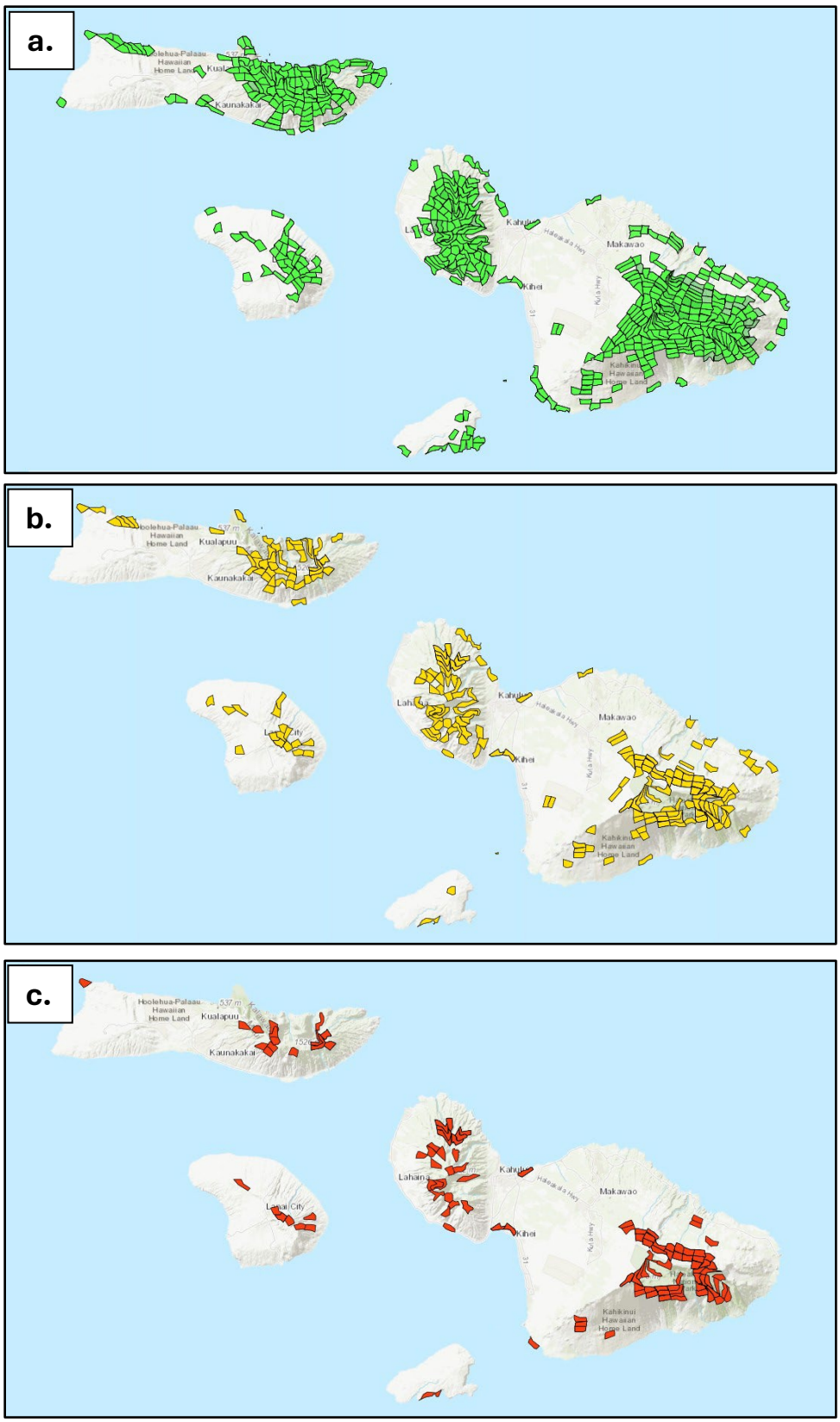


Figure 19. Combined recovery footprint for all taxonomic groups (a), priority footprint indicating especially important areas (b), and a narrowed footprint for initial implementation efforts (c).

iv. **Exceptional Areas.** The Conservation Planner can be used to identify areas with exceptionally high numbers of at-risk species to maximize efficiency or locate areas harboring species facing imminent extinction (Figure 20).

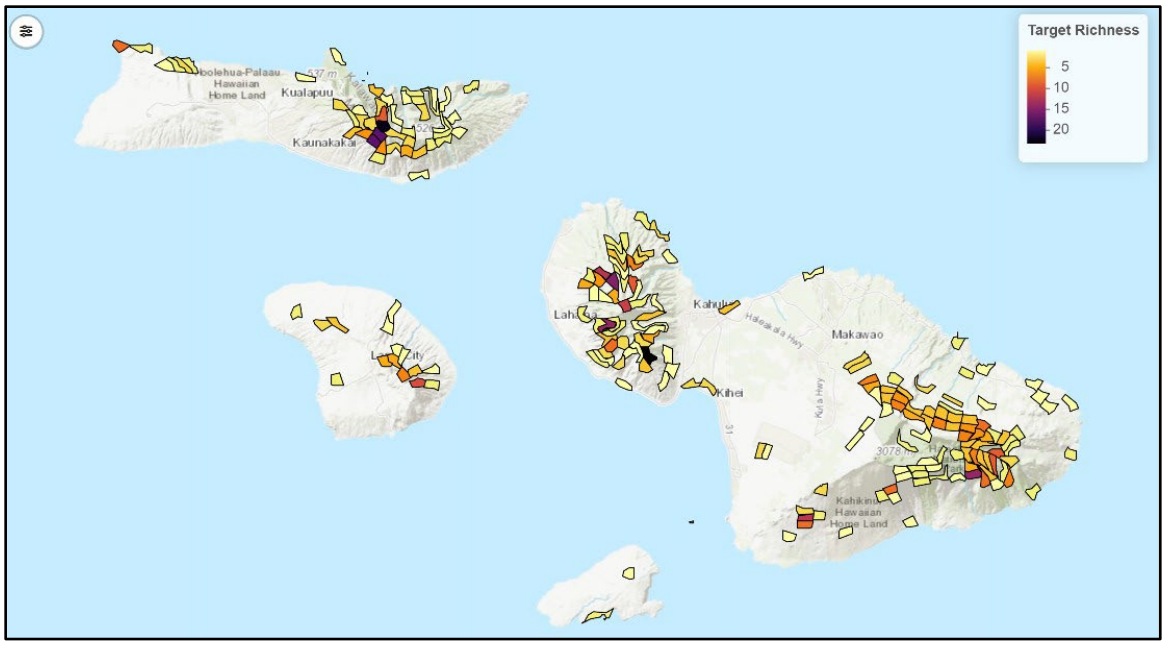


Figure 20. Heat map showing the target richness (number of at-risk species) in each of the planning units in the priority footprint.

j. Combined Priority Conservation Actions Across Taxonomic Groups

A modified Priority Threat Management process was used to help assess the conservation benefits of potential conservation actions. This effort also helped prioritize conservation actions (see Appendix 2). Table 25 shows the composition of the 17 primary action strategies and the 10 combination strategies used and Table 26 shows the expected benefits of the 27 strategies towards reaching persistence for each of the 17 species groups. Persistence is defined as the probability of species groups having viable, self-sustaining populations (over a 20-year period). No single strategy could get all species groups to a desired 50 percent persistence. Implementing both of the two highest-scoring strategies S25 and S27 came the closest to meeting all groups' needs, but all six of the plant subgroups and the freshwater snails failed to meet the desired 50 percent persistence probability (when adjusted for feasibility).

Table 25. The 17 individual action strategies (S1-S17) and 10 combination strategies (S18-S27) developed for use in the modified Priority Threat Management assessment. The color of individual strategies shows their primary impact (gray for most species, green mainly for plants and invertebrates, and blue for birds). The two most beneficial combination strategies are highlighted in pink (S25 plant- and invertebrate-focused and S27 bird-focused).

Individual Strategies	Description	Combination Strategies									
		S18	S19	S20	S21	S22	S23	S24	S25	S26	S27
S1	No Action										
S2	Current Ungulate Fences	X	X	X	X	X	X	X	X	X	X
S3	Future Ungulate Fences			X		X	X		X		X
S4	Fenceless Ungulate Control	X	X	X	X	X	X	X	X	X	X
S5	Habitat Management	X						X	X	X	X
S6	Wetland Management	X						X	X	X	X
S7	Vertebrate Predator Control	X			X	X	X	X	X	X	X
S8	Invertebrate Predator Control						X	X	X	X	X
S9	Current Predator Fences	X			X	X	X	X	X	X	X
S10	Future Predator Fences					X	X				X
S11	Landscape Rat Control					X	X		X		X
S12	Current Plant/Invert Tools	X						X	X		
S13	Future Plant/Invert Tools								X		
S14	Current Avian Disease	X								X	X
S15	Future Avian Disease										X
S16	Infrastructure Minimized										X
S17	Vertebrate Rehabilitation										X

Table 26. Benefits of conservation action strategies for the 17 species groups. The combination strategies generally produced more situations where a species group had 50 percent persistence levels (green) and produced the highest persistence levels for each species group (bold). Seven groups did not reach the 50 percent level with any strategy (yellow).

Strategy	C	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	S16	S17	S18	S19	S20	S21	S22	S23	S24	S25	S26	S27
Species Group	Only Current Actions	No Actions	Existing ungulate fences	Future ungulate fences	Ungulate control no fences	Terrestrial habitat mgmt	Stream / wetland habitat mgmt	Vertebrate predator control	Invertebrate pest control	Existing predator fences	Future predator fences	Landscape rodent control	Existing tools for plants & inverts	Future tools for plants & inverts	Avian disease prevention	Future tools for avian malaria	Modify infrastructure	Vertebrate health & rehabilitation										
Forest birds	11	8	13	14	10	15	9	19	9	12	15	17	8	8	28	39	8	11	36	15	17	22	25	23	21	25	37	**59**
Seabirds	36	17	32	30	23	29	20	46	25	31	48	42	20	20	18	17	38	35	47	31	32	48	54	54	50	53	50	**59**
Waterbirds	40	23	34	34	29	32	38	49	29	33	50	33	27	29	34	27	33	38	49	33	36	45	55	54	51	54	54	**58**
Plant Group 1	27	12	16	17	15	19	12	21	25	13	19	25	16	25	16	17	14	15	20	14	18	17	31	35	30	**48**	23	29
Plant Group 2	29	16	19	22	20	27	20	18	19	16	20	21	22	28	16	16	16	16	24	18	23	19	26	29	29	**45**	26	26
Plant Group 3	27	16	19	26	19	24	16	19	19	16	18	20	21	23	16	16	16	16	23	18	28	19	30	31	28	**44**	23	24
Plant Group 4	28	14	17	23	17	23	15	23	27	15	19	22	20	26	14	14	15	15	22	16	24	18	30	33	28	**42**	24	26
Plant Group 5	28	16	19	25	19	27	16	21	16	16	20	24	21	26	16	16	16	16	22	17	24	18	31	33	29	**46**	24	24
Plant Group 6	28	16	20	22	19	24	16	26	25	18	24	28	21	26	16	16	18	16	24	19	23	21	32	37	30	**43**	27	29
snails	26	14	20	21	18	23	17	30	34	24	39	28	28	37	15	15	17	17	37	21	24	29	41	48	48	**56**	39	46
Yellow-faced bees	23	19	25	27	23	34	24	25	40	21	22	20	36	41	19	19	19	19	39	27	31	34	37	45	50	**56**	45	46
Picture-winged flies	21	15	21	26	22	33	20	27	34	18	31	23	44	44	15	15	15	15	38	30	36	32	36	41	50	**57**	48	51
Blackburn's sphinx moth	41	36	41	41	39	47	39	40	46	39	39	39	42	51	38	38	43	39	50	41	42	42	46	50	56	**62**	52	53
Damselflies	35	27	31	33	30	35	37	39	37	28	31	30	44	51	27	27	30	27	39	34	37	39	41	43	49	**54**	41	43
Anchialine pool shrimp	36	29	32	33	31	31	41	31	31	30	30	30	30	33	30	30	33	30	38	31	32	33	35	35	49	52	**53**	**53**
Freshwater snails	24	20	23	25	22	27	30	30	33	20	20	24	28	33	20	20	24	25	33	26	29	32	36	38	39	**47**	38	41
Hawaiian hoary bat	40	37	41	41	38	41	44	46	40	41	43	46	38	40	37	37	47	40	50	42	43	48	51	51	52	53	51	**58**

Key Observations:

- Seabirds, waterbirds, and the Hawaiian hoary bat are not reliant on research or unproven future methods/technologies to reach a 50 percent persistence goal (Table 27). All other groups are at least somewhat dependent on future methods/technologies. However, it should be noted that while the experts assessing bat persistence were optimistic about bat conservation, the ability to numerically demonstrate that bat populations are trending in any direction is not currently available. Consequently, it is not possible to confirm that conservation actions are benefiting bat populations.
- Forest birds and terrestrial snails are both facing devastating singular threats. In the case of forest birds, introduced avian malaria is the most important threat. For snails, the introduced predatory rosy wolfsnail is the most significant threat. Proven measures to counteract these two threats at a landscape scale are currently lacking. Consequently, snail conservation is focused on *ex situ* conservation or quasi-*in situ* conservation (the kīpuka kāhuli snail exclosures) until landscape-level control measures for rosy wolfsnail (*Euglandina*) are available. Forest bird conservation is actively pursuing mosquito control measures and considering translocations to malaria-free areas.
- S13: Future tools for rare plants and invertebrate conservation is the single most important individual action. Thirteen of the 17 species groups had this in their top three individual actions—only the three bird groups and the bat did not rank it high. This strategy ranks as high as it does because it addresses three important needs:
    - Increase facilities to undertake captive rearing and propagation for plants and invertebrates (terrestrial arthropods, snails, and aquatic invertebrates),
    - Develops methods for cryopreservation and tissue culture, and
    - Develops landscape level control measures for currently difficult to manage or uncontrollable key threats, especially:
        - *Euglandina* and other snail predators (snails)
        - Slugs (plants)
        - Black twig borer (plants)
        - Coconut rhinoceros beetle (plants)
        - Naio thrips (plants)
        - Ants (plants, terrestrial arthropods, aquatic invertebrates)
        - Wasps (terrestrial arthropods)
        - Invasive aquatic species (aquatic invertebrates)

- S14 and S15: Tools that specifically target bird diseases are critical for forest birds (to address avian malaria) but not required for any other species group to reach 50 percent persistence.

Table 27. The distribution of the top three single action strategies across the species groups. The darkness of the green cells corresponds to the rank of the benefits from the action strategy.

Strategy	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	S16	S17
Taxonomic Group	Existing ungulate fences	Future ungulate fences	Ungulate control no fences	Terrestrial habitat mgmt	Stream / wetland habitat mgmt	Vertebrate predator control	Invertebrate pest control	Existing predator fences	Future predator fences	Landscape rodent control	Existing tools for plants & inverts	Future tools for plants & inverts	Avian disease prevention	Future tools for avian malaria	Modifiy infrastructure	Vertebrate health & rehabilitation
Forest birds						3							2	1		
Seabirds						2			1	3						
Waterbirds					3	2			1							
Plant Group 1							1			2		3				
Plant Group 2		3		2								1				
Plant Group 3		1		2								3				
Plant Group 4		3					1					3				
Plant Group 5		3		1								2				
Plant Group 6						2				1		3				
Terrestrial snails							3		1			2				
Yellow-faced bees							2				3	1				
Picture-winged flies							3				1	2				
Blackburn's sphinx moth				2			3					1				
Damselflies						3					2	1				
Anchialine pool shrimp		2			1							3				
Freshwater snails					3		1					2				
Hawaiian hoary bat						2				3					1	
Groups selecting	**0**	**5**	**0**	**4**	**3**	**6**	**7**	**0**	**3**	**4**	**3**	**13**	**1**	**1**	**1**	**0**

7. Conservation and Restoration of Native Terrestrial Ecosystems

a. Terrestrial Ecosystems of Maui Nui

As noted previously, the Hawaiian Islands exhibit an impressive range of environments, including tropical coastlines, alpine zones, rainforests, young lava flows and steep weathered landscapes. The islands of Maui Nui (Hawaiian for ‘Big Maui’) include Moloka‘i, Lāna‘i, Maui, and Kaho‘olawe and encompass essentially the entire span of environmental variation found in the greater archipelago. These four islands formed recurring connections into the single large island of Maui Nui, which at its maximum extent was larger than the present island of Hawai‘i (Price and Elliott-Fisk 2004). Through tectonic subsidence, the connections among the present-day islands gradually submerged, although during low sea stands of glacial periods, the islands reunited as recently as 20,000 years ago. Comprising a single large landmass for the majority of their history, these islands share diverse flora and fauna (Price 2004).

The MNLCP considers Hawaiian ecosystems in the context of the spatial configurations of major ecosystems, habitat quality, management potential, and principals of landscape ecology. Additionally, landscape-scale assessments are tied to species-based efforts in order to prioritize and coordinate management actions. Finally, specific targets were identified by considering several categories of management opportunity. The conservation targets identified here apply to near-term priorities related to stemming biodiversity loss and long term, aspiration goals associated with broader recovery of functional ecosystems.

b. Delineating Ecosystems

Ten mapped ecosystems are included in the MNLCP (Figure 21 and Appendix 8). These are zones mapped by Price et al. (2012), subdivided by elevation using a USGS digital elevation model (DEM): Lowland (0–1,000 m), Montane (1,000–2,000 m), Subalpine (2,000–3,000 m), and Alpine (>3,000 m). Because Wet and Mesic moisture zones extend only slightly into the Subalpine Zone, those areas are consolidated into the Montane Wet and Montane Mesic ecosystems, respectively. The result is nine climatically defined ecosystems. The lava pioneer communities are included as a distinct ecosystem due to unique edaphic conditions (Price et al. 2012).

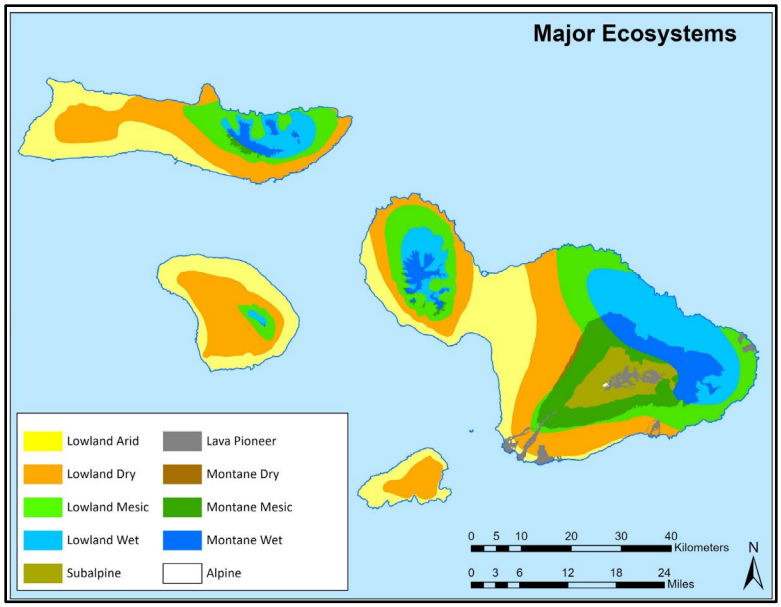


Figure 21. Major ecosystems on Maui Nui.

In addition to the landscape-scale ecosystems assessed for spatial characteristics, several local-scale ecological features are also recognized that represent special conditions and support unique flora and fauna. First, coastal areas represent a heterogeneous array of cliffs, sand dunes, and wetlands (Warshauer et al. 2009). Anchialine pools, defined as landlocked water bodies with subterranean connection to the ocean due to porous volcanic substrates, are found in only a few parts of the world and support unique fauna (Holthuis 1973); in Maui Nui, these are restricted to East Maui. Lava tube ecosystems similarly result from recent volcanic activity, representing another special feature supporting unique fauna (Howarth 1973); in Maui Nui, these are also restricted to East Maui. Cliffs have long represented refuges for unique Hawaiian flora (Gagné and Cuddihy 1990), and recent exploration using unmanned aerial vehicles (dones)has uncovered previously undocumented species (Nyberg et al. 2024). Finally, in upland wet climate regions with poor drainage, bogs represent localized, patchy habitats supporting unique biota (Gagné and Cuddihy 1990). For all of these, although they have definable coordinates in some cases, mapping of their extent and configuration has been limited, and therefore they were assessed only with respect to general conditions. See Table 28 for a list of Hawaiian ecosystems.

Table 28. Hawaiian ecosystems and unique communities. The amount of high-quality habitat remaining in each ecosystem, the MNLCP target for restoration and management, the island distribution of ecosystems, and whether the ecosystems are currently mapped.

Ecosystem	Original Extent (x1000 acres)	% Original Remaining	MNLCP Goal % Managed	Molokaʻi	Lānaʻi	Maui	Kahoʻolawe	Mapped
Lowland Arid	185.1	0.5	40	X	X	X	X	Yes
Lowland Dry	222.9	2.1	40	X	X	X	X	Yes
Lowland Mesic	120.2	21.3	40	X	X	X		Yes
Lowland Wet	95.2	59.1	59	X		X		Yes
Montane Dry	1.4	56.4	56			X		Yes
Montane Mesic	46.5	29.4	29	X		X		Yes
Montane Wet	41.1	91.6	91	X	X	X		Yes
Subalpine (Dry)	26	83.9	83			X		Yes
Alpine (Arid)	0.1	81.2	81			X		Yes
Lava Pioneer	12.4	75.2	75			X		Yes
Coastal	-	-	-	X	X	X	X	No
Anchialine Pools	-	-	-	X		X		No
Lava Tubes	-	-	-	X		X		No
Cliffs	-	-	-	X	X	X		No
Bogs	-	-	-	X	X	X		No

c. Summary of Ecosystems and Special Features

Lowland Arid — The most reduced and degraded ecosystem, with poor mapping and very little habitat left. Restoration (prioritized within the recovery footprint) is the main opportunity.

Lowland Dry — Once the most extensive ecosystem type, now down to roughly two percent remaining. A hotspot for rare species. Restoration is the key opportunity. Better high-resolution mapping is needed to locate remaining pockets of native vegetation.

Lowland Mesic — Fragmented ecosystem; areas mapped as "native dominated" often have extensive invasive plant understories. "Restore" is a more accurate designation than "Initiate" for much of this type. It holds the highest original plant diversity (and likely invertebrate diversity), making targeted restoration especially valuable.

Lowland Wet — Has the greatest percentage remaining and is the least invaded of the lowland ecosystems. Because there is substantial good-quality habitat on partner lands that has not yet been made ungulate-free, "Initiate" is the biggest opportunity here.

Montane Dry — Well represented on Hawaiʻi Island but confined to a narrow zone on East Maui, with little native-dominated vegetation left.

Montane Mesic — Historically converted to cattle pasture, so little remains, but a decline in ranching creates a strong restoration opportunity — many old pastures still have native trees and little invasive woody growth, making grass control plus native planting cost-effective.

Montane Wet — Retains a high percentage of original habitat and strong plant diversity, and is core habitat for most remaining forest birds. "Maintain" is the priority, though climate change may introduce new threats (like lowland-type invasive plant pressure) requiring more active management.

Subalpine — Similarly high-quality and largely intact, mostly protected within Haleakalā National Park and largely ungulate-free, so "Maintain" is again the priority. Climate change is a looming risk to its narrow climatic niches.

Alpine — A tiny area at the Haleakalā summit, with less original habitat remaining than expected due to roads, parking lots, and telescope facilities. Climate change is the dominant concern, given the ecosystem's small area and species' inability to migrate elsewhere.

Lava Pioneer — Occurs at low elevations but is comparatively more native-dominated than other lowland ecosystems.

Special Ecological Features — A diverse group of communities unified by the importance of hydrology: anchialine pools depend on groundwater-driven salinity and are vulnerable to pollution, bogs can be damaged by ungulate rooting that alters drainage and destroys rare plants, and caves are damaged by ungulates, aboveground disturbances, and groundwater changes. Managing these features effectively often requires watershed-level protection well beyond the features' immediate boundaries.

d. Determining Habitat Quality

Hawaiian vegetation spans a spectrum of conditions from mostly native to entirely dominated by invasive species. Using cover classes from Jacobi et al. (2017), three habitat quality zones were defined: High (native dominated), Low (non-native dominated), and Developed/Cropland (Figure 22). Although the source GIS layer did not designate “Sparsely Vegetated” cover types, those within the Lava Pioneer ecosystem were assigned to be High, and those outside of that were assigned to be Low based on various vegetation surveys (Price unpub. data).

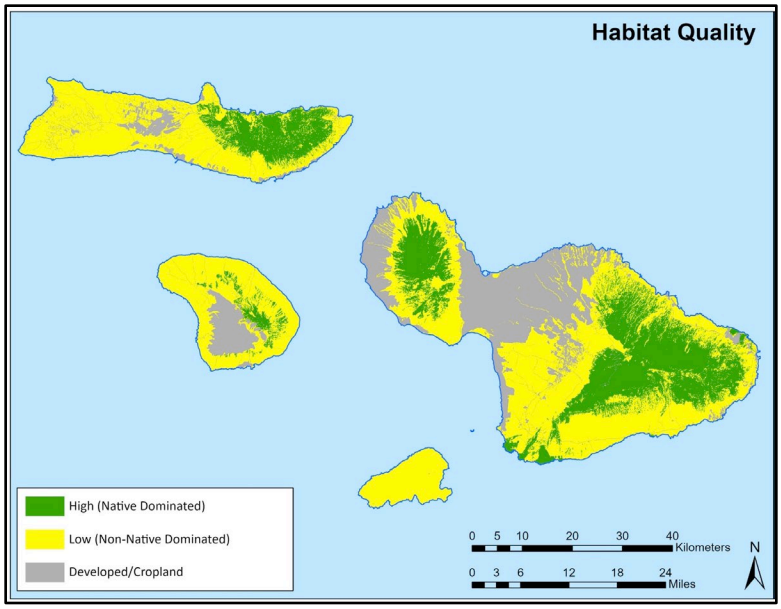


Figure 22. Remaining habitat quality of Maui Nui ecosystems.

e. Management Landscape

A previous species-based conservation planning effort, the Hawai‘i Gap Analysis Program (HIGAP), considered various conservation designations to go beyond traditional assessments of whether habitat was simply protected in some way from development, logging, etc. (Gon et al. 2006). This derived from the clear need for active management in the face of invasive species threats and identified areas that were not simply “protected” in the traditional sense but designated by policy to be actively managed. This concept was used to identify three tiers of “management potential” (Figure 23).

i. **Ongoing Ungulate Exclusion.** First, because feral ungulates are a primary cause of degradation to Hawaiian ecosystems (Leopold and Hess 2017), areas where feral ungulates have been excluded (through fencing and targeted removal) are considered to represent the highest level of management potential. In addition to being released from the direct effects of feral ungulates, these areas allow other conservation activities (for example, planting endangered plants) to proceed with a greater probability of success.

ii. **Conservation-Oriented Management Potential.** These are areas where managers have the authority to engage in conservation management (for example, fencing and removing ungulates) but these actions are not currently being pursued. These include certain agencies within the State of Hawai‘i (particularly the Department of Land and Natural Resources) or

where formal agreements exist, particularly Watershed Partnerships, which are composed of public and private lands co-managed to promote ecosystem health (Duffy and Martin 2019).

iii. **Other areas.** All areas that do not have ongoing ungulate exclusion or a conservation-oriented management goal.

By overlaying the habitat quality and management potential layers, the MNLCP identifies areas with different kinds of opportunities. The Habitat Quality and Management Potential layers were converted to raster format and intersected in ArcGIS Pro to determine the spatial extent of different combinations of characteristics. Table 29 outlines each possible combination of characteristics and defines the associated management opportunities. Areas with low-quality habitat that are outside partner lands and have no at-risk species offer more limited management opportunities.

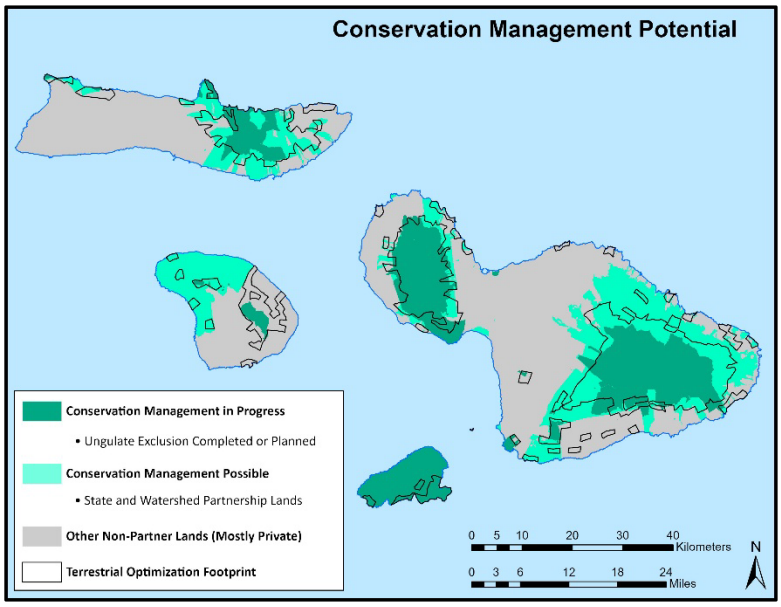


Figure 23. Management potential.

Table 29. Management opportunities. Habitat quality and management status of lands provide a framework for establishing opportunities for management.

Management Potential:	Low Quality Habitat	Medium Quality Habitat	High Quality Habitat
Managed (ungulate exclusion	Unlikely combination (roads & infrastructure within managed areas)	**Restore: within ungulate excluded areas**	**Maintain: Managed high quality habitat**
Unmanaged + HCA Partner Lands	Management or restoration unlikely (Agricultural Areas)	**Restore: within Recovery Footprint**	**Initiate: Management possible now**
Unmanaged + Other Ownership	Management or restoration unlikely (Urban/Agricultural Areas)	**Restore: within Recovery Footprint**	**Engage: Management possible if land is secured**

f. Biodiversity Hotspots

To add context to management opportunities, overall biodiversity patterns were explored. First, using plant species range maps by Price et al. (2012), an overall diversity layer was developed by assigning the range of each given species (out of a total of 1,087 species) a value of 1, and then adding all the values together for each 30 m grid cell (Figure 24). This was also done as a separate analysis for the 25 percent of species that have suffered the greatest degree of habitat loss (as measured by the percent of original habitat no longer in a high-quality state). While total original diversity represents processes relating to climate type, island history, and substrate age, species with the greatest habitat loss will be

concentrated in ecosystems that have been the most disrupted by human activity.

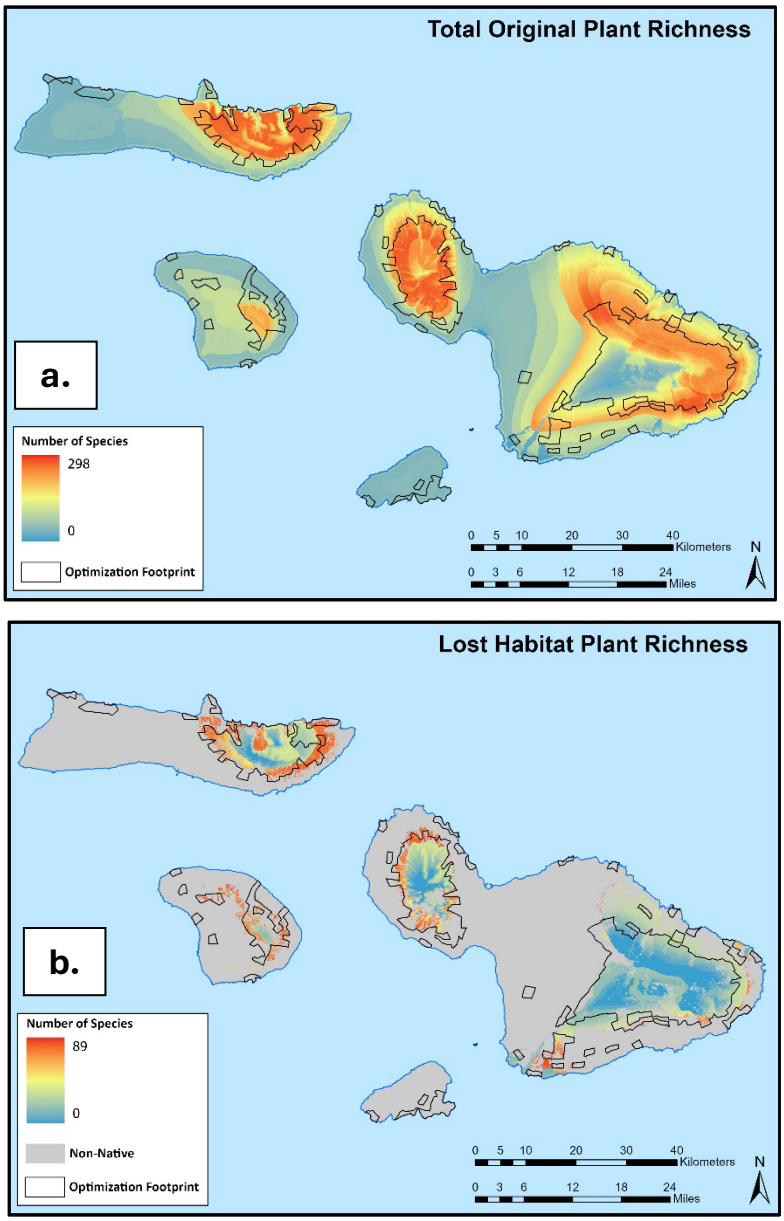


Figure 24. Plant species richness: (a) for all Maui Nui species and (b) for only plant species with significant habitat loss. Richness maps have the plant recovery footprint overlain.

g. Landscape-Level Patterns of Management Opportunities

Combining the key spatial layers reveals a landscape pattern of management opportunities (Figure 25), with different patterns depending on the ecosystem (Table 30). The MNLCP has a goal of restoring all ecosystems to a level that provides for all Maui Nui terrestrial and freshwater biodiversity. Some ecosystems have been degraded to the point that they have less than one percent of their original land area remaining as high-quality native ecosystems. The long-term goals for ecosystems are provided in Table 30.

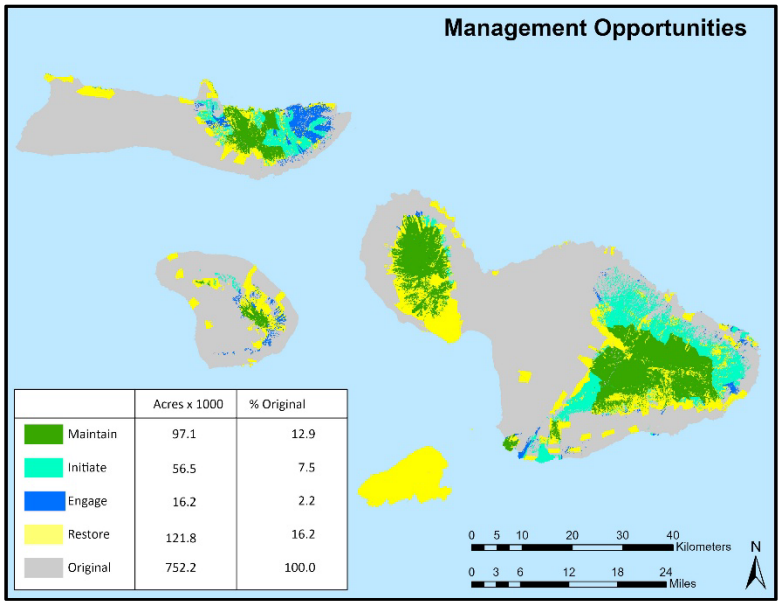


Figure 25. Management Opportunities. See Table 29 for details of analysis.

Table 30. Management opportunities and restoration targets by ecosystem type.

Opportunity:	Maintain		Initiate		Engage		Restore	
Ecosystem	Acres x 1000	% of Original	Acres x 1000	% of Original	Acres x 1000	% of Original	Acres x 1000	% of Original
Lowland Arid	0	0	0.4	0.2	0.4	0.2	27.4	14.8
Lowland Dry	1.1	0.5	1.3	0.6	2.3	1	35.1	15.7
Lowland Mesic	11.8	9.8	7.5	6.3	6.2	5.2	27.5	22.9
Lowland Wet	20.4	21.4	30.2	31.7	5.7	6	10.7	11.2
Montane Dry	0	0	0	0	0	0	0.6	41.6
Montane Mesic	10.4	22.4	3.2	6.8	0.1	0.1	13.5	29.1
Montane Wet	30.8	75	6.5	15.8	0.3	0.8	3	7.4
Subalpine	16.9	65	4.8	18.3	0	0	4	15.5
Alpine	0.1	81.2	0	0	0	0	0	0
Lava Pioneer	5.5	44.9	2.7	21.5	1.1	8.8	0.6	4.7

h. Overall Priorities for Managing Terrestrial Ecosystems.

Prioritizing the conservation of terrestrial ecosystems is challenging; however, there are some general guidelines:

i. **The highest priority** for terrestrial ecosystem conservation is generally the protection and management of existing high-quality habitat versus restoring degraded areas. This is especially important in those ecosystems with less than 40 percent of their original area remaining in high-quality habitat.
ii. **Managing high-quality areas** already in a conservation-oriented status can usually be accomplished more quickly than in other areas.
iii. **Restoring areas** that are important to species recovery, connectivity, high species diversity, or climate resilience is more important than restoring areas that only contribute to a target number of acres needed for ecosystems.

Table 31. Management opportunities. Habitat quality and management status of lands provide a framework for establishing opportunities for management strategies and priorities for conservation efforts.

Management Potential:	Low Quality Habitat	Medium Quality Habitat	High Quality Habitat
Managed (ungulate exclusion	Unlikely combination (roads & infrastructure within managed areas)	**Restore: within ungulate excluded areas**	**1: Maintain: Managed high quality habitat**
Unmanaged + HCA Partner Lands	Management or restoration unlikely (Agricultural Areas)	**Restore: within Recovery Footprint**	**2: Initiate: Management possible now**
Unmanaged + Other Ownership	Management or restoration unlikely (Urban/Agricultural Areas)	**Restore: within Recovery Footprint**	**3: Engage: Management possible if land is secured**

i. Climate Change-Related Patterns

Within the high-quality native ecosystems, climate change is an ongoing threat. The overlay of climate vectors (Figure 26) shows that while most of the quality

native habitat is changing more slowly than the Maui Nui average, it is still changing. Managers should monitor these changes and determine if management strategies need to be modified to mitigate these changes.

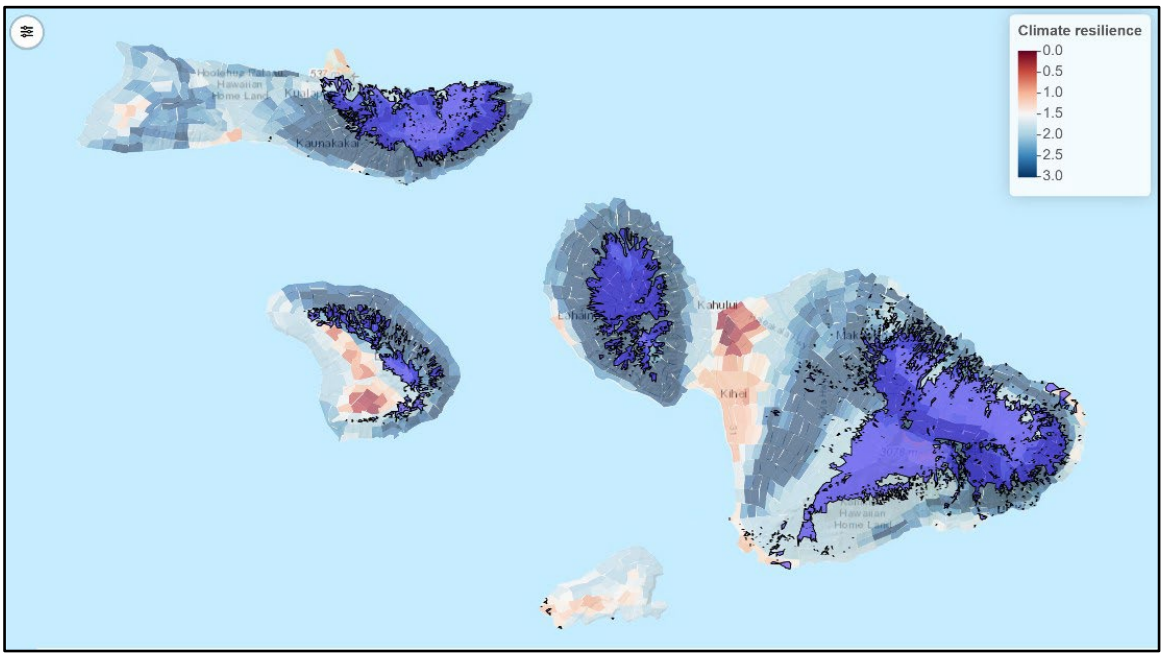


Figure 26. Overlay of climate change vectors with high-quality native habitat shows that it is changing more slowly (dark blue and medium blue) than other areas. Very little of the quality native forest is light brown or orange.

j. Prioritizing Management of Key Landscape Areas

Within the general priority framework in Table 31, the Conservation Planner can be used to highlight larger opportunity areas within the overall landscape. This focuses secondary evaluations to help prioritize within management categories, or even between them. For example, reviewing the spatial distribution of management categories, ungulate fencing, and native ecosystems highlights areas that may have higher conservation priority.

i. **Priority 1:** Maintain managed high-quality habitat that has ungulate fences. Existing management should continue.
ii. **Priority 2:** Initiate management on conservation lands. Identify existing native ecosystems that are in conservation status but not yet fenced. Maui Nui has extensive areas in conservation status that are not protected from ungulate damage. Some of those key areas are important from an ecosystem

perspective as well as an at-risk species recovery perspective (Figure 27; numbered areas on map). These include windward Haleakalā, two areas on Moloka‘i, and Lāna‘ihale.

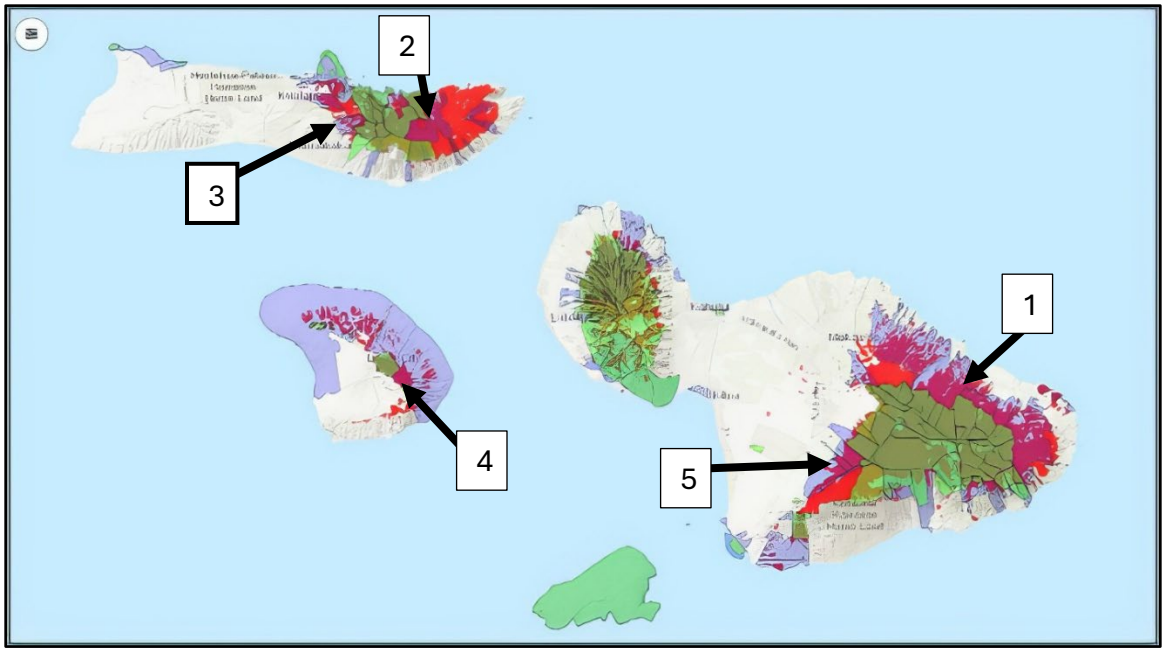


Figure 27. Mapping of opportunity areas for ecosystem conservation. Overlaying the high-quality native habitat layer with layers for conservation areas and ungulate exclusion shows where high-quality native habitat occurs outside of conservation areas and is unfenced (bright scarlet), where it occurs inside conservation areas but is unfenced (deep red), where it is inside conservation areas and fenced (greenish brown), and where it is fenced but not in conservation-oriented status (brownish yellow). Numbers show key areas to potentially prioritize for ungulate control.

iii. **Priority 3:** Engage with potential partners to implement management or include areas in conservation status. Several key areas to focus on engagement are noted (Figure 28, numbered areas). These include eastern Moloka‘i, above Makawao on windward Haleakalā, and the Kahikinui area.

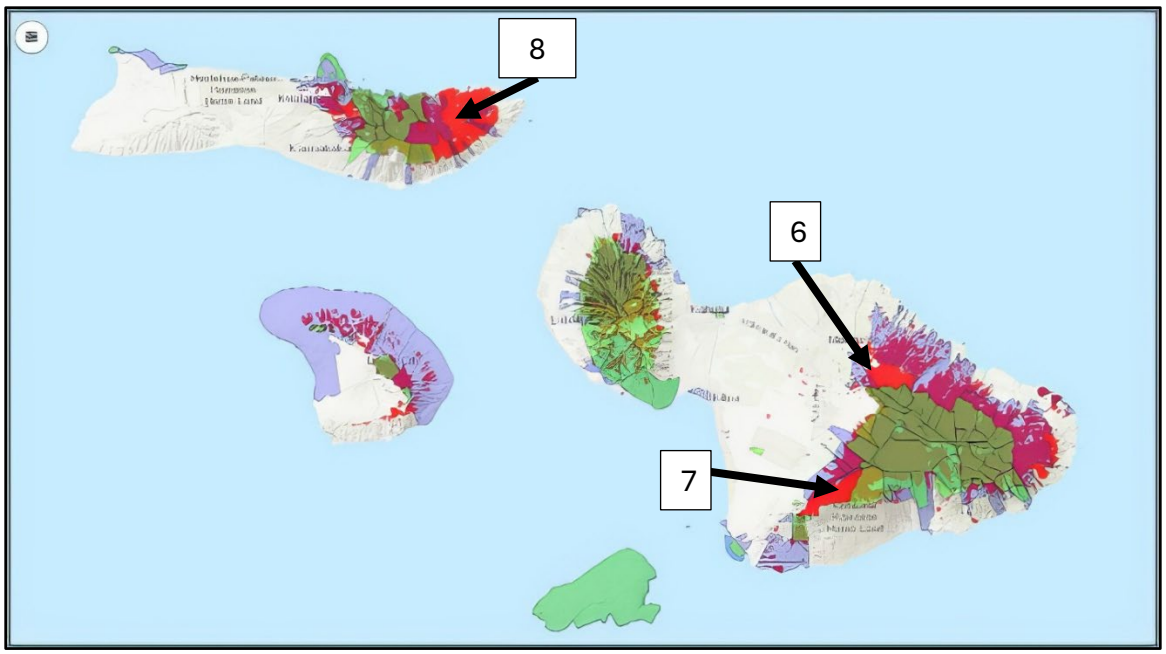


Figure 28. Mapping of opportunity areas for ecosystem conservation. Overlaying the high-quality native habitat layer with layers for conservation areas and ungulate exclusion shows where high-quality native habitat occurs outside of conservation areas and is unfenced (bright scarlet), where it occurs inside conservation areas but is unfenced (deep red), where it is inside conservation areas and fenced (greenish brown), and where it is fenced but not in conservation-oriented status (brownish yellow). Numbers show key areas to potentially prioritize engagement with land managers to promote conservation efforts.

iv. **<u>Priority 4:</u>** Undertake restoration of medium quality habitats to help meet ecosystem targets for area, as well as other values, especially to support the recovery of at-risk species.

8. Monitoring and Evaluation of Management Actions

a. Introduction.

The recovery of at-risk species and the restoration of native ecosystems is an exceedingly complex endeavor. The MNLCP provides an essential foundation for these efforts on Maui Nui by identifying priority species and ecosystems, the populations most in need of emergency intervention, the threats that most affect each species and ecosystem, and the management actions expected to move them toward recovery. Planning alone is not enough, however; managers also need to know whether the actions being taken are producing a positive ecological response and in turn be able to respond to circumstances and outcomes that require change in strategies and methods.

Uncertainty is inherent in conservation work. Monitoring has shown, for example, that mosquito control is essential to protecting forest birds from avian malaria, that fencing and ungulate removal benefit at-risk plants when weeds are also controlled and fences are maintained, and that the success of outplanting and rodent control likewise depends on continued assessment. Uncertainty also arises in detecting change itself; populations and ecosystems are dynamic and spatially variable and respond differently to historic and current conditions, and a changing climate adds further unpredictability to future threats such as fire (Mulder et al. 1999).

Recognizing these uncertainties, rather than being discouraging, is valuable — it creates space for managers to pause, review, and refine their approach, leading to better decisions about which actions to take and how. This is the origin of adaptive management; a structured, sequential learning process that reduces uncertainty and leads iteratively to more effective actions. It has three components: collecting monitoring data, evaluating that data, and using the evaluation to guide management decisions. Monitoring alone is not enough; the data must be evaluated and shared to influence decisions.

Adaptive management is essential to natural resource management (Sutter and Rutledge 2017). As an approach, it allows recovery actions to proceed despite uncertainty. As a learning process, it uses available knowledge and results to confirm or revise strategies. As a tool for achieving results, it leads to more effective, efficient, and lasting outcomes by prompting changes to management when an action is not working. Importantly, adaptive management requires the ability to change course — monitoring is not a wise use of resources if there is no opportunity to act on what is learned.

A well-run adaptive management program structures data collection over the long time frames needed to detect real change; produces tangible evidence of progress that supports institutional and funding decisions; allows assessment of return on investment; improves understanding of the effects of management and of species and ecosystem dynamics; and establishes benchmarks that staff and partners can recognize as success.

Integrating adaptive management into a monitoring program is nonetheless challenging. Institutionally, it requires sustained commitment, leadership, funding, and partner involvement. For those implementing it, monitoring must be focused on realistic objectives, fully integrated into management, and paired with analysis that is shared with colleagues, partners, and decision-makers. Determining the right monitoring intensity for each population or ecosystem is equally demanding, and at a personal level, adaptive management requires humility and curiosity.

The MNLCP offers a unique opportunity to build a programmatic approach to adaptive management focused on the highest-priority species and management questions. The plan addresses 442 at-risk species across eight taxonomic groups, with management actions prioritized spatially across Maui Nui. Because resources are limited, the MNLCP establishes priorities that help ensure species do not go extinct and that actions lead to recovery.

It also provides an important framework for focusing and evaluating monitoring. For each at-risk species, the MNLCP establishes quantitative population targets in the common currency of distribution and abundance and allocates those targets spatially among Planning Units (PUs). These spatially explicit targets define not simply

Steps to Initiate a Monitoring Plan

- **Specify the management goal and monitoring objective** — define the desired outcome, the species/ecosystem, location, attributes to measure, method, and time frame, and confirm the target can reliably be identified and detected.
- **Select the attributes to measure** — pick indicators (abundance, condition, reproduction, composition/structure, etc.) that are relevant, reliable, easy to measure, and easy to communicate to managers and funders.
- **Determine the appropriate monitoring intensity** — weigh vulnerability, uncertainty of the management action, opportunities, and return on investment to decide how much monitoring effort a given population or ecosystem warrants.
- **Choose a monitoring design** — decide between a complete census, an index sample, or probability-based sampling, and work through the sample-unit, spatial-allocation, sample-size, and permanence questions that design requires.
- **Write a monitoring protocol** — document the purpose, methods, data management approach, evaluation process, and roles/responsibilities so monitoring stays consistent across staff and time and update it annually.
- **Set up data management before data collection starts** — establish how data will be captured, stored, quality-checked (verified, validated, certified), archived, and made accessible for future analysis.
- **Build in a regular evaluation cycle** — schedule structured, impartial reviews (e.g., annual for effectiveness, every 5–10 years for recovery status) that translate monitoring data into management recommendations.
- **Establish a communication plan** — identify audiences (managers, funders, partners, the public) and products (e.g., an annual accomplishment report, a status dashboard) to share results and sustain support.
- **Secure institutional commitment and capacity** — line up leadership buy-in, dedicated funding, trained staff, and clear roles across planning, data collection, analysis, and reporting, since without the ability to act on findings, monitoring isn't worth doing.
- **Foster an adaptive management culture** — build staff comfort with uncertainty and a shared expectation that management will change in response to what monitoring shows.

how many individuals are needed for recovery, but how many are needed and where they are needed across the landscape. The Adaptive Conservation Planning software tool makes these targets available to managers for individual species and PUs.

Monitoring conducted under the MNLCP should therefore be designed and evaluated in relation to these spatially explicit population targets. Whether monitoring consists of complete censuses, index monitoring, probability-based sampling, demographic measures, or other approaches described below, the fundamental question is the same: is the population within each targeted PU progressing toward the abundance and distribution identified by the MNLCP? In this way, the MNLCP provides the quantitative and spatial framework, while the monitoring approaches described in this section provide the methods for measuring progress toward those targets. Monitoring results can then be used through the adaptive management process to determine whether current actions are sufficient or whether the location, type, or intensity of management should be modified.

A successful MNLCP Adaptive Management Program would demonstrate the value of the Integrated Multi-scale Conservation Planning (IMCP) process, show an accelerated trend toward preventing extinction and restoring ecosystems, build institutional capacity for adaptive, active learning, and serve as a model for other islands and programs.

This section provides an overview of the elements needed to establish and sustain such a program: key terms and the five types of monitoring objectives; guidance for developing monitoring plans; the process of moving from monitoring data to adaptive decisions; and a programmatic path forward.

b. Key Adaptive Management Terms

Monitoring is the process of acquiring information — through repeated measurements over time and comparable data from multiple samples — for the purpose of detecting change, interpreting results, and improving management decisions. It includes both a process (repeated measurement) and a purpose (improving decisions). Monitoring assesses the status of populations and ecosystems relative to a desired condition, the effectiveness of management actions, the intensity and impact of threats, and the efficiency of management and monitoring itself, while also helping to identify emerging threats.

Because populations take years to expand and ecosystems take decades to recover, monitoring requires time to yield meaningful information. Whether qualitative or quantitative, monitoring should be focused on the information

needed to assess management effectiveness; precise enough to detect ecologically significant change; repeatable at the same locations using the same methods; built on quality data that is accurate and available to managers; and sustained for as long as needed to understand outcomes. Monitoring differs from research in that it targets a change or trend tied to specific management actions, whereas research investigates the underlying causes of change; research can serve as good monitoring but is typically more resource intensive.

Evaluation is the systematic, objective assessment of monitoring data — examining whether the monitoring design meets its objectives, whether implementation was effective and efficient, and whether results are valid and ecologically sound (Kusek and Rist 2004). Evaluation follows monitoring and is required before monitoring data can inform management decisions.

Data Management is the systematic, comprehensive, and consistent approach to managing monitoring data: ensuring its quality, accuracy, and consistency and maintaining it in an accessible database for future analysis (Sutter et al. 2015). Data Management includes ensuring the accuracy and consistency of how data is collected and compiled, describing and maintaining data in an accessible and analyzable database, ensuring the quality of the data through all steps of data management, and establishing an infrastructure that allows the data to be available for future analyses and assessments.

Adaptive Management combines all these components into a structured, sequential learning process that increases knowledge, reduces uncertainty, and leads iteratively to more effective restoration, conservation, and recovery outcomes.

c. Five Types of Monitoring Objectives

Monitoring can focus on one or several of five objectives:

- Implementation Monitoring verifies that a planned management action was completed; this includes both the completion of a management action and documenting how the action was implemented (where, when, how, who). This monitoring asks if and how the management actions were implemented.
- Effectiveness Monitoring measures whether a completed action has achieved or is achieving its intended results — such as reduced threats or improved population and ecosystem condition, an increased number of individuals, higher survival and reproductive rates, improved health of individuals in a population, or a restored native understory. This monitoring asks if the implemented interventions result in improved ecological conditions for the population or ecosystem.

- Outcome Monitoring is a longer-term measure of whether a population or species is becoming healthy and self-sustaining, or an ecosystem is gaining ecological integrity, often combining data across all targeted populations of a species or ecosystem. Outcome monitoring often measures the same attributes used for effectiveness monitoring, but with additional attributes, such as successful establishment of new plants, increases in distribution and range, production of fledglings, and recruitment of key ecosystem species, reflecting the self-sustaining nature of the population and ecosystem, and combining data from all targeted populations of a species and ecosystems. Outcome monitoring is a longer-term process. This monitoring asks whether a suite of management actions is preventing extinction and advancing recovery of a species or ecosystem.
- Impact Monitoring is a program-wide assessment of progress toward preventing extinction, advancing recovery, and restoring ecosystems, often summarized in a colorful and well-designed dashboard. This monitoring addresses the question whether the program is successfully progressing toward recovery and restoration.
- Validation Monitoring**,** more akin to research, compares management methods to determine which are most effective and efficient, and explores the relationships and causes behind management responses.

A comprehensive adaptive management program should include the first four monitoring types; validation monitoring is applied where there is a specific need to better understand a management action.

d. Developing a Monitoring Plan

A well-designed monitoring plan follows five key steps, described below at a level of detail sufficient to identify what information is needed; full development of a monitoring plan will require additional guidance from qualified experts.

i. Specify the Management Goal and Monitoring Objectives

The first step is to establish clear, concise statements that focus both management and monitoring. Good questions come from understanding the resource and its threats, involving appropriate expertise, and sound critical thinking; they will evolve as knowledge grows.

A management goal describes the desired outcome of management over a specific time frame: what actions will be taken, and the desired impact — either a direction of change (e.g., more individuals, less invasive cover) or a

quantitative threshold (e.g., increase the population above 400, or reduce invasive cover below 10 percent).

A monitoring objective is a statement with five components: the species or ecosystem being managed and monitored; the location; the attributes to be measured; the method for collecting data; and the time frame for both management and monitoring.

This step should also address species and ecosystem identification and detectability — how to correctly distinguish the target from similar species or ecosystems, and how detectability may be affected by an organism's rarity, crypsis, low density, or setting. Detectability is rarely perfect; in some cases, fewer than 70 percent of individuals present may be detected (Kellner and Swihart 2014). It can be improved by quantifying detection rates or refining monitoring methods, such as adjusting how close transects need to be.

The MNLCP already provides the initial elements of a monitoring objective — the priority species or ecosystem and its location — while manager expertise supplies the rest, informed by existing programmatic documents, prior studies, and expert knowledge.

ii. **Select Attributes to Measure**

Attributes (also called indicators, metrics, or parameters) are the characteristics measured to assess management effectiveness and change in a population or ecosystem. For species, these can include number of individuals, measures of condition (survival, health, vigor, damage), and measures of reproduction; which attribute is most useful depends on how quickly it reflects change — long-lived species may show slow changes in abundance but faster changes in reproduction. For ecosystems, attributes can include composition, vertical and horizontal structure, connectivity, species richness, and reproduction of key species such as canopy or pollinator plants, as well as processes like light-gap formation or fire and rain events.

Indirect indicators (such as canopy openness as a proxy for understory vigor, burrow counts for seabirds, or water quality for aquatic invertebrates) can also be useful when a direct relationship is known or reasonably assumed. Covariates — factors such as habitat, soils, rainfall, or predation that influence how an attribute responds — should also be measured to help interpret patterns and trends.

Good attributes are relevant (biologically meaningful and tied to expected management impacts), reliable (precise and sensitive to management

effects), easily measured (quickly, consistently, and inexpensively), easily evaluated, and easily understood and communicated to managers, experts, and administrators.

iii. **Determine the Time and Resource Investment in Monitoring**

Determining the appropriate level of monitoring intensity is essential, particularly given that the MNLCP encompasses 442 taxa — from invertebrates to birds — across 492 planning units. Few projects will support a fully robust statistical sampling design, but all can provide useful insight into management success if resources are focused well. Criteria for setting the appropriate monitoring level include:

- Vulnerability of the population, species, or ecosystem (e.g., probability of extinction). Plant vulnerability, for instance, is reflected in PEPP listing status, with plants with few populations (POP)- or rare-on-island plants (ROI)- listed species representing the next tier of priority; among forest birds, kiwikiu and ʻākohekohe are considered most vulnerable. Species not facing imminent extinction can still be vulnerable when threats such as predators or disease are having an especially severe impact.
- Uncertainty in the outcomes of a management action — greater uncertainty warrants more monitoring, while proven approaches (e.g., fencing, ungulate control) may need less.
- Opportunities such as seed or plant availability, land access, funding, partnerships, or synergy with other management actions.
- Return on investment — the more a management action costs, the more valuable it is to measure whether it is succeeding.

Ultimately, the appropriate monitoring level for each population or ecosystem is a judgment call for leaders, managers, and experts, potentially supported by a multi-criteria scoring method that integrates these factors.

iv. **Develop a Monitoring Design**

The MNLCP establishes a framework that enables managers to apply the appropriate monitoring approach to the spatially explicit population targets identified in the Conservation Planner. Within that framework, managers will develop monitoring plans to detect not only whether a species is increasing, but whether its abundance and distribution within targeted PUs are progressing toward the MNLCP recovery targets. The monitoring design determines how these data will be collected and is central to ensuring precision, reliability, and defensibility. There are three general approaches.

- Complete Census. A complete census counts all individuals (or population units, such as burrows) or fully measures the composition and structure of an ecosystem within a defined area. It is generally quick, easy to evaluate and communicate, and requires no statistical analysis (Elzinga et al. 2001) — an efficient and accurate option for many at-risk species with small populations. Its intensity depends on population size, density, distribution, and site accessibility. A reliable complete census requires clearly and repeatably defined boundaries, a consistent survey method, and a consistently recognized counting unit (e.g., individuals, clonal units, burrows); where units are difficult to count consistently, cover or frequency measures may work better. A census can also be expanded to record condition (age class, sex, evidence of activity or threats, and, for ecosystems, site condition and structural change). This adds useful information at little extra cost, though it introduces observer variability; standardized field sheets, defined size and condition criteria, smaller subunits, and photo methods (photoplots or photopoints) can improve consistency, and drones or satellite imagery may help where a species or ecosystem is visible from above.
- Index Sample. An index sample uses one or more nonrandomly placed sample areas to track population or condition trends. Results are extrapolated to the whole population but cannot be used to statistically estimate it, since the representativeness of the sample is unknown. Index samples work well when locations are chosen thoughtfully by experienced ecologists, but users must understand potential biases, such as oversampling accessible or high-density areas.
- Sampling a Population. When a population or ecosystem is too large or resource-intensive to census completely, probability-based sampling — drawing a representative subset and using statistics to estimate the whole — is the appropriate, well-documented approach (Elzinga et al. 2001, Jean et al. 2005). There is no single best sampling design; each must fit the specific monitoring objective and ecological context. Designing a sample requires a team with relevant expertise to answer several interrelated questions:
    - What is the target population? — the full set of individuals or ecosystem features the sample is meant to represent, which may be a full biological population or natural community, or a defined geographic subset of one.
    - What is the appropriate sample unit? — options include individual organisms, quadrats, transects, points, or plotless (distance-based)

methods, chosen based on the attribute being measured. Where a dimensional unit such as a plot is used, its size and shape should reflect trade-offs among setup versus measurement time, organism size and abundance, spatial distribution, edge effects, ease of sampling, and disturbance from monitoring itself; more heterogeneous populations generally call for larger or longer plots to capture within-unit variability.

- How will sample units be spatially allocated? — units should be randomly placed, well interspersed across the sampled area and its natural variability, and spaced independently enough to avoid spatial correlation. Common random-sampling designs include simple random, systematic, stratified random, cluster, and two-stage sampling; the best choice depends on spatial distribution, accessibility, and sensitivity to sampling disturbance (e.g., trampling) (Jean et al. 2005 and Elzinga et al. 2001).
- How many sample units are needed? — more units generally increase precision and the power to detect change, though returns diminish beyond roughly 20–30 samples (Elinga et al. 2001). Precision also improves when variability is captured within, rather than between, sample units. Because appropriate sample size depends on natural variability that is not known in advance, a year of pilot sampling — or adaptive refinement during the first few years — can help resolve this circularity. Numerous sample-size calculators are available for different monitoring objectives (Elzinga et al. 2001).
- Will sample units be permanent or temporary? — permanent units reduce variability across sampling events (especially valuable for perennial or site-faithful species) and increase statistical power, but require durable marking and relocation effort, and may become less representative as a population's distribution changes over time. Despite these trade-offs, permanent plots are widely used.
- When will sampling occur? — timing (time of year and day) should align with when identification and detectability are easiest and **consistent.**
- **How will the data be analyzed? — an analysis plan, developed alongside** the sampling design, should identify the statistical tests to be used, incorporate covariates, and anticipate the form the results will take.

A more detailed discussion of the components of developing a sampling design is found in Appendix 9.

v. **Compile a Monitoring Protocol**

A monitoring protocol is a living document, ideally housed online and updated annually, that provides written guidance on how monitoring is implemented: its purpose, study design, sampling methodology, data management and analysis approach, evaluation process, reporting, and roles and responsibilities. Its primary value is ensuring monitoring is done consistently over time, across locations, and among different staff. Developing and updating the monitoring protocol also sharpens thinking about management goals and design. Best practice includes documenting standard operating procedures and testing the protocol with someone unfamiliar with the site or study design.

A complete protocol typically includes: purpose, goals, and objectives (including species or ecosystem description, identification guidance, status, location, ecology, and threats); permits and qualifications; management goals and monitoring objectives with associated attributes; the monitoring design and methodology, including personnel, equipment, and schedule; data management procedures (acquisition, verification, validation, version control, archiving, and distribution); roles and responsibilities for maintaining and implementing the protocol; and a document history recording revisions, review, and approval. A sample list of the components of a monitoring protocol can be found in Appendix 9.

e. From Monitoring Data to Adaptive Decision-Making

Monitoring data has value only when it is available for analysis, evaluation, and decision-making across both short- and long-time frames.

i. **Managing and Analyzing Monitoring Data**

Detecting the effectiveness of management and change in populations or ecosystems typically requires data spanning many sampling cycles, which in turn requires maintaining the integrity of long-term datasets (Lindenmayer and Likens 2010). Ensuring data remain usable over time requires detailed procedures for collecting and compiling data; describing it with metadata and keeping it in an accessible, analyzable database; ensuring data quality; building infrastructure for future analysis; producing reports and analysis products; and using lossless, nonproprietary file formats with clear retention and archiving practices. Data management is best understood as an extension of monitoring itself, sharing its early planning stages.

Data management itself proceeds through several stages: planning it from a project's outset; preparing infrastructure, procedures, and roles; collecting data with a defined capture system and equipment calibration; compiling and managing data; assuring data quality through verification (completeness), validation (scientific soundness), and certification (readiness for use); describing data through metadata; analyzing and evaluating it to inform management; preserving it through archiving and access policies; and sharing it through reporting appropriate to the audience (USEPA 2019, Sutter et al. 2015, Figure 29).

For qualitative monitoring (complete census, index plots), data can typically be plotted and reviewed across years, with attention paid to the limitations of what the data represent. For quantitative monitoring, the analytical approach should be identified alongside the sampling design; a power analysis — calculating the minimum sample size needed to detect a change of a given magnitude — helps clarify what the results can and cannot show. Different statistical approaches apply depending on whether the goal is estimating a population or ecosystem parameter, comparing a sample to a threshold (e.g., whether native cover exceeds 50 percent), or comparing values across time.

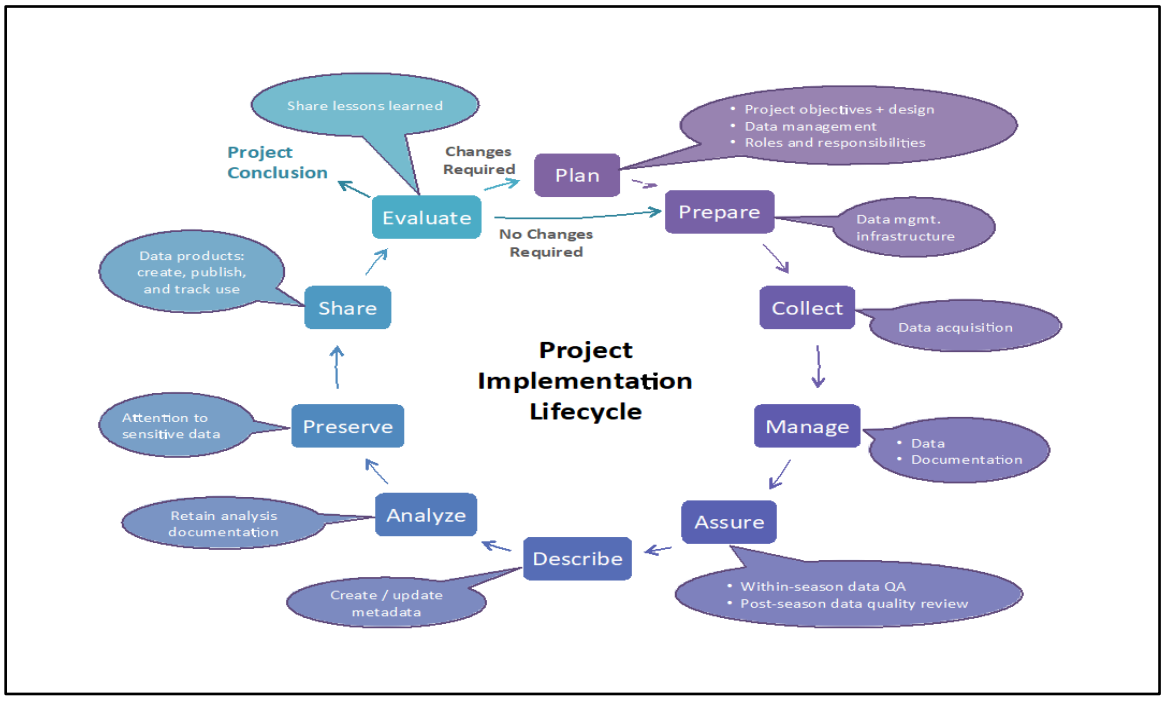


Figure 29. Stages of a monitoring project. Stages that involve data management planning are shown in boxes, with the corresponding data management components in the bubbles. The stages are presented as a cycle to reflect the recurring nature of monitoring at intervals. Arrows represent decision points to implement or conclude the project, or to revise project plans based on feedback or internal evaluation. (From Sutter et al. 2015)

ii. **Evaluation and Reporting**

Evaluation is the systematic, objective review of monitoring data to determine whether management actions are working (Kusek and Rist 2004), occurring at intervals suited to the question — annually for effectiveness, and every five to ten years for recovery status. Evaluation should assess both the type and implementation of management actions, gauge effectiveness through population or habitat response, identify emerging threats, and guide resource allocation. It should happen as soon as possible after data becomes available, while field details are still fresh — in Hawai‘i's varied growing and breeding seasons, this may mean quarterly evaluation meetings. Effective evaluation is structured (with a facilitated process covering all relevant information), impartial (including both field staff and independent reviewers), and inclusive of relevant expertise and manager involvement.

iii. **Communicating Results**

Communicating monitoring results and evaluation insights is essential to adaptive management. Managers and decision-makers are the primary audience, needing clear recommendations backed by supporting data so they can, in turn, report progress to administrators and funders. Other interested audiences include conservation practitioners, political leaders, institutions, and the public; a formal communication plan can identify these audiences, their key messages, and the best channels (written materials, social media, web content, video, presentations) to reach them, with the overarching goal of translating complex recovery data into simple, clear, understandable messages.

Two communication products have proven successful elsewhere: an annual accomplishment report summarizing management, monitoring, and protection actions by taxonomic group, species, and geographic area — including metrics such as miles of fencing built, acres of managed habitat, predator control coverage, populations monitored, species propagated, and acres newly protected; and a one- or two-page dashboard that visually communicates progress toward preventing extinction, recovering species, and restoring ecosystems, aimed particularly at political leaders and the public. See https://americaslongleaf.org/news-events/news/2026/june/america-s-longleaf-

highlights-2025-accomplishments-with-annual-report/ for an example of an annual accomplishment report.

f. A Programmatic Path Forward for Monitoring and Evaluation

The development of the MNLCP and the Adaptive Conservation Planner, together with DOFAW's commitment to protecting biodiversity, provides a unique opportunity to build a comprehensive, enduring adaptive management program. Integrating all of the elements above into one efficient, responsive program will be a challenge: it must be clearly linked to the missions of DOFAW and partner agencies, address monitoring across all resources (species, taxonomic groups, and ecosystems), and remain continuous over the long term to keep pace with how species and ecosystems respond to management.

Critical components for sustaining such a program, adapted from Kusek and Rist (2004), include:

i. **Institutional understanding,** through informational and training resources that help staff, agencies, and partners understand the program and build support and collaboration.
ii. **Establishing the continuous need for monitoring data,** through programmatic requirements for monitoring and evaluation of all management actions, supported by structured review with expert teams and clear timelines.
iii. **Establishing clear roles, responsibilities, and expertise** across every step of the process — leadership, planning, monitoring design, data collection, protocol development, analysis and data management, evaluation, and communication.
iv. **Establishing an annual accomplishment report, a current status dashboard, and regular social media updates,** to showcase adaptive management work and build support among staff, partners, and the public.
v. **Ensuring capacity** for data collection, monitoring design expertise, and technical skills in analysis and data management.
vi. **Establishing incentives that** recognize success across all adaptive management tasks.
vii. **Developing an adaptive management culture among staff —** one that embraces the uncertainty inherent in resource management and understands the role monitoring plays in reducing it and enabling recovery.

With the MNLCP as its foundation, and with sustained leadership among participating agencies, dedicated managers, and rigorous evaluation of management actions, this program is positioned for substantial success in preventing extinction, advancing recovery, and restoring Maui Nui's native ecosystems.

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Terms and Definitions

Action Prioritization Team: UH Mānoa–led team that addressed the “what to do” portion of the MNLCP. Using a modified Priority Threat Management process, the team worked with roughly 100 land managers and scientists to identify key threats, evaluate conservation actions, and estimate the costs and benefits of management strategies for Maui Nui’s at-risk species.

Adaptive Conservation Planner (Conservation Planner): Referred to in the plan as the “Conservation Planner,” this is the spatially oriented, adaptive analytic tool that houses the MNLCP’s final spatial products—including recovery footprints and species, habitat, and threat metadata—and allows managers to revise strategies, develop annual work plans, and explore management scenarios.

Algorithmic Pathway: One of the two process pathways used in spatial conservation planning, alongside the Expert-Led Pathway. Applied to taxonomic groups with numerous species and complex habitat or management considerations (plants, terrestrial arthropods, terrestrial snails), it used the OptimTFE algorithm to generate candidate footprints across Maui Nui’s 1,400 planning units, which experts and managers then compared against criteria such as habitat quality, accessibility, and climate resilience.

At-Risk Species: A species identified by the Taxonomic Teams and other experts—using existing endangered species lists, Hawai‘i’s State Wildlife Action Plan, and other sources—as likely to become extinct within the next 20 years without significant management intervention. The MNLCP identifies 442 at-risk species of birds, plants, invertebrates, and a bat, driven by factors such as low remaining population numbers, climate change, invasive species, and habitat loss.

Chapter 195D of the Hawai‘i Revised Statutes: Titled “Conservation of Aquatic Life, Wildlife, and Land Plants.” It is the state’s primary legal framework for protecting endangered and threatened species and is administered by the Hawai‘i Department of Land and Natural Resources (DLNR).

Communications Team: Develops and implements communication strategies aimed at informing the public and others about the MNLCP and its accomplishments.

Core Team: A DOFAW-led team that oversees MNLCP development, scheduling, and coordination of other MNLCP teams.

Endangered Species: A species formally listed under federal or State law as being in danger of extinction (denoted “E” in the plan’s ESA status tables), as distinct from a threatened species.

Endangered Species Act (ESA): The federal law under which species are formally listed as endangered or threatened. The plan tracks each at-risk species' ESA status (E = endangered, T = threatened) throughout its taxonomic group tables.

Ex situ: Conservation actions carried out outside a species' natural habitat, such as captive rearing, captive breeding, or seed storage/banking.

Expert-Led Pathway: One of the two process pathways used in spatial conservation planning, alongside Algorithmic Pathways. Used for taxonomic groups with fewer focal species or where expert knowledge was more informative than algorithmic methods (forest birds, seabirds, waterbirds, aquatic invertebrates), it had specialists identify priority areas directly through iterative web-map review.

Hawai‘i State Wildlife Action Plan (SWAP): One of the existing State planning documents the MNLCP draws on as a data source — alongside endangered species lists, conservation databases, and the literature — to help identify Maui Nui's at-risk species.

Implementation Team: DOFAW-led team of land managers, planners, and decision-makers responsible for the "when to do it" guidance—translating the results of the spatial prioritization, action prioritization, and ecosystem prioritization efforts into a final, cohesive implementation plan, and overseeing development of the Managers Handbook.

Integrated Multi-Scale Conservation Planning (IMCP): The MNLCP's overarching planning approach, combining spatial prioritization, action prioritization, and ecosystem planning into a single, adaptive decision-support system. IMCP is state-led and expert-engaged, integrates conservation across biological, spatial, and temporal scales, and is oriented toward decision support and partnership rather than prescriptive mandates.

In situ: Conservation actions carried out within a species' natural habitat or wild population, such as predator-proof exclosures or direct protection and management of existing wild populations, as distinguished from ex situ actions performed away from the wild.

Maui Nui: The landscape covered by the plan, encompassing the islands of Maui, Moloka‘i, Lāna‘i, and Kaho‘olawe.

OptimTFE: The site-selection spatial prioritization algorithm used within the TFE-SCP process. It generated hundreds of thousands of candidate recovery footprints, each meeting species-specific recovery or extinction-prevention targets, for comparison by species experts and land managers.

Persistence: The probability that a species group will have viable, self-sustaining populations over a 20-year period; the central metric used to evaluate the expected

benefit of different conservation action strategies under the Priority Threat Management analysis.

Planning Unit (PU): The basic spatial unit used in the MNLCP's spatial prioritization process. The Maui Nui landscape was divided into a network of 1,400 medium-sized planning units, each roughly 2 km², delineated using watersheds, natural barriers, and elevational bands. Recovery and priority footprints are built up as collections of planning units, and each unit is sized to support a single population of an at-risk species; finer-scale, within-PU siting of specific actions (like exactly where to outplant or fence) is then worked out by field teams using additional tools.

Priority Footprint: The subset of a recovery footprint identifying the highest-priority planning units for near-term conservation action, which can be further narrowed into more focused annual work objectives. For example, the Terrestrial Plant recovery footprint's 317 planning units include a priority footprint of 155 planning units.

Priority Threat Management: A (modified) structured decision-making process used by the Action Prioritization Team to evaluate the expected costs, feasibility, and conservation benefits of management strategies. It assessed 27 conservation strategies (17 single-focus, 10 combinations) to identify the most cost-effective ways to raise species groups' likelihood of persistence.

Recovery: Restoring an at-risk species to a healthy, self-sustaining population level. A species is considered by the MNLCP to no longer be at risk of extinction once it reaches its established recovery targets, even if it remains reliant on some ongoing conservation actions ("conservation reliant").

Recovery Footprint: The spatially explicit planning units identified by the Spatial Prioritization and Taxonomic Teams as containing existing or restorable habitat needed for an at-risk species, or taxonomic group, to reach its recovery goals.

Spatial Prioritization: The comprehensive planning process—implemented through the TFE-SCP process—that identifies where on the landscape to focus conservation efforts to meet explicit species recovery targets; it addresses the plan's "where to manage" question.

Spatial Prioritization Team: The USGS-led team responsible for the spatial prioritization effort, addressing "where to manage" for the recovery of Maui Nui's at-risk species using the TFE-SCP process.

Taxonomic Team: DOFAW-led teams of species specialists that formed the backbone of the planning effort, providing qualitative input on species distributions, habitat quality, conservation needs, and management feasibility. Seven Taxonomic Teams were established (forest birds, seabirds, waterbirds, plants, snails, aquatic invertebrates,

terrestrial arthropods); the Hawaiian hoary bat is an eighth taxonomic group without its own dedicated team.

Transparent, Flexible, Expert-engaged Systematic Conservation Planning (TFE-SCP) Process: The USGS-developed spatial planning process used to implement spatial prioritization in the MNLCP. It is a comprehensive process with distinct stages, each supported by custom-built tools and involving input from biological experts and land managers, emphasizing transparency in decision-making, flexibility to adapt to different data and management contexts, and deep expert engagement throughout.

Threatened Species: A species formally listed under federal or State law as likely to become endangered in the foreseeable future (denoted “T” in the plan’s ESA status tables), as distinct from a species already listed as endangered. For example, ‘i‘iwi is federally listed as threatened, and nēnē is federally threatened while being state-listed as endangered.

Appendices

1: Spatial Prioritization

2: Priority Threat Management Action Prioritization

3: At-Risk Species List

4: Plant Species List

5: Adaptive Conservation Planner and other tools

6: Terrestrial Snail List

7: Terrestrial Arthropod List

8: Ecosystem Conservation

9: Monitoring and Evaluation of Management Actions

Appendix 1: Spatial Prioritization

This appendix summarizes the spatial prioritization for the Maui Nui Landscape Conservation Plan. The approach is described in Berio Fortini et al. (2024). Methods and results for the Maui Nui application are in preparation for publication (Berio Fortini et al. in prep. a, b; Leopold et al. in prep.; Wilhite et al. in prep. a, b).

Why TFE-SCP: Rationale for the Spatial Prioritization Approach

Maui Nui has more than 400 species at risk of extinction, including federally listed species and many others that are rare, declining, or poorly known. These species face overlapping threats (invasive species, ungulates, wildfire, disease, habitat loss, and climate change) across shared landscapes. Planning that addresses species individually or focuses on single sites does not show managers where limited resources would benefit the most species.

The Maui Nui Landscape Conservation Plan addresses this limitation with Transparent, Flexible, Expert-engaged Systematic Conservation Planning (TFE-SCP; Berio Fortini et al. 2024), the spatial prioritization component of the Integrated Multi-Scale Conservation Planning (IMCP) approach. IMCP also includes action prioritization through Priority Threat Management and ecosystem prioritization; together, these components show where and how to prevent extinctions and recover species.

TFE-SCP generates many candidate conservation footprints with a flexible algorithm and selects among them by weighing trade-offs in expert-agreed criteria (Berio Fortini et al. 2024). DOFAW coordinated taxon expert teams who participated in collaborative review sessions throughout the process. The resulting conservation footprints are maintained in the Adaptive Conservation Planner (Conservation Planner) so they can be updated as knowledge and conditions change (Berio Fortini et al. in prep. a).

Conservation footprints were produced for seven focal taxonomic groups. They account for habitat quality, accessibility, climate resilience, and management feasibility, and can be coordinated across taxa and with action and ecosystem prioritization.

TFE-SCP Process Overview

TFE-SCP is a comprehensive planning process with distinct stages, each supported by purpose-built tools and involving expert and manager input (Berio Fortini et al. in prep. a). The process is state-led and practitioner-driven: DOFAW coordinates expert teams and will implement the resulting conservation priorities, while USGS provided the technical methodology, prioritization tools, and spatial analysis support. The process follows one of two pathways.

i. Algorithmic pathway (Plants, Terrestrial Arthropods, Terrestrial Snails)

- Data gathering and vetting. Species occurrence records, population locations, habitat suitability models, and planning-unit data are compiled through the Expert Data Platform, and taxon experts validate inputs and flag errors or gaps (Leopold et al. in prep.).

- Candidate footprint generation. The OptimTFE R package generates a large set of candidate conservation footprints, each meeting species-specific targets (Berio Fortini et al. 2024).
- Criteria exploration and trade-off analysis. Experts and managers compare candidate footprints across multiple conservation criteria and select a draft footprint that balances competing values (Berio Fortini et al. 2024).
- Planning-unit review. Planning units and species targets in poor habitat are reviewed and adjusted before and after draft footprint selection (Berio Fortini et al. in prep. a).
- Expert fine-tuning. Experts review the draft footprint by species and planning unit in the Conservation Planner and propose target changes with documented rationales through a review and approval workflow (Berio Fortini et al. in prep. a, b).
- Conservation footprint finalization. The reviewed footprint becomes the official conservation footprint, a versioned product for reporting and implementation, while the Conservation Planner remains a living platform for future updates.

ii. Expert-led pathway (Forest Birds, Seabirds, Waterbirds, Aquatic Invertebrates)

For groups with fewer focal species, specialists directly delineated priority areas through iterative web-map review, using the same Expert Data Platform and Conservation Planner but without OptimTFE candidate footprint generation (Leopold et al. in prep.).

Input Data for Spatial Prioritization

The spatial prioritization draws on data layers that characterize each planning unit's biological value, habitat condition, management context, and implementation feasibility. Experts reviewed and refined these inputs before they were used.

i. Species occurrence and population data

For plants, delineated populations were used where available and vetted occurrence records otherwise (Berio Fortini et al. in prep. b). For terrestrial snails, occurrence records include modern and historical observations, though many taxa have sparse data. For terrestrial arthropods, modern occurrence records were supplemented by host-plant associations for data-deficient species. Taxon experts vetted all occurrence data in collaborative review sessions, flagging errors, outdated records, and taxonomic issues.

ii. Habitat suitability and climate envelopes

Habitat suitability surfaces estimate where each species could occur based on environmental conditions. For plants, these were derived from habitat suitability models, as in Berio Fortini et al. (2024). Terrestrial snails had too few spatial records for conventional species distribution modelling, so expert-reviewed climate-similarity models were used as a proxy (Wilhite et al. in prep. a). For terrestrial arthropods, host-plant distributions and broad climate envelopes served as proxy suitability layers where occurrence data were likewise insufficient (Wilhite et al. in prep. b).

iii. Habitat quality layers

Native habitat condition, canopy cover, vegetation structure, and invasive species impacts were mapped across Maui Nui to distinguish high-quality native forest from degraded or converted areas.

iv. Biodiversity reserves

Lands with a durable conservation nexus, primarily the Forest Reserve System, were mapped because they have existing management infrastructure and long-term conservation commitment.

v. Ungulate management areas

Areas with active ungulate control (fencing and/or removal) were mapped because ungulate exclusion is a prerequisite for effective native species management in many habitats in Hawai‘i (Leopold et al. 2024).

vi. Accessibility

Accessibility layers characterize how easily a planning unit can be reached for management, using distance to roads and trails, helicopter landing zones, and terrain difficulty.

vii. Fire risk

Fire risk surfaces reflect the probability and potential impact of wildfire.

viii. Climate resilience

Climate resilience was estimated from the pace of historical changes in temperature and rainfall, to avoid basing estimates of climate risk on uncertain future projections (Berio Fortini et al. in prep. c).

ix. Additional group-specific inputs

Group-specific rules included representation across their historic range in Maui Nui, limits on the number of target species per planning unit, and constraints against co-locating incompatible species (Berio Fortini et al. in prep. a).

Selection Criteria and Trade-off Analysis

The final conservation footprint was not selected by a single metric or by the algorithm alone. Candidate footprints that all met species targets were compared across multiple criteria, and a draft footprint that balanced them was selected (Berio Fortini et al. 2024). The criteria were:

i. **Footprint area.** The number of planning units in the footprint. Smaller footprints concentrate management resources, but not at the cost of habitat quality or accessibility.

ii. **Habitat quality.** The native habitat condition of planning units in the footprint. Higher habitat quality increases the likelihood of successful species recovery and reduces the management effort needed to maintain viable populations.

iii. **Mean habitat suitability.** The mean habitat suitability of target species across planning units in the footprint.

iv. **Biodiversity reserves.** The proportion of the footprint within lands that have a durable conservation nexus (primarily the Forest Reserve System), and where existing management infrastructure can be used.

v. **Ungulate management.** The proportion of the footprint within areas with active ungulate control.

vi. **Accessibility.** How easily planning units can be reached for management. More accessible sites reduce operational costs and make sustained management more feasible.

vii. **Fire risk.** The exposure of planning units to wildfire. Lower fire risk reduces the chance of losing conservation investments.

viii. **Climate resilience.** The relative historical stability of temperature and rainfall in planning units.

ix. **Cross-taxon overlap.** The degree to which a footprint overlaps footprints of other taxonomic groups. Higher overlap creates opportunities for shared management.

Experts compared how candidate footprints ranked under different criteria weightings before selecting a draft footprint (Berio Fortini et al. in prep. a). The emphasis given to each criterion differed among taxonomic groups.

Conservation Footprints

The main outputs of the spatial prioritization are conservation footprints: maps of the planning units selected for species recovery or preventing extinction across Maui Nui. Footprints were produced for each of the seven focal taxonomic groups and can be viewed individually or combined as a multi-taxon “mega-footprint.” Three types of conservation footprints were produced:

i. Recovery footprints identify the planning units needed to meet species recovery targets in Maui Nui, where active management can support population persistence and growth toward recovery goals.

ii. Priority footprints are nested subsets of recovery footprints that identify priority areas where both preventing extinction and recovery goals can be advanced.

iii. Implementation subsets further prioritize within the recovery footprint to guide near-term action staging.

All conservation footprints retain species-level information: each planning unit records which species it was selected for, each species’ habitat suitability, and whether the selection was backed by occurrence records or model predictions (Berio Fortini et al. 2024).

Viewing tools

All group footprints can be viewed individually or combined in the Conservation Planner, which also displays the species-level information for each planning unit.

Expert-Led Selection for Taxonomic Groups

The spatial prioritization covers seven focal taxonomic groups. Plants, terrestrial arthropods, and terrestrial snails followed the algorithmic pathway. Forest birds, seabirds, waterbirds, and aquatic invertebrates followed the expert-led pathway, in which specialists delineated priority areas directly through iterative collaborative review rather than algorithmic footprint generation. Expert-led delineation was used because these groups have few focal species and strong existing expert knowledge (Leopold et al. in prep.).

Seabirds

The seabird footprint covers three federally listed species: Hawaiian petrel, Newell's shearwater, and band-rumped storm-petrel. Seabird specialists reviewed species data and draft recovery areas through the Expert Data Platform over several rounds of map edits, producing recovery areas that incorporate updated occurrence data, predator-proof fence locations, and expert-derived targets (Leopold et al. in prep.).

Forest Birds

Forest bird prioritization covered four species, three of them T&E, through web-map review.

Waterbirds

Waterbird prioritization covered five T&E species through web-map and spreadsheet review.

Aquatic Invertebrates

Aquatic invertebrate prioritization covered 11 species, four of them T&E, with delineation at the watershed scale. Planning units were then linked to named streams and diversions to support implementation.

Cross-group integration

All seven group footprints, algorithmic and expert-led, can be viewed together in the Conservation Planner, which supports concurrent viewing and management of multiple group footprints. Where footprints overlap, actions for one group can benefit others occupying the same landscape.

OptimTFE and the Algorithmically Prioritized Groups

The OptimTFE R package is the quantitative prioritization engine of TFE-SCP. Its flexible greedy algorithm builds many candidate conservation footprints that each meet all species-specific recovery or preventing-extinction targets but differ in which planning units are selected, and therefore in their values across the criteria used to select a draft footprint (Berio Fortini et al. 2024). The MNLCP applied OptimTFE to the three most species-rich taxonomic groups:

a. **Plants.** The reviewed recovery footprint contains 317 planning units. A priority footprint of 181 planning units selects up to three recovery populations per species where feasible, drawing on the best available rare plant data (Berio Fortini et al. in prep. b).

b. **Terrestrial Snails.** The exclosure network comprises 32 in-situ predator-proof exclosure sites with three targets per species (Wilhite et al. in prep. a).

c. **Terrestrial Arthropods.** Taxa with adequate occurrence data received an interim-recovery footprint, and data-deficient taxa were assigned to Focal Area Surveys or Basic Species-specific Surveys (Wilhite et al. in prep. b).

Plants: Conservation Footprint Detail

Plants are the largest and most data-rich taxonomic group in the spatial prioritization.

Approach

Inputs were habitat suitability models, delineated populations, and occurrence records; delineated populations were preferred as representation units where available, and vetted occurrence records were used otherwise. Candidate footprints met species-specific recovery targets on each mauna. The plant team emphasized habitat quality, biodiversity reserves, ungulate management, and compact footprint area. Experts then refined the draft footprint in the Conservation Planner, relocating, adding, and removing species targets with documented rationales (Berio Fortini et al. in prep. b).

Results

The reviewed recovery footprint contains 317 planning units. A priority footprint of 181 planning units selects up to three recovery populations per species where feasible, drawing on the best available rare plant data, while maintaining biogeographic representation of each species across the islands and mountain masses where it occurs. Filtering the priority footprint to Plant Extinction Prevention Program (PEPP) species identifies 110 planning units, an initial estimate of where near-term recovery staging and plant extinction prevention priorities overlap. Some species remain below target because suitable sites or known populations are lacking; these gaps are documented with rationales.

How to use the plant footprint

The recovery footprint identifies where active management should be concentrated for long-term species recovery, and the species targets in each planning unit indicate which species benefit from management there. The Conservation Planner shows the target species and expert notes for each planning unit, including why it was selected. The priority footprint supports implementation staging.

Terrestrial Snails: Conservation Footprint Detail

Terrestrial snails were prioritized algorithmically, but their primary management unit is a managed predator-free refuge enclosed by a predator-proof fence. Because native land snails continue to decline rapidly statewide and safe habitat outside exclosures is dwindling, the snail effort focuses on preventing extinction rather than full recovery. It designs a feasible, climate-resilient exclosure network to protect the most critically endangered Maui Nui snails

from invasive predators while preserving options for future recovery planning (Wilhite et al. in prep. a).

Approach

The prioritization uses Operational Taxonomic Units (OTUs) because some taxa are taxonomically uncertain or represent species complexes. Focal OTUs were drawn from the Snail Extinction Prevention Program and expert-provided lists, including federally listed and other rare and declining taxa. Records include modern and historical observations, though many taxa have sparse data. Because most taxa had too few records for conventional species distribution models, habitat compatibility was estimated with expert-reviewed climate-similarity models.

Unlike the plant and arthropod footprints, the snail network is built from candidate exclosure sites screened for biological suitability and constructability rather than from standard planning units, and existing and planned exclosures were carried forward. Each OTU has three targets within biodiversity reserves and its historical range, with limits on the number of OTUs per exclosure and constraints against co-locating incompatible taxa. The snail team emphasized exclosure count, accessibility, fire risk, and climate resilience. Experts then refined the draft network in the Conservation Planner, resolving species incompatibilities, packing, climate fit, and access constraints (Wilhite et al. in prep. a).

Results

The exclosure network comprises 32 exclosure sites, with three targeted exclosures for each OTU. A priority network of 14 sites provides one target for each of the 49 at-risk snail species.

How to use the snail footprint

The exclosure network identifies priority locations for exclosure construction and maintenance. For each site, the Conservation Planner shows the targeted OTUs, their habitat compatibility scores, occurrence evidence, and expert notes. The priority network identifies where to begin construction so that each species has at least one protected population.

Limitations

- Occurrence data are sparse for many snail taxa. The network reflects the best available information and should be updated as surveys find additional populations.
- The network is designed to prevent extinction and enable future recovery; it does not address landscape-scale threats to snails.

Terrestrial Arthropods: Conservation Footprint Detail

Terrestrial arthropods were prioritized algorithmically, but their workflow is defined by severe distribution-data limitations. The group includes federally listed and other rare or vulnerable taxa such as native *Drosophila* (picture-winged flies), *Hylaeus* (yellow-faced bees), *Manduca* (hawk moths), and leafhoppers.

Approach

Many species lack modern occurrence records, so two tiers of distributional information were used. Occurrence-driven taxa have modern records sufficient for place-based expert review. Host-plant and climate-envelope taxa rely on where their host plants occur or where environmental conditions are broadly suitable, which produces wide and uncertain ranges. Candidate footprints targeted five locations per species as an interim recovery target since we lacked information for most species to plan for full spatial recovery. Targets were places where feasible, and distributed across each species historic range. The arthropod team emphasized habitat quality, known rare arthropod occurrences, and alignment with the plant conservation footprint (Wilhite et al. in prep. b).

Products

The arthropod output comprises two products:

- A conservation footprint for taxa with enough modern occurrence evidence for place-based interim recovery planning.
- Focal Area Survey areas, which show where host-plant associations, habitat quality, accessibility, and other criteria suggest surveys should be focused. A species was assigned to Focal Area Surveys when it lacked enough targets to meet its interim recovery goal but had usable spatial information (host plants, a climate envelope, or occurrence points); species with no usable spatial information were assigned to Basic Species-specific Surveys.

How to use the arthropod footprint

For occurrence-backed species, the conservation footprint works like the plant and snail footprints, identifying planning units with target species and expert-reviewed justifications that can be viewed in the Conservation Planner. For data-deficient species, Focal Area Survey auxiliary layers indicate where additional survey effort could turn up additional species in areas suitable or convenient for conservation. The arthropod footprint overlaps substantially with the plant footprint, so many arthropod sites benefit from habitat protection and restoration already planned for plant recovery.

Limitations

Some species may have no targets until surveys provide defensible recovery locations.

Appendix 2: Priority Threat Management Action Prioritization

Introduction: The report used to develop the Action Prioritization efforts follows:

Development of Methods to Help State natural resource managers optimally allocate conservation effort on the four islands of Maui Nui

Authors: Kristen Harmon, Melissa Price

Prepared for Scott Fretz, Division of Forestry and Wildlife, Maui Nui

Abstract

With hundreds of species on the brink of extinction, conservation practitioners must decide which conservation actions to implement given limited funds. The State of Hawai‘i’s Division of Forestry and Wildlife is developing the Maui Nui Landscape Conservation Plan for Species at Risk for over 300 endangered or declining species native to Maui Nui (the islands of Maui, Moloka‘i, Lāna‘i, and Kaho‘olawe). As part of this effort, we modified a Priority Threat Management approach to guide resource allocation decisions for the conservation of biodiversity in Maui Nui. Data were elicited from land managers and scientists from state, federal, and private entities regarding the expected cost, feasibility, and benefit of management strategies to prevent extinction of 374 species likely to go extinct in the next 20 years in the absence of conservation actions. Elicited data were analyzed to identify cost-efficient strategies that would provide optimal gains in species persistence, and the total costs of preventing extinction across 374 species. We find that management actions such as invasive predator control, invasive ungulate control, and habitat management are important components of strategies with the greatest benefits across species, and for average annual cost of $26M efficiencies may be gained across groups by increasing management collaboration for geographically overlapping species. However, an upfront investment of $67M for actions that only benefit one or a few groups, such as predator exclusion fencing for snails and seabirds, mosquito control for avian malaria, and propagation facilities for plants and invertebrates, is critical to prevent species extinctions in those groups. The insights produced from this project will help managers make decisions about how to maximize conservation outcomes with limited conservation funds. Importantly, we have outlined a process that can be applied to other islands in the state of Hawai‘i and is a first step toward

transforming the Hawaiian Islands from the “Extinction Capital of the World” to the “Recovery Capital of the World”.

Background

Global biodiversity is being lost at rates that are unprecedented in human history. This loss of biodiversity threatens ecosystem services, national economies, food security, and the sustainable health and well-being of societies. Degradation of native ecosystems has been extensive in the Hawaiian Islands. Hundreds of species have become extinct and hundreds more are at risk, including more than 300 species in Maui Nui (the islands of Maui, Moloka‘i, Lāna‘i, and Kaho‘olawe; Figure 1). The Division of Forestry and Wildlife (DOFAW) is charged with the statutory responsibility for conservation of biodiversity in Hawaii (Chapter 195D, Hawaii Revised Statutes) and collaborates broadly with public and private conservation partners to carry out that mission on lands set aside for that purpose. Effective planning is essential to the success of efforts to conserve those species and the ecosystems upon which they depend. However, with more than 300 species at risk of extinction, including plants, invertebrates, and vertebrates, and novel threats that emerge and persist across landscapes that are ecologically and geologically complex and difficult to access, effective implementation of even the best plans can be challenging. Essential conservation actions are costly and often technically and logistically complex. The conservation of at-risk Hawaiian native species is further hampered by insufficient funding and exacerbated by a changing climate. For success in these efforts, it is critical that managers make well-informed decisions on how best to allocate the limited funds and

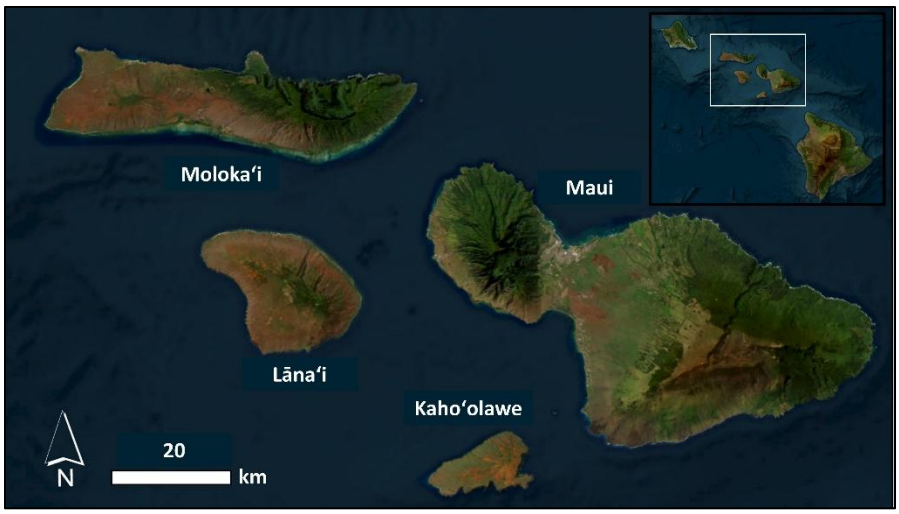
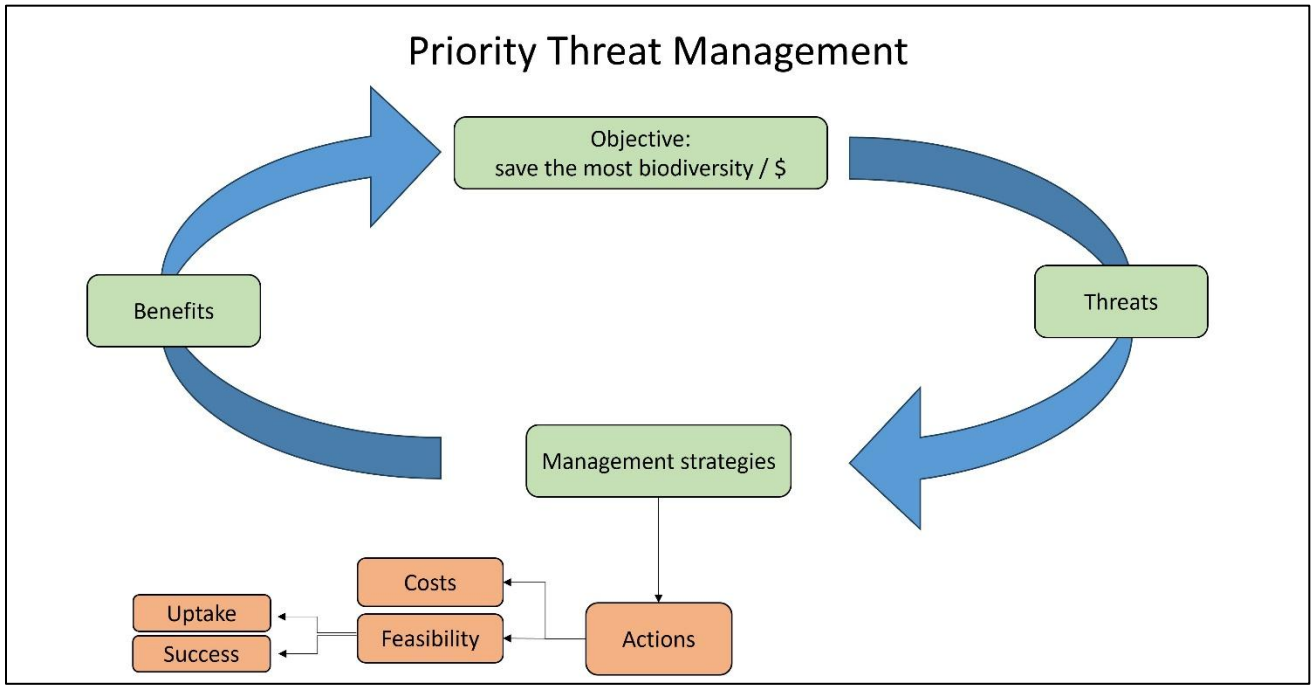


FIGURE 1: Four islands of Maui Nui

operational resources available for the work.

Recovery plans and related documents are available for many of Maui Nui's species at risk. While these documents provide excellent guidance for much of the essential work to be done, they are not generally implementation-ready for managers without further planning. It is highly impractical for managers to work off of dozens of separate plans simultaneously when multiple species and actions in the recovery plans potentially occur in the same place. To improve success and cost effectiveness in that scenario, managers must integrate species plans to identify the minimum reserve area needed to conserve *all* of the species at risk, make informed decisions about which areas to prioritize within that reserve, and determine which conservation actions are the most important ones to implement first. The State of Hawai'i's Division of Forestry and Wildlife is developing the Maui Nui Landscape Conservation Plan for Species at Risk for over 300 endangered or declining species native to Maui Nui to address these planning needs. The multispecies plan, which includes species of plants, vertebrates, and invertebrates, seeks to help managers make informed decisions on how to maximize conservation returns on financial investments at relevant spatial scales, to effectively communicate a blueprint for successful conservation, and to guide changes to policy and law essential for successful conservation of biodiversity in Maui Nui.

As part of this conservation plan, we utilized an adaptable decision framework (Figure 2), known as Priority Threat Management (PTM), to identify a spatially-explicit set of actions that



maximizes gains in species persistence per dollar spent on management of the Maui Nui target species over the next 20 years. is a quantitative decision support framework used to assess the

complementarity of alternative management strategies for recovering biodiversity within a region (Carwardine et al., 2019). The PTM approach has been applied to a wide range of resource management needs at different spatial scales (Kehoe et al. 2020; Camaclang et al. 2020). Results from this project will be integrated with the other components of the Maui Nui Landscape Conservation Plan for Species at Risk.

FIGURE 2: Priority Threat Management (PTM) framework (Carwardine et al. 2019)

Methods

Data Collection

The PTM approach uses a structured elicitation approach to identify components. Roughly 100 experts—individuals with expertise on the ecology and management of species of conservation concern in Maui Nui—participated in a series of focus groups and workshops held from August 2022 – March 2023. Experts were from scientific research institutions federal and state government agencies, and non-profit organizations. During the workshops, experts identified 374 species at risk of extinction within the next 20 years across the islands of Maui Nui (Appendix Table 1). These were grouped into 17 species groups, based on similarity of responses to threats and management actions (see Appendix Table 1), to facilitate the elicitation of the benefits of proposed management strategies. For plant species, we performed a separate Cluster Analysis to identify distinct groups of plant species that share major threats and would thus be expected to benefit similarly from management actions. Our analysis identified six groups of plants (see Appendix Table 1 for list of threats to each plant group). Experts agreed on a time period of 20 years over which to evaluate the costs and benefits of management strategies—long enough to detect the benefits of management, yet still within relevant management timeframes.

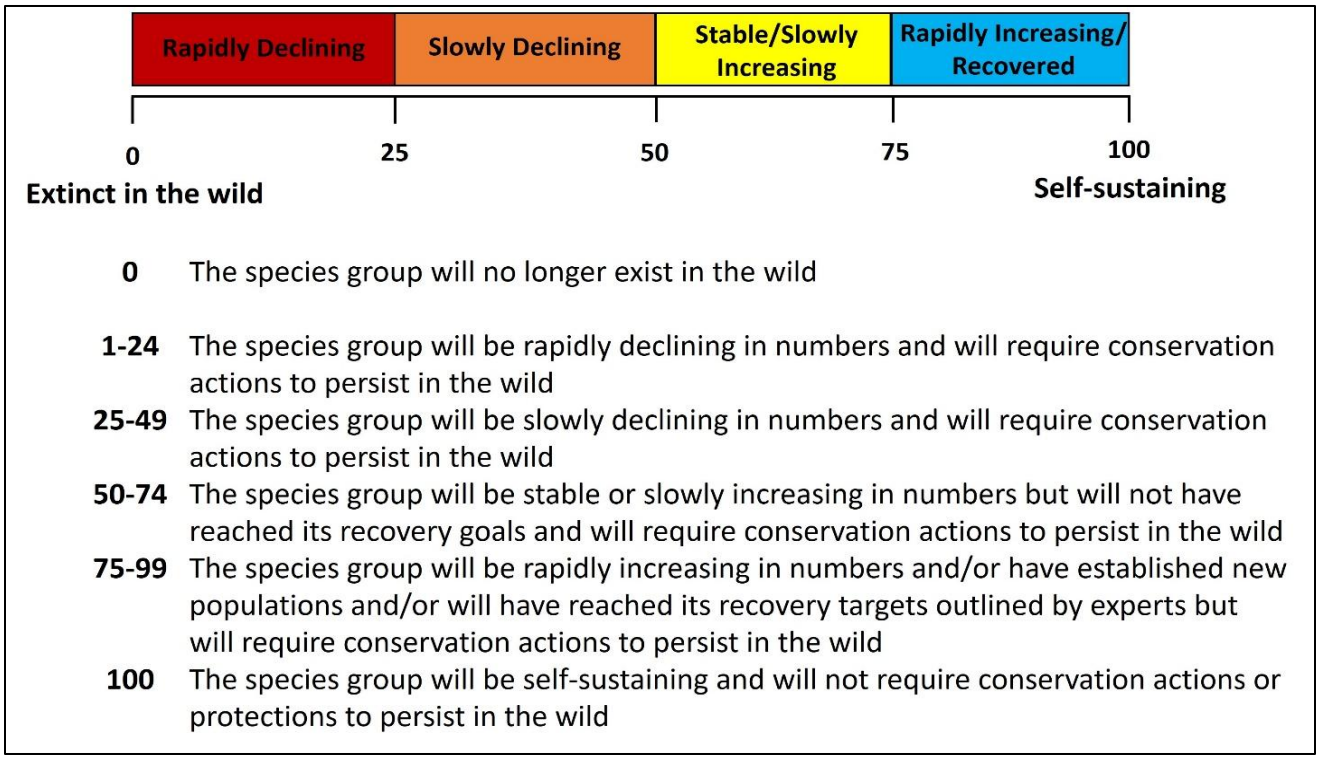
Using their expert judgment and information compiled from published literature and from species status assessments, recovery strategies, action plans, and/or management plans, experts worked in small groups to identify key threats and develop conservation actions to recover at-risk species. Actions were grouped into 17 management strategies such that each strategy could be implemented independently, with the assumption that if a strategy is selected, all actions within that strategy would be implemented (see Appendix Table 2 for a list of all actions within each strategy). None of the other actions were included in more than one management strategy. In addition, experts identified 10 combination strategies, multiple strategies that are anticipated to have synergistic effects when implemented together (see Appendix Table 2).

In small working groups, organized based on their knowledge and experience of a given strategy, we asked experts to estimate the total annual cost—inclusive of materials, equipment, labor, overheads, or costs associated with planning, consultations, or monitoring—of implementing each action in the strategy over the 20-year time period. For actions with uncertain costs, we verified or revised the estimates from the workshop based on information from literature and consultations with additional experts. Experts also had the opportunity to review and provide feedback on the complete set of cost estimates. For some management actions, a cost range was given. For these strategies, an average of the minimum and maximum was used as the "most likely" cost estimate.

We asked experts to provide estimates of the probability that each action (h) in each management strategy (i) will be implemented (U_{hi}), assuming that adequate funds are available, as well as the probability that the action will be successful (S_{hi}). These values were used to estimate a feasibility score for each action, as follows:

$$F_{hi} = U_{hi}S_{hi}, \quad (1)$$

where F_{hi} is the feasibility of action h in strategy i. We then calculated the total feasibility of each



strategy (F_i) as the average feasibility of the individual actions under that strategy. To determine the feasibility for combination strategies, we averaged the feasibilities of individual strategies included in the combination. Experts were asked to provide estimates of the probability of persistence (as a %)

of each species group under (a) a baseline (current status of species group); (b) a counterfactual scenario (no management action), and (c) each individual and combination strategy. We defined persistence as the probability of species groups having viable, self-sustaining populations over a 20-year time period. With the input of experts, we developed a relative scale bar from which experts could derive their estimates (Figure 3).

Figure 3: Scalebar used by experts to estimate the probability of persistence of each species group under the counterfactual and each management strategy.

We used a modified Delphi elicitation protocol (Hemming et al., 2018; Carwardine et al., 2019) to elicit estimates, which provides a structured approach to expert elicitation by asking experts to provide their individual "best guess" estimate (i.e., the most likely value) and the lowest and highest plausible estimates, which represent estimates under the most pessimistic and the most optimistic scenarios, respectively. We facilitated a group discussion among experts prior to the independent elicitation of estimates to ensure a common understanding of the questions and values being estimated. Experts had the opportunity to review and discuss the initial estimates as a group before finalizing their estimates. This approach allowed each expert to benefit from information about the diversity of responses among the wider group of experts, while avoiding many of the pitfalls of unstructured group elicitation methods, such as "groupthink" (Martin et al., 2012). We used the finalized estimates to determine the benefit of implementing each strategy.

Data Analysis

A total of 40 experts provided species persistence predictions. For each individual and combination management strategy, we calculated the benefit for each species group as the difference between the experts' best guess estimates of the performance of the strategy and of the performance of the counterfactual scenario, averaged over the number of experts who provided estimates for the strategy and species group:

$$B_{ij} = \frac{\sum_{k=1}^{K_{ij}} (P_{ijk} - P_{0jk})}{K_{ij}}, \quad \textbf{(2)}$$

where P_{ijk} is the best guess of the probability of persistence of species group j under strategy i as estimated by expert k, P_{0jk} is the best guess of the probability of persistence of species group j under

the counterfactual scenario estimated by expert k, K_{ij} is the number of experts who provided estimates for species group j under strategy i, and B_{ij} is the average benefit of strategy i for species group j.

To determine whether species groups exceed a minimum threshold probability of persistence under each individual or combination strategy, we estimated the expected probability of persistence of species group j under strategy i (P_{ij}) by weighting the average benefit (B_{ij}) of strategy i for species group j by the feasibility (F_i) of strategy i, then adding this expected benefit value to the average estimate of the persistence of species group j under the counterfactual scenario. The following equation was used:

$$P_{ij} = \frac{\sum_{k=1}^{K_{0j}} P_{0jk}}{K_{0j}} + B_{ij}F_i \ , \qquad \textbf{(3)}$$

where K_{0j} is the number of experts who provided estimates of the probability of persistence of species group j under the counterfactual scenario.

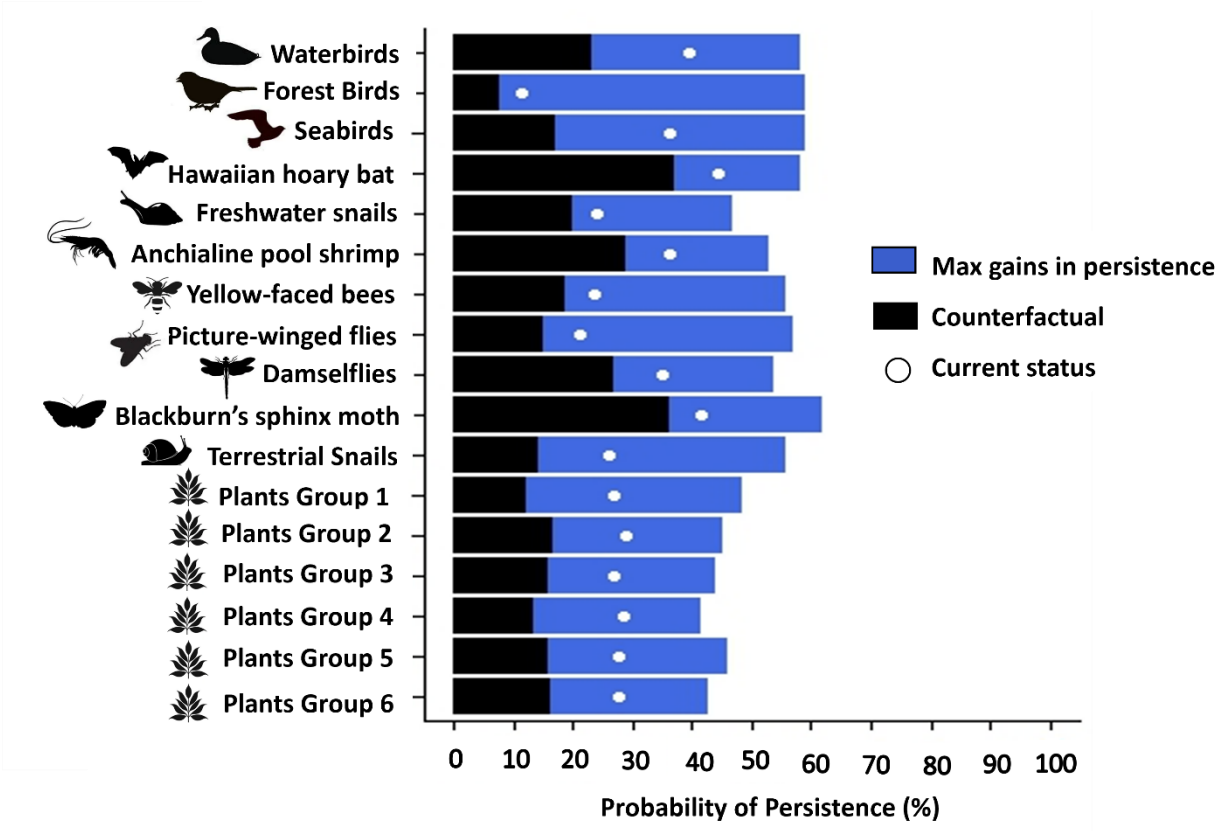
We then performed a complementarity analysis, in which we considered species groups with a probability of persistence greater than a selected threshold to be "secure." We performed this analysis for all species groups combined, as well as for individual taxonomic groups (e.g., vertebrates, invertebrates, and plants), for multiple threshold values—30–60% probability of persistence over the 20-year time period—based on the range of expert-elicited benefit estimates and solved the following integer linear programming problem for each threshold:

$$max \sum_{i=1}^{S} \sum_{j=1}^{G} T_{ij} x_{ij}, \qquad \textbf{(4)}$$

where $T_{ij} = 1$ and $x_{ij} = 1$ if species group j exceeds the threshold under strategy i, and 0 otherwise, C_i is the total cost of implementing strategy i, y_i is a decision variable that indicates whether strategy i is selected (1) or not (0), S is the total number of strategies, and G is the total number of species groups. To examine the effect of uncertainty in benefit estimates on the optimal sets of complementary strategies, we performed the same complementarity analysis using the highest and lowest estimates elicited from experts.

Uncertainty

We address uncertainty in our project by examining the differences in persistence predictions and projected costs of management strategies across experts. We also compared the selection of complementary sets of strategies for optimistic (upper bound), best guess (most likely), and pessimistic estimates (lower bound). This step allows us to check how robust our results are to experts' benefits estimates under the best and worst-case scenarios.



Results

Benefits of Management Strategies to Species Persistence

Not considering feasibility of actions, the average probability of persistence under the counterfactual scenario of no action over the next 20 years is expected to be <30% for 15 out of 17

and <20% for 11 species groups (Figure 4). Under the baseline, or current status, average persistence is predicted to be between 11-44%, with 11 species groups predicted to have <30% chance of persistence (Figure 4). If all management strategies are implemented in Maui Nui over the next 20 years, experts predict that all species groups (374 species) will have at least a 40% chance of persistence, with 11 species groups predicted to reach a 60% or greater chance of persistence (Figure 4; see Appendix Table 2 for full list of management actions within each strategy).

FIGURE 4: Average probability of persistence estimates across all experts for each of 17 species groups under the: a) counterfactual scenario of no management action (black bar); (b) baseline, or current status (white dot); c) all management actions (blue bar). Note: Estimates have not been adjusted for feasibility.

Accounting for feasibility of actions, all species groups still have at minimum a 40% chance of persistence; however, only one species group (Blackburn's sphinx moths) is predicted to have a >60% chance of persistence, and for 10 species groups the maximum probability of persistence is predicted to be between 50-60% (Table 1). Maximum benefits for plant and invertebrate species groups are predicted under combination strategy 25, and the maximum benefits predicted for vertebrate species groups are predicted under combination strategy 27, both of which are "kitchen sink" approaches that contain management actions to address all major threats to plants/invertebrates and vertebrates, respectively (Table 1). For most species groups there are one or two individual management strategies that are predicted to substantially increase probability of persistence when considered on their own, but greater gains in persistence come from implementing combination strategies. Probability of persistence for both seabirds and waterbirds are expected to be at or near 50% under strategy 10 alone (Table 1), which is the construction of new predator exclusion fences; however, probability of persistence is >50% for both groups under management strategies that combine predator control, ungulate control, and habitat management (strategies 22-26; Table 1). Across all insect species groups probability of persistence is predicted to be between 40-50% just by implementing strategy 13 on its own (Table 1). This strategy includes continuing current management actions, as well as investing in additional actions, specific to at-risk invertebrates, such as captive rearing facilities, artificial breeding habitat, pest control, population surveys, and propagation facilities for host plants. For Yellow-faced bees, Picture-winged flies, and Blackburn's sphinx moths several combination strategies that include actions to address feral ungulates, invasive vertebrates and invertebrates, and habitat management (strategies 23, 26, and 27) are expected to perform as good or better than strategy 13 alone (Table 1).

Strategies 10 and 13, when considered on their own, provide moderate gains in probability of persistence for terrestrial snails (near 40%), but predictions are close to 50% under combination strategies that include actions for future predator exclusion fences, invasive invertebrate pest control, and current management actions aimed at native snails (strategies 23, 24, and 27; Table 1). Under management strategy 15, the development of new tools to address avian malaria, forest birds are predicted to have close to 40% chance of persistence (Table 1). While combination strategies that include predator control, ungulate control, and habitat management (strategies 22-26) are predicted to benefit forest birds, these strategies do not provide an additional increase in persistence above strategy 15 (Table 1). For all plant groups, combination strategy 25 is the only strategy that is predicted to increase probability of persistence to above 40% (Table 1). Under all other strategies, probability of persistence is predicted to be close to the current status of each plant group (Table 1). Similarly, for freshwater snails, the majority of management strategies do not provide substantial gains above the current status of the species, with the exception of strategy 25 (Table 1). Anchialine pool shrimp are one of 6 species groups with a current status >30%. Probability of persistence is predicted to increase to 41% under strategy 6 alone, which is wetland and stream habitat management, and the greatest gains in persistence are predicted under strategies 24-27 (Table 1). While the greatest gain in persistence for the Hawaiian hoary bat is predicted under strategy 27, all but three strategies are predicted to increase probability of persistence above 40% (Table 1).

Complementarity of Management Actions

Considering complementarity of management actions among individual taxonomic groups produces different results from considering complementarity among all species groups. Among vertebrate species groups (forest birds, seabirds, waterbirds, and bats), strategy 27 provides the greatest gains in persistence per dollar; all vertebrate groups can be secured with a >50% chance of persistence for a cost of ~$40M per year (Appendix Figure 1). Strategy 27 also provides the most gains per dollar for the invertebrate's groups (Yellow-faced bees, Picture-winged flies, Damselflies, Blackburn's sphinx moth, freshwater snails, anchialine pool shrimp, and terrestrial snails) and will also cost ~$40M per year; however, the threshold at which all species groups can be secured is >40% (Appendix Figure 2). For all plant species groups, strategy 25 is the only strategy that gets any plant species to at least a 40% chance of persistence, and it will cost ~ $45M per year (Appendix Figure 3). Considering complementarity of actions across all species groups can get all groups to at least a 40% chance of

persistence for ~$55M per year (Figure 5), saving ~$10M per year compared to the cost of only considering complementarity among one taxonomic group at a time.

Cost of Securing All Species

Preventing the extinction of all species in Maui Nui over the next 20 years will, on average, be ~$55M per year (Figure 5). However, the actual cost per year varies, with a large portion of the overall costs projected in year one of implementation (Figure 6a). Year one includes an upfront investment in new field vehicles, propagation and rehabilitation facilities, research and development, and construction of ungulate and predator exclusion fences. The remainder of non-recurring costs are for continued construction of ungulate and predator exclusion fences in years 2-10 and field vehicle replacements in years 11 and 16 (Figure 6b). The majority of recurring costs, which are incurred annually, are for salaries and project materials (e.g., predator bait, traps, staff PPE (Figure 6c).

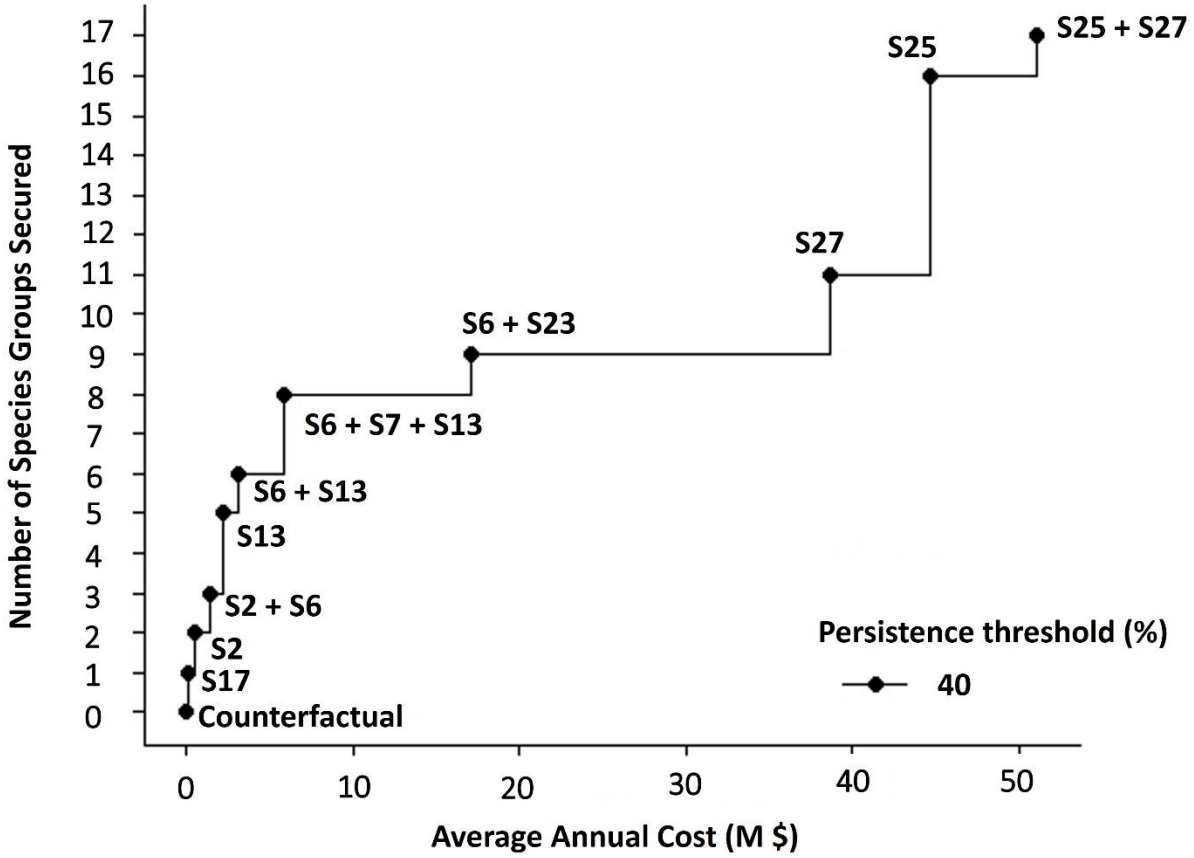


FIGURE 5: The number of species groups that can be secured with >40% probability of persistence by implementing the optimal set of complementary strategies (as represented by the strategy numbers), plotted against the total costs of those strategies.

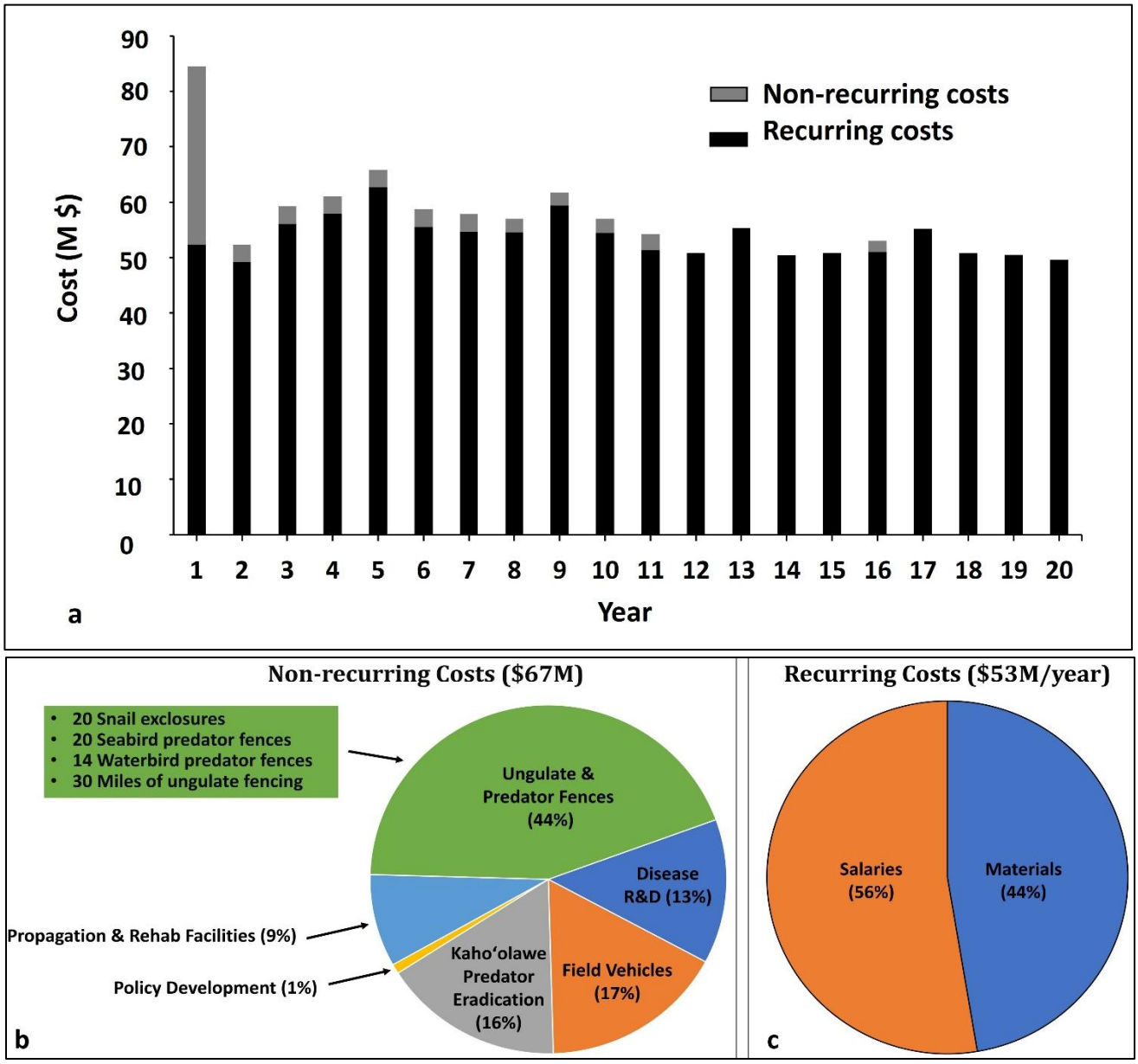


FIGURE 6: a) Non-recurring and recurring costs per year of implementing all management strategies; b) Breakdown of all non-recurring costs by management action; c) breakdown of all recurring costs by cost category.

Uncertainty

We found uncertainty in our results related to differences in individual expert estimates (Appendix Figure 4). It is important to note that while under the "best guess" estimates strategy 7, invasive

vertebrate predator control outside of exclusion fences, is predicted to benefit terrestrial snails, experts estimated that under the "most pessimistic" scenario this strategy could decrease the probability of persistence to close to the counterfactual (Appendix Figure 4). We also found differences between optimistic (upper bound) and pessimistic (lower bound) estimates of persistence. Under the complementarity analysis using the "most pessimistic" estimates all species groups can be secured with a threshold of >30% chance of persistence over the next 20 years, and under the "most optimistic" estimates all species groups can be secured with a >60% chance of persistence.

Table 1: Probability of persistence for each species group and strategy under a "best guess" most likely scenario ((benefit × feasibility) + counterfactual (Equation (4)), where B is the baseline scenario, or current status, and C is the counterfactual scenario of no management action. Maximum persistence estimates for each species group are in bold.

Strategy	B	C	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	S16	S17	S18	S19	S20	S21	S22	S23	S24	S25	S26	S27
Total Cost (M $)	-	0	10	24	49	302	18	55	43	60	82	56	205	44	84	28	15	1	730	59	82	120	281	325	689	895	568	773
Species group																												
Waterbirds	40	23	34	34	29	32	38	49	29	33	50	33	27	29	34	27	33	38	49	33	36	45	55	54	51	54	54	**58**
Forest Birds	11	8	13	14	10	15	9	19	9	12	15	17	8	8	28	39	8	11	36	15	17	22	25	23	21	25	37	**59**
Seabirds	36	17	32	30	23	29	20	46	25	31	48	42	20	20	18	17	38	35	47	31	32	48	54	54	50	53	50	**59**
Yellow-faced bees	23	19	25	27	23	34	24	25	40	21	22	20	36	41	19	19	19	19	39	27	31	34	37	45	50	**56**	45	46
Picture-winged flies	21	15	21	26	22	33	20	27	34	18	31	23	44	44	15	15	15	15	38	30	36	32	36	41	50	**57**	48	51
Damselflies	35	27	31	33	30	35	37	39	37	28	31	30	44	51	27	27	30	27	39	34	37	39	41	43	49	**54**	41	43
Blackburn's sphinx moth	41	36	41	41	39	47	39	40	46	39	39	39	42	51	38	38	43	39	50	41	42	42	46	50	56	**62**	52	53
Freshwater snails	24	20	23	25	22	27	30	30	33	20	20	24	28	33	20	20	24	25	33	26	29	32	36	38	39	**47**	38	41
Anchialine Pool Shrimp	36	29	32	33	31	31	41	31	31	30	30	30	30	33	30	30	33	30	38	31	32	33	35	35	49	52	53	**53**
Terrestrial Snails	26	14	20	21	18	23	17	30	34	24	39	28	28	37	15	15	17	17	37	21	24	29	41	48	48	**56**	39	46
Plants Group 1	27	12	16	17	15	19	12	21	25	13	19	25	16	25	16	17	14	15	20	14	18	17	31	35	30	**48**	23	29
Plants Group 2	29	16	19	22	20	27	20	18	19	16	20	21	22	28	16	16	16	16	24	18	23	19	26	29	29	**45**	26	26
Plants Group 3	27	16	19	26	19	24	16	19	19	16	18	20	21	23	16	16	16	16	23	18	28	19	30	31	28	**44**	23	24
Plants Group 4	28	14	17	23	17	23	15	23	27	15	19	22	20	26	14	14	15	15	22	16	24	18	30	33	28	**42**	24	26
Plants Group 5	28	16	19	25	19	27	16	21	16	16	20	24	21	26	16	16	16	16	22	17	24	18	31	33	29	**46**	24	24
Plants Group 6	28	16	20	22	19	24	16	26	25	18	24	28	21	26	16	16	18	16	24	19	23	21	32	37	30	**43**	27	29
Hawaiian hoary bat	40	37	41	41	38	41	44	46	40	41	43	46	38	40	37	37	47	40	50	42	43	48	51	51	52	53	51	**58**

Discussion

We identified a set of cost-effective strategies to prevent extinctions of all at-risk species across Maui Nui. Using a modified Priority Threat Management decision framework, we find that implementing all management strategies and addressing all major threats to at-risk species over the next 20 years is necessary to prevent the extinction of all 374 species. All species can be secured with a >40% probability of persistence for a cost of ~55M per year, and more than half of those species have a maximum probability of persistence >50%. Without management action, we find that all species have a 70% chance or greater of going extinct. Furthermore, the current level of threat management varies widely by species, but under the "status quo" most species still have <30% chance of persistence over the next 20 years.

We find that landscape-scale management actions, such as invasive predator and ungulate control and habitat management provide some benefit to all species groups, but preventing extinction of all species groups is only possible with the addition of species-specific actions. For example, predator exclusion fences are critical to preventing extinctions of seabirds, waterbirds, and terrestrial snails. Hawaiian forest birds will not persist without substantial and immediate investment in captive breeding facilities, application of techniques to address avian malaria, and translocations to mosquito free areas. Rare plants and invertebrates require investments in captive rearing and propagation facilities, research and development of techniques to control invasive pests, and increased capacity for population surveying and monitoring in order to persist across Maui Nui. These stopgap measures will require an upfront investment of ~$67M, mostly for building fences and facilities, and ~$53M per year to maintain fences and facilities and implement landscape-scale actions. Considering complementarity of management actions across all species groups provides a cost savings of ~$10M per year. Managers could gain efficiencies in species persistence and cost by collaborating on landscape-scale management actions for species that geographically overlap.

While most management strategies were predicted to increase the probability of species persistence, some strategies are predicted to have harmful impacts on some species groups in some instances. For example, implementing invasive vertebrate predator control outside of exclusion fences where invasive predatory snails, such as *Euglandina* spp., are present, may reduce the probability of persistence for native terrestrial snails, as invasive rodents are predators

of *Euglandina* spp. Furthermore, experts highlighted the importance of conducting surveys for rare plants and invertebrates prior to clearing areas for building fences or removing invasive vegetation. If found, these species may need to be relocated before continuing management in the area. Thus, the salaries budgeted for in our process include species specialist positions, as well as general biologist positions with training in native species identification.

We gained several major insights through our expert elicitation process. First, despite our best efforts, we had more contributed predictions for vertebrate species groups than for plants and invertebrates. This is likely due to the need for more plant and invertebrate expertise in the state, particularly within DLNR, which is consistent with the actions and salaried positions budgeted for in this project. Similarly, the gaps in knowledge about the distribution and abundance of native plants and invertebrates, due to the lack of current funding for these programs, made it challenging for experts to make confident predictions about the persistence of these species groups. A further challenge for plant species experts was the fact that unlike most other species groups, plants face a suite of threats that must be addressed to prevent extinctions. This likely explains why the only optimal management strategy for preventing extinctions of plants is a "kitchen sink" approach. Furthermore, it was challenging for experts to estimate geographic scope and costs for landscape-scale actions, particularly removal of invasive vegetation and control of invasive invertebrates, invasive vertebrates, and feral ungulates. In the future, combining results from this project with outputs from the spatial optimization component of the larger Maui Nui Landscape Conservation Plan will help provide insights into priority areas to implement these management actions.

Several management actions were identified in our process that are expected to provide gains in species persistence but need further development or are outside of the authority of the Hawai‘i Department of Land and Natural Resources (DLNR) to implement. For example, gene drives were brought up when discussing management actions to address invasive mammalian predators and avian malaria. As gene drives are still in the infancy stages of development elsewhere, and the research to develop them is not being conducted in Hawai‘i, experts agreed that this action should be left out of our formal process. Furthermore, strategy 11, landscape-scale suppression of rodents, does not include other invasive mammals, such as cats and mongoose, as the regulations for aerial rodenticides for these species have not yet been developed for Hawai‘i. Furthermore, our process does not include actions for biosecurity actions

on a state-wide scale, as this would be outside of DLNR's authority to implement; however, we did include actions to reduce the transport of invasive species within most of the strategies in our process.

As so many of our at-risk species are on the brink of extinction our process focuses on management actions that will prevent extinctions over the next 20 years. Future processes will be needed that identify optimal management actions to achieve species recovery. That said, our process provides insights into actions that need further consideration and development, as they may be necessary for species recovery (e.g., landscape-scale suppression of cats and mongoose) or they may provide substantial cost savings (e.g., state-wide biosecurity). Further, we have provided a decision framework that if applied to other islands in the Hawaiian archipelago can help us transform Hawai'i from the "Extinction Capital of the World" to the "Recovery Capital of the World".

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Appendix

Table 1: List of At-risk Species in Maui Nui

Species Group	Species
Waterbirds	*Anas wyvilliana*
	Fulica alai
	Branta sandvicensis
	Himantopus mexicanus knudseni

Forest Birds	*Drepanis coccinea*
	Palmeria dolei
	Paroreomyza montana
	Pseudonestor xanthophrys
Seabirds	*Oceanodroma castro*
	Pterodroma sandwichensis
	Puffinus newelli
Yellow-faced bees	*Hylaeus anthracinus*
	Hylaeus assimulans
	Hylaeus facilis
	Hylaeus flavipes
	Hylaeus hilaris
	Hylaeus longiceps
	Hylaeus mauiensis
	Hylaeus melanothrix
	Hylaeus psammobius
	Hylaeus satelles
Picture-winged flies	*Drosophila differens*
	Drosophila neoclavisetae
Damselflies	*Megalagrion jugorum*
	Megalagrion molokaiense
	Megalagrion nesiotes
	Megalagrion pacificum
	Megalagrion xanthomelas
Blackburn's Sphinx Moth	*Manduca blackburni*
Freshwater Snails	*Lymnaeidae spp.*
Anchialine Pool Shrimp	*Anchialine pool shrimp*
Terrestrial Snails	*Auriculellan spp.*
	Elasmias spp.
	Newcombia spp.
	Partulina spp.

	Perdicella spp.
	Tornatellaria spp.
	Tornatellides spp.
	Amastra spp.
	Laminella spp.
	Leptachatina
	Hiona spp.
	Philonesia spp.
	Striatura spp.
	Pleuropoma spp.
	Punctum spp.
	Catinella spp.
	Succinea spp.
	Columella spp.
	Lyropupa spp.
	Nesopupa spp.
	Pronesopupa spp.
	Vertigo spp./
Plants Group 1 Threatened by: Invasive invertebrates Invasive vertebrates Feral ungulates Loss of avian pollinators	*Cyanea dunbariae*
	Cyanea duvalliorum
	Cyanea gibsonii
	Cyanea glabra
	Cyanea grimesiana subsp. grimesiana
	Cyanea hamatiflora subsp. hamatiflora
	Cyanea heluensis
	Cyanea horrida
	Cyanea kauaulaensis
	Cyanea kunthiana
	Cyanea lobata subsp. baldwinii
	Cyanea lobata subsp. lobata
	Cyanea magnicalyx

	Cyanea mannii
	Cyanea maritae
	Cyanea mauiensis
	Cyanea mceldowneyi
	Cyanea munroi
	Cyanea obtusa
	Cyanea procera
	Cyanea profuga
	Cyanea solanacea
	Delissea undulata
	Geranium arboreum
	Geranium multiflorum
	Hibiscadelphus stellatus
	Lobelia dunbariae subsp. dunbariae
	Stenogyne calycosa
	Stenogyne kauaulaensis
	Stenogyne angustifolia
	Stenogyne bifida
	Hesperomannia arborescens
	Hibiscus arnottianus subsp. immaculatus
	Kokia cookei
Plants Group 2 Threatened by: Invasive invertebrates feral ungulates	*Achyranthes splendens var. rotundata*
	Argyroxiphium virescens
	Asplenium dielerectum
	Asplenium peruvianum var. insulare
	Canavalia pubescens
	Cyperus pennatiformis var. pennatiformis
	Cyrtandra halawensis
	Cyrtandra hematos
	Deparia kaalaana
	Diplazium molokaiense

	Hypolepis hawaiiensis var. mauiensis
	Kadua cordata subsp. remyi
	Kadua coriacea
	Menisciopsis boydiae
	Microlepia strigosa var. mauiensis
	Nototrichium humile
	Phyllostegia glabra var. glabra
	Phyllostegia macrophylla
	Pteris lidgatei
	Scaevola hobdyi
	Tetramolopium arenarium subsp. arenarium
	Tetramolopium arenarium subsp. laxum
	Tetramolopium lepidotum subsp. lepidotum
Plants Group 3 Threatened by: Feral ungulates	*Adenophorus periens*
	Bidens micrantha subsp. kalealaha
	Calamagrostis hillebrandii
	Canavalia molokaiensis
	Cenchrus agrimonioides var. agrimonioides
	Ctenitis squamigera
	Cyperus fauriei
	Cyperus neokunthianus
	Cyperus prescottianus
	Cyperus trachysanthos
	Festuca hawaiiensis
	Geranium hillebrandii
	Gouania vitifolia
	Ischaemum byrone
	Marsilea villosa
	Peperomia degeneri
	Phlegmariurus stemmermanniae
	Phyllostegia glabra var. lanaiensis

	Portulaca villosa
	Pseudognaphalium sandwicensium var. molokaiense
	Ranunculus hawaiensis
	Schenkia sebaeoides
	Schiedea salicaria
	Silene lanceolata
	Spermolepis hawaiiensis
	Tetramolopium capillare
	Tetramolopium remyi
	Tetramolopium sp. Nov 2
	Wollastonia kamolensis
	Wollastonia populifolia
Plants Group 4 Threatened by: Invasive invertebrates Invasive vertebrates Feral ungulates	*Abutilon eremitopetalum*
	Abutilon menziesii
	Acaena exigua
	Bidens campylotheca subsp. waihoiensis
	Bonamia menziesii
	Argyroxiphium sandwicense subsp. macrocephalum
	Bidens campylotheca subsp. pentamera
	Bidens wiebkei
	Brighamia rockii
	Cyrtandra ferripilosa
	Cyrtandra filipes
	Cyrtandra oxybapha
	Cyrtandra munroi
	Dubautia plantaginea subsp. humilis
	Geniostoma tinifolium var. lanaiense
	Geniostoma triflorum
	Gouania hillebrandii
	Gynochthodes trimera
	Haplostachys haplostachya

	Hibiscus brackenridgei subsp. brackenridgei
	Hibiscus brackenridgei subsp. molokaiananus
	Isodendrion pyrifolium
	Kadua laxiflora
	Kanaloa kahoolawensis
	Lysimachia maxima
	Melicope adscendens
	Melicope balloui
	Melicope mucronulata
	Melicope multiflora
	Melicope munroi
	Melicope oppenheimeri
	Melicope ovalis
	Melicope reflexa
	Melicope wailauensis
	Mezoneuron kavaiense
	Mucuna persericea
	Myrsine vaccinioides
	Neraudia sericea
	Panicum fauriei var. carteri
	Peperomia subpetiolata
	Peucedanum sandwicense
	Phlegmariurus mannii
	Phyllostegia bracteata
	Phyllostegia brevidens
	Phyllostegia haliakalae
	Phyllostegia hispida
	Phyllostegia mannii
	Phyllostegia parviflora var. parviflora
	Phyllostegia pilosa
	Phyllostegia stachyoides

	Pittosporum halophilum
	Plantago princeps var. laxifolia
	Platanthera holochila
	Portulaca sclerocarpa
	Ranunculus mauiensis
	Remya mauiensis
	Sanicula purpurea
	Sanicula sandwicensis
	Santalum haleakalae var. lanaiense
	Schiedea diffusa subsp. difusa
	Schiedea haleakalensis
	Schiedea hookeri
	Schiedea jacobii
	Schiedea laui
	Schiedea lydgatei
	Schiedea nuttallii
	Schiedea pubescens
	Schiedea sarmentosa
	Sesbania tomentosa
	Silene alexandri
	Vigna o-wahuensis
	Wikstroemia villosa
	Lysimachia lydgatei
	Scaevola coriacea
	Viola lanaiensis
Plants Group 5 Threatened by: Invasive vertebrates Feral ungulates	*Bidens conjuncta*
	Calamagrostis expansa
	Exocarpos menziesii
	Festuca molokaiensis
	Geranium hanaense
	Joinvillea ascendens subsp. ascendens

	Portulaca molokiniensis
	Sicyos hillebrandii
	Sicyos macrophyllus
	Solanum incompletum
	Solanum nelsonii
	Tetramolopium rockii var. calcisabulorum
	Tetramolopium rockii var. rockii
Plants Group 6 Long-lived species	*Dracaena fernaldii*
	Eugenia koolauensis
	Alectryon macrococcus var. auwahiensis
	Alectryon macrococcus var. macrococcus
	Alphitonia ponderosa
	Colubrina oppositifolia
	Flueggea neowawraea
	Gardenia brighamii
	Gardenia remyi
	Nothocestrum latifolium
	Ochrosia haleakalae
	Pritchardia munroi
	Zanthoxylum hawaiiense
Hawaiian hoary bat	*Lasiurus semotus*

Table 2: Management Strategies List

Individual Strategies	
S1. Counterfactual	No conservation actions are being taken; existing ungulate and predator fences are no longer being maintained.
S2. Existing Ungulate Fences **Goal:** Remove ungulates from inside fences to minimize impacts to sensitive species from wild ungulates	1. Conduct maintenance checks twice per year and after major storm events on currently existing ungulate fences that are complete or will be complete by Dec. 2023 (see M2. Map). Perform maintenance as needed, including repairing, replacing, or retrofitting/improving fence design, as well as conducting fence corridor vegetation removal. 2. Conduct ungulate control via trapping or aerial shooting within and adjacent to existing fences (see M2. Map). Deploy snares and warning signs regarding snares along fencelines to minimize incursions and remove animals that enter the fenced areas. Maintain and check traps in areas adjacent to fences with high ungulate activity to minimize pressure on fences.
S3. Future Ungulate Fences **Goal:** Remove ungulates from inside fences to minimize impacts to sensitive species from wild ungulates	1. Build and maintain two new ungulate fences per year across Maui Nui (see M3. Map) that are not yet funded and are not currently under construction. Conduct maintenance checks twice per year and after major storm events on newly-built ungulate fences (see M3. Map). Perform maintenance as needed, including repairing, replacing, or retrofitting/improving fence design, as well as conducting fence corridor vegetation removal. 2. Conduct ungulate control via trapping or aerial shooting within and adjacent to newly-built fences (see M3. Map). Deploy snares and warning signs regarding snares along fencelines to minimize incursions and remove animals that enter the fenced areas. Maintain and check traps in areas adjacent to fences with high ungulate activity to minimize pressure on fences.

	1. Conduct education and outreach on management of wild ungulates on private lands.
	2. Work with private landowners to create new ungulate control units to increase hunting opportunities on private lands.
	3. Continue to promote public hunting by conducting hunter training courses for hunting permits, maintaining the "Go Hunt Hawai'i" website & newsletter, and conducting road access maintenance. Evaluate the number of permits issued, animals removed, population numbers, and vital rates of ungulate populations using hunter data and camera trap and/or drone data. Based on evaluation, revise bag limits and open/closed seasons as needed.
	4. Issue permits for commercial wildlife removal services.
	5. Create a landowner incentive program that targets private landowners and incentivizes access and hunting opportunities and removal of axis deer from private lands. Funds would be disbursed directly to landowners based on the number of animals removed (deer tails). Funds are capped at 10% of land acreage for lands greater than 1500 acres and capped at 100 animals for lands less than 1500 acres.
	6. Create an ungulate control implementation guide that guides ungulate control efforts for private and public lands owners. Guide will include zone designation for ungulate tolerance (zero tolerance areas vs public hunting areas vs other lands), based on carrying capacity and land designation goals.
S4. Ungulate Control Outside of Fences **Goal:** Minimize impacts to sensitive species from wild ungulates	7. Deploy and maintain corral traps to remove ungulates outside of ungulate fences in areas where fences will not be built within the next 20 years but need immediate ungulate control (see M4. Map).
	8. Conduct aerial shooting 6-12 hours per week outside of ungulate fences within designated shoot zones in areas where fences will not be built within the next 20 years but need immediate ungulate control (see M4. Map).
	1. Outplant non-coastal matrix (common) plants and seeds from propagation or the wild across Maui Nui within currently known habitat for sensitive species, as well as habitat being prepared for sensitive species, in areas without predator exclusion fences (see M5. map).
S5. Terrestrial Habitat Management to Support Sensitive Species **Goal:** Improve habitat for sensitive species	2. High-volume broadcast seeds of non-coastal matrix plants across Maui Nui within currently known habitat for sensitive species, as well as habitat being prepared for sensitive species, in areas without predator exclusion fences (see M5. map).
	3. Remove invasive localized stands of monoculture species (e.g., strawberry guava) across Maui Nui within currently known habitat for sensitive species, as well as habitat being prepared for sensitive species, in areas without predator exclusion fences (see M5. map).

	4. Remove invasive grasses (biofuel) across Maui Nui within currently known habitat for sensitive species, as well as habitat being prepared for sensitive species, in areas without predator exclusion fences (see M5. map).
	5. Remove invasive understory shrubs and herbs across Maui Nui within currently known habitat for sensitive species, as well as habitat being prepared for sensitive species, in areas without predator exclusion fences (see M5. map).
	6. Remove invasive canopy species across Maui Nui within currently known habitat for sensitive species, as well as habitat being prepared for sensitive species, in areas without predator exclusion fences (see M5. map).
	7. Create firebreaks (fuel breaks and/or 'green' breaks) in areas across Maui Nui with a high risk of wildfire that have at-risk species and in areas adjacent to areas with at-risk species.
	8. Identify and address incipient or potentially explosive populations of invasive species (i.e. pines, albizia, African tulip) in currently known habitat for sensitive species, as well as habitat being prepared for sensitive species (see M5. map).
S6. Wetland/Stream Habitat Management **Goal:** Improve habitat for sensitive species	1. Modify stream barriers/diversions in areas across East Maui where stream flow is reduced and there is potential for lateral entrainment (see M6. Map 1).
	2. Remove invasive species from lowland streams below 1,200 ft. across Maui Nui. Conduct continuous monitoring to maintain invasive species-free streams (see M6. Map 1).
	3. Remove invasive plants in wetland habitats across Maui Nui outside of predator exclusion fences that provide essential habitat for the entire life cycle of native waterbirds (see M6 Map).
	4. Identify and remove sediment from wetlands across Maui Nui that have high rates of sediment accumulation that impact the natural flow of water within critical waterbird habitat (see M6. Map 2 for potential wetland sites).
	5. Create forested bioshields around critical waterbird habitat across Maui Nui (see M6. Map 2).
	6. Identify and protect (through planning zoning or other mechanisms) future waterbird habitat from urban development and climate change impacts.
	7. Build pumping & diking systems for wetlands and drill new wells for extracting groundwater, monitoring salinity, and maintaining hydrological conditions at appropriate levels to avoid or minimize impacts of outbreaks of disease (e.g. botulism) in areas across Maui Nui that provide critical habitat to waterbirds (see M6. Map 2).

	8. Support restoration and management of loʻi for waterbird habitat, sustainable food and ecosystem services through partnerships with lo'i farmers.
	9. Protect anchialine ponds across Maui Nui (see M6. Map 3) by formally including anchialine ponds within protected areas or preclude them from development by zoning or other means, as well as removing invasive species within anchialine ponds.
S7. Invasive Vertebrate Predator Control **Goal:** Minimize impacts to sensitive species from predators	1. Perform localized removal of invasive rodents across Maui Nui within currently known habitat for sensitive species, as well as habitat being prepared for sensitive species, that does not have a predator exclusion fence (see M7. Map). Deploy A24 traps and tracking tunnels. Check and maintain traps and tunnels every 3 months. Establish two new trap grids each year. Perform continuous monitoring to evaluate efficacy of trapping effort and to modify trapping grids as needed.
	2. Trap & remove feral dogs on state, federal, and private lands across Maui Nui managed for sensitive species (see M7. Map) when feral dogs become an issue.
	3. Use a combination of trapping methods (A24, Doc250, Timms traps, and leg hold traps) to remove invasive rats, cats, and mongoose locally on Maui, Molokai, and Lanai within currently known habitat for sensitive species, as well as habitat being prepared for sensitive species, that does not have a predator exclusion fence (see M7. Map). Establish one new trap site each year in these areas. Check and maintain traps for one week every other week. Eradicate cats from the island of Kahoolawe using a combination of trapping and pesticides (see M7. Map).
	4. Control cattle egrets as roosts are identified across Maui Nui in wetlands managed for seabirds and waterbirds (see M7. Map).
	5. Control barn owls as individuals are identified across Maui Nui in areas managed for sensitive species (see M7. Map).
	6. Maintain rodent-free offshore islets by conducting annual trapping surveys for one week, checking traps every 2 days.
	7. Create a predator control response team to provide technical assistance and trapping support for private landowners, nonprofits, and farmers.
S8. Invasive Invertebrate Pest Control **Goal:** Minimize impacts to sensitive species from invertebrate pests	1. Survey sensitive species habitat across Maui Nui (see M8. Map), as well as habitat being prepared for sensitive species, to detect and eradicate little fire ants.
	2. Survey sensitive species habitat across Maui Nui (see M8. Map), as well as habitat being prepared for sensitive species, to detect and control bighead ants.

	3. Survey sensitive species habitat across Maui Nui (see M8. Map), as well as habitat being prepared for sensitive species, to detect and control Argentine ants.
	4. Survey sensitive species habitat across Maui Nui (see M8. Map), as well as habitats being prepared for sensitive species, to detect and control Western Yellowjackets.
S9. Existing Predator Exclusion Fences **Goal:** Minimize impacts to sensitive species from predators	1. Maintain currently existing predator exclusion fences for snails that are complete or will be complete by Dec. 2023 (see M9. Map). Conduct yearly fence maintenance, including checking fences for damage and repairing, replacing, or retrofitting/improving fence design. Conduct fence corridor vegetation removal as needed. Deploy & maintain traps & bait boxes for rats, mice, mongoose, and cats inside existing fences.
	2. Maintain currently existing predator exclusion fences for seabirds that are complete or will be complete by Dec. 2023 (see M9. Map). Conduct yearly fence maintenance, including checking fences for damage and repairing, replacing, or retrofitting/improving fence design. Conduct fence corridor vegetation removal as needed. Deploy & maintain traps & bait boxes for rats, mice, mongoose, and cats inside existing fences.
	3. Translocate seabird chicks into currently existing seabird fences (see M9. Map) and conduct yearly surveys to determine recruitment rates of translocated individuals.
	4. Use social attraction equipment to encourage adult seabirds to nest inside existing seabird fences (see M9. Map) and conduct yearly surveys to determine recruitment rates of translocated individuals.
	5. Translocate rare invertebrates into currently existing predator exclusion fences (see M9. Map) and conduct yearly surveys to determine recruitment rates of translocated individuals.
	6. Remove invasive plants inside existing predator exclusion fences (see M9. Map) when identified & replace with common native plants.
	7. Construct and deploy soft-release snail boxes/grow boxes in unpopulated snail exclosures in Maui Nui that need regular maintenance.
S10. Future Predator Exclusion Fences **Goal:** Minimize impacts to sensitive species from predators	1. Build and maintain 4 new predator exclusion fences per year for snails within the next 5 years (20 fences total) that are each ~1/4-1/2 acre. Conduct yearly fence maintenance, including checking fences for damage and repairing, replacing, or retrofitting/improving fence design. Conduct fence corridor vegetation removal as needed. Deploy & maintain traps & bait boxes for rats, mice, mongoose, and cats inside newly built fences.

	2. Build and maintain two new predator exclusion fences per year for seabirds within the next ten years (20 fences total) that each support 500 adult nesting seabirds. Conduct yearly fence maintenance, including checking fences for damage and repairing, replacing, or retrofitting/improving fence design. Conduct fence corridor vegetation removal as needed. Deploy & maintain traps & bait boxes for rats, mice, mongoose, and cats inside newly-built fences.
	3. Build and maintain two new predator exclusion fences per year for waterbirds within the next ten years (20 fences total). Conduct yearly fence maintenance, including checking fences for damage and repairing, replacing, or retrofitting/improving fence design. Conduct fence corridor vegetation removal as needed. Deploy & maintain traps & bait boxes for rats, mice, mongoose, and cats inside newly-built fences.
	4. Build and maintain new cat-proof (but not rodent proof) fences for waterbirds that will be complete within the next 10 years in habitats that have primarily feral cat predators and/or are difficult to build full predator exclusion fences.
	5. Translocate rare invertebrates into newly-built predator exclusion fences and perform yearly surveys to determine recruitment rates of translocated individuals.
	6. Remove invasive plants inside newly-built predator exclusion fences when identified & replace with common native plants.
S11. Landscape-scale Rodent Suppression **Goal:** Minimize impacts to sensitive species from rodents	1. Eradicate rodents (rats and mice) on the entire island of Kahoolawe (see M11. Map).
	2. Suppress rats regionally on a landscape-scale on Maui, Lanai, and Molokai by deploying aerial rodenticide one per year in remote areas outside of predator exclusion fences that are difficult to access, away from urban areas, and managed for sensitive species, as well as areas where localized predator control is currently taking place (see M11. Map).
S12. Existing Tools for Rare Plants and Invertebrates **Goal:** Increase populations of rare plants & invertebrates	1. Outplant rare plants and seeds from propagation or the wild (see M12. Map) inside ungulate-free areas, including ungulate fences or areas that ungulates cannot access, as well as unfenced coastal terrestrial areas, but outside of predator exclusion fences.
	3. Maintain captive plant propagation facilities built by Dec 31, 2023.
	4. Maintain existing plant pathology diagnostic facilities built by Dec 31, 2023.
	5. Maintain captive invertebrate propagation facilities built on Oahu by Dec 31, 2023.

	6. Create and maintain artificial habitat for insects (e.g., bee blocks, pond liners) in areas outside predator exclusion fences that provide habitat for rare insects (see M12. Map).
	7. Reintroduce arthropods outside of predator exclusion fences in dry forest and coastal habitats up to 2,000 ft.
S13. Future Tools for Rare Plants and Invertebrates **Goal:** Increase populations of rare plants & invertebrates	1. Develop cryopreservation techniques to preserve rare plants at a university in Hawaii.
	2. Develop cryopreservation techniques to preserve rare invertebrates at a university in Hawaii.
	3. Conduct genomics research to develop genetic bar-code readers & examine phylogenetics for rare plants and invertebrates at a university in Hawaii.
	4. Conduct research to improve plant tissue culture techniques to maintain or grow plant cells at a university in Hawaii.
	5. Develop pollination & pollen storage techniques for rare plants at a university in Hawaii.
	6. Build & maintain one captive invertebrate propagation facility on Maui by Dec 31, 2025.
	7. Build & maintain one rare plant propagation facility on Maui, Molokai, and Lanai by Dec 31, 2025.
	8. Develop effective methods to control invasive pests (e.g., naio thrips, slugs, black twig borers, and coconut rhinoceros beetles).
S14. Avian Disease Prevention **Goal:** Prevent the spread of avian diseases	1. Maintain existing captive forest bird breeding facilities built by Dec 31, 2023.
	2. Translocate 50 pairs of ʻĀkohekohe and ʻAlauahio from habitats within Maui Nui (see M14. Map 1 & Map 2) into forest bird habitat safe from avian disease on other Hawaiian Islands.
	2. Continue to release Wolbachia IIT mosquitoes (or SIT in the future) in locations identified by the East Maui environmental assessment (see M14. Map 3).
	3. Expand existing forest bird breeding facilities on land already owned by the state within the next 10 years.
	4. Aerial broadcast BTI in streams near forest bird habitat where mosquitoes are breeding (see M14. Map 3).
	5. Create a response network to survey and remove bird carcasses in wetlands and report potential outbreaks of avian botulism.

	1. Administer a one-time preventative vaccine or probiotic to forest birds across their current ranges (see M15. Map 1, Map 2, & Map 3) to prevent the spread of avian malaria.
S15. Future Tools for Avian Malaria **Goal:** Prevent the spread of avian malaria	2. Collaborate with partners outside of Hawaii to develop genetically engineered mosquitoes (via gene drive or gmo). Release genetically modified mosquitoes in critical forest bird habitat on Maui (see M15. Map 4) or perform island-wide population replacement if regional suppression does not work.
S16. Modification & Minimization of Infrastructure **Goal:** Reduce impacts to native birds from energy infrastructure and car strikes	1. Draft and introduce new legislation (for state-wide policy) to reduce anthropogenic lighting from all state-owned facilities (including streetlights, buildings, schools, etc.), including in Maui Nui. After legislation has passed, hire contractors to perform assessment and planning for all state buildings in Maui Nui.
	2. Advocate and conduct outreach for reduced lighting on city and county, federal, state, and commercial businesses across Maui Nui.
	3. Conduct public outreach to reduce lighting in residential areas across Maui Nui. Educate the public about impacts of lighting on bird and bat species through flyers. Work with vendors to make changes attainable to the public by providing products at reasonable costs.
	4. Work with Hawaiian Electric Company to construct diverters to minimize take of waterbirds and seabirds from collisions with powerlines in areas designated by the Habitat Conservation Plan, as well in all areas where T&E species may be impacted (see M16. Map 1).
	5. Work with wind energy companies to modify wind turbines to reduce avian collisions in areas designated by the Habitat Conservation Plan.
	6. Construct speed bumps within Haleakala to minimize car strikes to seabirds (see M16. Map 2) and Nene (see M16. Map3). Monitor efficacy of speed bumps every year for the first three years and then every three years thereafter. Work with Department of Transportation to modify how they manage roadsides to decrease attraction by Nene.
	7. Conduct community outreach to raise awareness about car strikes/roadkill in residential and commercial areas across Maui Nui. Educate public on the importance of not feeding wildlife. Work with Department of Transportation to reduce appealing roadside habitat for sensitive bird species.
	8. Dedicate staffing for reviewing and updating existing Habitat Conservation Plans.
S17. Vertebrate Health Recovery & Rehabilitation	1. Provide financial support to the Hawaii Wildlife Center on Hawaii island for transporting and rehabilitating injured birds from Maui Nui.

Goal: Recovery downed and injured wildlife	2. Construct satellite rehabilitation facilities on Maui and Molokai to provide temporary facilities for examining and stabilizing downed or injured birds.

Combination Strategies	
S18: Current Tools (M2 + M4 + M5 + M6 + M7 + M9 + M12 + M14)	Existing Ungulate Fences + Ungulate Control Outside Fences + Terrestrial Habitat Management + Stream/Wetland Habitat Management + Invasive Vertebrate Predator Control + Existing Predator Exclusion Fences + Existing Tools for Rare Plants & Invertebrates + Avian Disease Prevention
S19: Existing Ungulate Control (M2 + M4)	Existing Ungulate Fences + Ungulate Control Outside Fences
S20: Existing & Future Ungulate Control (S2 + M3)	Existing Ungulate Fences + Ungulate Control Outside Fences + Future Ungulate Fences
S21: Existing Ungulate Control + Existing Predator Control (S2 + M7 + M9)	Existing Ungulate Fences + Ungulate Control Outside Fences + Invasive Vertebrate Predator Control + Existing Predator Exclusion Fences
S22: Existing & Future Ungulate Control + Existing & Future Predator Control (S4 + M3 + M10 + M11)	Existing Ungulate Fences + Ungulate Control Outside Fences + Future Ungulate Fences + Invasive Vertebrate Predator Control + Existing Predator Exclusion Fences + Future Predator Exclusion Fences + Landscape-scale Rodent Suppression
S23: Existing & Future Ungulate Control + Existing & Future Predator Control + Invertebrate Pest Control (S5+ M8)	Existing Ungulate Fences + Ungulate Control Outside Fences + Future Ungulate Fences + Invasive Vertebrate Predator Control + Existing Predator Exclusion Fences + Future Predator Exclusion Fences + Landscape-scale Rodent Suppression + Invasive Invertebrate Pest Control
S24: Existing Ungulate Control + Existing Predator Control + Habitat Management + Invertebrate Pest Control + Existing Tools for Rare Plants & Invertebrates (S4 + M5 + M6 + M8 + M12)	Existing Ungulate Fences + Ungulate Control Outside Fences + Terrestrial Habitat Management + Stream/Wetland Habitat Management + Invasive Vertebrate Predator Control + Existing Predator Exclusion Fences + Invasive Invertebrate Pest Control + Existing Tools for Rare Plants & Invertebrates
S25: Existing & Future Ungulate Control + Existing & Future Predator Control + Invertebrate Pest Control + Habitat Management + Existing & Future Tools for Rare Plants & Invertebrates (S7 + S3 + M11 + M13)	Existing Ungulate Fences + Ungulate Control Outside Fences + Terrestrial Habitat Management + Stream/Wetland Habitat Management + Invasive Vertebrate Predator Control + Existing Predator Exclusion Fences + Existing Tools for Rare Plants & Invertebrates + Future Ungulate Fences + Future Predator Exclusion Fences + Invasive Invertebrate Pest Control + Landscape-

	scale Rodent Suppression + Future Tools for Rare Plants & Invertebrates
S26: Existing Ungulate Control + Existing Predator Control + Habitat Management + Invertebrate Pest Control + Avian Disease Prevention (S4 + M5 + M6 + M8 + M14)	Existing Ungulate Fences + Ungulate Control Outside Fences + Terrestrial Habitat Management + Stream/Wetland Habitat Management + Invasive Vertebrate Predator Control + Existing Predator Exclusion Fences + Avian Disease Prevention
S27: Existing & Future Ungulate Control + Existing & Future Predator Control + Invertebrate Pest Control + Habitat Management + Avian Disease Prevention + Future Tools for Avian Malaria + Minimization & Modification of Infrastructure + Vertebrate Health Recovery & Rehabilitation (S9 + M3 + M15 + M16 + M17)	Existing Ungulate Fences + Ungulate Control Outside Fences + Terrestrial Habitat Management + Stream/Wetland Habitat Management + Invasive Vertebrate Predator Control + Existing Predator Exclusion Fences + Avian Disease Prevention + Future Ungulate Fences + Invasive Invertebrate Pest Control + Future Predator Exclusion Fences + Landscape-scale Rodent Suppression + Future Tools for Avian Malaria + Minimization & Modification of Infrastructure + Vertebrate Health Recovery & Rehabilitation

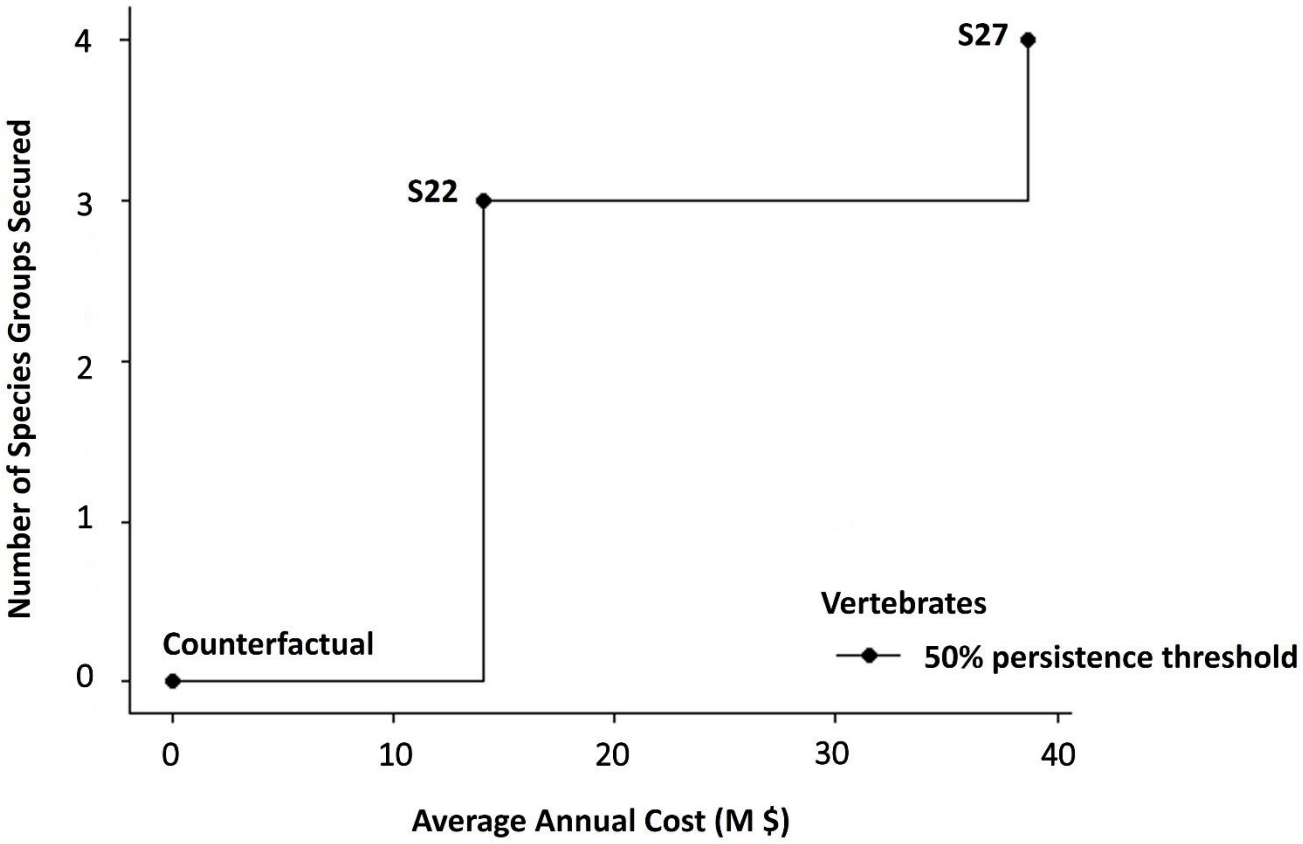


Figure 1: The number of vertebrate species groups that can be secured with >50% probability of persistence by implementing the optimal set of complementary strategies (as represented by the strategy numbers), plotted against the total costs of those strategies.

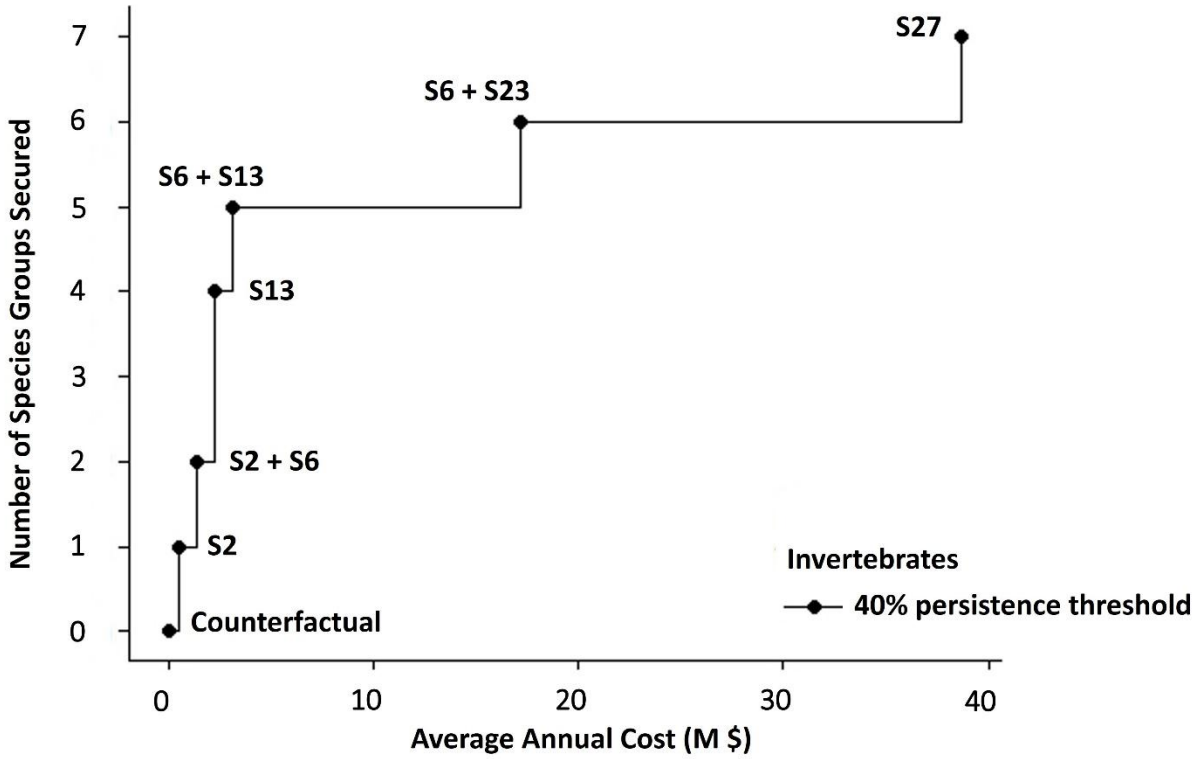


Figure 2: The number of invertebrate species groups that can be secured with >40% probability of persistence by implementing the optimal set of complementary strategies (as represented by the strategy numbers), plotted against the total costs of those strategies.

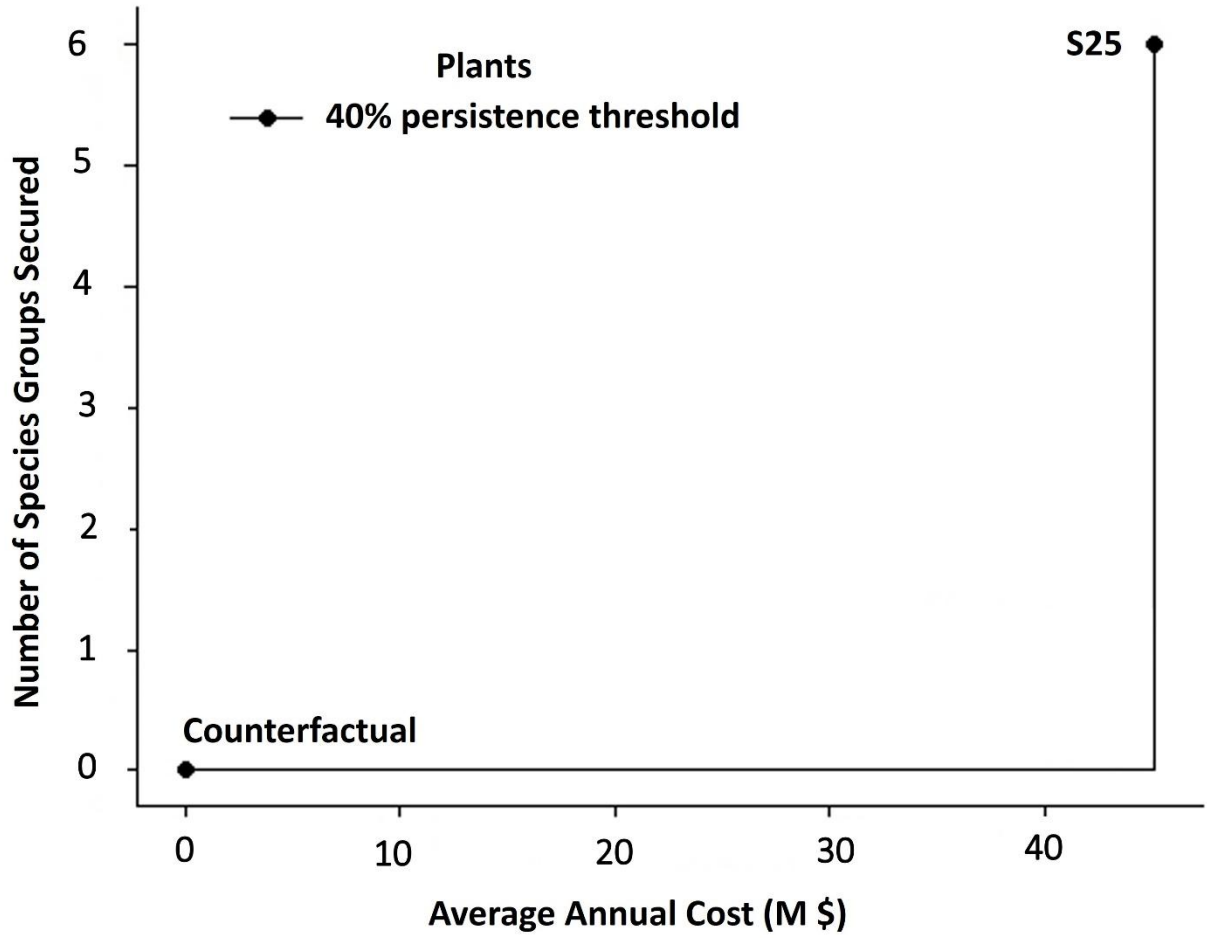


Figure 3: The number of plant species groups that can be secured with >40% probability of persistence by implementing the optimal set of complementary strategies (as represented by the strategy numbers), plotted against the total costs of those strategies.

Appendix 3: At-Risk Species in MNLCP

Taxonomic Group	Species	Common Name
Aquatic Invertebrates	Megalagrion nesiotes	Flying earwig Hawaiian damselfly
Aquatic Invertebrates	Megalagrion pacificum	Pacific Hawaiian damselfly
Aquatic Invertebrates	Megalagrion xanthomelas	Orangeblack Hawaiian damselfly
Aquatic Invertebrates	Procaris hawaiana	Anchialine pool Shrimp
Aquatic Invertebrates	Lymnaeidea ssp	
Aquatic Invertebrates	Antecaridina lauensis	
Aquatic Invertebrates	Calliasmata pholidata	
Aquatic Invertebrates	Halocaridina rubra	
Aquatic Invertebrates	Metabetaeus lohena	
Aquatic Invertebrates	Palaemonella burnsi	
Aquatic Invertebrates	Pele ramseyi	
Bats	Lasiurus semotus	Hawaiian hoary bat
Forest Birds	Drepanis coccinea	`I`iwi
Forest Birds	Palmeria dolei	`Akohekohe (crested honeycreeper)
Forest Birds	Paroreomyza montana newtoni	
Forest Birds	Pseudonestor xanthophrys	Maui parrotbill (Kiwikiu)
Plants	Abutilon eremitopetalum	No common name
Plants	Abutilon menziesii	Ko`oloa`ula
Plants	Acaena exigua	Liliwai
Plants	Achyranthes splendens var. rotundata	Round-leaved chaff-flower
Plants	Adenophorus periens	Palai la`au
Plants	Alectryon macrococcus var. auwahiensis	Mahoe
Plants	Alectryon macrococcus var. macrococcus	Mahoe
Plants	Alphitonia ponderosa	
Plants	Argyroxiphium sandwicense subsp. macrocephalum	`Ahinahina
Plants	Argyroxiphium virescens	
Plants	Asplenium dielerectum	No common name
Plants	Asplenium peruvianum var. insulare	No common name
Plants	Bidens campylotheca subsp. pentamera	Ko`oko`olau
Plants	Bidens campylotheca subsp. waihoiensis	Ko`oko`olau
Plants	Bidens conjuncta	Ko`oko`olau
Plants	Bidens micrantha subsp. kalealaha	Ko`oko`olau
Plants	Bidens wiebkei	Ko`oko`olau
Plants	Bonamia menziesii	No common name
Plants	Brighamia rockii	Pua `ala
Plants	Calamagrostis hillebrandii	Hillebrand's reedgrass
Plants	Canavalia molokaiensis	`Awikiwiki
Plants	Canavalia pubescens	`Awikiwiki
Plants	Cenchrus agrimonioides var. agrimonioides	Kamanomano
Plants	Clermontia grandiflora subsp. maxima	

Taxonomic Group	Species	Common Name
Plants	Clermontia hanaulaensis	
Plants	Clermontia lindseyana	`Oha wai
Plants	Clermontia oblongifolia subsp. brevipes	`Oha wai
Plants	Clermontia oblongifolia subsp. mauiensis	`Oha wai
Plants	Clermontia peleana subsp. singuliflora	`Oha wai
Plants	Clermontia samuelii subsp. hanaensis	`Oha wai
Plants	Clermontia samuelii subsp. samuelii	`Oha wai
Plants	Colubrina oppositifolia	Kauila
Plants	Ctenitis squamigera	Pauoa
Plants	Cyanea asplenifolia	Haha
Plants	Cyanea copelandii subsp. haleakalaensis	Haha
Plants	Cyanea dunbariae	haha
Plants	Cyanea duvalliorum	haha
Plants	Cyanea gibsonii	haha
Plants	Cyanea glabra	Haha
Plants	Cyanea grimesiana subsp. grimesiana	Haha
Plants	Cyanea hamatiflora subsp. hamatiflora	Haha
Plants	Cyanea heluensis	
Plants	Cyanea horrida	haha nui
Plants	Cyanea kauaulaensis	No common name
Plants	Cyanea kunthiana	Haha
Plants	Cyanea lobata subsp. baldwinii	Haha
Plants	Cyanea lobata subsp. lobata	Haha
Plants	Cyanea magnicalyx	haha
Plants	Cyanea mannii	Haha
Plants	Cyanea maritae	haha
Plants	Cyanea mauiensis	haha
Plants	Cyanea mceldowneyi	Haha
Plants	Cyanea munroi	haha
Plants	Cyanea obtusa	Haha
Plants	Cyanea procera	Haha
Plants	Cyanea profuga	Haha
Plants	Cyanea solanacea	Popolo
Plants	Cyperus fauriei	No common name
Plants	Cyperus neokunthianus	No common name
Plants	Cyperus pennatiformis var. pennatiformis	No common name
Plants	Cyperus prescottianus	
Plants	Cyperus trachysanthos	Pu`uka`a
Plants	Cyrtandra ferripilosa	haiwale
Plants	Cyrtandra filipes	Ha`iwale
Plants	Cyrtandra hematos	Ha`iwale
Plants	Cyrtandra munroi	Ha`iwale
Plants	Cyrtandra oxybapha	Ha`iwale

Taxonomic Group	Species	Common Name
Plants	Deparia kaalaana	No common name
Plants	Diplazium molokaiense	No common name
Plants	Dracaena fernaldii	Hala pepe
Plants	Dubautia plantaginea subsp. humilis	Na`ena`e
Plants	Eugenia koolauensis	Nioi
Plants	Exocarpos menziesii	Heau
Plants	Festuca hawaiiensis	No common name
Plants	Festuca molokaiensis	No common name
Plants	Flueggea neowawraea	Mehamehame
Plants	Gardenia brighamii	Hawaiian gardenia
Plants	Gardenia remyi	Nanu
Plants	Geniostoma tinifolium var. lanaiense	Kamakahala
Plants	Geniostoma triflorum	Kamakahala
Plants	Geranium arboreum	Nohoanu
Plants	Geranium hanaense	Nohoanu
Plants	Geranium hillebrandii	Nohoanu
Plants	Geranium multiflorum	Nohoanu
Plants	Gouania hillebrandii	No common name
Plants	Gouania vitifolia	No common name
Plants	Greeneochloa expansa	Maui reedgrass
Plants	Gynochthodes trimera	
Plants	Haplostachys haplostachya	Honohono
Plants	Hesperomannia arborescens	No common name
Plants	Hibiscadelphus stellatus	
Plants	Hibiscus arnottianus subsp. immaculatus	Koki`o ke`oke`o
Plants	Hibiscus brackenridgei ssp. molokaianus	(=Native yellow hibiscus) ma`o hau hele
Plants	Hibiscus brackenridgei subsp. brackenridgei	(=Native yellow hibiscus) ma`o hau hele
Plants	Hypolepis hawaiiensis var. mauiensis	olua
Plants	Ischaemum byrone	Hilo ischaemum
Plants	Isodendrion pyrifolium	wahine noho Kula
Plants	Joinvillea ascendens subsp. ascendens	`Ohe
Plants	Kadua cordata subsp. remyi	kopa
Plants	Kadua coriacea	Kio`ele
Plants	Kadua laxiflora	pilo
Plants	Kanaloa kahoolawensis	Kohe malama malama o Kanaloa
Plants	Kokia cookei	Koki`o
Plants	Lobelia dunbariae subsp. dunbariae	
Plants	Lysimachia lydgatei	No common name
Plants	Lysimachia maxima	No common name
Plants	Marsilea villosa	Ihi`ihi
Plants	Melicope adscendens	Alani
Plants	Melicope balloui	Alani
Plants	Melicope mucronulata	Alani

Taxonomic Group	Species	Common Name
Plants	Melicope multiflora	Alani
Plants	Melicope munroi	Alani
Plants	Melicope oppenheimeri	
Plants	Melicope ovalis	Alani
Plants	Melicope reflexa	Alani
Plants	Melicope wailauensis	
Plants	Menisciopsis boydiae	Kupukupu makali`i
Plants	Mezoneuron kavaiense	Uhiuhi
Plants	Microlepia setosa var. mauiensis	No common name
Plants	Mucuna persericea	sea bean
Plants	Myoporum degeneri	
Plants	Myrsine vaccinioides	Kolea
Plants	Neraudia sericea	No common name
Plants	Nothocestrum latifolium	`Aiea
Plants	Nototrichium humile	Kulu`i
Plants	Ochrosia haleakalae	Holei
Plants	Panicum fauriei var. carteri	Carter's panicgrass
Plants	Peperomia degeneri	
Plants	Peperomia subpetiolata	`Ala `ala wai nui
Plants	Peucedanum sandwicense	Makou
Plants	Phlegmariurus mannii	Wawae`iole
Plants	Phlegmariurus stemmermanniae	No common name
Plants	Phyllostegia bracteata	No common name
Plants	Phyllostegia brevidens	No common name
Plants	Phyllostegia glabra var. glabra	
Plants	Phyllostegia glabra var. lanaiensis	No common name
Plants	Phyllostegia haliakalae	No common name
Plants	Phyllostegia hispida	No common name
Plants	Phyllostegia macrophylla	
Plants	Phyllostegia mannii	No common name
Plants	Phyllostegia parviflora var. parviflora	No common name
Plants	Phyllostegia pilosa	No common name
Plants	Phyllostegia stachyoides	No common name
Plants	Pittosporum halophilum	Hoawa
Plants	Plantago princeps var. laxifolia	Kuahiwi laukahi
Plants	Platanthera holochila	No common name
Plants	Portulaca molokiniensis	
Plants	Portulaca sclerocarpa	Po`e
Plants	Portulaca villosa	Ihi
Plants	Pritchardia munroi	loulu
Plants	Pseudognaphalium sandwicensium var. molokaiense	`Ena`ena
Plants	Pteris lidgatei	No common name
Plants	Ranunculus hawaiensis	Makou

Taxonomic Group	Species	Common Name
Plants	Ranunculus mauiensis	Makou
Plants	Remya mauiensis	Maui remya
Plants	Sanicula purpurea	No common name
Plants	Sanicula sandwicensis	No common name
Plants	Santalum haleakalae var. lanaiense	Lanai sandalwood (=`iliahi)
Plants	Scaevola coriacea	Dwarf naupaka
Plants	Scaevola hobdyi	
Plants	Schenkia sebaeoides	Awiwi
Plants	Schiedea diffusa subsp. diffusa	No common name
Plants	Schiedea haleakalensis	No common name
Plants	Schiedea hookeri	No common name
Plants	Schiedea jacobii	No common name
Plants	Schiedea laui	No common name
Plants	Schiedea lydgatei	No common name
Plants	Schiedea nuttallii	No common name
Plants	Schiedea pubescens	Ma`oli`oli
Plants	Schiedea salicaria	No common name
Plants	Schiedea sarmentosa	No common name
Plants	Sesbania tomentosa	Ohai
Plants	Sicyos hillebrandii	
Plants	Sicyos macrophyllus	`Anunu
Plants	Silene alexandri	No common name
Plants	Silene lanceolata	No common name
Plants	Solanum incompletum	Popolo ku mai
Plants	Solanum nelsonii	Popolo
Plants	Spermolepis hawaiiensis	No common name
Plants	Stenogyne angustifolia	No common name
Plants	Stenogyne bifida	No common name
Plants	Stenogyne calycosa	
Plants	Stenogyne kauaulaensis	No common name
Plants	Tetramolopium arenarium subsp. arenarium	No common name
Plants	Tetramolopium capillare	Pamakani
Plants	Tetramolopium lepidotum subsp. lepidotum	No common name
Plants	Tetramolopium remyi	No common name
Plants	Tetramolopium rockii var. calcisabulorum	No common name
Plants	Tetramolopium rockii var. rockii	No common name
Plants	Tetramolopium sp. Nov 2	
Plants	Vigna owahuensis	No common name
Plants	Viola lanaiensis	No common name
Plants	Wikstroemia villosa	No common name
Plants	Wollastonia kamolensis	Nehe
Plants	Wollastonia populifolia	
Plants	Zanthoxylum dipetalum var. dipetalum	

Taxonomic Group	Species	Common Name
Plants	Zanthoxylum hawaiiense	A`e
Seabirds	Hydrobates castro	Band-rumped Storm-petrel
Seabirds	Pterodroma sandwichensis	Hawaiian petrel
Seabirds	Puffinus newelli	Newell"s shearwater
Terrestrial Arthropods	Acrolepia beardsleyi	
Terrestrial Arthropods	Agrotis cremata	
Terrestrial Arthropods	Blackburnia deroderus	
Terrestrial Arthropods	Blackburnia gracilis	
Terrestrial Arthropods	Blackburnia lenta	
Terrestrial Arthropods	Caconemobius howarthi	
Terrestrial Arthropods	Cerotrioza bivittata	
Terrestrial Arthropods	Chersodromia hawaiiensis	
Terrestrial Arthropods	Coleotichus blackburniae	
Terrestrial Arthropods	Colobopyga pritchardiae	
Terrestrial Arthropods	Crawforda triopsyllina	
Terrestrial Arthropods	Dictyophorodelphax praedicta	
Terrestrial Arthropods	Dictyophorodelphax usingeri	
Terrestrial Arthropods	Drosophila affinidisjuncta	
Terrestrial Arthropods	Drosophila balioptera	
Terrestrial Arthropods	Drosophila bostrycha	
Terrestrial Arthropods	Drosophila cilifera	
Terrestrial Arthropods	Drosophila clavisetae	
Terrestrial Arthropods	Drosophila differens	Hawaiian picture-wing fly
Terrestrial Arthropods	Drosophila disjuncta	
Terrestrial Arthropods	Drosophila grimshawi	
Terrestrial Arthropods	Drosophila gymnobasis	
Terrestrial Arthropods	Drosophila hamifera	
Terrestrial Arthropods	Drosophila hirtipalpus	
Terrestrial Arthropods	Drosophila lanaiensis	
Terrestrial Arthropods	Drosophila lasiopoda	
Terrestrial Arthropods	Drosophila limitata	
Terrestrial Arthropods	Drosophila liophallus	
Terrestrial Arthropods	Drosophila nalowale	
Terrestrial Arthropods	Drosophila neoclavisetae	Hawaiian picture-wing fly
Terrestrial Arthropods	Drosophila nukea	
Terrestrial Arthropods	Drosophila obscuripes	
Terrestrial Arthropods	Drosophila odontophallus	
Terrestrial Arthropods	Drosophila oreas	
Terrestrial Arthropods	Drosophila orphnopeza	
Terrestrial Arthropods	Drosophila orthofascia	
Terrestrial Arthropods	Drosophila paenehamifera	
Terrestrial Arthropods	Drosophila peniculipedis	
Terrestrial Arthropods	Drosophila pihulu	

Taxonomic Group	Species	Common Name
Terrestrial Arthropods	Drosophila pilatisetae	
Terrestrial Arthropods	Drosophila planitibia	
Terrestrial Arthropods	Drosophila prostopalpis	
Terrestrial Arthropods	Drosophila sodomae	
Terrestrial Arthropods	Drosophila spectabilis	
Terrestrial Arthropods	Drosophila toxochaeta	
Terrestrial Arthropods	Drosophila truncipenna	
Terrestrial Arthropods	Drosophila varipennis	
Terrestrial Arthropods	Drosophila vesciseta	
Terrestrial Arthropods	Drosophila villitibia	
Terrestrial Arthropods	Ectemnius haleakalae	
Terrestrial Arthropods	Eidoleon perjurus	
Terrestrial Arthropods	Eopenthes arduus	
Terrestrial Arthropods	Eopenthes auratus	
Terrestrial Arthropods	Eopenthes celatus	
Terrestrial Arthropods	Eopenthes deceptor	
Terrestrial Arthropods	Eopenthes plebeius	
Terrestrial Arthropods	Eopenthes politus	
Terrestrial Arthropods	Euodynerus nigripennis	
Terrestrial Arthropods	Genophantis leahi	
Terrestrial Arthropods	Glyphodes cyanomichla	
Terrestrial Arthropods	Glyphodes exaula	
Terrestrial Arthropods	Gonitis vulpicolor	
Terrestrial Arthropods	Helicoverpa confusa	
Terrestrial Arthropods	Hylaeus angustulus	
Terrestrial Arthropods	Hylaeus anthracinus	Anthricinan yellow-faced bee
Terrestrial Arthropods	Hylaeus assimulans	Assimulans yellow-faced bee
Terrestrial Arthropods	Hylaeus coniceps	
Terrestrial Arthropods	Hylaeus connectens	
Terrestrial Arthropods	Hylaeus facilis	Easy yellow-faced bee
Terrestrial Arthropods	Hylaeus flavipes	
Terrestrial Arthropods	Hylaeus fuscipennis	
Terrestrial Arthropods	Hylaeus haleakalae	
Terrestrial Arthropods	Hylaeus hilaris	Hilaris yellow-faced bee
Terrestrial Arthropods	Hylaeus kukui	
Terrestrial Arthropods	Hylaeus laetus	
Terrestrial Arthropods	Hylaeus longiceps	Hawaiian yellow-faced bee
Terrestrial Arthropods	Hylaeus mauiensis	
Terrestrial Arthropods	Hylaeus melanothrix	
Terrestrial Arthropods	Hylaeus niloticus	
Terrestrial Arthropods	Hylaeus paumako	
Terrestrial Arthropods	Hylaeus psammobius	
Terrestrial Arthropods	Hylaeus satelles	

Taxonomic Group	Species	Common Name
Terrestrial Arthropods	Hylaeus specularis	
Terrestrial Arthropods	Hylaeus unicus	
Terrestrial Arthropods	Hylaeus volatilis	
Terrestrial Arthropods	Ithamar annectans	
Terrestrial Arthropods	Ithamar hawaiiensis	
Terrestrial Arthropods	Kalania hawaiiensis	
Terrestrial Arthropods	Kirkaldiella euphorbiae	
Terrestrial Arthropods	Leptogryllus deceptor	
Terrestrial Arthropods	Manduca blackburni	Blackburn's sphinx moth
Terrestrial Arthropods	Megalagrion jugorum	
Terrestrial Arthropods	Megalagrion molokaiense	
Terrestrial Arthropods	Megatrioza mauiensis	
Terrestrial Arthropods	Megatrioza molokaiensis	
Terrestrial Arthropods	Micromus cookeorum	
Terrestrial Arthropods	Micromus haleakalae	
Terrestrial Arthropods	Micromus lobipennis	
Terrestrial Arthropods	Micromus vulcanius	
Terrestrial Arthropods	Neomachilis heteropus	
Terrestrial Arthropods	Neseis haleakalae	
Terrestrial Arthropods	Nesocryptias villosa	
Terrestrial Arthropods	Nesosydne acuta	
Terrestrial Arthropods	Nesosydne argyroxiphii	
Terrestrial Arthropods	Nesosydne bridwelli	
Terrestrial Arthropods	Nesosydne cyrtandrae	
Terrestrial Arthropods	Nesosydne geranii	
Terrestrial Arthropods	Nesosydne leahi	
Terrestrial Arthropods	Nesosydne longipes	
Terrestrial Arthropods	Nesosydne sulcata	
Terrestrial Arthropods	Nesotocus munroi	
Terrestrial Arthropods	Nesotocus spp.	
Terrestrial Arthropods	Oechalia grisea	
Terrestrial Arthropods	Oliarus hevaheva	
Terrestrial Arthropods	Oliarus priola	
Terrestrial Arthropods	Omiodes anastrepta	
Terrestrial Arthropods	Omiodes asaphombra	
Terrestrial Arthropods	Omiodes monogona	
Terrestrial Arthropods	Omiodes musicola	
Terrestrial Arthropods	Orthotylus diospyricola	
Terrestrial Arthropods	Orthotylus diospyrivorus	
Terrestrial Arthropods	Orthotylus diospyroides	
Terrestrial Arthropods	Orthotylus ureraphila	
Terrestrial Arthropods	Orthotylus xylosmoides	
Terrestrial Arthropods	Pentarthrum obscurum	

Taxonomic Group	Species	Common Name
Terrestrial Arthropods	Philodoria hesperomanniella	
Terrestrial Arthropods	Philodoria hesperomanniella sp. n	
Terrestrial Arthropods	Philodoria keahii	
Terrestrial Arthropods	Philodoria lipochaetaella	
Terrestrial Arthropods	Philodoria platyphylliella	
Terrestrial Arthropods	Philodoria platyphylliella sp. n	
Terrestrial Arthropods	Philodoria sp.	
Terrestrial Arthropods	Philodoria sp. 2	
Terrestrial Arthropods	Philodoria spilota	
Terrestrial Arthropods	Philodoria touchardiella	
Terrestrial Arthropods	Philodoria wilkesiella	
Terrestrial Arthropods	Phyllococcus oahuensis	
Terrestrial Arthropods	Plagithmysus	
Terrestrial Arthropods	Plagithmysus alani	
Terrestrial Arthropods	Plagithmysus bidensae	
Terrestrial Arthropods	Plagithmysus dubautianus	
Terrestrial Arthropods	Plagithmysus forbesii	
Terrestrial Arthropods	Plagithmysus fractus	
Terrestrial Arthropods	Plagithmysus fugitivus	
Terrestrial Arthropods	Plagithmysus geranii	
Terrestrial Arthropods	Plagithmysus koae	
Terrestrial Arthropods	Plagithmysus lanaiensis	
Terrestrial Arthropods	Plagithmysus longicollis	
Terrestrial Arthropods	Plagithmysus pipturicola	
Terrestrial Arthropods	Plagithmysus pulvillatus	
Terrestrial Arthropods	Plagithmysus rubi	
Terrestrial Arthropods	Plagithmysus simillimus	
Terrestrial Arthropods	Plagithmysus speculifer	
Terrestrial Arthropods	Plagithmysus swezeyanus	
Terrestrial Arthropods	Plagithmysus sylvai	
Terrestrial Arthropods	Proterhinus bridwelli	
Terrestrial Arthropods	Proterhinus fuscicolor	
Terrestrial Arthropods	Proterhinus sp.	
Terrestrial Arthropods	Rhynchephestia rhabdotis	
Terrestrial Arthropods	Rhyncogonus lahainae	
Terrestrial Arthropods	Rhyncogonus molokaiensis	
Terrestrial Arthropods	Rhyncogonus sharpi	
Terrestrial Arthropods	Rhyncogonus simplex	
Terrestrial Arthropods	Rynchogonus montygorum	
Terrestrial Arthropods	Rynchogonus sordidus	
Terrestrial Arthropods	Rynchogonus sp.	
Terrestrial Arthropods	Rynchogonus vestitus	
Terrestrial Arthropods	Rynchogonus zeta	

Taxonomic Group	Species	Common Name
Terrestrial Arthropods	Saicella smithi	
Terrestrial Arthropods	Spheterista oheoheana	
Terrestrial Arthropods	Thyrocopa apatela	
Terrestrial Arthropods	Udea dryadopa	
Terrestrial Snails	Amastra goniops	
Terrestrial Snails	Amastra montana	
Terrestrial Snails	Auriculella lanaiensis	
Terrestrial Snails	Laminella aspera	
Terrestrial Snails	Laminella citrina	
Terrestrial Snails	Laminella venusta	
Terrestrial Snails	Leptachatina nitida	
Terrestrial Snails	Leptachatina vitreola	
Terrestrial Snails	Lyropupa striatula	
Terrestrial Snails	Newcombia canaliculata	
Terrestrial Snails	Newcombia cumingi	Newcomb's Tree snail
Terrestrial Snails	Newcombia cumingi sp. 1	
Terrestrial Snails	Newcombia pfeifferi	
Terrestrial Snails	Partulina anceyana	
Terrestrial Snails	Partulina crocea	
Terrestrial Snails	Partulina dolei	
Terrestrial Snails	Partulina induta	
Terrestrial Snails	Partulina kaaeana	
Terrestrial Snails	Partulina marmorata	
Terrestrial Snails	Partulina mighelsiana bella	
Terrestrial Snails	Partulina mighelsiana mighelsiana	
Terrestrial Snails	Partulina mighelsiana polita	
Terrestrial Snails	Partulina mighelsiana ssp.	
Terrestrial Snails	Partulina perdix ssp.	
Terrestrial Snails	Partulina perdix undosa	
Terrestrial Snails	Partulina porcellana ssp.	
Terrestrial Snails	Partulina porcellana wailuaensis	
Terrestrial Snails	Partulina proxima schauinslandi	
Terrestrial Snails	Partulina proxima ssp.	
Terrestrial Snails	Partulina redfieldi	
Terrestrial Snails	Partulina semicarinata	Lanai tree snail
Terrestrial Snails	Partulina splendida	
Terrestrial Snails	Partulina tappaniana ampulla	
Terrestrial Snails	Partulina tappaniana dubiosa	
Terrestrial Snails	Partulina tappaniana eburnea	
Terrestrial Snails	Partulina terebra lignaria	
Terrestrial Snails	Partulina terebra ssp.	
Terrestrial Snails	Partulina tessellata	
Terrestrial Snails	Partulina undosa	

Taxonomic Group	Species	Common Name
Terrestrial Snails	Partulina ustulata	
Terrestrial Snails	Partulina variabilis	Lanai tree snail
Terrestrial Snails	Partulina virgulata	
Terrestrial Snails	Perdicella helena balteata	
Terrestrial Snails	Perdicella helena ssp.	
Terrestrial Snails	Perdicella kuhnsi	
Terrestrial Snails	Perdicella ornata	
Terrestrial Snails	Pleuropoma laciniosa piliformis	
Terrestrial Snails	Pleuropoma laciniosa ssp. 1	
Terrestrial Snails	Pleuropoma laciniosa ssp. 2	
Waterbirds	Anas wyvilliana	Hawaiian duck
Waterbirds	Branta sandvicensis	Hawaiian goose
Waterbirds	Fulica alai	Hawaiian coot (alae ke`oke`o)
Waterbirds	Gallinula galeata sandvicensis	Hawaiian common gallinule
Waterbirds	Himantopus mexicanus knudseni	Hawaiian stilt

Appendix 4: Plant Species List

At-Risk Plant Species	Occupied PUs for Species	Federal and State Status	PEPP Species	Desired Recovery Populations	Recovery PUs in MNLCP	Occupied PUs in Recovery Footprint
Abutilon eremitopetalum	2	E	PEPP	10	9	2
Abutilon menziesii	11	E	ROI	6	6	5
Acaena exigua	1	E	PEPP	5	5	1
Achyranthes splendens var. rotundata	2	E	ROI	4	4	1
Adenophorus periens	0	E	PEPP	3	3	0
Alectryon macrococcus var. auwahiensis	7	E	PEPP	10	10	5
Alectryon macrococcus var. macrococcus	17	E	PEPP	6	6	5
Alphitonia ponderosa	6		ROI	5	5	3
Argyroxiphium sandwicense ssp. macrocephalum	27	T		1	10	10
Argyroxiphium virescens	5		PEPP	10	4	2
Asplenium dielerectum	4	E	PEPP	6	6	2
Asplenium peruvianum var. insulare	10	E		5	5	5
Bidens campylotheca ssp. pentamera	14	E		10	10	9
Bidens campylotheca ssp. waihoiensis	12	E	POP	10	10	9
Bidens conjuncta	10	E		10	10	10
Bidens micrantha ssp. kalealaha	17	E		10	10	9
Bidens wiebkei	8	E	POP	10	10	6
Bonamia menziesii	9	E	POP	8	8	4
Brighamia rockii	7	E	PEPP	11	10	6
Calamagrostis hillebrandii	4	E	POP	10	10	4
Canavalia molokaiensis	8	E	PEPP	10	10	4
Canavalia pubescens	34	E	POP	10	10	8
Cenchrus agrimonioides var. agrimonioides	2	E	ROI	5	5	2
Clermontia grandiflora ssp. maxima	0		PEPP	10	3	0
Clermontia hanaulaensis	1			10	7	1
Clermontia lindseyana	1	E	POP	5	5	1
Clermontia oblongifolia ssp. brevipes	7	E	POP	10	10	5
Clermontia oblongifolia ssp. mauiensis	28	E	PEPP	10	10	9
Clermontia peleana ssp. singuliflora	0	E	PEPP	3	3	0
Clermontia samuelii ssp. hanaensis	14	E		10	10	8

Appendix 4: Plant Species List

At-Risk Plant Species	Occupied PUs for Species	Federal and State Status	PEPP Species	Desired Recovery Populations	Recovery PUs in MNLCP	Occupied PUs in Recovery Footprint
Clermontia samuelii ssp. samuelii	13	E	PEPP	10	10	9
Colubrina oppositifolia	3	E	ROI	3	3	2
Ctenitis squamigera	12	E	POP	9	10	8
Cyanea asplenifolia	12	E	POP	10	10	10
Cyanea copelandii ssp. haleakalaensis	20	E		10	10	10
Cyanea dunbariae	3	E	PEPP	10	10	2
Cyanea duvalliorum	11	E	PEPP	10	10	7
Cyanea gibsonii	5	E	PEPP	5	6	4
Cyanea glabra	3	E		10	10	3
Cyanea grimesiana ssp. grimesiana	0	E	PEPP	5	5	0
Cyanea hamatiflora ssp. hamatiflora	19	E		10	10	10
Cyanea heluensis	1		PEPP	10	7	1
Cyanea horrida	15	E	POP	10	10	10
Cyanea kauaulaensis	5	E	POP	10	10	5
Cyanea kunthiana	45	E		15	15	14
Cyanea lobata ssp. baldwinii	2	E	PEPP	10	5	1
Cyanea lobata ssp. lobata	13	E	PEPP	10	10	7
Cyanea magnicalyx	14	E	PEPP	10	10	10
Cyanea mannii	9	E		10	10	8
Cyanea maritae	25	E	PEPP	10	10	10
Cyanea mauiensis	2	E		10	10	2
Cyanea mceldowneyi	14	E	POP	10	10	10
Cyanea munroi	1	E	PEPP	10	8	1
Cyanea obtusa	6	E	PEPP	10	10	6
Cyanea procera	5	E	PEPP	10	10	5
Cyanea profuga	3	E	PEPP	10	10	3
Cyanea solanacea	10	E	PEPP	10	10	7
Cyperus fauriei	4	E	ROI	7	7	4
Cyperus neokunthianus	1	E	PEPP	10	10	1
Cyperus pennatiformis var. pennatiformis	3	E	POP	7	7	3

Appendix 4: Plant Species List

At-Risk Plant Species	Occupied PUs for Species	Federal and State Status	PEPP Species	Desired Recovery Populations	Recovery PUs in MNLCP	Occupied PUs in Recovery Footprint
Cyperus prescottianus	1		PEPP	6	6	1
Cyperus trachysanthos	0	E		3	2	0
Cyrtandra ferripilosa	9	E	POP	10	10	7
Cyrtandra filipes	10	E	POP	10	10	7
Cyrtandra hematos	4	E	PEPP	10	10	4
Cyrtandra munroi	11	E	POP	10	10	10
Cyrtandra oxybapha	0	E	POP	10	10	0
Deparia kaalaana	2	E	PEPP	6	6	2
Diplazium molokaiense	9	E	PEPP	10	10	5
Dracaena fernaldii	11	E		10	10	10
Dubautia plantaginea ssp. humilis	3	E	PEPP	10	8	3
Eugenia koolauensis	0	E	PEPP	2	2	0
Exocarpos menziesii	0	E		2	2	0
Festuca hawaiiensis	0	E		3	3	0
Festuca molokaiensis	0	E	PEPP	10	3	0
Flueggea neowawraea	2	E	PEPP	5	5	1
Gardenia brighamii	8	E	PEPP	9	9	3
Gardenia remyi	19	E	PEPP	7	7	5
Geniostoma tinifolium var. lanaiense	2	E	PEPP	5	5	2
Geniostoma triflorum	7	E	PEPP	10	10	7
Geranium arboreum	15	E	PEPP	10	10	8
Geranium hanaense	2	E		10	4	2
Geranium hillebrandii	6	E		10	9	6
Geranium multiflorum	21	E		11	11	11
Gouania hillebrandii	5	E	POP	10	10	5
Gouania vitifolia	0	E	PEPP	3	3	0
Greeneochloa expansa	19	E	ROI	6	6	6
Gynochthodes trimera	0		ROI	3	3	0
Haplostachys haplostachya	0	E		2	2	0
Hesperomannia arborescens	4	E	PEPP	10	10	3

Appendix 4: Plant Species List

At-Risk Plant Species	Occupied PUs for Species	Federal and State Status	PEPP Species	Desired Recovery Populations	Recovery PUs in MNLCP	Occupied PUs in Recovery Footprint
Hibiscadelphus stellatus	2		PEPP	10	5	2
Hibiscus arnottianus ssp. immaculatus	4	E		10	10	4
Hibiscus brackenridgei ssp. brackenridgei	16	E	PEPP	8	8	5
Hibiscus brackenridgei ssp. molokaianus	0	E		10	10	0
Hypolepis hawaiiensis var. mauiensis	4	E	PEPP	10	10	4
Ischaemum byrone	6	E		5	7	5
Isodendrion pyrifolium	0	E	PEPP	4	4	0
Joinvillea ascendens ssp. ascendens	13	E	POP	6	6	4
Kadua cordata ssp. remyi	4	E	PEPP	10	3	3
Kadua coriacea	0	E	POP	1	1	0
Kadua laxiflora	8	E	PEPP	10	10	6
Kanaloa kahoolawensis	1	E	PEPP	8	8	1
Kokia cookei	0	E	PEPP	10	3	0
Lobelia dunbariae ssp. dunbariae	1		PEPP	10	3	1
Lysimachia lydgatei	8	E	PEPP	10	10	7
Lysimachia maxima	4	E	PEPP	10	10	4
Marsilea villosa	3	E		5	5	3
Melicope adscendens	3	E	PEPP	10	10	3
Melicope balloui	4	E	POP	10	10	4
Melicope mucronulata	4	E	PEPP	10	10	4
Melicope multiflora	1			10	10	1
Melicope munroi	5	E	PEPP	10	10	5
Melicope oppenheimeri	1		PEPP	10	8	1
Melicope ovalis	8	E	POP	10	10	6
Melicope reflexa	8	E	POP	10	10	7
Melicope wailauensis	1		PEPP	10	4	1
Menisciopsis boydiae	0	E		8	8	0
Mezoneuron kavaiense	2	E	PEPP	3	3	1
Microlepia setosa var. mauiensis	3	E	ROI	4	4	3
Mucuna persericea	4	E	POP	10	10	3

Appendix 4: Plant Species List

At-Risk Plant Species	Occupied PUs for Species	Federal and State Status	PEPP Species	Desired Recovery Populations	Recovery PUs in MNLCP	Occupied PUs in Recovery Footprint
Myoporum degeneri	0			10	0	0
Myrsine vaccinioides	9	E		12	9	9
Neraudia sericea	15	E	PEPP	10	10	8
Nothocestrum latifolium	19	E	POP	7	10	7
Nototrichium humile	1	E	NO	2	2	0
Ochrosia haleakalae	9	E	NO	5	5	5
Panicum fauriei var. carteri	3	E	NO	7	7	3
Peperomia degeneri	0		PEPP	10	3	0
Peperomia subpetiolata	2	E	PEPP	10	4	2
Peucedanum sandwicense	1	T	ROI	5	6	1
Phlegmariurus mannii	11	E	POP	7	7	5
Phlegmariurus stemmermanniae	1	E	PEPP	5	5	1
Phyllostegia bracteata	18	E	PEPP	10	10	6
Phyllostegia brevidens	3	E	PEPP	5	5	3
Phyllostegia glabra var. glabra	3		PEPP	5	5	2
Phyllostegia glabra var. lanaiensis	0	E	NO	10	3	0
Phyllostegia haliakalae	10	E	PEPP	10	10	6
Phyllostegia hispida	8	E	PEPP	10	10	6
Phyllostegia macrophylla	12		PEPP	6	6	5
Phyllostegia mannii	3	E	PEPP	10	10	2
Phyllostegia parviflora var. parviflora	0	E	PEPP	5	5	0
Phyllostegia pilosa	3	E	PEPP	10	10	3
Phyllostegia stachyoides	22	E	PEPP	7	7	7
Pittosporum halophilum	5	E	PEPP	10	10	3
Plantago princeps var. laxifolia	13	E	PEPP	8	9	6
Platanthera holochila	6	E	PEPP	7	7	2
Portulaca molokiniensis	2		PEPP	10	8	2
Portulaca sclerocarpa	0	E	NO	2	2	0
Portulaca villosa	7	E	ROI	8	8	4
Pritchardia munroi	8	E	PEPP	10	10	6

Appendix 4: Plant Species List

At-Risk Plant Species	Occupied PUs for Species	Federal and State Status	PEPP Species	Desired Recovery Populations	Recovery PUs in MNLCP	Occupied PUs in Recovery Footprint
Pseudognaphalium sandwicensium var. molokaie	5	E	NO	7	7	3
Pteris lidgatei	6	E	PEPP	7	7	5
Ranunculus hawaiensis	0	E	PEPP	3	3	0
Ranunculus mauiensis	10	E	POP	7	7	4
Remya mauiensis	9	E	NO	10	10	8
Sanicula purpurea	4	E	ROI	5	5	4
Sanicula sandwicensis	7	E	POP	5	5	3
Santalum haleakalae var. lanaiense	24	E	NO	12	13	10
Scaevola coriacea	14	E	PEPP	8	8	3
Scaevola hobdyi	1		PEPP	10	4	1
Schenkia sebaeoides	6	E	NO	7	7	6
Schiedea diffusa ssp. diffusa	12	E	PEPP	8	8	5
Schiedea haleakalensis	2	E	PEPP	10	9	2
Schiedea hookeri	0	E	PEPP	2	2	0
Schiedea jacobii	4	E	PEPP	10	8	3
Schiedea laui	4	E	PEPP	10	6	3
Schiedea lydgatei	5	E	NO	10	10	4
Schiedea nuttallii	0	E	PEPP	4	4	0
Schiedea pubescens	18	E	PEPP	10	10	6
Schiedea salicaria	3	E	NO	10	10	3
Schiedea sarmentosa	3	E	NO	10	8	3
Sesbania tomentosa	15	E	ROI	9	9	6
Sicyos hillebrandii	1		PEPP	10	4	0
Sicyos macrophyllus	0	E	POP	2	2	0
Silene alexandri	2	E	PEPP	10	8	2
Silene lanceolata	2	E	NO	4	4	2
Solanum incompletum	0	E	PEPP	4	4	0
Solanum nelsonii	4	E	ROI	4	4	3
Spermolepis hawaiiensis	11	E	ROI	9	9	7
Stenogyne angustifolia	0	E	NO	3	3	0

Appendix 4: Plant Species List

At-Risk Plant Species	Occupied PUs for Species	Federal and State Status	PEPP Species	Desired Recovery Populations	Recovery PUs in MNLCP	Occupied PUs in Recovery Footprint
Stenogyne bifida	4	E	PEPP	10	10	4
Stenogyne calycosa	9		PEPP	10	10	5
Stenogyne kauaulaensis	4	E	PEPP	10	10	2
Tetramolopium arenarium ssp. arenarium	0	E	NO	3	3	0
Tetramolopium capillare	3	E	PEPP	10	3	3
Tetramolopium lepidotum ssp. lepidotum	1	E	PEPP	3	3	0
Tetramolopium remyi	2	E	PEPP	10	10	1
Tetramolopium rockii var. calcisabulorum	2	T	NO	5	10	2
Tetramolopium rockii var. rockii	2	T	NO	5	10	1
Tetramolopium sp. nov. 2 (Mauna Kahalawai)	2		PEPP	10	3	2
Vigna owahuensis	4	E	ROI	8	11	4
Viola lanaiensis	6	E	ROI	10	10	6
Wikstroemia villosa	11	E	POP	10	10	8
Wollastonia kamolensis	1	E	PEPP	10	5	1
Wollastonia populifolia	0		PEPP	10	0	0
Zanthoxylum dipetalum var. dipetalum	1		ROI	4	4	1
Zanthoxylum hawaiiense	5	E	ROI	8	8	5
TOTAL (197 species)			197			
Endangered		170				
Threatened		4				
PEPP Plant Extinction Prevention Program species			102			
POP species with few populations			30			
ROI species that are rare on an island			19			
Species with insufficient data to ID Recovery Areas					2	

Appendix 5: The Adaptive Conservation Planner and other Tools

Introduction: Only the Conservation Planner and Find My Habitat are currently available. The other tools will be released in the near future.

- Conservation Planner (Section 1)
- Find My Habitat (Section 2)
- Potential Recovery Sites (TBA)
- Climate Vectors (TBA)
- Footprint Summarizer (TBA)

1. The Adaptive Conservation Planner (Conservation Planner)

The Adaptive Conservation Planner (Conservation Planner) is the TFE-SCP platform that carries conservation footprints (called Recovery Footprints in the MNLCP) from draft prioritization outputs into expert fine-tuning and then into long-term plan stewardship. It is not a public web map or implementation-task tracker; it is the internal, editable interface where DOFAW staff, taxon experts, and managers can inspect, refine, and maintain conservation footprints as living products.

What it does

The Conservation Planner) allows authorized users to:

- View conservation footprints for any of the seven taxonomic groups, individually or overlaid
- Select a planning unit and see which species are targeted there, their habitat suitability scores, occurrence evidence, and whether targets are met
- Propose edits: relocate species targets to different planning units, add new target locations, delete target assignments, or modify preventing-extinction status
- Attach notes to specific taxa or planning units, documenting expert observations, access concerns, land-tenure information, or future revision needs
- Save draft edits privately, publish proposed edits for team review, and submit edits for admin approval
- Export footprint data in multiple formats (CSV, GeoPackage, GeoJSON) for use in GIS or other planning systems
- View nested footprint products (recovery, preventing extinction, implementation subsets) within a single interface

How it supports the "living plan" concept

- A core argument of TFE-SCP is that conservation plans should not be static documents that become outdated. The Conservation Planner) operationalizes this by maintaining both:

- A versioned official footprint - a stable, dated snapshot for reporting, handoff, and formal implementation reference (e.g., the 2026 official footprint)
- A living platform - where ongoing edits, annual reviews, new population discoveries, and management-condition updates can be incorporated without losing the audit trail of how and why changes were made

This dual structure means DOFAW can deliver a stable product for planning while also keeping the footprint current as knowledge evolves. Future annual reviews can alter specific planning units, and the change history remains traceable.

Current status

The Conservation Planner has been used operationally for plant recovery footprint fine-tuning (the most mature use case), snail exclosure review, and arthropod target review. It supports concurrent management of multiple taxonomic-group footprints and has been tested through full edit-review-approve cycles.

Planned improvements include better occurrence-data display for some groups, conflict detection between recovery and preventing-extinction edits, custom species-specific overlays, clearer save/publish behavior, and documentation for deployment beyond Maui Nui.

Relationship to other systems

The preferred long-term architecture envisions the Conservation Planner handling specialized TFE-SCP footprint logic (species targets, criteria, notes, footprint versions), while DOFAW's Esri environment handles broader implementation tracking (field surveys, work orders, fence status, effort records). The Conservation Planner is not intended to replace Esri-based field systems but to complement them with the species-level, multi-criteria footprint logic that standard GIS platforms do not natively support.

- The Conservation Planner will be described in more detail in the Managers Handbook being developed for DOFAW staff.

Access

- https://icculus.shinyapps.io/MauiNui-ConservationFootprintManager/

2. Find My Microhabitat Tool

Find My Microhabitat is a TFE-SCP support tool that bridges the gap between planning-unit-scale conservation footprints and implementation-scale site selection. Conservation footprints identify priority planning units - landscape-level areas where management should be concentrated - but field managers need to identify specific microsites within those planning units where restoration, outplanting, predator control, or other actions will be most effective.

What it does

Find My Microhabitat helps managers and field biologists zoom in from a planning-unit selection to the actual places on the ground where species recovery actions should occur. Within a selected planning unit, the tool integrates high-resolution spatial data - including vegetation structure, canopy cover, topographic position, moisture gradients, and other locally relevant habitat variables - to identify the most suitable microsites for target species.

This is particularly useful for

- **Outplanting decisions:** identifying specific locations within a footprint planning unit where rare plant reintroductions are most likely to succeed
- **Exclosure siting:** refining candidate kipuka kahuli locations within broader priority areas
- **Survey targeting:** directing field surveys to the most promising microsites for data-deficient species
- **Restoration planning:** identifying where habitat restoration actions will yield the greatest benefit for target species

Why it matters

Conservation footprints operate at a landscape scale - they answer the question "where should we focus?" But implementation happens at a much finer scale. A planning unit might be hundreds of hectares, while an outplanting site, survey transect, or exclosure is measured in fractions of a hectare. Without a bridge between these scales, managers are left to make fine-scale siting decisions without the spatial data support that informed the broader prioritization.

- **Find My Microhabitat fills this gap.** It takes the "where" from the conservation footprint and adds "exactly where within that area" using the best available fine-resolution data.

Current status

The tool is already being used by Plant Extinction Prevention Program (PEPP) partners for outplanting site selection within plant conservation footprint planning units. It is functional but needs documentation and generalization for broader

deployment. The current priority is to document the tool so it can be used by additional DOFAW staff and partners without requiring developer support for each application.

3. **Potential Recovery Site (PRS)**
4. **Climate Vectors**
5. **Footprint Summarizer**

Appendix 6: At-Risk Snail Species

Terrestrial Snails	Current Populations	ESA Listing	Facing Imminent Extinction	Recovery		Insufficient Data	
				Interim Recovery	Full Recovery	Area Surveys	Species Surveys
Amastra montana	1		X	3			
Auriculella lanaiensis	2		X	3			
Laminella aspera	0		X	3			
Laminella citrina	1		X	3			
Laminella venusta	5		X	3			
Leptachatina nitida	1		X	3			
Leptachatina vitreola	4		X	3			
Lyropupa striatula	2		X	3			
Newcombia canaliculata	1		X	3			
Newcombia cumingi	1	E	X	3			
Newcombia cumingi sp. 1	1		X	3			
Newcombia pfeifferi	6		X	3			
Partulina anceyana	3		X	3			
Partulina crocea	0		X	3			
Partulina dolei	2		X	3			
Partulina induta	1		X	3			
Partulina kaaeana	1		X	3			
Partulina marmorata	1		X	3			
Partulina mighelsiana bella	5-10		X	3			
Partulina mighelsiana mighelsiana	5-10		X	3			
Partulina mighelsiana polita	5-10		X	3			
Partulina mighelsiana ssp.	5-10		X	3			
Partulina perdix ssp.	1		X	3			
Partulina perdix undosa	2		X	3			
Partulina porcellana ssp.	1		X	3			
Partulina porcellana wailuaensis	3		X	3			
Partulina proxima schauinslandi	3		X	3			
Partulina proxima ssp.	1		X	3			
Partulina redfieldi	15+		X	3			
Partulina semicarinata	3	E	X	3			
Partulina splendida	4		X	3			
Partulina tappaniana ampulla	1		X	3			
Partulina tappaniana dubiosa	1		X	3			
Partulina tappaniana eburnea	1		X	3			
Partulina terebra lignaria	2		X	3			
Partulina terebra ssp.	1		X	3			
Partulina tessellata	5		X	3			
Partulina undosa	2		X	3			
Partulina ustulata	1		X	3			
Partulina variabilis	1	E	X	3			

Terrestrial Snails	Current Populations	ESA Listing	Facing Imminent Extinction	Recovery		Insufficient Data	
				Interim Recovery	Full Recovery	Area Surveys	Species Surveys
Partulina virgulata	5-10		X	3			
Perdicella helena balteata	5-10		X	3			
Perdicella helena ssp.	5-10		X	3			
Perdicella kuhnsi	5-10		X	3			
Perdicella ornata	5-10		X	3			
Pleuropoma laciniosa piliformis	1		X	3			
Pleuropoma laciniosa ssp. 1	7		X	3			
Pleuropoma laciniosa ssp. 2	2		X	3			
Total (49 species)							
Endangered		3					
Facing Imminent Extinction			47				
Interim Recovery				49			
Full Recovery					0		

Appendix 7: Terrestrial Arthropod List

Terrestrial Arthropods	All Known Occupied PUs	ESA Listing	Facing Imminent Extinction	Recovery		Insufficient Data	
				Interim Recovery	Full Recovery	Area Surveys	Species Surveys
Acrolepia beardsleyi						X	
Agrotis cremata							X
Blackburnia deroderus							X
Blackburnia gracilis							X
Blackburnia lenta							X
Caconemobius howarthi							X
Cerotrioza bivittata						X	
Chersodromia hawaiiensis							X
Coleotichus blackburniae						X	
Colobopyga pritchardiae						X	
Crawforda triopsyllina						X	
Dictyophorodelphax praedicta						X	
Dictyophorodelphax usingeri						X	
Drosophila affinidisjuncta	8			5			
Drosophila balioptera	13			5			
Drosophila bostrycha	8			5			
Drosophila cilifera	11			5			
Drosophila clavisetae	6			3			
Drosophila differens	4	E				X	
Drosophila disjuncta	13			5			
Drosophila grimshawi	39			5			
Drosophila gymnobasis	2					X	
Drosophila hamifera	4					X	
Drosophila hirtipalpus	6						X
Drosophila lanaiensis	13			5			
Drosophila lasiopoda	3						X
Drosophila limitata	21			5			
Drosophila liophallus	6			5			
Drosophila nalowale	0					X	

Terrestrial Arthropods	All Known Occupied PUs	ESA Listing	Facing Imminent Extinction	Recovery		Insufficient Data	
				Interim Recovery	Full Recovery	Area Surveys	Species Surveys
Drosophila neoclavisetae	4	E				X	
Drosophila nukea	1					X	
Drosophila obscuripes	4					X	
Drosophila odontophallus	11			5			
Drosophila oreas	3					X	
Drosophila orphnopeza	18			5			
Drosophila orthofascia	14			5			
Drosophila paenehamifera	4					X	
Drosophila peniculipedis	6			5			
Drosophila pihulu	4			5			
Drosophila pilatisetae							X
Drosophila planitibia	8			5			
Drosophila prostopalpis	2						X
Drosophila sodomae	6			5			
Drosophila spectabilis	10			5			
Drosophila toxochaeta	3					X	
Drosophila truncipenna	4			5			
Drosophila varipennis	3					X	
Drosophila vesciseta	5			5			
Drosophila villitibia	5						X
Ectemnius haleakalae							X
Eidoleon perjurus							X
Eopenthes arduus							X
Eopenthes auratus							X
Eopenthes celatus							X
Eopenthes deceptor							X
Eopenthes plebeius							X
Eopenthes politus							X
Euodynerus nigripennis							X
Genophantis leahi						X	

Terrestrial Arthropods	All Known Occupied PUs	ESA Listing	Facing Imminent Extinction	Recovery		Insufficient Data	
				Interim Recovery	Full Recovery	Area Surveys	Species Surveys
Glyphodes cyanomichla						X	
Glyphodes exaula						X	
Gonitis vulpicolor						X	
Helicoverpa confusa							X
Hylaeus angustulus	3			5			
Hylaeus anthracinus	15	E	X	5			
Hylaeus assimulans	17	E		5			
Hylaeus coniceps	6			5			
Hylaeus connectens	6			5			
Hylaeus facilis	1	E	X			X	
Hylaeus flavipes	4		X	5			
Hylaeus fuscipennis	10			5			
Hylaeus haleakalae	6			5			
Hylaeus hilaris	1	E	X			X	
Hylaeus kukui	1					X	
Hylaeus laetus	7		X	5			
Hylaeus longiceps	7	E	X	5			
Hylaeus mauiensis	0						X
Hylaeus melanothrix	0						X
Hylaeus niloticus							X
Hylaeus paumako	1		X			X	
Hylaeus psammobius	3		X			X	
Hylaeus satelles	0						X
Hylaeus specularis	1					X	
Hylaeus unicus	10			5			
Hylaeus volatilis	1		X			X	
Ithamar annectans						X	
Ithamar hawaiiensis						X	
Kalania hawaiiensis							X
Kirkaldiella euphorbiae						X	

Terrestrial Arthropods	All Known Occupied PUs	ESA Listing	Facing Imminent Extinction	Recovery		Insufficient Data	
				Interim Recovery	Full Recovery	Area Surveys	Species Surveys
Leptogryllus deceptor							X
Manduca blackburni	15	E			6		
Megalagrion jugorum						X	
Megalagrion molokaiense						X	
Megatrioza mauiensis						X	
Megatrioza molokaiensis						X	
Micromus cookeorum							X
Micromus haleakalae							X
Micromus lobipennis							X
Micromus vulcanius							X
Neomachilis heteropus							X
Neseis haleakalae							X
Nesocryptias villosa							X
Nesosydne acuta						X	
Nesosydne argyroxiphii						X	
Nesosydne bridwelli						X	
Nesosydne cyrtandrae						X	
Nesosydne geranii						X	
Nesosydne leahi						X	
Nesosydne longipes						X	
Nesosydne sulcata						X	
Nesotocus munroi						X	
Nesotocus spp.						X	
Oechalia grisea							X
Oliarus hevaheva							X
Oliarus priola							X
Omiodes anastrepta						X	
Omiodes asaphombra						X	
Omiodes monogona						X	
Omiodes musicola							X

Terrestrial Arthropods	All Known Occupied PUs	ESA Listing	Facing Imminent Extinction	Recovery		Insufficient Data	
				Interim Recovery	Full Recovery	Area Surveys	Species Surveys
Orthotylus diospyricola						X	
Orthotylus diospyrivorus						X	
Orthotylus diospyroides						X	
Orthotylus ureraphila						X	
Orthotylus xylosmoides						X	
Pentarthrum obscurum							X
Philodoria hesperomanniella						X	
Philodoria hesperomanniella sp. n						X	
Philodoria keahii						X	
Philodoria lipochaetaella						X	
Philodoria platyphylliella						X	
Philodoria platyphylliella sp. n						X	
Philodoria sp.							X
Philodoria sp. 2							X
Philodoria spilota						X	
Philodoria touchardiella						X	
Philodoria wilkesiella						X	
Phyllococcus oahuensis							X
Plagithmysus							X
Plagithmysus alani						X	
Plagithmysus bidensae						X	
Plagithmysus dubautianus						X	
Plagithmysus forbesii						X	
Plagithmysus fractus							X
Plagithmysus fugitivus							X
Plagithmysus geranii						X	
Plagithmysus koae						X	
Plagithmysus lanaiensis						X	
Plagithmysus longicollis							X
Plagithmysus pipturicola						X	

Terrestrial Arthropods	All Known Occupied PUs	ESA Listing	Facing Imminent Extinction	Recovery		Insufficient Data	
				Interim Recovery	Full Recovery	Area Surveys	Species Surveys
Plagithmysus pulvillatus						X	
Plagithmysus rubi						X	
Plagithmysus simillimus						X	
Plagithmysus speculifer							X
Plagithmysus swezeyanus						X	
Plagithmysus sylvai						X	
Proterhinus bridwelli							X
Proterhinus fuscicolor							X
Proterhinus sp.							X
Rhynchephestia rhabdotis						X	
Rhyncogonus lahainae						X	
Rhyncogonus molokaiensis							X
Rhyncogonus sharpi							X
Rhyncogonus simplex							X
Rynchogonus montygorum						X	
Rynchogonus sordidus							X
Rynchogonus sp.							X
Rynchogonus vestitus						X	
Rynchogonus zeta							X
Saicella smithi							X
Spheterista oheoheana						X	
Thyrocopa apatela							X
Udea dryadopa						X	
Total (172)							
Endangered		8					
Facing Imminent Extinction			9				
Recovery Goal					1		
Interim Recovery Goal				31			
Focal Area Surveys						82	
Basic Species-specific Surveys							58

Appendix 8: Conservation and Restoration of Native Terrestrial Ecosystems

Reserved for the final version.

Appendix 9: Monitoring and Evaluation of Management Actions

A. Developing a Monitoring Design to Take Samples from a Population or Ecosystem

When a population or ecosystem is too large and resource-intensive to completely count or measure its entire spatial extent, then taking a sample is an appropriate and well-documented approach (Elzinga et al. 2001, Jean et al. 2005). A sample is a subset of individuals or other features selected from the target population that is used to draw conclusions about the larger group. In many cases, taking a sample is more efficient and accurate; taking data from a limited number of locations to estimate the whole.

The concept of sampling a population has a range of terms and concepts are important in its discussion and implementation.

Definition of Terms

Terms related to the physical units and space being monitored.

- ***Target Population*** is the set of all individuals in a population or all features of an ecosystem for which sampling is designed to make an inference (defined below).
- ***Population Unit or Observational Unit*** refers to the individual entity being measured or counted within a specific population, such as a plant, bird, snail, nest, burrow, or for an ecosystem: composition, structure, and species richness. It is the basic building block for calculating population totals, densities, or for conducting statistical surveys.
- ***Sample Units*** are the individual units within population that are sampled. Examples: individuals, plots, burrows.

Terms related to the statistical aspects of monitoring.

- ***Inference*** is a conclusion derived from the sampled information about the target population. Those conclusions are usually with calculated degrees of certainty (precision).
- ***Sample*** is the selected (measured) subset of the total possible number of sample units in the target population at one point in time.
- ***Sample Statistic*** is a specific numerical value calculated from a sample. It serves as an estimate to make inferences about the larger, often unmeasurable population. The sample statistic includes values such as the mean, standard deviation, medium, and range.

- ***Sampling Error*** is the deviation between a sample statistic and the target population parameter, caused by observing only a subset of the population rather than the whole. It is a natural characteristic of sampling data and does not represent mistakes (although mistakes can be made!). It is reduced by increasing sample size (improving representation) and adjusting the sampling design (improving how the population is sampled to reduce variability in the data).
- ***Precision*** refers to the spread of data points, indicating how close measurements in one sample are to one another. A precise set of measurements has a small standard deviation and low variability, indicating high repeatability. It is distinct from accuracy, which measures closeness to the true value
- ***Power*** is the risk of failing to detect a true change in the target population. A goal of a sampling design is to maximize the ability of detecting a change.
- ***Covariates*** are measurements (attributes) that are not the primary focus of monitoring but can influence the dependent (primary) attribute. These could include habitat factors such as cover of invasive species and indications of soil disturbance and climate factors such as precipitation and temperatures. It is an independent variable that "covaries" with the outcome and is often used to control for confounding factors, reduce error variance, and improve the precision of statistical models, and understand change.

Considerations in the Development of a Sampling Design

There are complex interrelated decisions that need to be made for developing a sampling design. A sampling design is the approach used to select the sample to make inferences about the target population. Its purpose is to ensure that the data collected can be used to detect the status of the target population and/or changes in the target population. There is no best or standardized sampling design, they are developed to fit specific monitoring objectives and the ecological context of the species and ecosystem. As Kenkel et al (1989) stated: "…the decision-making process in sampling must be viewed as a flexible exercise, dictated not by generalized recommendations but by specific objectives; there is no panacea in ecological sampling".

Sampling design decisions can be grouped into seven questions.

- What is the target population?
- What is the appropriate sample unit?
- How will the sample units be spatially allocated (positioned) within the target population?
- How many sample units are needed to detect a desired change?
- Will the sample units be permanent or non-permanent?

- When will the samples be temporal allocated?
- How will the data be analyzed?

The complexity and interrelatedness of these questions require that a sampling design be developed with a team that has expertise in the species, the management actions, and sampling design.

1. ***What is the target population?***

The target population is the set of all individuals in a population or all features of an ecosystem for which sampling is designed to make an inference. Often this is equivalent to a biological population (the group of individuals of the same species occurring in an area in which they can interbreed) or a natural community of a native ecosystem (assemblage of plants, animals, and other organisms that occur together in the same area and under similar environmental conditions). In some situations, the target population is a subset of the biological population or native ecosystem, defined by a geographic area such as the individuals of a species nesting on a cliff face.

Circumscribing the target population is essential for sampling and reporting results.

2. ***What is the appropriate sample unit?***

The type of sampling unit used for monitoring depends primarily on the attribute that is being measured (Elzinga et al. 2001). There are several types of sampling units.

- Individual plants or animals
- Quadrats (plots)
- Transects (lines)
- Points
- Plotless (distance)

The first decision in the selection of the sample unit is whether data is most efficiently collected individual by individual or by individuals in a specified area. Each individual of the species is measured for occurrence and condition (for plants: height, flowers, seeds, damage; for snails: size). The attributes of density, cover, and frequency, if a selected attribute, are measured by a dimensional sample unit, such as plots or transects.

Once the dimensional sampling unit is selected then the size and shape of the sampling unit need to be considered. A commonly used sample unit is the one-meter square plot. They are easily established and easy to sample, which allow many one-meter square plots to be set up in a target population. They work best in populations that are randomly or regularly distributed. If a population or ecosystem has high variability in density or species

composition or includes an ecotonal gradient, then a larger-square or longer-rectangular plots results in more precise data. That is due to the amount of variability in the population that is captured by within each sample unit versus between them.

Elzinga et al. (2001) lists the considerations for selecting the most appropriate size and shape sampling unit. They include:

- Set up time versus measuring time
- Size of individuals
- Abundance of target population
- Spatial distribution of the individuals in the population
- Edge Effects
- Ease in sampling
- Disturbance caused by monitoring

In many cases the selection of the sample unit and its size and shape is clear, but at times a more systematic thoughtful assessment is needed. An excellent overview of this can be found in Elzinga et al. 2001.

3. ***How will the sample units be spatially allocated (positioned) with the target population?***

There are three requirements for sampling units to obtain the best estimates of the true population. The sampling units need to be:

- Randomly (unbiased) placed.
- Well interspersed: sample units are positioned throughout the area to be sampled and across the natural variability found in the population or ecosystem.
- Independent: sample units are far enough apart so that measurements are not spatially correlated.

Probability-based sampling ensures that the sample is representative of the population of interest and that conclusions of the larger population are valid. It also reduces the bias in the positioning of sample units.

There are several common spatial designs that incorporate random sampling

- Simple Random Sampling
- Systematic Sampling
- Stratified Random Sampling
- Cluster Sampling
- Two Stage Sampling

The most appropriate design is determined by the spatial distribution of the population or observational unit, accessibility within the population of interest, sensitivity of the target species to effects of sampling (trampling), and how these relate to the precision of the data.

See Jean (2005) and Elzinga et al. (2001) for detailed descriptions of each of these spatial designs.

4. How many sample units are needed to detect a desired change?

Detecting a meaningful change in the population or ecosystem in relationship to human mediated management is the goal of a sampling design. The difficulties in detecting change arise from the natural variability in the population and ecosystems and the dynamic nature of that variability over time.

The number of sample units (sample size) is the most direct method of increasing the precision of the data and thus the ability of detect change. In general, precision increases with larger sample sizes. More samples capture more of the variability in the population. But this is not proportional, as the sample size increases beyond 20 or 30, the increase in precision diminishes (Elzinga et al. 2001). Precision is also increased when the variability within the population is captured within the sample units versus across the sample units. This reduces the standard deviation of the data – the data is more similar across sample units.

There is an obvious circularity to determining the sample size to detect an identified level of change. Sample size should be based on the amount of variability in the actual data. Yet that variability is not known until data is collected. One can do a year of pilot sampling to measure the variability or adaptively modify the sampling design during the first few years of sampling to maximize the precision of the data.

The relationship of sample size and precision is illuminated by sample size calculations. There are numerous sample size calculators available with different sample size calculations for different sampling objectives (e.g. estimating a value at one point in time, detecting differences between two values using either permanent or temporary plots, and detecting differences between two values using either permanent or temporary plots). For information on calculating sample sizes themselves, equations are provided in Elzinga et al. 2001.

5. Will the sample units be permanent or temporary?

Another sampling design decision is determining whether the sampling units are permanent over time or temporary being reestablished at each sampling event. The default

usually is permanent sampling units, but a consideration of which to use is valuable as a sampling design is developed.

The advantages of permanent sample units are that they reduce variability across sampling events, especially true for species that remain or come back to the same location each year (perennial plants, nesting seabirds). Reduced variability results in increased statistical power, thus fewer sample units to obtain a desired level of precision and detect an important change

The disadvantages of permanent sample units are the initial marking and annual relocating of sample units - which can be difficult and time consuming, their loss from environmental or organismal damage, the impact of sampling in the same location from sampling period to sampling period, and that over time the sample units become less representative of the population as the population changes in distribution, density, and/or condition. But even with these disadvantages, permanent plots are used extensively to estimate changes in populations and ecosystems.

6. ***When will the sampling be temporally allocated?***

The scheduling of the annual or seasonal sampling of a species or ecosystem should be when identification and detectability is easiest. This includes both the time of year and time of day.

7. ***How will the data be analyzed?***

The development of a sampling method includes selecting the statistical tests for analyzing the data. Data analysis develops sample statistics for each attribute and assesses the precision of the data and its power to detect change in the target population. Covariates are included in the analysis to help explain patterns and trends in the data. Developing an analysis plan makes one think ahead about the output of the monitoring data; how it is analyzed and the format of the results.

B. Components of a Monitoring Protocol

Monitoring protocols provide written guidance on how to implement the monitoring of a population, habitat, or ecosystem.

The primary purpose of a monitoring protocol is to ensure that the monitoring is done consistently over time and among different field staff, ensuring the value of the data collected.

The following is a list of the components for a monitoring protocol.

- **Purpose, Goals, Objectives**
    - Description of the species or ecosystem
        - Guide for identification and differentiation from related species or ecosystems, including detectability
    - Summary of status and priority
    - Location(s) and directions to the population(s) or ecosystem being managed and monitored
    - Ecology of the species and the habitat it occurs in or the ecosystem and its variability across environmental gradients
        - In text form with or without a conceptual model (See box in appendix for discussion of conceptual models)
    - Description and impact of threats and disturbances
    - Conservation and management concerns
- **Permits, Authorizations, Permissions, and Qualifications**
- **Management Goals and Monitoring Objectives**
    - Management Goal
    - Monitoring Objective(s)
    - Attributes to be Measured for each Monitoring Objective
- **Monitoring Design**
    - Monitoring or Sampling Methodology
        - Instructions on how to sample:
            - Species and/or ecosystem parameters
            - Covariate parameters
        - Personnel and equipment need
    - Monitoring Schedule
        - Based on identification (life history), access, other priority species in PU
    - Surveys in Adjacent and Travel Areas

- **Data Management**
    - Data Acquisition Quality
        - Tabular data
        - Spatial data
        - Photographic data
        - Maintenance and calibration of equipment
    - Data Verification and Validation
        - Verification: Process to assess that all required data have been collected and recorded
        - Validation: Process to review the validity of the data (see next section)
    - Version Control of Data
    - Archiving Data
    - Data Distribution and Publication
- **Roles and Responsibilities**
    - Maintaining the Monitoring Protocol
    - Implementing the Monitoring Protocol
- **Document History**
    - Development of Protocol
    - Recommendations for Modifications and Lessons Learned
        - Table by date and submitter
    - Review, approval, and updates
        - See example below

Revision Date and Version	Author	Changes Made	Reason for Change	Review	Approval

C. Stages of Data Management

The following is a brief overview of the stages of data management. More detail can be found in Sutter et al. 2015 and EPA 2019.

- Plan
    - Start designing the data management process at the initiation of a monitoring project
- Prepare
    - Design the data management infrastructure
        - Procedures and training for data collection and data quality
        - Roles and Responsibilities
        - Workspace and File Management
- Collect
    - Develop Data Capture System
    - Calibration and Quality Control for Instruments and Equipment
- Manage
    - Compilation of data: how, where, who
- Assuring Data Quality – Key Step in the process
    - During data collection
        - Review as collected, remeasurement, debriefing
    - After data collection: 3 steps to ensure data quality review
        - Verification – Assesses completeness
        - Validation – Assesses content and that the data make scientific sense (data outside of realistic realm - range checks, logical consistency)
        - Certification – Ensures the data ready for use, analysis, and sharing
- Describing Data
    - Metadata describes what a data set user needs to successfully use the data set, including
        - What the data include
        - Why data was collected
        - How data was collected
        - Data quality checks
        - Who to contact for questions
- Analysis and Evaluation of Monitoring
    - Know how the data will be analyzed before collection
    - Evaluate the analyzed data and how it will modify and improve management
- Preserve
    - Archiving where and who

    - Access policies: use of data and legal confidentiality
- Share
    - Report to selected audiences with appropriate messaging and communication vehicles
- Evaluate Data Management
    - Review the data management process